



**GREEN  
CLIMATE  
FUND**

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15 – 18 July 2024

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Provisional agenda item 10

**GCF/B.39/02/Add.16/Rev.01**

**15 July 2024**

# Consideration of funding proposals – Addendum XVI

## Funding proposal package for FP241

### Summary

This addendum contains the following six parts:

- a) A funding proposal summary titled “Financing Mitigation and Adaptation Projects (FMAP) in Indian MSMEs” submitted by Small Industries Development Bank of India (SIDBI);
- b) No-objection letter(s) issued by the national designated authority(ies) or focal point(s);
- c) Environmental and Social report(s) disclosure;
- d) Independent Technical Advisory Panel’s assessment;
- e) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- f) Gender documentation of the funding proposal.

*These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled “Financing Mitigation and Adaptation Projects (FMAP) in Indian MSMEs” submitted by Small Industries Development Bank of India (SIDBI) is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.*

### **Disclaimer:**

***The boundaries and names shown and the designations used including maps, references and citations in this Funding Proposal do not imply any official endorsement or acceptance by the Green Climate Fund. The final status of Jammu and Kashmir is yet to be decided.***

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# Funding Proposal

Project/Programme title: *Financing Mitigation and Adaptation Projects (FMAP) in Indian MSMEs*

Country(ies): *India*

Accredited Entity: *Small Industries Development Bank of India*

Date of first submission: *[2024/03/18]*

Date of current submission *[2024/06/22]*

Version number *[V.12]*



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### *NOTE TO ACCREDITED ENTITIES ON THE USE OF THE FUNDING PROPOSAL TEMPLATE*

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

**Please submit the completed proposal to:**

[fundingproposal@gcfund.org](mailto:fundingproposal@gcfund.org)

**Please use the following name convention for the file name:**

“FP-[SIDBI]-[India/South Asia]-[2024/05/01]”

## PROJECT/PROGRAMME SUMMARY

|                                                            |                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                              |                                                                                                                     |                                                                                        |
|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>A.1. Project or programme</b>                           | Programme                                                                                                                                                                                                                                                                                                                                                             | <b>A.2. Public or private sector</b>                                                                                                                                                                                         | Private                                                                                                             |                                                                                        |
| <b>A.3. Request for Proposals (RFP)</b>                    | <p>If the funding proposal is being submitted in response to a specific GCF <a href="#">Request for Proposals</a>, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.</p> <p><u>Not applicable</u></p>                                                                               |                                                                                                                                                                                                                              |                                                                                                                     |                                                                                        |
| <b>A.4. Result area(s)</b>                                 | <p>Check the applicable <a href="#">GCF result area(s)</a> that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of <b>GCF and Co-financers' contribution</b> devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.</p> |                                                                                                                                                                                                                              |                                                                                                                     |                                                                                        |
|                                                            |                                                                                                                                                                                                                                                                                                                                                                       | <b>GCF contribution</b>                                                                                                                                                                                                      | <b>Co-financers' contribution<sup>1</sup></b>                                                                       |                                                                                        |
|                                                            | <b>Mitigation total</b>                                                                                                                                                                                                                                                                                                                                               | 60 %                                                                                                                                                                                                                         | 60 %                                                                                                                |                                                                                        |
|                                                            | <input checked="" type="checkbox"/> Energy generation and access                                                                                                                                                                                                                                                                                                      | 25 %                                                                                                                                                                                                                         | 25 %                                                                                                                |                                                                                        |
|                                                            | <input checked="" type="checkbox"/> Low-emission transport                                                                                                                                                                                                                                                                                                            | 25 %                                                                                                                                                                                                                         | 25 %                                                                                                                |                                                                                        |
|                                                            | <input checked="" type="checkbox"/> Buildings, cities, industries and appliances                                                                                                                                                                                                                                                                                      | 10 %                                                                                                                                                                                                                         | 10 %                                                                                                                |                                                                                        |
|                                                            | <input type="checkbox"/> Forestry and land use                                                                                                                                                                                                                                                                                                                        | <u>Enter number</u> %                                                                                                                                                                                                        | <u>Enter number</u> %                                                                                               |                                                                                        |
|                                                            | <b>Adaptation total</b>                                                                                                                                                                                                                                                                                                                                               | 40 %                                                                                                                                                                                                                         | 40 %                                                                                                                |                                                                                        |
|                                                            | <input checked="" type="checkbox"/> Most vulnerable people and communities                                                                                                                                                                                                                                                                                            | 15 %                                                                                                                                                                                                                         | 15 %                                                                                                                |                                                                                        |
|                                                            | <input checked="" type="checkbox"/> Health and well-being, and food and water security                                                                                                                                                                                                                                                                                | 15 %                                                                                                                                                                                                                         | 15 %                                                                                                                |                                                                                        |
|                                                            | <input checked="" type="checkbox"/> Infrastructure and built environment                                                                                                                                                                                                                                                                                              | 10 %                                                                                                                                                                                                                         | 10 %                                                                                                                |                                                                                        |
|                                                            | <input type="checkbox"/> Ecosystems and ecosystem services                                                                                                                                                                                                                                                                                                            | <u>Enter number</u> %                                                                                                                                                                                                        | <u>Enter number</u> %                                                                                               |                                                                                        |
|                                                            | <b>A.5. Expected mitigation outcome</b><br><br><i>(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)</i>                                                                                                                                                                                                                                     | <p>Indicate climate house gas (GHG) emission reductions or removals in tCO<sub>2</sub>eq over total lifespan of the project/programme<sup>2</sup></p> <p>35.34 million tCO<sub>2</sub> eq.<br/>(Over equipment lifetime)</p> | <b>A.6. Expected adaptation outcome</b><br><br><i>(Core indicator 2: direct and indirect beneficiaries reached)</i> | <p>Indicate total number of direct and indirect beneficiaries</p> <p>10.80 million</p> |
|                                                            | <p>Indicate number of direct beneficiaries</p> <p>0</p>                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                              |                                                                                                                     | <p>Indicate number of indirect beneficiaries</p> <p>10.80 million</p>                  |
|                                                            | <p>Indicate % of direct beneficiaries vis-à-vis total population</p> <p>0%</p>                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                              |                                                                                                                     | <p>Indicate % of indirect beneficiaries vis-à-vis total population</p> <p>0.0076%</p>  |
| <b>A.7. Total financing (GCF + co-finance<sup>3</sup>)</b> | 3792.6 million USD (including sub-participation)                                                                                                                                                                                                                                                                                                                      | <b>A.9. Project size</b>                                                                                                                                                                                                     | Large (Over USD 250 million)                                                                                        |                                                                                        |
| <b>A.8. Total GCF funding requested</b>                    | 215.6 million USD<br><i>For multi-country proposals, please fill out annex 17.</i>                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                              |                                                                                                                     |                                                                                        |

|                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                         |                                                                                                                                                                                           |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A.10. Financial instrument(s) requested for the GCF funding</b>       | <p>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</p> <div> <input checked="" type="checkbox"/> Grant      USD 15.6 million         <input type="checkbox"/> Equity      <u>Enter number</u> </div> <div> <input checked="" type="checkbox"/> Loan      USD 200 million         <input type="checkbox"/> Results-based payment      <u>Enter number</u> </div> <div> <input type="checkbox"/> Guarantee      <u>Enter number</u> </div> |                                                                         |                                                                                                                                                                                           |
| <b>A.11. Implementation period</b>                                       | <p>Indicate the number of years and months the project/programme is expected to be implemented.</p> <p>13 years</p>                                                                                                                                                                                                                                                                                                                                                                                 | <b>A.12. Total lifespan</b>                                             | <p>Indicate the maximum number of years over which the outcomes of the investment are expected to be effective, i.e. to lead to adaptation and/or mitigation results.</p> <p>15 years</p> |
| <b>A.13. Expected date of AE internal approval</b>                       | <p>This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/programme, if available.</p> <p>8/31/2024</p>                                                                                                                                                                                                                                                                                                                                         | <b>A.14. ESS category</b>                                               | <p>Refer to the AE's safeguard policy and <a href="#">GCF ESS Standards</a> to assess your FP category.</p> <p>I-2</p>                                                                    |
| <b>A.15. Has this FP been submitted as a CN before?</b>                  | <p>Yes <input checked="" type="checkbox"/>    No <input type="checkbox"/></p>                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>A.16. Has Readiness or PPF support been used to prepare this FP?</b> | <p>Yes <input type="checkbox"/>    No <input checked="" type="checkbox"/></p>                                                                                                             |
| <b>A.17. Is this FP included in the entity work programme?</b>           | <p>Yes <input type="checkbox"/>    No <input checked="" type="checkbox"/></p>                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>A.18. Is this FP included in the country programme?</b>              | <p>Yes <input type="checkbox"/>    No <input checked="" type="checkbox"/></p>                                                                                                             |
| <b>A.19. Complementarity and coherence</b>                               | <p>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</p> <p>Yes <input checked="" type="checkbox"/>    No <input type="checkbox"/></p>                                                                                                                                                                                                                                                                        |                                                                         |                                                                                                                                                                                           |
| <b>A.20. Executing Entity information</b>                                | <p>If not the Accredited Entity, please indicate the full legal name of the Executing Entity(ies) and provide its country of registration and ownership type. Note that there can be more than one Executing Entity. Also indicate if an Executing Entity is the National Designated Authority. Refer to the definition of Executing Entity in the Accreditation Master Agreement.</p> <p>AE is the Executing Entity for this programme</p>                                                         |                                                                         |                                                                                                                                                                                           |
| <b>A.21. Executive summary (max. 750 words, approximately 1.5 pages)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                         |                                                                                                                                                                                           |

### **Climate change problem**

India is highly vulnerable to climate change impacts and ranks 7th on the Climate Risk Index and 31st on the INFORM Risk Index, incurring losses worth \$68 billion, annually, due to extreme weather events.

1. In 2016 India's total GHG emissions, excluding Land Use Land-Use Change and Forestry (LULUCF) were 2,838.89 million tonne CO<sub>2</sub>e and 2,531.07 million tonne CO<sub>2</sub>e with the inclusion of LULUCF. Carbon dioxide continues to remain the most emitted GHG. Increased burning of fossil fuels, methane emissions from livestock and increasing aluminium and cement production are attributed to this. MSMEs generate around 110 million tonnes of carbon dioxide, equivalent to the energy usage of up to 50 million metric tonnes of oil per year from 200 energy-intensive manufacturing clusters in India, according to a 2018 report<sup>4</sup>. MSMEs also face several barriers to become climate resilient owing to reasons such as usage of conventional processes and technologies with unskilled / semi-skilled manpower, leading to higher resource consumption and associated GHG emissions. The technology advancement in these units is usually low due to resource constraints including availability of finance, access to technology, and skilled manpower. This provides an opportunity to reduce their resource consumption including electricity, fossil fuels, water, etc. in a cost-effective manner.

### **Proposed interventions**

2. Considering the above, the proposed project is an attempt to sensitise the MSMEs (towards adopting climate /cleaner production technologies and develop/deploy adaptation technologies) and to sensitize the banks/FIs towards having dedicated climate finance instruments and to lend with increased confidence to meet the demand from MSMEs to adopt/develop the necessary climate technologies for mitigation and adaptation interventions. Through, SIDBI's various climate initiatives and the learnings gained thereof, it is strongly felt that MSMEs in India can be an important contributor to both the mitigation and adaptation efforts of the Government of India from the demand side as well as supply side. The project titled 'Financing Mitigation and Adaptation Project in Indian MSMEs' is targeted at the following:
  1. **Develop and deploy an ecosystem to promote Climate Financing to MSMEs-** Utilizing the past learning of SIDBI, the programme will set up a blended finance facility targeting the reduction of carbon emissions and enhancing the resilience to the impacts of climate change, in line with the requirements of MSMEs. The aim is to provide loans at concessional terms to MSMEs either directly or through Participating Financial Institutions (PFIs) to develop and deploy technologies for mitigation and adaptation interventions.
  2. **Undertake Technology Need Assessments for MSMEs-** To facilitate the MSMEs to undertake mitigation interventions within their systems and operations, it is imperative to provide handholding support for technology needs assessment.
  3. **Awareness Generation and Capacity Building of MSMEs-** The technology assessment will help to build awareness on the climate technologies and create niche market demand for its adoption among the MSMEs. Further, the technology need assessment would pave the way for awareness raising and training activities for the MSMEs to disseminate the information about low carbon and climate-resilient technologies, as well as available financing amongst MSMEs.
  4. **Awareness Generation and Capacity Building of Participating Financial Institutions-** SIDBI will sensitise India's financial sector to the importance and benefits of climate finance with special emphasis on capacity building and creating awareness of climate risk assessment to tackle the challenges posed by climate change.
  5. **Implement knowledge management and sharing on climate change adaptation and mitigation for MSMEs-** The learning of the program will be disseminated amongst stakeholders through knowledge products to enable large scale replication of technologies for mitigation and adaptation interventions within MSME sector.
  6. **Enabling environment for a low-emission and climate-resilient MSME sector-** The project seeks to support MSMEs operating in the ecosystem of climate adaptation technologies in the various sectors of economy including agriculture, agroforestry, water resources, disaster response services, etc.
  7. **Develop and implement MSME sectoral adaptation and mitigation benchmarking tools-** The program will contribute to the upgrading and further development of the online assessment / benchmarking tools for MSMEs.

### **Climate results/benefits**

- **Reduction of GHG emissions-** The programme will reduce 35.34 million tCO<sub>2</sub>eq of GHG emissions over equipment lifecycle with the adoption of low carbon climate technologies
- **Promoting Industrial Decarbonization-** The exclusive investment in climate friendly Clean Technologies in MSMEs will accelerate industrial decarbonization trajectory of the country through (a) New innovations in climate change domain; and (b) Acceleration of the market for climate technologies.
- **Promotion of climate financing-** A total of ~10000 climate loans will be provided to the eligible MSMEs.

- **Energy Saving-** The Project will enable saving of more than 27 lakhs MWh energy. Studies indicate that the value of the productivity and operational benefits derived can be up to 2.5 times the value of the energy savings.
- **Impact on beneficiaries-** The programme will impact a total of 10.80 million beneficiaries. Additionally, the project will create ~ 420,000 new climate jobs directly related to climate change Mitigation.
- **Leveraging Private Investment in climate financing-** GCF fund of USD 215.6 million as loan and TA grant will potentially leverage USD 3577 million (Co finance of USD 800 million mobilization of 2400 million plus MSME promoter's contribution of USD 377 million) it is associated public and private sector sources, hence making a total programme outlay of USD 3792.6 million. The fund will be utilized towards a dedicated investments in MSME sector for carbon emission reduction and promoting MSMEs for accelerating adaptation actions. Higher concessional finance would help MSMEs in overcoming the barriers of high capital cost as well as increased compliance cost.
- **Promoting Innovation-** The support from the programme to new and young companies and entrepreneurs who often exploit technological or commercial opportunities that have been neglected by more established enterprises or even challenge the business models of existing companies will unleash climate innovations, and thereby facilitate climate low carbon industrialization. This will be done especially in the case of adaptation interventions which have to be locally suited and context specific
- **Promoting women-based entrepreneurship-** Various TA activities including capacity building and financial literacy of women workers/civil society representatives will be conducted so that they can become an independent entrepreneur. These initiatives will help in increasing the existing women employment ratio in MSMEs.

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<sup>4</sup> [Greening of MSMEs key in India's journey towards climate-resilience \(downtoearth.org.in\)](https://downtoearth.org.in/)



## PROJECT/PROGRAMME INFORMATION

### B.1. Climate context (max. 1000 words, approximately 2 pages)

#### *Country context and climate profile*

The Indian subcontinent encompasses varied climatic zones including the north covered by Himalayan tundra, the North-west covered by arid desert and the north-west, central and North-eastern parts covered by humid climate. The monsoons are crucial for the Indian economy and are often regarded as the backbone of the agricultural sector, which employs over half of India's population. They also play a pivotal role in the management of water resources and the generation of hydro-based clean energy. However, the variability of the Indian monsoons and the associated wet and dry conditions have direct impacts on socio-economic aspects of the country, including the gross domestic product (GDP) (Gadgil and Gadgil 2006; Wang, Gadgil, and Kumar 2006; Gulati, Saini, and Jain 2013).

#### **Observed and projected climate change**

##### *Temperature*

##### Observed:

1. India's average annual mean temperature during 1901-2020 showed an increasing trend of 0.62°C/100 years, with significant increasing trend in maximum temperature (0.99°C/100 years) and relatively lower increasing trend (0.24°C/100 years) in minimum temperature. Considerably greater warming of 0.14–0.25°C per decade has been experienced over the last 30 years. This warming trend has been highest during the post-monsoon season (0.88°C/100 years) followed by winter season (0.68°C/100 years). India's average temperature has risen by around 0.7°C during 1901–2018.<sup>5</sup>

##### Future:

2. Temperatures in India are projected to increase by ~4°C by 2080-2099 under the RCP8.5 emissions scenario, and ~1.1°C under the RCP2.6 emissions scenario. By mid-century, projections indicate increases in annual average temperatures of 1.5–2.5°C in subtropical regions and 1.2–2.2°C in tropical regions in the country. Warming is stronger in annual minimum and maximum temperatures than in the average. Under all emissions scenarios, the rise in annual minimum temperatures is 18-21% higher than the rise in average temperatures (Figure 1). Such temperature increases are also supported by regional models used for the development of the IPCC's Sixth Assessment Report (AR6)<sup>6</sup> Under the SSP2 4.5 scenario, mean temperature in the South Asia region is expected to increase by 0.6–1.3°C by 2040, relative to a baseline period of 1980–2010 (Figure 2). Warming on higher emissions pathways is strongly biased towards the winter and pre-monsoon months. Such conditions will see an increase in the duration of heat waves and 'hot' (when temperatures exceed the current average maximums and minimums by 10%) days and nights.

<sup>5</sup> <https://www.nature.com/articles/s41612-023-00494-0>

<sup>6</sup> <https://www.ipcc.ch/report/ar6/wg1/>

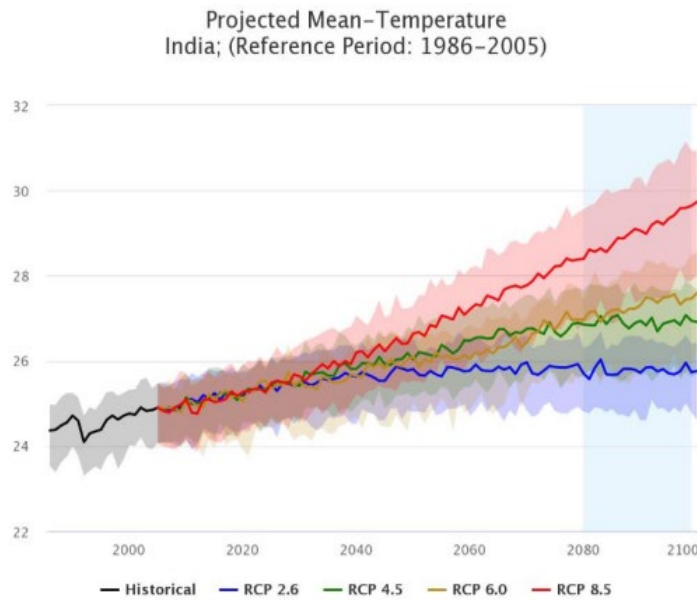


Figure 1: Projected mean temperature for India for all emissions scenarios relative to a reference period of 1986 to 2005

3. As per a World Bank study, unusual and unprecedented spells of hot weather are expected to occur far more frequently and cover much larger areas. Long-term predictions show that the heatwaves will affect more than 300 million people by 2050 and lower the quality of life for almost 600 million Indians by 2100. Under 4°C warming, the west coast and southern India are projected to shift to new, high-temperature climatic regimes with significant impacts on agriculture.

#### Precipitation

##### Observed:

4. While the Indian monsoons, by nature, show high variability through space and time, current climate change trends indicate that we are likely to breach global warming records earlier than expected (WMO 2023). A decline in monsoon rainfall since the 1950s has already been observed. The frequency of heavy rainfall events has also increased. A very recent study by Council for Energy, Environment and Water a think tank based in India has highlighted the following:

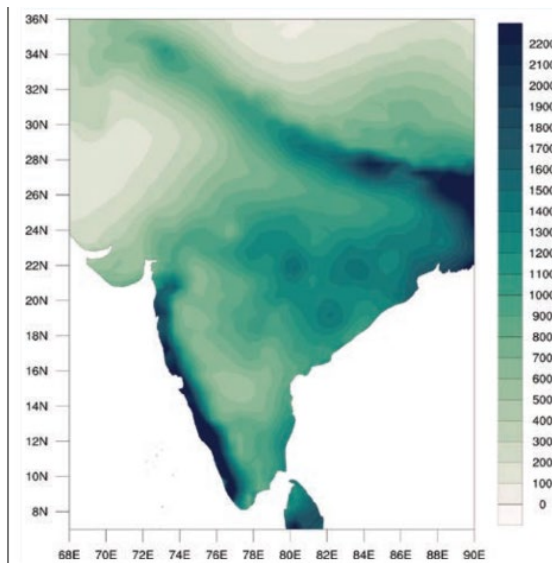


Figure 2: Annual Mean Rainfall (mm) over the period 1991-2020<sup>7</sup>

<sup>7</sup> [https://climateknowledgeportal.worldbank.org/sites/default/files/country-profiles/15503-WB\\_India%20Country%20Profile-WEB.pdf](https://climateknowledgeportal.worldbank.org/sites/default/files/country-profiles/15503-WB_India%20Country%20Profile-WEB.pdf)

5. In the past 40 years during the southwest monsoon, it was observed that India as a whole experienced 29 'normal', 8 'above-normal', and 3 'below-normal' monsoon years. However, analysis of these trends at the district level showed that approximately 30 per cent of India's districts witnessed a high number of deficient rainfall years and 38 per cent witnessed a high number of excessive rainfall years..
6. 55 per cent of tehsils witnessed an increase in southwest monsoon rainfall in the past decade (2012-2022), by more than 10 per cent compared to the climatic baseline (1982-2011)
7. While the decreasing trends in southwest monsoon were not statistically significant continuously over forty years, we found that nearly 11 per cent of the Indian tehsils (blocks) witnessed a decrease particularly in the past decade (2012-2022), by more than 10 per cent compared to the climatic baseline (1982- 2011). These are in the Indo-Gangetic plains, which contribute to more than half of India's agricultural production, northeastern India, and the Indian Himalayan region. These regions also host fragile but highly diverse ecosystems.
8. In our study of localised wet rainfall extremes, we found that nearly 64 per cent of Indian tehsils experienced an increase in the frequency of heavy rainfall days by 1-15 days per year in the past decade during the southwest monsoon. This pattern is prominent in the tehsils of states with the highest GDPs.

#### Future:

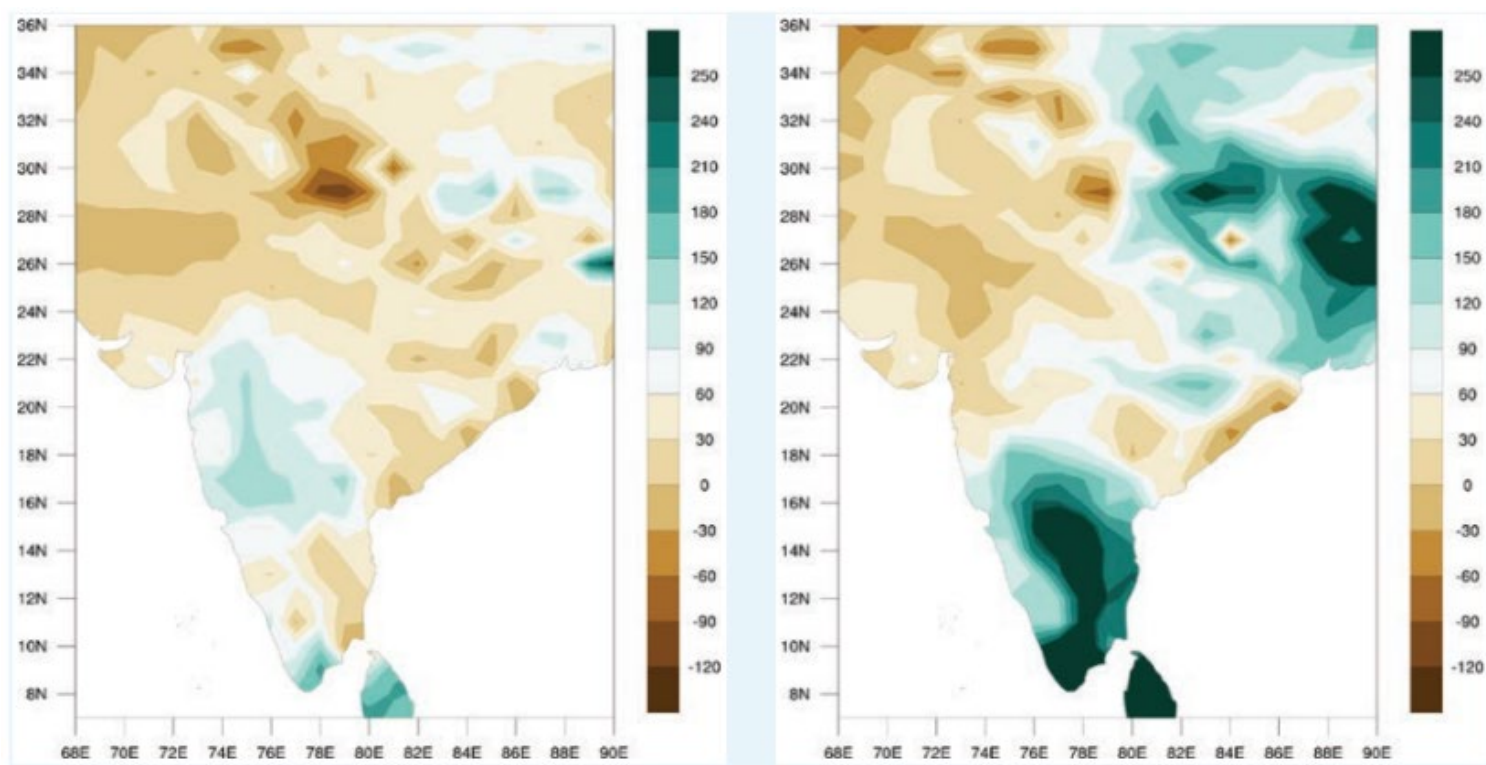


Figure 3: Projected Change in Annual Precipitation by 2040-2059 (left) and by 2080-2090 (right) vs 1986-2005

9. A 2°C rise in the world's average temperatures will make India's summer monsoon highly unpredictable. At 4°C warming, an extremely wet monsoon that currently has a chance of occurring only once in 100 years is projected to occur every 10 years by the end of the century. An abrupt change in the monsoon could precipitate a major crisis, triggering more frequent droughts as well as greater flooding in large parts of India. India's northwest coast to the south-eastern coastal region could see higher than average rainfall.

#### *Climate hazards and impacts*

10. In 2022 alone, Asia witnessed over 81 natural hazards, with 83 per cent of them being hydro-meteorological. India, in particular, incurred significant losses, primarily due to floods caused by the monsoons (WMO 2023).

#### Droughts

11. In monsoon-dominated regions like India, droughts have been recurrent and caused major famines in the 19th and 20th centuries. The southwest monsoon rainfall in India is the primary source of agricultural water and groundwater recharge, accounting for 80% of the total annual rainfall. Droughts in India due to the weakening of the southwest monsoon are

closely linked to Indian Ocean warming and El Nino/Southern Oscillation (ENSO)<sup>8</sup>. India is highly vulnerable to drought with about two-thirds of its area prone to drought. Being an agricultural-dominant country and home to 1.4 billion people, droughts in India profoundly impact agricultural productivity, water resource management, and socio-economic well-being. India has witnessed a rise in the frequency, severity, and duration of droughts over the recent decades, which is projected to be further exacerbated by climate change. With the increasing food demand due to rising population and urbanization the impact of droughts is expected to become more severe in the future.

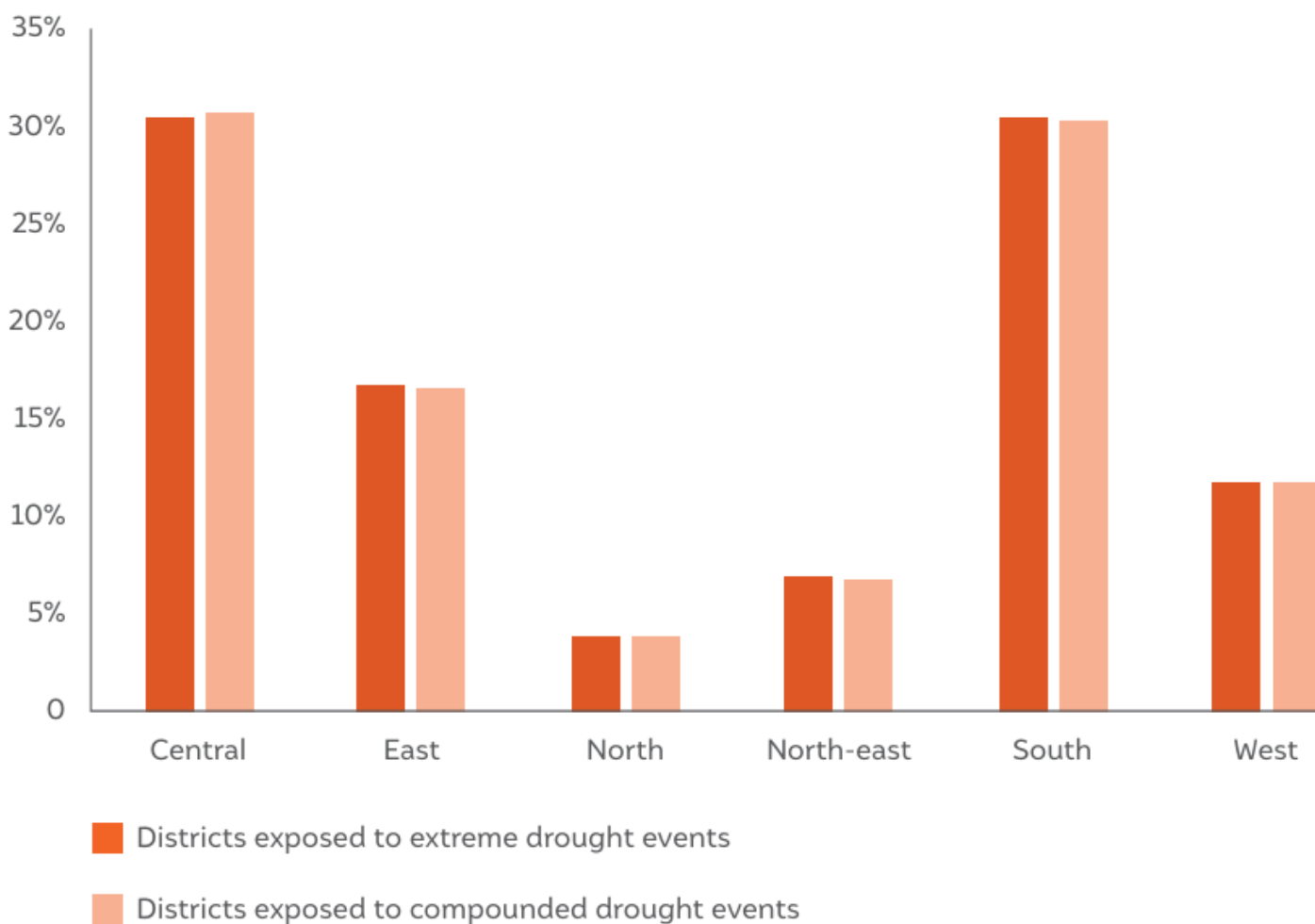


Figure 4: Regions in India exposed to extreme drought events and those exposed to compounded drought events

12. Evidence indicates that parts of South Asia have become drier since the 1970s with an increase in the number of droughts. Droughts have major consequences. In 1987 and 2002-2003, droughts affected more than half of India's crop area and led to a huge fall in crop production. Droughts are expected to be more frequent in some areas, especially in north-western India. Crop yields are expected to fall significantly because of extreme heat by the 2040s.<sup>9</sup>

#### Floods

13. Extreme precipitation events in India in the past caused flooding, affected economy, and resulted in the loss of lives (Wallemacq et al., 2015). Several extreme precipitation events occurred during the recent decades are noteworthy for their impacts on society and infrastructure. For instance, extreme precipitation (940 mm in 18 h) event in one of the south-western states (July 2005) was a disaster that led to flooding affecting 20 million people and causing around 1200 deaths

<sup>8</sup> <https://www.nature.com/articles/s41597-023-02856-y>

<sup>9</sup> <https://www.nature.com/articles/s41597-023-02856-y>

(Gupta and Nair, 2011). Similarly, the same state experienced flooding in September 2017 due to extreme precipitation (330 mm in 24 h), which largely affected road transportation. (Boyaj et al., 2017; van Oldenborgh et al., 2016).

14. Other than localized flooding, which is mostly caused by extreme precipitation, the river basin scale flooding can be due to extreme precipitation and other factors associated with catchment characteristics (Shah and Mishra, 2016; Mishra and Lilhare, 2016; Sharma et al., 2018). For instance, extreme precipitation resulted in a large flooding in north India in June 2013 claiming more than 6000 lives and an estimated loss of \$3.8 billion (Kumar, 2013). More recently, extreme precipitation in August 2018 in south India caused a widespread flooding and death of more than 440 people (Mishra and Shah, 2018; Mishra et al., 2018a). A preliminary estimate of economic loss due to this flood exceeds over \$3 billion (Gulf News, 30 August 2018).
15. Both flooding and extreme precipitation events are likely to increase under the warmer climate in India (Mukherjee et al., 2018) posing more risk of economic loss and damage to infrastructure. Dottori et al. (2018) showed that 1.5 °C increase in global mean temperature from the pre-industrial level will lead to an increase in human losses from flooding by 70–80% with a higher risk in South Asia. Rapid and unplanned urbanization further increases the risks of sea water intrusion.

#### Melting of Glaciers

16. According to a recent report, the glaciers in Asia's Hindu Kush Himalaya region are melting at an alarming rate due to global warming, which causes both dangerous flooding and water shortage. According to the assessment conducted by the International Centre for Integrated Mountain Development (ICIMOD), the glaciers in the region shed ice about 65% faster during the 2010s compared to the preceding decade<sup>10</sup>. The study predicts that by the end of the century, these glaciers could lose up to 75% of their volume, leading to significant flooding and water shortages for the approximately 240 million people residing in the mountainous area. Glaciers in the northwestern Himalayas and in the Karakoram range - where westerly winter winds are the major source of moisture - have remained stable or even advanced. On the other hand, most Himalayan glaciers - where a substantial part of the moisture is supplied by the summer monsoon - have been retreating over the past century.
17. At 2.5°C warming, melting glaciers and the loss of snow cover over the Himalayas are expected to threaten the stability and reliability of northern India's primarily glacier-fed rivers, particularly the Indus and the Brahmaputra. The Ganges will be less dependent on melt water due to high annual rainfall downstream during the monsoon season. The Indus and Brahmaputra are expected to see increased flows in spring when the snows melt, with flows reducing subsequently in late spring and summer. Alterations in the flows of the Indus, Ganges, and Brahmaputra rivers could significantly impact irrigation, affecting the amount of food that can be produced in their basins as well as the livelihoods of millions of people (209 million in the Indus basin, 478 million in the Ganges basin, and 62 million in the Brahmaputra basin in the year 2005).

#### Cyclones

18. Since the middle of the 20th century (1951-2018), the frequency of very severe cyclonic storms during the post-monsoon season has increased significantly. During 2000-2018, the frequency of severe cyclones has been increasing at the rate of 1 event per decade. A zone wise analysis of cyclone hotspots conducted by Council for Energy, Environment and Water Resources highlights that the eastern part of India is highly exposed to extreme cyclone events. Further, the southern and western areas are more exposed than the northern and north-eastern zones. Various studies show that the warming of oceans causes sea levels to rise, causing thermal-hydro expansion, which intensifies the strength and frequency of cyclones in the coastal regions of India (Mimura N., 2013). A total of 283 cyclones hit the Indian coastline between 1877 and 2005; as many as 106 of these were extreme cyclonic events that affected a 50 km-long strip on the east coast of India, and 35 hit the west coast (ADRC, 2012). Human-induced climate change is causing more frequent and intense cyclones on India's west coast, finds a new study<sup>11</sup>.

A study by National Disaster Management Institute (NDMI)<sup>12</sup> highlights the following:

<sup>10</sup> <https://economictimes.indiatimes.com/news/new-updates/himalayan-glaciers-risk-losing-75-of-their-volume-by-2100-due-to-global-warming/articleshow/101130703.cms?from=mdr>

<sup>11</sup> <https://indianexpress.com/article/technology/science/climate-change-frequent-intense-cyclones-india-west-coast-8969486/>

<sup>12</sup> <https://www.downtoearth.org.in/news/climate-change/cyclones-getting-fiercer-account-for-nearly-half-of-indias-deaths-due-to-climate-related-disasters-85587>



19. Cyclones affect 11 states and Union territories, found an NDMI study *Mapping Climatic and Biological Disasters in India*. There are 96 districts along the coasts officially declared cyclone-prone.
20. "Cyclones accounted for 48% of India's overall human life loss due to climate-related disasters, followed by heat waves (26 per cent), 18 per cent due to floods and 8 per cent due to cold waves," - NDMI study.
21. Due to increase in sea surface temperature and ocean heat content, intensity of cyclones is increasing along Indian coasts. "Annual frequency of cyclones significantly reduced from 1951-2018; however, frequency of Very Severe Cyclonic Storms during post-monsoon season increased significantly," according to the study by the researchers from the University Of Central Lancashire, UK, the Central University of Tamil Nadu and the University of the West of England, UK.
22. Frequency is rising in Arabian Sea and reducing in the Bay of Bengal, overall. Pre-monsoon frequency is coming down on both sides.

#### Heat waves

23. In parts of Asia, there is moderate evidence of a rising trend in everyday temperature extremes<sup>13</sup>. India is amongst the most affected countries in Asia. In summer 2019, half of the country was hit by severe heat waves, with temperatures reaching above 48 degrees Celsius in numerous regions and 6,100 individuals reported to have died since 2010<sup>14</sup>. Over the last 50 years, the length of heat waves in central and northwestern India has increased by about five days<sup>15</sup>. According to the Ministry of Earth Science, the occurrence of heat waves in northwest India has increased by one per 20 years. The average total span of heat waves grew by two days every ten years, whereas the maximum duration climbed by roughly 1.5 days. By the end of the 21st century, the heat waves are projected to be 3 to 4 times more frequent, under the RCP8.5 scenario, as compared to 1976-2005 levels<sup>16</sup>.

#### Climate change vulnerability and impacts

24. In 2019, India was the 7th most affected country<sup>17</sup> due to climate change led extreme weather events – both in terms of the fatalities (2,267 people) as well as the economic losses (USD 66,182 million). As per a study by Council for Energy, Environment and Water, extreme weather events due to climate change has led to **17 out of 20 people** in India being vulnerable to extreme hydrological and meteorological (or "hydromet") disasters like flood, drought and cyclone. The same study highlights that 75% of Indian states and Union Territories are vulnerable to extreme weather events.

| Zone                                                | Component of vulnerability | Exposure<br>(People, livelihood, assets, etc. that could be adversely affected)     |                                                                                     |                                                                                     | Sensitivity<br>(The degree to which a system is affected)                           |                                                                                     |                                                                                       | Adaptive capacity <sup>3</sup><br>(The ability of a system to adjust to climate change or to cope with the consequences) |
|-----------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|                                                     |                            |  |  |  |  |  |  |                                                                                                                          |
| Northern                                            |                            | Medium                                                                              | Low                                                                                 | Low                                                                                 | High                                                                                | Medium                                                                              | Medium                                                                                | Low                                                                                                                      |
| Southern                                            |                            | Medium                                                                              | High                                                                                | High                                                                                | High                                                                                | High                                                                                | High                                                                                  | Low                                                                                                                      |
| Eastern                                             |                            | High                                                                                | Medium                                                                              | High                                                                                | High                                                                                | Low                                                                                 | High                                                                                  | Medium                                                                                                                   |
| North – Eastern                                     |                            | High                                                                                | Low                                                                                 | Low                                                                                 | Medium                                                                              | Low                                                                                 | Low                                                                                   | Low                                                                                                                      |
| Western                                             |                            | Low                                                                                 | Low                                                                                 | Medium                                                                              | Medium                                                                              | High                                                                                | High                                                                                  | Low                                                                                                                      |
| Central                                             |                            | Medium                                                                              | High                                                                                | None                                                                                | Low                                                                                 | Medium                                                                              | Low                                                                                   | Low                                                                                                                      |
| High: (0.41-1) Medium: (0.21-0.40) Low: (0.00-0.20) |                            |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                       |                                                                                                                          |

Figure 5: Region wise climate vulnerability in India

<sup>13</sup> [https://www.ipcc.ch/site/assets/uploads/2018/03/SREX\\_Full\\_Report-1.pdf](https://www.ipcc.ch/site/assets/uploads/2018/03/SREX_Full_Report-1.pdf)

<sup>14</sup> <https://www.wri.org/insights/buildings-are-hidden-source-indian-cities-extreme-heat>

<sup>15</sup> [https://unfccc.int/sites/default/files/resource/INDIA\\_%20BUR-3\\_20.02.2021\\_High.pdf](https://unfccc.int/sites/default/files/resource/INDIA_%20BUR-3_20.02.2021_High.pdf)

<sup>16</sup> <https://reliefweb.int/sites/reliefweb.int/files/resources/Assessment%20of%20climate%20change%20over%20the%20Indian%20region%20-%20A%20report%20of%20the%20Ministry%20of%20Earth%20Sciences%20%28MoES%29%2C%20Government%20of%20India.pdf>

<sup>17</sup> <https://www.unicef.org/india/what-we-do/climate-change>

25. India ranks 31st on the INFORM Risk Index, incurring losses worth \$68 billion, annually, due to extreme weather events. The country has a very high absolute exposure to flooding (ranking joint 8th), including riverine, flash and coastal. Exposure to tropical cyclones and related hazards (ranking 13th), along with drought is also high (ranking 13th). India's disaster risk is also driven by its vulnerability (ranking joint 31st), which is driven by high levels of social deprivation.
26. According to ND Global Adaptation Initiative Index, the key components comprising vulnerability in India are: Ecosystem services (18% of vulnerability), food (19% of vulnerability), human habitat (15% of vulnerability), health (20% of vulnerability), infrastructure (10% of vulnerability) and water (17% of vulnerability). Further, India's socio-economic characteristics further add to its vulnerabilities. For example, the country accounts for approximately one third of the global poor with ~2 million people being without shelter, ~92 million people lacking access to safe drinking water, and approximately one third of the population depending on biomass for their cooking and heating requirements. Accordingly, a large proportion of the country is dependent on climate-sensitive agricultural and ecosystem-based livelihoods. According to the World Bank, it is estimated that climate change is likely to impact 600 million people across India if the growth in GHG emissions is not contained. Therefore, adapting to impacts of climate change has been a high priority for India, as rightly outlined in India's NDCs submitted to the UNFCCC.

### Climate house Gas (GHG) emissions profile

27. As per the Third National Communication (TNC) and Initial Adaptation Communication submitted by India to the United Nation's Framework Convention on Climate Change (UNFCCC) in December 2023, India's net national emissions in 2019 stood at 2.6 billion tonnes of carbon dioxide equivalent (CO<sub>2</sub>e), marking a 4.56 per cent increase from 2016 levels and a 115 per cent increase since 1994. Carbon dioxide continues to remain the most emitted GHG, comprising 79 per cent of total emissions in 2019. Increased burning of fossil fuels, methane emissions from livestock and increasing aluminium and cement production are attributed to this.

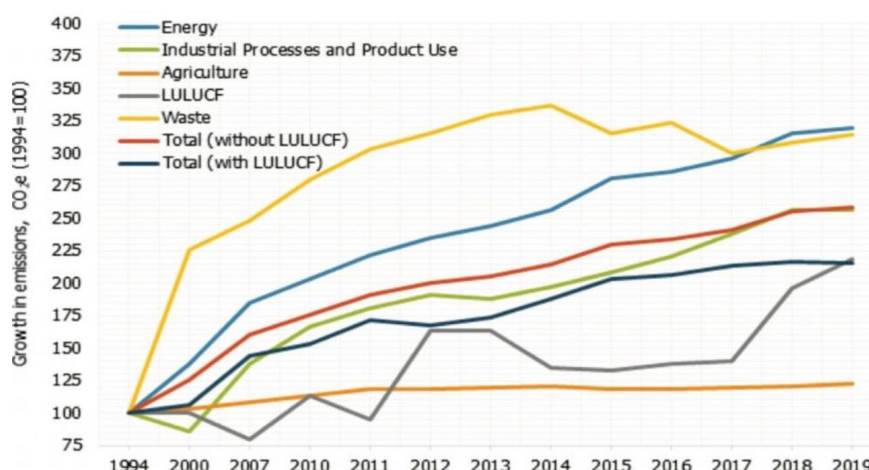


Figure 6: Contribution of various sectors to GHG emissions of India.

28. As per India's GHG inventory 2019, the contribution of energy sector towards the overall GHG emission stands at 75.80 per cent, followed by Agriculture sector (13.44 per cent), Industrial Process and Product Use (IPPU) sector accounts for 0.25 per cent and waste sector accounts for 2.33 per cent.
29. The Energy sector, which includes the consumption of fossil fuels in the Country and their associated fugitive emissions, is the main GHG emitting sector in the Country, contributing to 75.81 per cent of the total GHG emissions in 2019 (excluding LULUCF). The total emissions from the energy sector were 23,74,330 Gg CO<sub>2</sub>e in 2019, increasing by 11.5 per cent since 2016, mainly due to the steady increase in the Country's energy consumption, including the consumption of coal and natural gas for power generation and the consumption of liquid fuels for transportation.

30. The IPPU sector, which includes GHG emissions produced by a variety of industrial activities that transform raw materials by chemical or physical means, represented 8.41 percent of GHG emissions (without LULUCF) in 2019. The same year, their emissions reached 2,63,540 GgCO<sub>2</sub>e, increasing by 16 per cent since 2016 mainly due to the increase in the production of cement, aluminium, and lime.
31. The main GHG emissions from the agriculture sector are methane from livestock's enteric fermentation and rice cultivation and nitrous oxide from manure management and agriculture soil. The agriculture sector represented 13.44 percent of the total GHG emissions (4,20,968 GgCO<sub>2</sub>e) in 2019, a decrease of 1 percent compared to 2016. Even though in absolute terms, the emissions from agriculture sector increased by 3.22 percent over 2016, the share of emissions from agriculture sector is progressively reducing over the years.
32. Land Use and Land-Use Change and Forestry Sector removed 20 per cent of the Country's carbon dioxide emissions in 2019. In the same year, its emissions reached -4,85,472 GgCO<sub>2</sub>e, increasing its status as a sink by 58 per cent since 2016. 5.3.10 In 2019, the waste sector, which includes GHG emissions accounted for only 2.34 per cent of total GHG emissions.

### Groundwater

Climate change influences groundwater systems in several ways. In terms of the hydrological cycle, climate change can affect the amounts of soil infiltration, deeper percolation, and hence groundwater recharge. Also, rising temperature increases evaporative demand over land, which limits the amount of water to replenish groundwater. More than 60% of India's agriculture is rain-fed, making the country highly dependent on groundwater. Even without climate change, 15% of India's groundwater resources are overexploited. The comparison of groundwater extraction between the years 2020 and 2022 (as assessed by the Central Ground Water Board (CGWB) and States) indicates decrease in extraction (on an average for the entire country) from around 244.92 Billion Cubic Meter (BCM) to 239.16 BCM.

According to the Central Groundwater Board of India, 17% of groundwater blocks are overexploited, meaning that the rate at which water is extracted is higher than the rate at which the aquifer can recharge. Groundwater resources, which support over 60% of irrigated agriculture and around 85% of domestic water supply in rural areas and 45% of it in urban areas, are already under pressure from groundwater pollution and the effects of climate change.

### Impact on Food Security

The World Food Summit in 1996 defined food security thus: "Food security exists when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food which meets their dietary needs and food preferences for an active and healthy life." According to this definition, there are three main dimensions to food security: food availability, access to food, and food absorption. Food security is one of the leading concerns associated with climate change. Climate change affects food security in complex ways. It impacts crops, livestock, forestry, fisheries and aquaculture, and can cause grave social and economic consequences in the form of reduced incomes, eroded livelihoods, trade disruption and adverse health impacts. However, it is important to note that the net impact of climate change depends not only on the extent of the climatic shock but also on the underlying vulnerabilities. According to the Food and Agriculture Organization (2016), both biophysical and social vulnerabilities determine the net impact of climate change on food security.<sup>18</sup> Even without climate change, world food prices are expected to increase due to growing populations and rising incomes, as well as a greater demand for biofuels. While overall rice yields have increased, rising temperatures with lower rainfall at the end of the growing season have caused a significant loss in India's rice production. Without climate change, average rice yields could have been almost 6% higher (75 million tons in absolute terms). Recent studies shows that wheat yields peaked in India and Bangladesh around 2001 and have not increased since despite increasing fertilizer applications. Observations show that extremely high temperatures in northern India - above 34°C - have had a substantial negative effect on wheat yields, and rising temperatures can only aggravate the situation. Seasonal water scarcity, rising temperatures, and intrusion of sea water would threaten crop yields, jeopardizing the country's food security. Should current trends persist, substantial yield reductions in both rice and wheat can be expected in the near and medium term. Under 2°C warming by the 2050s, the country may need to import more than twice the amount of food-grain than would be required without climate change.

<sup>18</sup> <https://www.orfonline.org/research/climate-change-and-food-security-in-india>



### Impact on Energy Security

Climate-related impacts on water resources can undermine the two dominant forms of power generation in India - hydropower and thermal power generation - both of which depend on adequate water supplies to function effectively. To function at full efficiency, thermal power plants need a constant supply of fresh cool water to maintain their cooling systems. Climate-related impacts on water resources can undermine the two dominant forms of power generation in India - hydropower and thermal power generation - both of which depend on adequate water supplies to function effectively. To function at full efficiency, thermal power plants need a constant supply of fresh cool water to maintain their cooling systems. The increasing variability and long-term decreases in river flows can pose a major challenge to hydropower plants and increase the risk of physical damage from landslides, flash floods, glacial lake outbursts, and other climate-related natural disasters. Decreases in the availability of water and increases in temperature will pose major risk factors to thermal power generation.

### Climate Change and Impact on Health

Climate change is expected to have major health impacts in India- increasing malnutrition and related health disorders such as child stunting - with the poor likely to be affected most severely. Child stunting is projected to increase by 35% by 2050 compared to a scenario without climate change. Malaria and other vector-borne diseases, along with and diarrheal infections which are a major cause of child mortality, are likely to spread into areas where colder temperatures had previously limited transmission. Heat waves are likely to result in a very substantial rise in mortality and death, and injuries from extreme weather events are likely to increase. Heatwaves could slow or reverse India's progress towards Sustainable Development Goals (SDGs) without heat impact assessments. Deadly heatwaves fuelled by climate change in 2022 made almost 90 percent of Indians more vulnerable to public health issues, food shortages and increased risks of death<sup>19</sup>.

### Summary of climate hazards, its impacts and indicative adaptation solutions:

| Hazards <sup>20</sup>                                                                                                     | Impact <sup>21</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Adaptation solutions                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Increased temperatures and heat waves<br>This increase is expected to be more pronounced in the northern regions of India | <ul style="list-style-type: none"> <li>Increased heat stress-related mortality and morbidity, particularly in urban areas and among agricultural workers</li> <li>Reduced grain yields and quality due to heat stress; reduced plantation productivity where temperature thresholds are exceeded</li> <li>Diminished milk production due to heat stress</li> <li>Reduced and shifted ranges for species, leading to biodiversity loss</li> <li>Increased forest degradation from wildfires, impacting biodiversity and livelihoods</li> </ul> | <ul style="list-style-type: none"> <li><b>Heat-Health Action Plans:</b> Cities like Ahmedabad have developed heat preparedness plans and early warning systems. These include public awareness campaigns, training for healthcare professionals, and adjustments to urban planning to reduce heat impacts (<a href="#">Knowlton et al., 2014</a>).</li> </ul>      |
|                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li><b>Example of specific technological intervention:</b> Financing early warning systems in MSMEs located in vulnerable regions</li> </ul>                                                                                                                                                                                    |
|                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li><b>Architectural and Urban Planning Improvements:</b> Implementing building codes that enhance natural cooling and use reflective materials can help mitigate indoor heat. Urban planning that includes more green spaces and better airflow can also reduce urban heat island effects (Mahadevia et al., 2020).</li> </ul> |
|                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li><b>Example Specific technological intervention:</b> Financing various climate proofing products, such as improved roofing and insulation.</li> </ul>                                                                                                                                                                        |
|                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li><b>Agricultural Adaptation:</b> Farmers in heat-prone areas have begun to adopt new practices and crop varieties that are more resistant to heat stress. Modifying irrigation practices to reduce water use and increase efficiency is also critical (<a href="#">Taraz, 2018</a>).</li> </ul>                              |
|                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li><b>Examples specific technological intervention:</b> Financing technologies that reduce water use, increase resilience of crops etc</li> </ul>                                                                                                                                                                              |
|                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li><b>Enhanced Surveillance and Healthcare:</b> Improved surveillance systems can track heat-related illnesses more effectively, ensuring timely medical interventions. Increasing the capacity of</li> </ul>                                                                                                                  |

<sup>19</sup> <https://www.cam.ac.uk/stories/india-heatwaves>

<sup>20</sup> [USAID](#); [WorldBank](#); [IPCC AR6](#)

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                  | <p>dependent on forests</p> <ul style="list-style-type: none"> <li>Increased glacial melt in the short term, leading to flooding and waterlogging</li> <li>Reduced contribution of glacial melt to Himalayan rivers in the long term, reducing water available for irrigation and hydropower</li> <li>exacerbate heat-related illnesses and increase the prevalence of diseases such as diarrhea and malnutrition</li> </ul> | <p>healthcare systems to deal with heat-related emergencies is also vital (<a href="#">Mazdiyasni et al., 2017</a>).</p> <ul style="list-style-type: none"> <li><b>Community-Based Approaches:</b> Community engagement and education about the risks of heat waves and the importance of hydration and cooling methods are important for increasing resilience, especially in vulnerable populations (<a href="#">Kotharkar &amp; Ghosh, 2021</a>).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Increased frequency and intensity of heavy precipitation events. | <ul style="list-style-type: none"> <li>Increased flood-related mortality and morbidity including an increase in gastrointestinal disease from degraded water quality</li> <li>Degraded crop and pasture land from flooding, erosion</li> <li>by the 2040s, extreme river floods could affect an additional 13 to 34 million people</li> <li>Increased incidence of landslides</li> </ul>                                     | <ul style="list-style-type: none"> <li><b>Urban Planning and Infrastructure Improvements:</b> Enhancing drainage systems in urban areas to handle larger volumes of water can prevent flooding. Additionally, redesigning buildings to withstand heavy rainfall and using permeable materials in urban infrastructure can reduce runoff and increase water absorption (<a href="#">Kishtawal et al., 2010</a>).</li> <li><b>Example of specific technological intervention:</b></li> <li><b>Early Warning Systems and Forecasting:</b> Developing advanced weather forecasting systems can help predict extreme rainfall events more accurately, providing crucial lead time for preparedness and response. This includes the use of high-resolution models to better predict monsoon patterns and rainfall intensity (Pai et al., 2015).</li> <li><b>Agricultural Practices:</b> Farmers are encouraged to adopt rainwater harvesting and water-efficient irrigation techniques to manage water resources better. Crop diversification and the use of drought-resistant crop varieties can also help mitigate the effects of both excess and insufficient rainfall (<a href="#">Sarkar &amp; Maity, 2022</a>).</li> <li><b>Community-Based Approaches:</b> Engaging communities in resilience-building activities, such as constructing flood barriers and raising awareness about flood risk management, is crucial. This includes training local communities to respond effectively to flooding and heavy rainfall events (<a href="#">Mukherjee et al., 2018</a>).</li> <li><b>Policy and Regulation:</b> Implementing policies that integrate climate change adaptation into development planning is critical. This includes enforcing building codes that require flood resilience and managing land use to prevent deforestation, which exacerbates flood risk (<a href="#">Maharana et al., 2020</a>).</li> </ul> |
| Drought                                                          | <ul style="list-style-type: none"> <li>Reduced yields due to drought and water stress, particularly for rainfed agriculture</li> </ul>                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li><b>Water Resource Management:</b> Implementing integrated water management practices, including the construction of small-scale water harvesting structures, rejuvenation of traditional water bodies, promoting efficient irrigation techniques, and enhancing groundwater recharge. This helps in sustaining water availability during drought periods (<a href="#">Samui &amp; Kamble, 2011</a>).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | <ul style="list-style-type: none"> <li>• Degraded crop and pasture land from erosion and drying</li> <li>• Drought-induced water stress for domestic, agricultural and hydropower use</li> </ul>                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• <b>Agricultural Adaptations:</b> Promoting drought-resistant crop varieties, changing planting dates, and adopting new farming techniques such as micro-irrigation systems to optimize water use efficiency. (<a href="#">BIRTHAL et al., 2015</a>).</li> <li>• <b>Community-Based Approaches:</b> Strengthening community-based adaptation strategies, including capacity building for local adaptation planning and management. Engaging communities in drought risk assessment and in the design and implementation of adaptation measures helps enhance local resilience (<a href="#">Prabhakar &amp; Shaw, 2008</a>).</li> <li>• <b>Policy and Governance:</b> Developing policies that support sustainable land management, promote water conservation, and provide economic incentives for adopting sustainable agricultural practices. Policies should also focus on improving drought monitoring and early warning systems to ensure timely responses (Udumale et al., 2014).</li> <li>• <b>Technological Innovations:</b> Utilizing technology for better water management, including the use of satellite imagery and remote sensing for monitoring drought progression and assessing water resources. Technology can also aid in disseminating climate information and advisories to farmers to help them make informed decisions (<a href="#">Palanisami et al., 2010</a>).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Sea level rise                                              | <ul style="list-style-type: none"> <li>• Saltwater inundation and intrusion and storm surges, impacting coastal aquaculture and agricultural land productivity</li> <li>• Degradation/loss of coastal wetland habitats and mangroves</li> <li>• Increased flooding and saline intrusion, threatening water quality, agricultural land and infrastructure</li> <li>• By the 2070s to the end of the century, coastal flooding could impact an additional 5 to 18 million people</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Infrastructure Enhancement and Protection:</b> Strengthening coastal defenses such as sea walls, levees, and storm surge barriers to protect against high tides and storm surges exacerbated by rising sea levels. This also includes elevating roads and bridges to maintain functionality even during floods (<a href="#">Nicholls, 2011</a>).</li> <li>• <b>Ecosystem-Based Adaptation:</b> Preserving and restoring mangroves, coral reefs, and other natural buffers that absorb the impact of storm surges and high waves. These ecosystems not only protect the coastline but also provide habitat for biodiversity and support fisheries (<a href="#">Byravan, Rajan, &amp; Rangarajan, 2010</a>).</li> <li>• <b>Relocation and Retreat:</b> Implementing planned relocation for communities most at risk from sea level rise, ensuring that new settlements are built in safer areas. This strategy is often considered as a last resort but is becoming increasingly necessary in highly vulnerable areas (<a href="#">Khristodas et al., 2023</a>).</li> <li>• <b>Policy and Regulation Enhancements:</b> Updating coastal zone management policies to restrict development in high-risk areas, promoting sustainable land use practices, and ensuring that all new constructions are resilient to future sea level rise. Policies should also ensure that adaptation efforts are integrated into broader urban and regional planning frameworks (<a href="#">Almeida &amp; Mostafavi, 2016</a>).</li> <li>• <b>Community Engagement and Capacity Building:</b> Educating and involving local communities in adaptation planning to ensure they have the knowledge and resources to cope with the impacts of sea level rise. Community-driven approaches can lead to more sustainable and accepted solutions (<a href="#">Natesan &amp; Parthasarathy, 2010</a>).</li> </ul> |
| Changes in ocean temperature, dissolved oxygen, and acidity | <ul style="list-style-type: none"> <li>• Impacts plankton productivity with cascading effects up the food web</li> </ul>                                                                                                                                                                                                                                                                                                                                                                  | <ul style="list-style-type: none"> <li>• <b>Enhancing Monitoring and Research:</b> Improving monitoring systems to track changes in ocean temperature, dissolved oxygen, and acidity more accurately. This includes deploying more sophisticated sensors and conducting regular surveys to</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|                | <ul style="list-style-type: none"> <li>loss of coral reefs and the diminishing of their role in the marine ecosystem, fishery productivity and marine biodiversity are threatened</li> </ul>                                                                                                                                 | <p>gather comprehensive data, which can inform adaptive management strategies (<a href="#">Mitra et al., 2009</a>).</p> <ul style="list-style-type: none"> <li><b>Ecosystem-Based Management:</b> Promoting ecosystem resilience by protecting and restoring mangroves, coral reefs, and seagrasses that can buffer the adverse effects of acidity and temperature changes. These ecosystems play critical roles in carbon sequestration and in providing habitats that may be more resilient to changing conditions (<a href="#">George &amp; Lugendo, 2023</a>).</li> <li><b>Community Engagement and Livelihood Diversification:</b> Engaging coastal communities in monitoring efforts and providing them with the tools and knowledge to adapt. This includes diversifying livelihoods to reduce dependence on fisheries, which may be impacted by changes in ocean conditions, thus enhancing community resilience (<a href="#">Iwasaki, Razafindrabe, &amp; Shaw, 2009</a>).</li> <li><b>Technological Interventions:</b> Implementing advanced technologies to enhance water quality and control pollution inputs that exacerbate the effects of ocean acidification and oxygen depletion. This includes treatment facilities and stricter regulations on pollutants entering the ocean (<a href="#">Mandal &amp; Das, 2021</a>).</li> <li><b>Policy and Regulatory Frameworks:</b> Developing and implementing policies that integrate climate change adaptation into marine and coastal management plans. Policies should support sustainable fishing practices, pollution control, and habitat protection to enhance the resilience of marine ecosystems and human communities dependent on them (<a href="#">Das et al., 2020</a>).</li> </ul>                                                                                                                           |
| Water scarcity | <ul style="list-style-type: none"> <li>Reduced contribution of glacial melt to Himalayan rivers in the long term, reducing water available for irrigation and hydropower</li> <li>The potential loss of glaciers as a result of climate change is expected to be a significant threat to water resources in India</li> </ul> | <ul style="list-style-type: none"> <li><b>Rainwater Harvesting and Storage:</b> Implementing rainwater harvesting systems to capture and store rainfall for use during dry periods. This approach includes building tanks, reservoirs, and other storage facilities to collect rainwater from rooftops and land surfaces, which can be used for agriculture, drinking, and other domestic purposes (<a href="#">Roll &amp; Andrade, 2016</a>).</li> <li><b>Improving Irrigation Efficiency:</b> Adopting modern irrigation technologies such as drip and sprinkler systems, which significantly reduce water waste and increase water use efficiency in agricultural practices. These systems deliver water directly to the plant roots, minimizing evaporation and runoff (<a href="#">Kumar et al., 2023</a>).</li> <li><b>Grey Water Recycling:</b> Implementing grey water treatment systems for recycling and reusing wastewater from domestic activities (except toilet flushing) for irrigation and industrial processes. This strategy reduces freshwater demand and pressure on existing water resources (Sushmitha, Chanakya, &amp; Khuntia, 2019).</li> <li><b>Community-Based Water Management:</b> Engaging local communities in managing water resources through participatory governance models. This involves training and empowering community members to manage local water resources sustainably, including decisions on water allocation and conservation practices (<a href="#">Varma et al., 2021</a>).</li> <li><b>Policy and Institutional Support:</b> Developing and implementing policies that support sustainable water management, including incentives for water-saving technologies, regulations on water use, and investments in water infrastructure to enhance supply and distribution systems (<a href="#">Basu &amp; Shaw, 2013</a>).</li> </ul> |

|                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <p>Glacial lake outburst floods (GLOFs)</p>                                                                                       | <ul style="list-style-type: none"> <li>• Significant GLOF risk exists in some regions of the India Himalayas, with projections of increased hazard levels due to glacial retreat and new lake formations. This includes a potential seven-fold increase in GLOF triggering probability and a three-fold increase in affected downstream areas in some regions (<a href="#">Allen et al., 2016</a>).</li> <li>• In the broader Indian Himalayas, hundreds of glacial lakes have been identified, with certain lakes assessed as very high-risk due to various factors such as potential avalanches and rockfalls. This underscores the need for targeted monitoring and mitigation strategies (<a href="#">Dubey &amp; Goyal, 2020</a>).</li> <li>• The 2014 GLOF in the Indian Himalayas highlighted the dynamic nature of these risks, with substantial glacier retreat leading to lake expansions and increased GLOF threats. This calls for ongoing risk assessments and resilience planning for vulnerable communities (<a href="#">Majeed et al., 2020</a>).</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Risk Assessment and Monitoring:</b> Extensive use of remote sensing and GIS technologies aids in monitoring glacial lakes and assessing their burst potential. This allows for timely updates and risk assessments of vulnerable areas (<a href="#">Dubey &amp; Goyal, 2020</a>).</li> <li>• <b>Infrastructure Adaptation:</b> Strengthening and designing infrastructure to withstand potential floods, including the reinforcement of existing dams and the strategic placement of new ones. This includes the integration of GLOF considerations into the design of hydroelectric projects, ensuring they account for possible outburst floods (<a href="#">Jain et al., 2012</a>).</li> <li>• <b>Early Warning Systems:</b> Implementation of early warning systems that can provide immediate alerts to downstream communities in the event of a GLOF, enhancing community response and evacuation procedures.</li> <li>• <b>Community Engagement and Preparedness:</b> Educating and involving local communities in disaster risk reduction strategies, including evacuation drills and the development of local response plans. This also involves enhancing local capacities to respond to emergencies.</li> <li>• <b>Artificial Lowering of Lake Levels:</b> Techniques such as siphoning or controlled drainage have been proposed to lower the water levels in lakes deemed to be at high risk of causing GLOFs, thus reducing the pressure on natural or moraine dam barriers and lowering the risk of sudden breaches (<a href="#">Kattelmann &amp; Watanabe, 1996</a>).</li> <li>• </li> </ul> |
| <p>Cyclones: climate change expected to potentially decrease the overall frequency of cyclones but increase the intensity and</p> | <ul style="list-style-type: none"> <li>• <b>Loss of Life and Property:</b> Cyclones in the Bay of Bengal have historically caused substantial loss of life and</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>• <b>Community-Based Disaster Preparedness:</b> Empowering local communities through awareness, training on rescue measures, and maintaining infrastructure such as cyclone shelters. Building local capacity is crucial for effective disaster response and recovery (<a href="#">Das, 2018</a>).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



frequency of the most extreme events

damage to property. These storms result in severe flooding, destruction of homes, and loss of life, primarily due to high winds, heavy rainfall, and storm surges

([Jyoteeshkumar Reddy et al., 2021](#)).

- **Impact on Livelihoods:** Cyclones significantly impact agricultural productivity, especially in coastal regions. The associated heavy rains and flooding can lead to crop failures, loss of livestock, and damage to fishing infrastructure, which are critical sources of livelihood for many communities ([Pandit et al., 2021](#)).
- **Ecosystem Damage:** Cyclones can cause extensive damage to coastal ecosystems, such as mangroves, coral reefs, and wetlands. The high winds, saltwater intrusion, and physical damage from debris can alter these habitats, affecting biodiversity and the ecological services they provide ([Mishra et al., 2021](#)).
- **Socio-Economic Impacts:** The aftermath of cyclones often sees a rise in poverty, displacement of communities, and long-term socio-economic disruptions. Recovery and rebuilding efforts can

- **Infrastructure Resilience:** Strengthening buildings, communication systems, and transportation networks to withstand cyclonic conditions. This includes constructing cyclone-resistant structures and improving early warning systems to ensure timely evacuation and response ([Kakumanu et al., 2018](#)).
- **Integrated Water Resources Management:** Implementing strategies to manage water resources effectively during cyclones, such as rainwater harvesting, controlled release of water from reservoirs, and ensuring the integrity of dams and levees ([Palanisami, 2010](#)).
- **Ecological and Coastal Management:** Protecting and restoring mangroves, coral reefs, and other natural barriers to mitigate the impact of storm surges and coastal erosion caused by cyclones. These ecosystems act as natural buffers against the force of cyclones ([Baccar et al., 2021](#)).
- **Policy and Governance:** Developing policies that prioritize climate adaptation and disaster risk reduction. This includes land use planning, investment in disaster risk reduction technologies, and community engagement in policy-making processes to ensure that adaptation strategies are comprehensive and inclusive ([Winchester, 2000](#)).



be costly and time-consuming, impacting economic development and social stability in the affected regions

([Christian et al., 2018](#)).

- **Impact on Marine Life:** Cyclones also affect marine life, including fish populations and seabirds, disrupting marine food chains and affecting fisheries, which are vital for the livelihoods of coastal communities ([Nicoll et al., 2017](#)).

### Financial Implication of Climate Change

33. A report revealed that in 2022, India suffered a loss of 8% in GDP compared to a hypothetical world without climate change<sup>22</sup>. Further, a latest report by The Reserve Bank of India predicts that up to 4.5 per cent of India's GDP could be at risk by 2030 due to climate change related impacts<sup>23</sup>.
34. The Reserve Bank of India (RBI) has focused on climate change as the primary theme in its Report on Currency and Finance (RCF) for 2022-23, a theme-based annual report on contemporary issues concerning central banking and macroeconomic issues in India. This year's report emphasizes climate goals as a policy priority and analyses the potential macro-financial consequences of climate change for India.
35. The theme for this report, published on May 3, 2023 is "Towards a Greener Cleaner India" and it claims that fiscal policy has a prominent role in driving the green transition, from fossil fuels to renewables, and highlights the urgent need for a green taxonomy, a classification system that determines whether an investment is sustainable or not.
36. The report looks into the physical indicators of climate change and policy action globally and in India. It goes on to discuss the macroeconomic effect of climate change in India, climate change and the financial sector, and policy options to mitigate climate risks. "India's goal of achieving the net zero target by 2070 would require an accelerated reduction in the energy intensity of GDP by around 5% annually and a significant improvement in its energy mix in favour of renewables to around 80% by 2070-71. India's climate financing requirement is estimated to be at least 2.5% of GDP annually till 2030," says the report, quantifying the scale of the effort needed towards climate action. Financial stability, the core mandate of central banks around the world, can face risks due to climate change, says the RCF report, as it can affect the valuation of financial assets by influencing investors' risk perceptions.
37. The pressure of currency depreciation in countries frequently affected by climate disasters can cause financial instability, higher import costs, and negative terms of trade. In this context, central banks around the globe are gradually broadening their objectives beyond the original scope or focus, the report says.
38. Giving other examples of how climate change can affect inflation and financial assets, the report says that climate change can affect price stability through supply shocks such as food and energy shortages.
39. Additionally, demand shocks can arise when firms and households lose wealth on account of frequent natural disasters. Physical and transition risks can affect the balance sheets of financial institutions and banks, limiting the flow of credit to

<sup>22</sup> [https://mst.org.br/wp-content/uploads/2023/11/Report-Loss\\_and\\_Damage\\_Today.pdf](https://mst.org.br/wp-content/uploads/2023/11/Report-Loss_and_Damage_Today.pdf)

<sup>23</sup> <https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=21071>

the real economy. The report says climate-induced uncertainty can make households save more for precautionary purposes, bringing down the real equilibrium interest rate, which is the interest rate at which the demand for money matches the supply.

#### MSMEs

40. The MSME sector plays a pivotal role from various economic and social perspectives in India. The share of gross value added (GVA) from the MSME sector in India's GDP was 30.0%<sup>24</sup> and the share of exports was 49.8% in 2019-20. The MSME sector provides the highest employment among the non-agriculture sectors in India, and over 90% of MSMEs in the country are micro-units. The informal sector (excluding the agriculture sector) employs 68.4%<sup>25</sup> of the employable workforce in India as per the Periodic Labour Force Survey (PLFS) for 2018-19.

#### 41. Definition and Categorisation of MSMEs in India:

| Enterprise Category | Criteria based on investment in plant & machinery / Equipment & turnover                                |
|---------------------|---------------------------------------------------------------------------------------------------------|
| Micro               | Investment up to INR 1 crore (~USD 125,000) and turnover up to INR 5 crore (~USD 625,000)               |
| Small               | Investment up to INR 10 crore (~USD 1.25 million) and turnover up to INR 50 crore (~USD 6.25 million)   |
| Medium              | Investment up to INR 50 crore (~USD 6.25 million) and turnover up to INR 250 crore (~USD 31.25 million) |

42. The MSME sector in India comprising of around 63.88 million units engaged in either manufacturing or services or trading activities, creating employment for about 110 million people, contributing about 45% to manufacturing output and about 40% of exports, is an energy intensive sector, which can contribute to achievement of NDC targets. There are over 7,000 products, ranging from traditional to high-tech, that are manufactured by Indian MSMEs today. The sector is the biggest employer in India outside the agriculture sector. Currently, MSME sector is characterized by resource intensive technology profile. The sector consumes about 25% of the total energy consumed by the industrial sector in India. Trend analysis in the sector from 2012-13 to 2019-20 indicates that 6.8% year on year growth has been observed in energy consumption including both thermal and electrical energy consumption. The projected energy consumption shows a steep increase (by more than 100%) from 30 million toe in 2016-17 to 72.17 million toe in 2029-30. This sector faces barriers like limited access to technology, capacity to adopt new technologies and financing. The barriers in MSME sector towards adopting climate resilient technologies is also highlighted in the Nationally Determined Contribution (NDC) submitted by Government of India to the United Nations Framework Convention on Climate Change (UNFCCC).

<sup>24</sup> <https://www.pib.gov.in/PressReleasePage.aspx?PRID=1744032>

<sup>25</sup> [https://cse.azimpremjuniuniversity.edu.in/wp-content/uploads/2019/06/Annual\\_Report\\_PLFS\\_2018\\_19\\_HL.pdf](https://cse.azimpremjuniuniversity.edu.in/wp-content/uploads/2019/06/Annual_Report_PLFS_2018_19_HL.pdf)

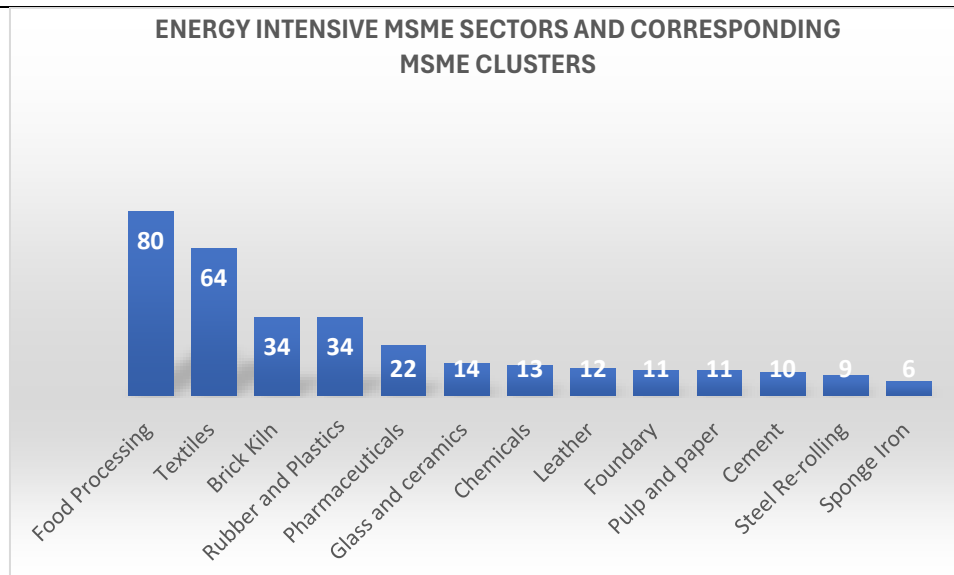


Figure 7: MSME clusters (the numbers indicated in the graph) in energy intensive sectors<sup>26</sup>

43. The major energy intensive MSME subsectors include: food processing, textile, pulp and paper, chemicals, glass and ceramics, brick kilns, foundry, sponge iron, steel re-rolling, forging, pharmaceuticals, rubber and plastic, leather, cement, etc.. The energy intensive MSME sub-sectors contribute significantly to India's economy.
44. Micro, Small and Medium Enterprises and the informal sectors will require considerable financial support, technology transfer and capacity building support to achieve a just transition to low-carbon growth and to be climate resilient. Thus, the proposed project aims to bridge the gap in investment and technical support for upscaling climate mitigation and adaptation technologies in MSME landscape by providing low-cost targeted capital to prepare the MSME sector to the challenges and opportunities posed by the India's low carbon and net-zero development trajectory. Although Indian MSMEs are a diverse and heterogeneous group, they face common problems related to, *inter alia*, lack of awareness, limited technical know-how, low technology levels, limited access to modern technologies, and high cost of capital.
45. Most Indian companies are yet to completely grasp the implications of climate change and how it will affect them. This is particularly true for the Indian MSME sector where managing the immediate challenges of everyday business remains a higher priority than addressing long term risks. At the same time, the MSME sector is also most vulnerable to climate risks since most within the sector lack the financial resources to endure even a single rupture in supply or demand, caused for instance by an extreme weather event or a new environmental or climate regulation.
46. The MSMEs located in vulnerable regions, prone to climate hazards are either not aware of the adaptation technologies that could make their units resilient or are perceived as riskier by the PFIs, thus restricting flow of capital to finance climate resilient interventions.

### MSMEs role in low carbon transition

47. MSMEs in energy intensive manufacturing sectors have high potential for emission reduction by adopting renewable energy sources, low carbon alternative sources of energy, energy and resource efficiency measures in the production value chain, battery storage, and other low carbon technologies. Adoption of clean energy sources in MSMEs will be critical for India's transition towards net-zero economy, in line with the announcement made by India at UNFCCC COP26. Thus, there has been momentum for accelerating decarbonisation of Indian industries of which MSME sector has been a key stakeholder by adopting appropriate technologies and solutions such as cleaner fuel, common combustion facilities and energy efficiency technologies.

<sup>26</sup> <https://fmc.org.in/wp-content/uploads/2012/10/MSMEReport-Enabling-Emission-Reductions-in-the-MSME-sector-in-India.pdf>

## Mitigation Needs

48. India will need to reduce its emissions to below 4,500 MtCO<sub>2</sub>e by 2030 and to below 3,200 MtCO<sub>2</sub>e by 2050 to be within its fair-share range compatible with global 1.5°C IPCC scenarios<sup>27</sup>. Through its Long-Term Low-Carbon Development Strategy, India has committed to a 45% reduction in the GHG emissions per unit of GDP by 2030 as compared to 2005 levels, with the country aiming to reach net zero by 2070<sup>28</sup>. It is estimated that USD 2.5 Trillion (at 2014-15 prices) will be required for meeting India's climate targets between 2015 and 2030. This implies USD 170 Billion per year for climate action, however the average climate finance flow was USD 19 Bn per annum from 2016-18<sup>29</sup>. In the country's energy sector, coal has continued to lead electricity generation, with the country having the second-most coal intensive electricity sector among the G20 in 2020. At sub-national level, the State Action Plan on Climate Change includes strict regulations toward reducing coal footprints. However, other government policies such as more than doubling the coal consumption limit and the one billion tonne coal production target could increase power sector emissions by an estimated 1,427 MtCO<sub>2</sub>e. Renewable energy is being promoted, and in 2015 India set a target to install 175 GW from renewable energy sources by 2022, of which 100 GW will be from solar photovoltaic (PV), 60 GW from wind, 10 GW from biomass, and 5 GW from small hydropower station<sup>30</sup>. However, the renewable energy sector added less than 60% of its targeted capacity in 2018-2019, highlighting the needs to strengthen policy framework and manufacturing capacity to ensure long-term growth of the sector. Furthermore, the cost of the solar PVs has been reduced drastically in the past decade and the generation cost is lesser than thermal power plants generation. The investment in large scale renewable projects is also slowly flowing in. However, investments for MSME sector in renewables are not forthcoming due to perceived investment risks, smaller scale investments, regulations for RE integrations<sup>31</sup>. Considering both aspects, banks and financing institution in India follows a restrictive and conservative approach. Thus, addressing investment gaps in renewables in MSME sector is critical for achieving the NDC targets of the country and facilitating the sector towards low carbon pathways. Civil society experts have noted that the next focus of India's emission mitigation policy, cantered largely on electricity generation, must be expanded to the industrial sector and particularly in the MSME sector and that it is critical to devise ways for the sector to reduce dependence on fossil energy while ensuring rapid growth in manufacturing and associated jobs and enhancing international competitiveness<sup>32</sup>. In the MSME sector, emissions per unit of production is generally higher than that of the corresponding large enterprises partly due to use of obsolete technologies and poor management practices. The sector faces further barriers like non-availability of technology or limited adaptability to changing technology. Moreover, MSMEs often find it hard to invest in latest technology due to various technical and non-technical limitations leading to limited priority for sustainability issues. As a result, the MSME sector consumes ~25% of the total energy utilised by the industrial sector while generating significant amount of total industrial emissions in India<sup>33</sup>.
49. Energy consumption (thermal and electrical) in the MSME sector has been projected to increase by more than 100% from 30 million tons of oil equivalents (mtoe) in 2016-17 to 72.17 mtoe in 2029-2030, necessitating the application of resource efficient technologies and practices in the sector to reduce correlated increases in GHG emissions. Detailed project reports and energy audits conducted under several SIDBI / BEE programs identify low-cost technologies, which have demonstrated significant energy savings and the potential to reduce both direct and indirect GHG emissions. Some of these Low cost technologies include: installation of APFC (for PF); energy efficient lighting; VFD motors and pumps; energy efficient pump and motors; energy efficient compressors; compressed air leakages; fan less cooling towers;

<sup>27</sup> These values exclude land use emissions. The 1.5°C fair share ranges for 2030 and 2050 are drawn from the Climate Action Tracker (CAT), which compiles a wide range of perspectives on what is considered fair, including considerations such as responsibility, capability, and equality. Countries with 1.5°C fair-share ranges reaching below zero, particularly between 2030 and 2050, are expected to achieve such strong reductions by domestic emissions reductions, supplemented by contributions to global emissions reduction efforts via, for example, international finance. On a global scale, negative emission technologies are expected to play a role from the 2030s onwards, compensating for remaining positive emissions. The CAT's evaluation of NDCs shows the resulting temperature outcomes if all other governments were to put forward emissions reduction commitments with the same relative ambition level. The 2030 projections of GHG emissions are from the CAT's August 2022 update and are based on implemented policies, expected economic growth or trends in activity and energy consumption.

<sup>28</sup> <https://moef.gov.in/wp-content/uploads/2022/11/Indias-LT-LEDS-2.pdf>

<sup>29</sup> <https://www.climatepolicyinitiative.org/wp-content/uploads/2020/09/Landscape-of-Green-Finance-in-India.pdf>

<sup>30</sup> <https://economictimes.indiatimes.com/industry/renewables/india-misses-re-capacity-target-due-to-low-solar-rooftop-wind-energy-project-installations-parliamentary-panel/articleshow/98870462.cms?from=mdr>

<sup>31</sup> Lack of DISCOM initiatives in (a) executing net metering regulations, (b) streamlining processes and (c) in being part of payment security mechanisms are barriers in the uptake of RE systems in MSMEs, with installation of a net-metering connection taking three to six months in some states. Therefore, growth in the market is usually slow unless DISCOMs have a stake in the RE system implementation. More information available at: <https://solarrooftop.gov.in/knowledge/file-62.pdf>

<sup>32</sup> <https://www.imf.org/-/media/Files/Publications/WP/2023/English/wp23218-print-pdf.ashx>

<sup>33</sup> [https://www.ceew.in/sites/default/files/CEEW\\_Sustainable\\_Development\\_Uncertainties\\_India\\_Climate\\_Policy\\_30Apr18.pdf](https://www.ceew.in/sites/default/files/CEEW_Sustainable_Development_Uncertainties_India_Climate_Policy_30Apr18.pdf)

energy efficient boilers; solar water heater; fibre-reinforced plastic (FRP) blades; condensate and waste heat recovery; installation of solar photovoltaic systems; energy saving by modification in process machine; thermal insulation; retrofitting for optimal fuel usage and increase and decrease in the contract demand. These energy saving technologies in energy intensive MSME sub-sectors can achieve GHG reductions in the range of 10%–50%<sup>34</sup>.

### MSME role in adoption of adaptation technologies

50. Adaptation actions have enormous potential for investment from private sector tapping through potential business opportunities. Adaptation actions have enormous potential for investment from private sector tapping through potential business opportunities. Although MSMEs are themselves vulnerable to climate change, they still have important roles to play in building community resilience. First, they can develop climate-resilient goods and services that are affordable to vulnerable populations. MSMEs are closely integrated into their communities, and as a result, have local knowledge of consumer demand and supply. This level of understanding can help MSMEs identify areas of need and respond accordingly. This may be especially useful for adaptation, since climate change impacts vary depending on where place-specific climatic, environmental, and socio-economic factors intersect. Second, MSMEs can upgrade their current products. In general, small and medium firms don't have the same governance structure as larger corporations, allowing them to adapt their products more rapidly to changes in the climate. Third, MSMEs can directly engage with community members to build adaptive practices. All of these actions can go a long way toward helping communities prepare for current and future climate change risks. MSMEs, with wide reaching presence across the market and employing up to 78 percent of the population, are in a better position to get products and services to hard-to-reach populations and contribute to making vulnerable populations more climate-resilient<sup>35</sup>. MSMEs being locally connected are better positioned to develop customized and innovative products and services in areas of agriculture and livestock sector including crop diversification, product value chains, and improved micro irrigation and pest management, afforestation, clean and affordable drinking water, water harvesting, recycling and Municipal solid waste management, etc. However, start-ups, innovators and MSMEs developing such climate-resilient affordable and customized goods and services with innovative business models require customized and dedicated financial support to meet their investment needs, which hitherto is not available in India.

### Adaptation Needs

51. A study by Climate Policy Initiative highlights the following barriers to adaptation financing in India (a). Limited knowledge of investment opportunities (b). Lack of capacity to develop commercially feasible projects (c) Unavailability of bankable projects and business models<sup>36</sup>. Also, the adaptation technologies need to be appropriate solution that are locally suited and in most of India the technology/service providers are usually start-ups that comes under MSMEs. Hence, the objective of this activity is to create a debt instrument tailored to adaptation needs for improved access to finance in selected MSMEs. This programme will further focus on regions in India with high climate vulnerability: coastal, eastern belt and Indian Himalayan regions and the focus sectors would be agriculture, water resources, agroforestry, disasters and services.
52. The potential impacts of climate change on businesses can be classified as either direct or indirect: Direct impacts of climate change affect the physical assets of a business, such as buildings, machinery and equipment or other vital infrastructure and Indirect impacts of climate change affect business operations through changes in the business environment, such as new regulations and market demand, changes in prices and availability of inputs and financing or effects on stakeholders. Companies that are not prepared to manage the direct and indirect impacts of climate change may face competitive disadvantages, especially if their competitors are better prepared or less affected by climate impacts. This is even more true for India's MSMEs: For most of them it is a higher priority to manage immediate challenges instead of long or term risks. At the same time, MSMEs are highly vulnerable to climate risks as they often lack the financial resources to endure even a single rupture in supply or demand, caused for instance by an extreme weather event or a new environmental or climate regulation<sup>37</sup>.

### Impact of Climate Change on businesses including MSMEs:

- **Rising temperatures:** Average temperatures as well as temperature extremes have been increasing and projected to further increase across India. For business, temperature rises are likely to have an indirect impact such as reduced productivity of workers or increased energy use, as more cooling is required in manufacturing processes. Rising temperatures can also negatively affect crop yield and harvests.

<sup>34</sup><https://www.giz.de/de/downloads/giz2013-en-climate-risk-study.pdf>

<sup>35</sup><https://www.wri.org/insights/micro-small-and-medium-enterprises-key-players-climate-adaptation>

<sup>36</sup>[https://www.climatepolicyinitiative.org/wp-content/uploads/2024/02/Financing-Adaptation-India\\_reportannexes.pdf](https://www.climatepolicyinitiative.org/wp-content/uploads/2024/02/Financing-Adaptation-India_reportannexes.pdf)

<sup>37</sup><https://www.giz.de/en/downloads/giz2013-en-climate-risk-study.pdf>



- **Precipitation changes:** Higher precipitation in some regions, and less rainfall in others have indirect impacts on yield, buildings, sewerage and ground water availability
- **Loss of freshwater:** A report by the Intergovernmental Panel on Climate Change (IPCC) states that water and its availability and quality will be the main pressures on and issues for societies and the environment under climate change. The industry demand for water in India is estimated to more than triple between 2000 and 2025. Decreasing availability, declining quality and growing demand for water are likely to have serious repercussions for the cost and productivity of companies. Furthermore, water being typically one of the most contentious issues with local stakeholder, reputational damages especially for highly water intensive industries are likely.
- **Price hike of energy:** Climate change affects both supply and demand of electricity: while higher temperatures lead to an increased demand for energy, e.g., for air conditioning and cooling, it also affects supply, e.g. by damaging business operations of energy providers. At the same time, energy needs in India are expected to triple between 2006 and 2025. For all these reasons, energy prices are likely to show greater volatility and gradually increase over the next decades.
- **Regulatory requirements:** Both the need to reduce greenhouse gas emissions for mitigating climate change and the need to adapt to the impacts of climate change has already triggered policy action, changing the framework conditions for business. More stringent regulation affecting business can be expected in the future, especially in the fields of emission reduction and resource consumption. Besides, national and state level regulation relevant to all the Indian business, exporting MSMEs need to also anticipate the policies of foreign countries.

### Leveraging Private Sector Investment for financing MSMEs

53. India needs substantial amounts of climate finance by 2050 to achieve its ambitious sustainability goals. An estimated USD 10.1 trillion will be required to get to net-zero by 2070. Public investment alone will be insufficient to tackle India's adaptation and mitigation targets, so financing that can bring in private sector investment is vital to scaling climate solutions with urgency. As per a recent consultation by SIDBI, the consulted stakeholders responded that Non-Banking Financial Companies (NBFCs) do not prominently incorporate the financing landscape for climate technologies into their stated objectives. Given the nascent nature of climate technologies, a considerable segment of Large and Medium NBFCs adopt a cautious "wait and watch" stance, refraining from explicit commitments to climate financing. As a result, the funding of climate initiatives by these NBFCs often takes place incidentally rather than through deliberate strategic planning<sup>38</sup>. As per the findings of the survey by RBI vide its notification dated July 27, 2022 and also as indicated in RBI's Report on Currency and Finance 2022-23 issued on May 03, 2023, it is observed that the financing flows from the banking sector towards climate financing are presently quite low and there is a need for training, sensitization and capacity building of the banking officials in this regard as also putting in place proper systems and processes towards climate risk assessments and integrating suitable framework for M&E, ESG & gender, etc. Further, RBI vide its draft disclosure framework on climate related financial risk, 2024 vide its notification dated February 28, 2024 has indicated glide path with timelines for putting in place suitable systems and processes for climate related risk disclosures by the banks and other financial institutions<sup>39</sup>. Moreover, in the very recent G20 Delhi declaration, the leaders unanimously accepted the recommendation of the Sustainable Finance Working Group (SFWG) to scale up climate financing through innovating finance models such as blended finance, to leverage private finance and multilateral climate funds. SIDBI has been historically working with the PFIs under its institutional finance vertical and realizes the importance to tapping these to scale up climate financing for MSMEs. Further, many of these PFIs have better outreach, especially in rural areas, thus ensuring that units located in the rural parts of the country are able to maximize from the current intervention.

### B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

*Present the theory of change (ToC) that contains a goal statement and describes how the proposed project/programme will contribute towards the goal statement by using results chain links from activities, outputs, to outcomes. By referring to the sample ToC diagram template available in the guidance note, present a ToC diagram (approximately 1 page) which visually represents the same logic in the narrative description. The ToC diagram and narrative may include a wide range of co-benefits<sup>40</sup> as applicable in the context of the project/programme.*

<sup>38</sup> <https://www.ifc.org/en/insights-reports/2023/blended-finance-for-climate-investments-in-india>

<sup>39</sup> <https://rbidocs.rbi.org.in/rdocs/Publications/PDFs/CLIMATERISK46CEE62999A4424BB731066765009961.PDF>

<sup>40</sup> GCF categorizes co-benefits into six areas which are: environmental, social, economic, gender, adaptation (relevant for pure mitigation projects) and mitigation (relevant for pure adaptation projects). Further guidance is available in the funding proposal (FP) guidance note.



Note all co-benefits will need to be further elaborated in section D.3 (sustainable development potential) and correspondent co-benefit indicators should be provided under section E.5 (project/programme specific indicators).

The theory of change should also include any relevant barriers (social, gender, fiscal, regulatory, technological, financial, ecological, institutional, etc.) that need to be addressed as well as risks and assumptions. Note that the assumptions can be elaborated further in sections E.3 (GCF outcome level: reduced reduced emissions and increased resilience) and E.5 (project/programme specific indicators) for each relevant indicator, as appropriate.

54. The objective of the programme is to develop and demonstrate an ecosystem to promote Climate Financing to MSMEs in mitigation and adaptation interventions. To achieve this objective, GCF funding is sought in the form of non-reimbursable funds (grant) of up to USD 15.6 million for technical assistance, and in the form of reimbursable funds (loan) in the amount of up to USD 200 million GCF reimbursable funds will be blended with co-finance from SIDBI of up to USD 800 million. Out of total financing, SIDBI will on-lend (i) USD 200 million directly to MSMEs, and (ii) USD 800 million to NBFC/FIs as on lending / refinance support. To achieve results at a scale, the project has potential to leverage private sector investment worth USD 2400 million. The concessional funds are expected to trigger the market towards climate investments.
55. The proposed programme's activities have been structured as two components, which collectively contribute to the achievement of three outcomes, with corresponding outputs and activities presented below.

| Component                                                                                                     | Output                                                                                                                                                                                                                                           | Activity                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component 1 — Term loans to support climate change adaptation and mitigation in MSMEs.                        | <b>Output 1.1:</b> MSME mitigation technologies financed and implemented. Increased access to mitigation finance for MSMEs                                                                                                                       | <b>Activity 1.1.1:</b> Establish a loan instrument to provide concessional loans for MSME mitigation needs.                                               |
|                                                                                                               | <b>Output 1.2:</b> MSME adaptation technologies financed and implemented. Increased access to adaptation finance for MSMEs                                                                                                                       | <b>Activity 1.2.1:</b> Establish a loan instrument to provide concessional loans for MSME adaptation needs.                                               |
| Component 2 — Technical support for climate change adaptation and mitigation in MSMEs through a dedicated PMU | <b>Output 2.1:</b> Capacity built and knowledge management improved to support climate change adaptation and mitigation in MSMEs and PFIs. Capacities of selected MSMEs and PFIs are enhanced to rapidly scale their business and climate impact | <b>Activity 2.1.1:</b> Undertake baseline studies for 500 MSMEs                                                                                           |
|                                                                                                               |                                                                                                                                                                                                                                                  | <b>Activity 2.1.2:</b> Undertake awareness raising and trainings on climate change adaptation and mitigation for MSMEs and Financial Institutions         |
|                                                                                                               |                                                                                                                                                                                                                                                  | <b>Activity 2.1.3:</b> Implement knowledge management and sharing on climate change adaptation and mitigation for MSMEs                                   |
|                                                                                                               | <b>Output 2.2:</b> Enabling environment for a low-emission climate-resilient MSME sector                                                                                                                                                         | <b>Activity 2.2.1:</b> Develop and implement MSME sectoral adaptation and mitigation benchmarking tools                                                   |
|                                                                                                               |                                                                                                                                                                                                                                                  | <b>Activity 2.2.2:</b> Digital Green Credit Rating Tool (DGCRT) includes integration of Green parameters (Energy and Environment) with Credit rating tool |

56. Accordingly, the programme outputs of financing MSME uptake of mitigation and adaptation technologies will achieve the outcome of the adoption of low-emission climate-resilient technologies across MSME sectors, ultimately contributing to GCF results areas ARA 1 and 2, as well as MRA 1,2,3. These outputs will also generate several co-benefits, including:

i) improved water supplies; ii) enhanced job creation; iii) greater gender equality in MSMEs; iv) improved productivity and profitability; and v) reduced resource use and pollution. Moreover, the programme outputs of increased capacity and knowledge for MSMEs to upscale their climate change adaptation and mitigation activities — as well as an improved enabling environment for low-emission, climate-resilient initiatives in the sector — will contribute to the outcomes of: i) a strengthened market environment for upscaling commercialised adaptation and mitigation MSMEs technologies (PV units under this project for those MSMEs who want to install solar plants in their respective units); and ii) improved resilience of the most vulnerable populations, reduced GHG emissions in the MSME sector of India. These outcomes will ultimately contribute to GCF results areas ARA 1 and 2, as well as MRA 1,2,3. Similarly, as stated above under MRA 2, the proposed Program aims strengthen the learnings from existing the EVOLVE initiative of the AE geared towards promoting the use of low-carbon transport in the country. The contribution of GCF shall culminate and enable quick replicability and scalability in the country and would bring a paradigm shift towards the adoption and uptake of the low emission transport. It would simultaneously help the upstream supply chain of the EVs in the country.

57. The project's outcomes will contribute to the achievement of the overarching goal of the programme: **IF** MSME mitigation and adaptation technologies are financed and implemented, MSMEs gain improved access to finance, knowledge management is improved, capacities of selected MSMEs are enhanced to upscale their business and climate impact, and an enabling environment is created for a low-emission climate-resilient MSME sector; **THEN** investment for climate change projects in India's MSME sector will be unlocked and the country's industrial decarbonization trajectory will be accelerated; **BECAUSE** MSMEs will have access to finance, technology and capacity building support to become resilient and developing products promoting adaptation measures. The project's Theory of Change is described below, followed by a diagram summarising its causal pathways and results chains.

### Key barriers and challenges

58. Despite the urgent imperative and scope for low carbon transition of sector, MSMEs have limited financial capacity to bear the transition cost associated with replacing high emitting machineries and infrastructure and adopting climate mitigation technologies. The current incentive and financing schemes are limited from the perspective of MSMEs especially given the inherent challenges and barriers of financing the MSME sector in itself. As per the study of International Finance Corporation<sup>41</sup>, only 16 percent of MSMEs in India are financed through formal banking systems and the credit gap to MSMEs is estimated to be about USD 240 billion (about INR 16.66 trillion as per 2018 estimates). In addition to inadequate access to finance and high cost of capital, MSMEs also lack access to markets, insurance, climate-smart inputs and services, and knowledge about mitigation and adaptation technologies and options. A recent International Energy Agency study<sup>42</sup> indicates that technologies are readily available today to achieve most of the carbon emissions through 2030. However, MSMEs often use outdated technologies which are energy intensive due to inadequate access to finance and lack of technological expertise and knowledge. For example, small and medium sized boilers used in many MSMEs are powered by coal and are significantly inefficient, i.e., by up to 30-35 percent less efficient<sup>43</sup>. Due to the inherent small size and lack of skilled human resources and awareness of technical know-how of climate technologies, MSMEs also have limited adoption of climate resilient and low carbon technologies and practices. A summary of the key barriers and the corresponding programme activities that addresses the barriers are detailed below:

| Barrier                                                                                                                                                                                                                                 | Impact of the barrier                                                                                                                                                                                                                       | Project intervention                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Barrier 1:</b><br>MSMEs operating in climate-centric projects are riskier investments due to their smaller scale, limited financial resources, and potentially volatile market conditions. Additionally, MSMEs often lack sufficient | This leads to a reluctance to provide financing or offering unfavourable terms, such as higher interest rates or shorter repayment periods. Lack of collateral further makes it more challenging for MSMEs to access the necessary funding. | Provision of concessional loans to make financing of climate focussed projects fiscally viable for financial institutions |

<sup>41</sup> <https://www.ifc.org/wps/wcm/connect/dcf9d09d-68ad-4e54-b9b7-614c143735fb>

<sup>42</sup> <https://www.iea.org/reports/net-zero-by-2050>

<sup>43</sup> <https://theprint.in/opinion/indias-msme-sector-largest-after-chinas-but-no-one-is-talking-about-its-role-in-emissions/745629/>

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| collateral to secure loans for climate-centric projects.                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Barrier 2:</b><br>Perceived risk of lending to MSMEs by the PFIs, especially for those MSMEs in areas exposed to climate and environmental risks, and underdeveloped financial markets limiting the supply of finance for climate-resilient, low-emitting businesses. | Lack of capacity and awareness among PFIs are a deterrent for MSMEs to adopt climate technologies as they will have limited lenders providing them required financing to undertake these interventions. This holds true especially for those MSMEs located in climatic vulnerable regions                                                                                                                                                                                                                                                                                                                                                                                                                                                       | This barrier will be overcome by first sensitising and building the capacity of PFIs towards understanding climate financing, encouraging them to set up dedicated climate focused vertical. Further, the project will help Indian banking system to identify the risks associated with climate financing to MSMEs and will also support Indian banks/FIs to formulate / customize their climate financing schemes accordingly.                                                                                                                                                                                                                                                                                            |
| <b>Barrier 3:</b><br>Immature capital and financial markets which limits the supply of funding for clean technologies in MSMEs.                                                                                                                                          | Cost of capital and access for climate projects in MSME sector is one of the most significant barriers hindering transfer of necessary low-carbon and climate resilient technologies in India. The average ticket size for loans for climate investments in MSMEs is small, thereby increasing the transaction costs for lenders. There is an urgent need for innovative financial solutions to fund such climate projects at concessional & softer terms at larger scale. Also, capacity building of supply and demand sides are equally crucial.                                                                                                                                                                                              | The programme will help in complementing the efforts of the Reserve Bank of India in putting in place suitable systems and processes for a dedicated climate finance vertical in the NBFCs/PFIs which would aid in enhancing the confidence of these FIs for lending to the MSMEs for implementing adaptation and mitigation technologies. Further, SIDBI through its direct lending to MSMEs for climate financing also intends to showcase best practices to enable climate financing to MSMEs. SIDBI will also enable a diverse set of innovative financial instruments such as a risk sharing facility, financial facilities to aggregate smaller investment opportunities to pool risk and direct equity investments. |
| <b>Barrier 4:</b><br>Knowledge and capacity barriers for the expansion of low-emission, climate-resilient practices across MSME value chains.                                                                                                                            | MSMEs not having idea about the climate impact of their existing technologies, even if they have an understanding of their impact, lack of robust data on the available mitigation technologies will impede their uptake of these technologies.<br>As the country strides towards achieving net zero, lack of understanding among MSMEs about environmental policies would result in compliance burden which will have impact on their future operations<br>There is a growing consensus in various multilateral fora on the need of private sector investment focused on climate financing. Lack of understanding and capacities among PFIs will result in reduced lending to MSMEs to uptake climate resilient and low emission technologies. | This barrier will be overcome by sectoral baseline studies, awareness raising campaigns, and implementation of a knowledge management and sharing system for climate change adaptation and mitigation options.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Barrier 5:</b><br>Limited adoption of resilient and low-emission production technologies and management practices due to weak institutional                                                                                                                           | Due to the inherent small size and limited capacities, the MSME promoter is involved in looking after many day-to-day functions like production, marketing, sales, collection of receivables, etc. and hence, the MSMEs generally lack awareness and technical know-how on climate technologies. It is one of the most important factors hindering the uptake of climate projects in MSME sector in India. The current incentive schemes do not                                                                                                                                                                                                                                                                                                 | This barrier will be overcome by the development of sectoral adaptation and mitigation benchmarking tools, contributions to policy initiatives for Climate Financing of MSMEs, and recommendations for an energy savings certificate trading scheme for the MSME sector.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
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| environment and policy barriers. | <p>provide enough motivation to the borrowers for adoption / enhancement of their performance with respect to the resource efficiency parameters. The policies / standards available in India prescribed the minimum performance standards. In case of MSMEs, there are limited incentives to improve their performance on a continual basis. Availability of Market mechanism providing motivation to the Indian MSMEs such as trading of energy savings certificates can also be another self-enabler requires a policy change.</p> <p>It has been observed that generally the modern / latest technologies are costlier as compared with conventional or fabricated technologies. Further, MSMEs also face high transaction costs to invest in such low emission technologies and recover its investments from savings as it requires detailed energy calculations / engineering designs etc. Usually, such proposals take longer time, which leads to unexpected rejections from financial institutions and also delayed decisions of MSMEs to go for such interventions.</p> <p>MSMEs are slow to adopt new technologies and continue to use the outdated technologies which may be highly energy inefficient and polluting in nature. There is a need to build local sectoral expertise through feasibility studies, local/international collaborations, trainings, and knowledge transfers both in public and private sector. MSMEs need customized solutions due to their nature of operations. Not many reputed / large-scale Original Equipment Manufacturers (OEMs) consider the MSME sector as a potential customer for certain reasons including the business ticket size and unavailability of local channel partners. Exposure of MSMEs towards availability of such technological options in the market is often limited and depends on either the Industrial Associations or vendors or system integrators available in the local market.</p> |  |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

59. A series of stakeholder consultations conducted as a part of this programme, highlighted the key barriers across this segment on adopting climate technologies across adaptation and mitigation as indicated below:

| Sector             | Key Barriers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solar- Open Access | <ul style="list-style-type: none"> <li>Lack of partnership opportunities for large developers with MSMEs: Low corporate ratings &amp; small-scale/disaggregated opportunity leads to higher risk preventing large developers to partner with MSMEs.</li> <li>Opportunities available only to stable firms: Due to the risky nature of the segment, financing is only limited to small/medium sized businesses with large revenues (&gt;INR 10-20 Cr). Micro enterprises aren't able to leverage such financing.</li> </ul> |
| Roof Top Solar     | <ul style="list-style-type: none"> <li>Limited awareness (&gt;50% MSMEs across clusters weren't aware of RTS)</li> <li>Technical Challenges i.e., limited by availability of rooftop space</li> <li>Financing Challenge (Limited by ability to put capital upfront)</li> <li>RESCO model not developed given low offtaker rating and challenges in collection at disaggregated level</li> </ul>                                                                                                                            |
| Energy Efficiency  | <ul style="list-style-type: none"> <li>Inherent inability to access capex easily limits the degree to which they can adopt these interventions within their operations</li> </ul>                                                                                                                                                                                                                                                                                                                                          |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                  | <ul style="list-style-type: none"> <li>• Low uptake given lack of after sales support ecosystem: Installation of novel technology requires strong after sales support. However, the ecosystem currently lacks these services.</li> <li>• High transaction cost: One-time cost of shift to energy efficient technology/processes is high and difficult for MSMEs to bear. With a lack of collateral, a lot of them aren't able to finance this transition.</li> <li>• Information failure: Inadequate data on unit level energy consumption, energy savings achieved till date from different interventions, and inadequate dissemination of knowledge make the decision of investing in energy saving technology difficult.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Electric Vehicle | <ul style="list-style-type: none"> <li>• <b>Economic:</b> The price of electric vehicles is more than their ICE counterparts. In the case of electric buses and trucks, the initial cost is almost two to three times higher than their ICE counterparts. The higher upfront cost will translate into owning of lesser vehicles than ICE variants thereby reducing the earning potential of the operators.</li> <li>• <b>Access to finance:</b> At present, the financing terms are not viable for most of the operators willing to deploy electric variants. The scheduled commercial banks with better financing terms in terms of high loan-to-value ratio (LTV), longer tenure, and lower interest rates are reluctant to fund the operators due to many perceived risks. The low LTV is translating into higher upfront capital required for the purchase of e-vehicles especially in the case of e-bus and trucks, and shorter tenure are resulting in high monthly payments making the operations unviable.</li> <li>• <b>Technical:</b> At present, there are few products available in the market in comparison to diesel models. The adequate product is available in the two-wheeler segment and three-wheeler cargo segment but are missing from the other product categories. The buses are used for different buses such as school buses, employee transport, intracity, intercity etc. and requires models of different specifications with suitable price to match the business model. The same exists in the freight sector as well.</li> <li>• <b>Regulatory:</b> At present, the incentives for electric vehicles are available at the central government level only with a limited number of vehicles. The demand incentives are not available in every state with no clear mandate on the transition to electric vehicles.</li> <li>• <b>Behavioral:</b> The lower total cost of ownership is estimated for electric vehicles across different vehicle modes. However, due to the lack of trust in EVs and higher upfront cost requirements, entities are reluctant to transition to electric mobility. It is accentuated in the case of buses and trucks.</li> <li>• <b>Structural:</b> The vast network of diesel outlets across the country in comparison to meagre public charging stations is one of the major apprehensions among potential users. Since batteries come with a limited range, it is pertinent to have adequate charging stations to ensure freedom in the deployment of electric buses and trucks. The battery warranty available in the market is less than the vehicle's operational requirement for the useful life of the vehicle. The battery replacement cost is further increasing the capital cost of owning buses</li> </ul> |
| Adaptation       | <p><b>Narrative and information gaps:</b></p> <ul style="list-style-type: none"> <li>• Information gaps (i.e., poor data and analytics) around climate risks and the benefits of timely adaptation, deter investments</li> <li>• Narrative for action is weak: 'Cost of inaction' not well quantified, existing perspectives narrow and mono sectoral</li> </ul> <p><b>Lack of/limited Bankability</b></p> <ul style="list-style-type: none"> <li>• Financial return/ROI from A&amp;R projects often perceived as low or non-existent</li> <li>• Lack of linking A&amp;R projects to socioeconomics returns, thus limited bankability</li> <li>• Many projects are 'sub scale', have high transaction costs</li> <li>• Due to a lack of proven effectiveness, many A&amp;R technology solutions have garnered limited interest</li> </ul> <p><b>Project Structure and Complexity</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



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|  | <ul style="list-style-type: none"> <li>• Projects cut across traditional sectors, require multi stakeholder alignment (govts., communities, funders, etc.), increasing complexity</li> <li>• Projects often require multiple funders with different risk return impact expectations to come together, increasing structuring complexity</li> </ul> <p>Consultations with leading global DFIs has revealed a two-fold challenge in disbursing adaptive finance in India:</p> <ul style="list-style-type: none"> <li>▪ There is a need for credible intermediaries who can act as a bridge between the DFIs and the private sector is critical – an intermediary that can bring a strong pipeline of investible projects and help unlock 4x-10x private investments will be crucial</li> <li>▪ The biggest challenge to mobilize climate finance is not the availability of capital but the lack of innovative financial instruments</li> </ul> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Barriers to climate finance for Indian MSMEs:

60. India needs substantial amounts of climate finance by 2050 to achieve its ambitious sustainability goals. An estimated USD10.1 trillion will be required to get to net-zero by 2070. Public investment alone will be insufficient to tackle India's adaptation and mitigation targets, so financing that can bring in private sector investment is vital to scaling climate solutions with urgency. As per a recent consultation by SIDBI, the consulted stakeholders responded that Non-Banking Financial Companies (NBFCs) do not prominently incorporate the financing landscape for climate technologies into their stated objectives. Given the nascent nature of climate technologies, a considerable segment of Large and Medium NBFCs adopt a cautious "wait and watch" stance, refraining from explicit commitments to climate financing. As a result, the funding of climate initiatives by these NBFCs often takes place incidentally rather than through deliberate strategic planning<sup>44</sup>.
61. As per the findings of the survey by RBI vide its notification dated July 27, 2022 and also as indicated in RBI's Report on Currency and Finance 2022-23 issued on May 03, 2023, it is observed that the financing flows from the banking sector towards climate financing are presently quite low and there is a need for training, sensitization and capacity building of the banking officials in this regard as also putting in place proper systems and processes towards climate risk assessments and integrating suitable framework for M&E, ESG & gender, etc
62. Consultations with the wider MSME ecosystem comprising of leading financial institutions (including SIDBI's on-ground branch officers) revealed the need for financing instruments and assistance designed specifically to meet the adaptive needs of the Indian MSME sector as shown below:
  - Role of the financial institutions in climate resilience financing for MSMEs is to provide timely financial support, customized financial products, and services, holding workshops on climate resilience for MSMEs, etc.
  - Financial institutions felt that they will be comfortable to provide climate financing for the MSME sector if they get low-cost funds/ cheaper sources of finance, non-sovereign loans coupled with technical assistance so that lenders can develop climate financing products, forbearance in NPA classification of climate finance loans, blended finance, longer tenure loans, default guarantee system, knowledge of reliable/proven technology, etc.
  - To enhance climate resilience, the financial community should offer solutions which are demanded by the market, especially in the area of bringing out low cost MSME focused financial products
  - FIs also suggested that all types of climate financing be included in priority sector and inclusion of climate finance in CSR activities
  - In future, if financial institutions are mandated to disclose their emissions, including Scope 3 emissions, this could result in some difficulties in obtaining finance, which could be overcome with better preparedness and efforts on the part of MSMEs in curbing emissions
  - Most stakeholders advocated for more training, building greater awareness for MSMEs on how they can see the bigger picture better, and contribute to arrest adverse climate change - very few MSMEs were found to know what the impact of their actions on climate is and they need to be made aware of how to manufacture their products more responsibly and without causing harm to environment irrespective of the sectors they are active in

<sup>44</sup> <https://www.ifc.org/en/insights-reports/2023/blended-finance-for-climate-investments-in-india>



- MSMEs would require handholding and support through sector-wise intervention so that the specific issues relating to climate change can be identified and dealt with carefully

### Targeted approach to address key technological barriers and challenges

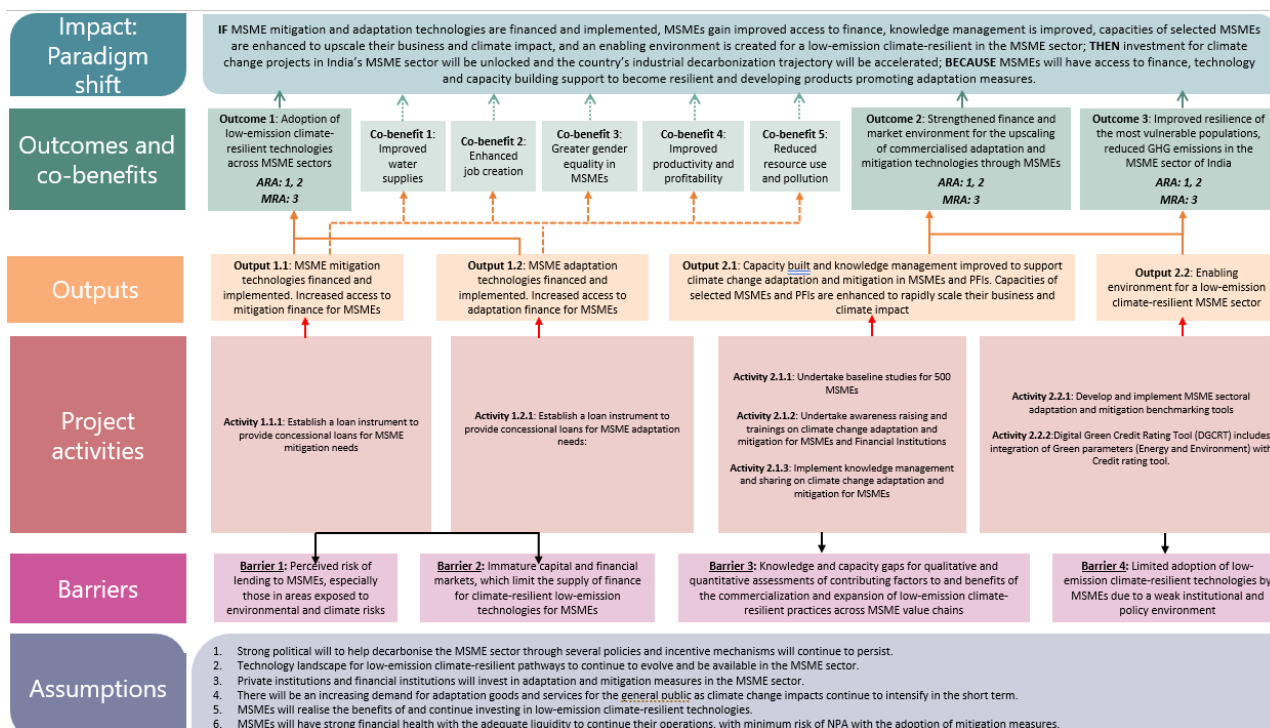
63. Basis the stakeholder consultations and SIDBI's past experience, targeted solutions have been identified to address the issue of financing climate technologies in MSMEs. There are detailed as below:

| Sector            | Solution Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | SIDBI expertise and experience                                                                                                                                                                                                                                                    |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solar-Open Access | <ul style="list-style-type: none"> <li>Designing mitigants for key risks involved such as low corporate rating: SIDBI has conducted a detailed exercise to identify key risks involved in the Open access market improving its ability to underwrite loans in this segment. For example, it has developed several parameters across technology, regulation, entrepreneur ability, etc. to determine the riskiness of a project.</li> <li>Aggregate demand at cluster level &amp; reduce risk via pooling: SIDBI can leverage its network of RMs to pool multiple MSMEs looking to invest in Open Access projects to reduce underwriting risk and enable partnership opportunities between large developers and MSMEs.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                        | Of the total investments in the solar roof top/open access market for MSMEs, SIDBI alone constitutes ~10% of all funding flow. SIDBI's direct lending portfolio for this segment stands at USD 220 million making it one of the largest banks operating in this segment in India. |
| Roof Top Solar    | <ul style="list-style-type: none"> <li><b>Co-Lending partnerships:</b> Increase awareness &amp; penetration via partnerships with green NBFCs, regional FIs under co-lending model, and panel developers/EPC contractors</li> <li><b>Customer experience:</b> Improve experience by developing integrated solutions &amp; end-to-end digital journeys</li> <li><b>Risk-sharing facility:</b> Develop RESCO model leveraging RSFs, payment guarantees and on-bill financing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                   |
| Energy Efficiency | <p>SIDBI, through the current programme intends to address all the aforementioned barriers from affordability (by offering concessional finance by blending sources) to technical assistance using the following mitigation measures:</p> <ul style="list-style-type: none"> <li>Develop new deployment models to scale adoption</li> <li>OEM dealer-linked point of sale financing (e.g., GST/cash flow based)</li> <li>Corporate supply chain partnership (e.g., Crompton, H&amp;M, etc.)</li> <li>Partner with progressive DISCOMs in power hungry states</li> <li>Strengthen the traditionally risky ESCO business model</li> <li>Address recovery challenges basis standardization of savings</li> <li>Rate ESCOs basis financial performance/pool ESCOs to reduce risk</li> <li>Technical risk assessment capability development for partner banks</li> <li>Improve payback period for technology (e.g., leveraging blended finance, decarb. monetization, etc.)</li> <li>Demand Aggregation: SIDBI can leverage its network of RMs to pool multiple MSMEs looking to invest in energy efficiency projects to reduce underwriting risk</li> </ul> | SIDBI has pioneered EE loans in India since early 2000s and has extensively invested in this segment forming various ties and partnerships. Total portfolio stands at around USD 700 million.                                                                                     |
| Electric Vehicle  | <ul style="list-style-type: none"> <li>The proposed programme will support mitigating the financial costs associated with the adoption of EVs. In this regard, the funding is proposed to achieve improved access to finance through favourable financing terms.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | SIDBI is operating several schemes for Non-Banking Financial Companies (NBFCs) and launched EV scheme for lending to NBFCs. Under the scheme, SIDBI is lending                                                                                                                    |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|            | <ul style="list-style-type: none"> <li>The primary goal of the project will be to alleviate the financing barriers by reducing out-of-pocket costs. This will prompt operators to start electrification as government policies increasingly incentivize electrification of the transport sector. The increase in uptake will encourage OEMs to come up with more products suiting the business requirements of consumers while also improving the network of public fast charging stations due to increased demand. Thus by addressing the technology and structural barriers, the programme shall unlock increased adoption of EVs across a range of MSME customers, especially fleet operators.</li> </ul> <p>Furthermore, the project findings will be used to identify the gaps at a regulatory level which will be used to formulate relevant policy solutions for the sector. These policy recommendations will be designed on the basis of project outcomes and will be driven to increase availability of products (old and new), and reduction in battery prices in the future.</p> | to unrated/unserved NBFCs who are active in EV financing. The total portfolio stands at around USD 25 million. |
| Adaptation | SIDBI aims to tackle these challenges by acting as the intermediary with the expertise and capabilities to unlock requisite financing for MSMEs in the adaptation space by offering access to, at scale (whether by itself or through other intermediaries such as NBFCs, NGOs, etc.), a range of financing instruments that can meet the varied needs of the various industries within the sector.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | The support in adaptation field for MSMEs is one of the key focus areas for SIDBI.                             |

### Theory of Change diagram

64. Based on SIDBI's experience and learnings from financing MSMEs in India, a detailed theory of change has been devised to address the various barriers and challenges mentioned above while catalysing financing for climate technologies directly as well indirectly through PFIs.



### Mapping of TOC assumption with project outputs

| Assumption                                                                                                                                                                                                                                             | Related project outputs               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Conducive policy environment will help to decarbonize the MSME sector through several policies and incentive mechanisms will continue to persist.                                                                                                      | Output 1.1 and Output 2.2             |
| Technology landscape for low emission climate resilient pathways will continue to evolve and be available in the MSME sectors.                                                                                                                         | Output 1.1 and Output 1.2             |
| Private sector and financial institutions will continue to invest in adaptation and mitigation measures in MSME sector.                                                                                                                                | Output 1.1 and Output 1.2             |
| An increased demand for adaptation goods and services for the general public will continue to evolve owing to intensification of climate change impacts.                                                                                               | Output 1.1, Output 1.2 and Output 2.1 |
| Capacity building and awareness generation activities envisaged in this project will help the SME's and Financial Institutions in realizing the benefits of low emission climate resilient technologies and they would continue investing in the same. | Output 2.1                            |
| MSME's will have strong financial capability in terms of adequate liquidity to continue their operations with minimum risk of NPA with the adoption of mitigation measures.                                                                            | Output 1.1 and Output 2.2             |

### B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

| Outcome number | GCF Mitigation Results Area (MRA 1-4) |                                     |                                                   |                                | GCF Adaptation Results Area (ARA 1-4)           |                                                      |                                               |                                            |
|----------------|---------------------------------------|-------------------------------------|---------------------------------------------------|--------------------------------|-------------------------------------------------|------------------------------------------------------|-----------------------------------------------|--------------------------------------------|
|                | MRA 1<br>Energy generation and access | MRA 2<br>Low-emission transport     | MRA 3<br>Building, cities, industries, appliances | MRA 4<br>Forestry and land use | ARA 1<br>Most vulnerable people and communities | ARA 2<br>Health, well-being, food and water security | ARA 3<br>Infrastructure and built environment | ARA 4<br>Ecosystems and ecosystem services |
| Outcome 1      | <input checked="" type="checkbox"/>   | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/>               | <input type="checkbox"/>       | <input checked="" type="checkbox"/>             | <input checked="" type="checkbox"/>                  | <input type="checkbox"/>                      | <input type="checkbox"/>                   |
| Outcome 2      | <input checked="" type="checkbox"/>   | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/>               | <input type="checkbox"/>       | <input checked="" type="checkbox"/>             | <input checked="" type="checkbox"/>                  | <input type="checkbox"/>                      | <input type="checkbox"/>                   |
| Outcome 3      | <input checked="" type="checkbox"/>   | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/>               | <input type="checkbox"/>       | <input checked="" type="checkbox"/>             | <input checked="" type="checkbox"/>                  | <input type="checkbox"/>                      | <input type="checkbox"/>                   |

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

| Co-benefit number | Co-benefit                          |                                     |                                     |                                     |                                     |                                     |
|-------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
|                   | Environmental                       | Social                              | Economic                            | Gender                              | Adaptation                          | Mitigation                          |
| Co-benefit 1      | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
| Co-benefit 2      | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Co-benefit 3      | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            |
| Co-benefit 4      | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Co-benefit 5      | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/>            | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |

### B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Define the project/programme. Describe the proposed set of components, outputs and activities that will be undertaken by the project/programme to attain the intended outcomes elaborated in Section B.2 (a). Please also elaborate how the project/programme activities will address the barriers described in Section B.2.

This section of the funding proposal should be consistent with sections C.2 (financing by component), E.5 (project/programme results level) and E.6 (project/programme activities and deliverables).

## Programme Description:

65. SIDBI through the programme 'Financing Mitigation and Adaptation Project in MSMEs' aims to develop and demonstrate an ecosystem to promote Climate Financing to MSMEs in mitigation and adaptation interventions. To achieve this GCF fund is sought in the form of GCF grant (USD 15.6 million) for technical assistance and GCF loan (USD 200 million) which will be blended with co-finance from SIDBI of USD 800 million to finance the MSMEs for their green interventions. Out of this blended facility, SIDBI will directly lend USD 200 million to the MSMEs whilst USD 800 million will be given to various PFIs as on-lending/refinance. To achieve results at a scale, private sector investment worth USD 2400 million will be potentially leverage from the market and the total indicative cost of the entire project is USD 3792.6 million.
66. The proposed activities in the programme emanates from SIDBI's more than a decade experience of ecosystem strengthening aligned to climate activities for MSMEs. Based on our experience, the following elements will be considered as part of the programme:
  - a. **Handholding:** Besides financing, MSMEs require technical handholding support in implementing climate interventions specific to adaptation and mitigation in their units.
  - b. **Market Creation:** There needs to be a market push for the MSMEs to adopt climate technologies. Since, MSMEs in India work on the neighbourhood theory, financing ~10000 loans, at a concessional rate, as a part of this programme, would help in demonstrating the climate technologies that MSMEs can adopt within their clusters.
  - c. **Concessional Finance:** Climate financing to MSMEs is a relatively new area, constrained with lack of awareness among MSMEs and FIs about the climate interventions required in MSMEs. To boost market demand, concessional financing based on innovative financing model will be very helpful. One of the examples of a blended facility model that has been very successful is that of the 4E scheme- The 4E scheme was set up with USD 5 million corpus (28% financed by The World Bank and 72% co-financed by SIDBI) with blended rates to provide concessional funding to MSMEs. The proceeds from this was further used as revolving fund to further finance the MSMEs for their climate interventions. This model was very successful and at present in FY 2024, the total disbursement stands at USD 987.9 Million. Based on numerous stakeholder conversations with several PFIs, it is felt that availability of lines of credit at concessional rates is likely to incentivize financial institutions to provide loans to MSMEs for adopting climate resilient measures. Adaptation measures suffer heavily from information gaps with unquantified risks leading to a higher risk perception for financiers. Further, financial return from adaptation projects is often considered to be low/ non-existent leading to their lack of bankability at the market rates. Thus, concessional financing can alleviate concerns about commercial viability of sector in given market
  - d. **Ease of Access:** Simplified end to end process would boost demand from MSMEs to demand for climate financing. For example, in the 4E schemes and in other running schemes of SIDBI, digitisation and seamless loan application process has proven to be very helpful in the expansion of the programme on the whole.
  - e. **Mainstreaming climate finance in Indian Banking:** There is a need to mainstream climate finance within the Indian Banking system, more focused on MSMEs. Currently, financing green technologies to MSMEs, especially those located in climate vulnerable regions is perceived as a riskier venture by the banks and hence there is a need to understand the climate space so as to enhance their lending portfolios focused towards climate financing to MSMEs.
  - f. **Capacity Building:** It was seen that training and capacity building of MSMEs, developing e-modules of the training and disseminating these through banking and industrial associations could help manifold in raising awareness among not just MSMEs but also the financiers. Additionally, stakeholders advocated for more training, building greater awareness for MSMEs on how they can see the bigger picture better, and contribute to arrest adverse climate change - very few MSMEs were found to know what the impact of their actions on climate is and they need to be made aware of how to manufacture their products more responsibly and without causing harm to environment irrespective of the sectors they are active in. Utilizing the lessons learnt by SIDBI in running pilot scale interventions in the area of energy efficiency as well as a blended finance facility, the proposed programme will set up a blended finance facility aimed at encouraging the Lending Institutions to take up financing of reduction of carbon emissions and enhancing the resilience to the impacts of climate change, in line with the requirements of MSMEs. The programme proposes to cover manufacturing as well as solution / service provider MSMEs addressing adoption of clean technologies towards emission reduction and providing solutions to enhance climate resilience of communities. The programme will build capacities of the select Lending Institutions for climate financing in MSMEs as also to facilitate networking of technology providers and facilitate technology customization as per the local needs and priorities of MSMEs.
67. A substantial component of the programme is focused on leveraging the strength of PFIs to finance the MSMEs for their low carbon and climate resilient intervention. The Accredited Entity, acting as the Executing Entity, will

refinance/on-lend the GCF Reimbursable Funds, together with the Co-financing, to the PFIs. The Programme will offer refinance against eligible assets created by the empanelled PFIs and originated no longer than six months prior to approval from the Accredited Entity for co-financing in accordance with the FAA. SIDBI has been working with banks and NBFCs under its institutional finance mandate. SIDBI's continued partnership with Indian Banking System through its institutional finance vertical is highlighted below:

**Refinance to banks:**

SIDBI expands resources of other banks/PFIs through refinancing activities thereby boosting the flow of credit to MSMEs. As of March 31, 2023, the outstanding portfolio was USD 35,924 million/INR 2,98,173 crores, the year on year growth was a significant 79%. The portfolio is in respect of 49 banks which comprises of 12 Public Sector Banks (PSBs), 17 Private Sector Banks (PvSBs), 6 Foreign Banks (FBs), 10 Small Finance Banks (SFBs), 3 Regional Rural Banks (RRBs), and 1 Urban Cooperative Bank (UCB).

**Refinance to Non-Banking Financial Companies (NBFCs):**

At the end of the fiscal year 2023, the Bank's NBFC portfolio stood at USD 4266.8 million/ INR 33,415 crores across 56 NBFCs, an increase of 86% in outstanding. During FY 2023, sanctions and disbursements showed a growth by 67% and 81%, respectively over FY 2022.

**Refinance to Micro Finance Institutions (MFIs):**

SIDBI extended financial support to MFIs in the form of equity/ quasi-equity, refinance, term loans, grant support, etc., enabling smooth flow of credit to MFIs/ Informal Enterprises. The net outstanding portfolio stood at USD 590.36 million/ INR 4,900 crore as in FY 2023 as against USD 375.6 million/ INR 3,118 crores during FY 2022, registering growth of 57% over previous FY. The disbursements during FY 2023 recorded a growth of 32% over the previous FY.

**NBFCs emerging interest towards green/climate financing:**

As per a recent stakeholder consultation conducted for NBFCs, the following are the highlights:

- o 17 NBFCs participated in the consultation
- o 92% of the NBFC leaders in the workshop agreed that Green Finance represents a significant opportunity for their business. The priority areas overwhelmingly emerged as (a). Electric Mobility (3W and 2W) (b). Solar Rooftops (Institutional solar and RESCOs)

• **Indian FI Interest in climate financing:**

There is increased interest among PFIs and NBFCs regarding climate financing. Hence, there is a need to sensitise the PFIs to increase their lending portfolio towards climate financing of MSMEs. Concessional loans to PFIs will further act as a booster to effect this objective. The programme will also build synergy and leverage from the experiences of previous projects undertaken on pilot demonstration scale in the country. The programme as catalyst would help scaling up of some of the earlier initiatives, taken by Govt of India and SIDBI towards creating momentum in climate financing landscape for MSMEs. Through this project, SIDBI will leverage its learning from serving MSMEs through various initiatives including successful implementation of GEF funded World Bank project "Financing Energy Efficiency at MSMEs in India"<sup>45</sup>, "Partial Risk Sharing Facility (PRSF) for Energy Efficiency Project", and other dedicated line of credits from Japan International Co-operation Agency (JICA), Kreditanstalt für Wiederaufbau (KfW), Agence Française de Development (AFD), the World Bank (WB), etc.

68. A joint approach of Mitigation and Adaptation is more effective than the focus only on energy efficiency. The success of climate/climate financing project, particularly for MSMEs require both technical handholding support and financial support by way of concessional loans. Further, in order to realize low hanging fruits, incentivizing the units at initial stage to create climate MSME ambassadors and pool of champion units helps in early investment decision by majority of MSMEs as they follow the "**Neighbourhood Theory**".
69. 'Neighbourhood theory' refers to the phenomena wherein new technologies or interventions are adopted/incorporated/mainstreamed within the larger industry landscape based on showcasing of individual success stories. This holds particularly true for the Indian MSME sectors where most constituents are reluctant first movers. In this regard, incentivisation of individual units to adopt newer technology/interventions primarily through financial mechanism followed by a real world on ground showcase of the technology/intervention's benefits is seen to catalyse the unit's 'neighbours' into adopting the same technology/intervention.

<sup>45</sup> <https://documents1.worldbank.org/curated/en/561461577729300064/pdf/India-Financing-Energy-Efficiency-at-MSMEs-Project.pdf>



70. Thus, utilising this phenomena SIDBI intends to catalyse large scale adoption of climate related technologies/interventions within the MSME sector by first focusing on specific set of clusters/industries and technologies
71. While concessional loans as an instrument for climate financing is commonly used, this project has identified several other avenues to tackle the risks associated with financing climate focussed projects. It has conducted deep sector level analysis, identified targeted technological interventions, and devised numerous financial tools to maximize the impact of this financing.
72. The integration of the learnings of SIDBIs previous programme has been at the centre of the design of this proposal. For instance, in Annexure 22 regarding the impact calculation we have clearly indicated how the previous lending portfolio of SIDBI has helped us to extrapolate cost allocation to each sector, energy savings in each sector and the GHG impact of the project on the whole.

| Component                                                                                                     | Output                                                                                                                                                                                                                                           | Activity                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component 1 — Term loans to support climate change adaptation and mitigation in MSMEs.                        | <b>Output 1.1:</b> MSME mitigation technologies financed and implemented. Increased access to mitigation finance for MSMEs                                                                                                                       | <b>Activity 1.1.1:</b> Establish a loan instrument to provide concessional loans for MSME mitigation needs.                                               |
|                                                                                                               | <b>Output 1.2:</b> MSME adaptation technologies financed and implemented. Increased access to adaptation finance for MSMEs                                                                                                                       | <b>Activity 1.2.1:</b> Establish a loan instrument to provide concessional loans for MSME adaptation needs                                                |
| Component 2 — Technical support for climate change adaptation and mitigation in MSMEs through a dedicated PMU | <b>Output 2.1:</b> Capacity built and knowledge management improved to support climate change adaptation and mitigation in MSMEs and PFIs. Capacities of selected MSMEs and PFIs are enhanced to rapidly scale their business and climate impact | <b>Activity 2.1.1:</b> Undertake baseline studies for 500 MSMEs                                                                                           |
|                                                                                                               |                                                                                                                                                                                                                                                  | <b>Activity 2.1.2:</b> Undertake awareness raising and trainings on climate change adaptation and mitigation for MSMEs and Financial Institutions         |
|                                                                                                               |                                                                                                                                                                                                                                                  | <b>Activity 2.1.3:</b> Implement knowledge management and sharing on climate change adaptation and mitigation for MSMEs                                   |
|                                                                                                               | <b>Output 2.2:</b> Enabling environment for a low-emission climate-resilient MSME sector                                                                                                                                                         | <b>Activity 2.2.1:</b> Develop and implement MSME sectoral adaptation and mitigation benchmarking tools                                                   |
|                                                                                                               |                                                                                                                                                                                                                                                  | <b>Activity 2.2.2:</b> Digital Green Credit Rating Tool (DGCRT) includes integration of Green parameters (Energy and Environment) with Credit rating tool |

#### Component 1 — Term loans to support climate change adaptation and mitigation in MSMEs.

73. Under this component, ~10000 number of loans will be provided to MSMEs as detailed below:
- Executing Entity (SIDBI) will lend around 20% of the loans directly and another 80% of the loans will be financed indirectly through PFIs. SIDBI will provide technical handholding support to the selected Participating Financial Institutions (PFIs) i.e. Non-Banking Finance Companies (NBFCs), Small Finance Banks (SFBs), Micro Finance Institutions (MFIs) etc. to design and implement climate lending products within their asset portfolio, in line with the climate financial instruments of SIDBI. SIDBI will select the PFIs based on their asset portfolio and experience.
  - The MSMEs manufacturing units and service providers on either supply or demand side seeking to adopt low carbon measures and promote climate resilient technologies/ products/ services shall be eligible under the project. MSMEs executing projects / providing services, related to the activities eligible under the Scheme, to

various government bodies / private entities on BOT, BOOT, BOOM, PPP, etc. models backed by suitable contract shall be eligible for finance for their CAPEX and or OPEX requirements. The unit should be in compliance with the Environment & Social Management Framework of SIDBI being utilised currently for climate projects MSMEs will own, operate, and manage the assets financed under the programme.

- c. The activities under this component will be carried out by engaging technical support staff/experts.

#### **Programme Outputs and Activities:**

***Output 1.1: MSME mitigation technologies financed and implemented. Increased access to mitigation finance for MSMEs.***

##### **Activity 1.1.1: Establish a loan instrument to provide concessional loans for MSME mitigation needs.**

74. Debt instrument tailored to mitigation needs for improved access to finance in selected MSMEs. The key intervention of technologies to be financed under FMAP are (a). renewable Energy- PV systems Mostly Captive projects, up to capacity of 10 Megawatt and wind turbine up to 5 Megawatt of capacity (b). Electric vehicles and charging infrastructure, all type of vehicles 2w,3w and 4w vehicles will be covered under this programme (c). Energy efficiency- Industrial processes and appliances, insulation, high performance glass, LED lighting, High-Efficiency Heat Pumps, solar water heater etc. SIDBI has established standard operating guidelines for selection of MSMEs involved in the Climate /sustainable projects. The same SOPs would be followed while assessing an intended borrower, i.e., MSMEs for providing debt financing. A detailed list of relevant mitigation measures and technologies that will be eligible for finance is attached as Annexure.
75. The programme has been designed to offer concessional loans to PFIs (NBFCs/SFBs/MFIs) and driving them to further leverage 3x the lending amount into climate technologies in order to ensure that results are achieved at a scale. Through this, the programme intends to encourage the PFIs to create an ecosystem of climate financing which would allow them to lend to the MSMEs with an increased understanding of the associated risks and underwriting capabilities required. SIDBI's direct lending to the MSMEs sets an example for the PFIs to follow to successfully lend to the MSMEs for adopting low emission climate resilient technologies.

Other TA elements will include:

- Carry out detailed technical assessment of potential MSME units.
- Carry out credit risk rating evaluation of potential MSME units.
- Selection of MSMEs according to Eligibility Criteria as well as technical and risk assessment results.
- Incentivize the units to create climate MSME ambassadors and a group of champion units.
- Online loan application and speedy appraisal process setup.
- End to end technical and handholding support to MSMEs, PFIs, NBFCs, and other stakeholders.
- Facilitate networking of technology providers.

76. Activity 1.1.1 and 1.2.1 will jointly address the barriers related to: i) lack of private investment at the MSMEs level due to immature capital market and high upfront capital costs for climate products; and ii) limited understanding of MSME sector by institutional financial players. This will be achieved by (i). Offering concessional finance to MSMEs directly encouraging them to adopt relevant climate resilient technologies and interventions (ii). Offering concessional lines of capital and blended financial products to FIs which will help in reducing their risk of lending towards MSMEs thus encouraging them to offer additional financing (ii). Handholding by SIDBI helping FIs understand the relevant challenges and risks of financing MSMEs and developing the requisite underwriting skills to finance climate technologies within the MSME space. Additionally, Activity 1.1.1 will address barriers related to limited climate financing provided by banks and the need for training bank officials, enhancing climate risk assessment processes, and integrating monitoring, ESG, and gender frameworks. This will be achieved by SIDBI directly lending to MSMEs to demonstrate climate financing, aiming to encourage other financial institutions to also engage in climate financing for MSMEs.

***Output 1.2: MSME adaptation technologies financed and implemented. Increased access to adaptation finance for MSMEs.***

##### **Activity 1.2.1: Establish a loan instrument to provide concessional loans for MSME adaptation needs.**

77. A study by Climate Policy Initiative highlights the following barriers to adaptation financing in India (a). Limited knowledge of investment opportunities (b). Lack of capacity to develop commercially feasible projects (c) Unavailability of bankable projects and business models<sup>46</sup>. Furthermore, the adaptation interventions are local in nature and within India it has been

<sup>46</sup> [https://www.climatepolicyinitiative.org/wp-content/uploads/2024/02/Financing-Adaptation-India\\_reportannexes.pdf](https://www.climatepolicyinitiative.org/wp-content/uploads/2024/02/Financing-Adaptation-India_reportannexes.pdf)

observed that a large number of providers of such adaptive technologies/interventions are MSMEs. Hence, the objective of this activity is to create a debt instrument tailored for improved access to finance in select MSMEs that offer adaptive technologies/interventions for sectors such as agriculture, water resources, agroforestry, disasters and services. This programme will further focus on regions in India with high climate vulnerability: coastal, eastern belt and Himalayan regions and the focus sectors would be agriculture, water resources, agroforestry, disasters and services. An indicative list of relevant adaptation measures and technologies that will be eligible for loan finance is presented in Annexure 22.

78. The adaptation intervention under this programme attempts to tackle both the demand and supply side concerns. On the demand side, the MSMEs located in vulnerable regions are to be targeted where financing for climate resilient technologies is considered risky. The concessional loans here would prove to be useful to these vulnerable MSMEs to increase their resilience to climate hazards which would have long term direct impact on their operations. At present these MSMEs have very limited knowledge of technological solutions to make them climate resilient.
79. On the supply side, the programme targets financing those units that are value chain members and suppliers to larger units manufacturing technologies related to adaptation technology. Herein, focus will also be on financing MSMEs working on innovative solutions that manufacture or service locally suited, decentralised adaptation technologies.
80. Adaptation projects are considered to have limited bankability due to limited financial returns, lack of linkages with socio-economic returns, etc. In order to address these challenges this programme has been designed to offer concessional debt financing to bankable MSMEs who provide relevant adaptive technologies and services to wider adaptation projects.
81. In addition to lending through concessional debt, SIDBI intends to bring its expertise of different innovative financial instruments for adaptation needs.
82. Indicative list of adaptation solutions are as detailed below. The adaptation technologies are ever evolving especially owing to the increased number of climate induced hazards and alternative measures/technologies will be considered if they are deemed fit going forward.

| S.N o. | Adaptive Measure                                                                                                                          | Sectors                        |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 1      | More effective water treatment facilities and a range of other water related effort, including recycling and technologies to reduce water | Water                          |
| 2      | Various climate proofing products such as improved roofing and insulation                                                                 | Disaster resilience            |
| 3      | Software applications on mobile devices concerning climate related services                                                               | Disaster Response              |
| 4      | New construction materials, technologies, and processes as well as new approach to design and planning                                    | Disaster resilience            |
| 5      | Climate-related insurance offerings, from building damage to crop losses and cover for perishable goods                                   | Disaster response              |
| 6      | Other financial products from simple loans to help others pay for adaptation work, to climate derivatives and investment funds            | Agriculture                    |
| 7      | Risk management services                                                                                                                  | Agriculture/ Disaster Response |
| 8      | Weather and other information services                                                                                                    | Agriculture/Disaster response  |
| 9      | Supplying alternate raw materials for those that are at risk from climate change                                                          | Disaster Response              |

83. These indicative technologies are relevant across regions in India as showcased in the table below:

| Region       | Hazards                                                 | Risk/Impact                                                                                                                                  | Adaptation Interventions                                                                                                        | Technological |
|--------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------|
| Coastal Belt | Cyclones, Sea water intrusion, droughts, coastal floods | <ul style="list-style-type: none"> <li>More than 95 per cent of coastal Indian districts are vulnerable to extreme event hotspots</li> </ul> | <ul style="list-style-type: none"> <li>Weather forecasting and early warning systems to identify the climate related</li> </ul> |               |

|              |                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                                                                                    | <ul style="list-style-type: none"> <li>Human-induced climate change is causing more frequent and intense cyclones on India's west coast</li> <li>Damage to infrastructure</li> <li>Impact on Public Health- Loss of lives and livelihoods</li> <li>Drinking water scarcity</li> <li>Impact on yield/agricultural productivity</li> <li>Impact on the economy for mitigating the impact of these disasters</li> </ul> | <p>natural hazards in advance and to prevent loss in lives and livelihood</p> <ul style="list-style-type: none"> <li>Weather and other information services to predict natural hazards and to enable people to take timely adaptive actions</li> <li>Software application- Consumer software applications and enterprise software applications on mobile devices concerning climate related services. For example; Soil moisture sensors, LiDAR, drones, IoT based sensors, wireless mesh network,</li> <li>Crop breeding programs, biofertilizers and microbial pesticides</li> <li>Flood mitigation and relocation,</li> <li>Flood and other disaster response services,</li> <li>Floating agriculture</li> <li>Climate-smart precision irrigation systems and solutions- MSME units to be supported for developing and commercial scale deployment of irrigation solutions that reduce water usage and enhance the resilience of crops to climate risks.</li> <li>Manufacturing of raw materials</li> <li>Manufacturing of Climate innovation related products</li> <li>Water Efficient technologies to reduce overuse of water in drought prone areas such as rainwater harvesting, surface water storage, bioswales for managing stormwater, aquifer recharge</li> <li>Supplying alternative raw materials for those that are at risk from climate change</li> <li>Glacier-covering tarpaulins to prevent melting of glaciers</li> <li>Technologies to cater to the impact of climate induced migration like establishing temporary camps, manufacturing of tents, solar boats etc</li> </ul> |
| Himalayan    | Melting of glaciers, flash floods, cloudbursts, glacial lake outbursts, landslides | <ul style="list-style-type: none"> <li>Damage to infrastructure</li> <li>Impact on Public Health- Loss of lives and livelihoods</li> <li>Water shortage</li> <li>Impact on agriculture productivity</li> </ul>                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Eastern Belt | Floods                                                                             | <ul style="list-style-type: none"> <li>Two out of twelve states have highest vulnerability to climate change. Eight other States have moderate to high vulnerability to climate change</li> <li>Impact on agricultural productivity</li> <li>Impact on Public Health- Loss of lives and livelihoods</li> <li>Displacement and migration</li> </ul>                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

84. In addition to offering concessional finances, a range of TA activities to determine the feasibility of the adaptive measures/product and the associated MSMEs will be carried out. As was the case within mitigation, these activities will comprise of assessing the health of MSMEs viability of product in question and assisting the wider FI network in financing these interventions.

## **Component 2 — Technical support for climate change adaptation and mitigation in MSMEs**

### ***Output 2.1: Capacity built and knowledge management improved to support climate change adaptation and mitigation in MSMEs and PFIs. Capacities of selected MSMEs and PFIs are enhanced to rapidly scale their business and climate impact***

85. The programme will extend technical support to MSMEs and PFIs to address knowledge gaps on low carbon and climate resilient technologies. MSMEs will also be facilitated to develop bankable business proposals and access finance through the Fund of the project.
86. Capacity building and strengthening of other financial institutions is a continuous process adopted by SIDBI under various programmes such as PRSF wherein 15 banks/FIs/NBFCs are empanelled with SIDBI to support financing through performance contracting routes. While SIDBI conducts regular knowledge/capacity building programmes for other banks/FIs to support them in enhancing their portfolios and to strengthen their capacities to appraise loans for MSMEs, focused climate specific capacity building for MSMEs is the need of the hour. Hence, large scale capacity building programmes supporting the wider FI ecosystem in India to understand MSME climate financing landscape will be a critical part of the TA activities in this programme. Successful implementation of these activities is expected to play a key role towards the success of the larger programmes.
87. The best practices and learning of the programmes will be shared with larger audiences including other MSMEs through e-modules on a digital platform. The TA component will support in development of a repository of knowledge and learning established under the proposed program in digital form which will be maintained by SIDBI beyond project period and will be accessible by all stakeholders.
88. The proposed programme has been designed incorporating and building on learnings acquired by SIDBI over time (as showcased in preceding sections). Further, the program will build on the existing knowledge landscape developed by SIDBI under its ongoing schemes.
89. The necessity of these activities has been determined through multiple stakeholder conversations with players across the climate financing ecosystems including MSMEs, PFIs, local communities etc. Basis their inputs, we have considered numerous TA related activities that have been included in our proposal which are designed to address climate concerns in the long term. The activities under this component will be carried out by engaging technical support staff/experts.

### **Activity 2.1.1: Undertake baseline studies for 500 MSMEs**

90. MSMEs have limited knowledge and understanding of the low carbon and climate resilient technologies and need to be persuaded on the potential benefits of adoption of such technologies in terms of energy saving, GHG emission reduction, enhanced operational efficiency, long-term monetary benefits, etc. These technologies need assessments will be undertaken to:
- Identify the baseline parameters and current consumption of energy, water, etc.
  - Identify the applicable low carbon and climate-resilient technologies.
  - Demonstrate the savings possibilities in terms of specific energy consumption, GHG emission reductions, cost savings, and operational efficiency, etc.
  - Encourage MSMEs to adopt/manufacture these low carbon and climate-resilient technologies
  - Create an ecosystem for climate related services as part of their routine activity.
91. The purpose of the need assessment exercise has been conceived to address the uncertainties where there is no awareness of the baseline and expected environmental impact of technologies to be adopted. The need assessment exercise would enable an MSME to ascertain which climate technology best fits their circumstances.



92. To reach out to MSMEs working innovatively in the space of mitigation and adaptation and having potential for large scale impact, one of the interventions under this activity will be to hold a challenge fund to select such projects through a competitive process. This would also help in building the pipeline of such innovative projects under FMAP.
93. The technology assessment studies include feasibility and technical studies of specific technologies identified for adoption in a particular MSME, energy audits of MSME units, lifecycle analysis of products or services that help the MSMEs to assess their current resource and emission footprints, identify the potential low carbon and climate-resilient technologies and their benefits. These baseline studies will help in building awareness on the climate technologies and create niche market demand for its adoption among the MSMEs.
  - These assessments will be built on SIDBI's previous work on sectoral technology gap mapping of MSME clusters<sup>47</sup>
  - The technology need assessment is imperative for those MSMEs in regions of high climate vulnerability prone to climate hazards such as droughts, cyclones, floods etc so as to equip them with the knowledge of those technologies that can make their respective units resilient. The specific technologies financed under this programme will help the MSMEs to equip themselves against the climate hazards that they are most vulnerable to.
94. Activity 2.1.1 will address barriers related to: i) lack of awareness of relevant climate resilient technologies within MSMEs and ii) climate related data scarcity in MSMEs.

**Activity 2.1.2: Undertake awareness raising and trainings on climate change adaptation and mitigation for MSMEs and Financial Institutions.**

95. This activity will address the skills and capacity gaps in MSMEs, financial institutions in adopting climate technologies, climate risk management and climate investments. Awareness raising and training activities will be informed by the results of the baseline studies undertaken through Activity 2.1.1 and will be carried out to create awareness and disseminate the information about low carbon and climate-resilient technologies, as well as available financing amongst MSMEs. SIDBI will design and operate cluster-specific climate technology pilot demonstration interventions to promote demand for adoption of climate technologies (low carbon and climate resilient) invoking the neighbourhood theory. SIDBI through its regional and branch offices and associated agencies spread across India will also organise promotional activities on climate technologies and their co-benefits periodically to generate demand for adoption of climate technologies. These activities will target a range of stakeholders including MSMEs, Industry Associations, Banks / FIs, Equipment Manufacturers / Suppliers / Vendors, etc. These trainings will be imparted on various topics, including:

**MSMEs / Industry associations:** Low carbon, water efficient technologies, climate resilient technologies and their life cycle benefits. MSME participants will be sensitized through 16 programmes on various technological options.

**Banks/FIs:** Banking / MFIs / NBFCs organisations will be trained on Technical and Financial Appraisals of such proposals (Low carbon, water efficient technologies, climate resilient), risk mitigation measures, etc. through 8 training programmes.

**Equipment Manufacturers / Suppliers / Vendors:** Empanelment of at least 200 Manufacturers / Suppliers / Vendors for their different Low carbon and water efficient technologies and connecting them with MSMEs/IAs through a minimum of 8 technology exhibitions. Quality of equipment provided through the project's activities will be ensured through appropriate QA/QC criteria as per national quality management and technology standards for the MSME sector<sup>48</sup>

96. In Indian context, the technology or machine supplier provides guarantee/ warranty of equipment along with annual maintenance services to MSMEs. The issues related with QA/QC are usually taken care of by MSME. Machines will be financed under the proposed project where QA/QC will be ensured at the supplier level. SIDBI and the empanelled PFIs will also ensure that the warranty coverage of the equipment is checked at the level of due diligence conducted by MSMEs/PFIs. Further, the technology due diligence exercise shall also ensure that only those OEMs with extensive experience in after sales services shall be empanelled to provide after sale services for these technologies. The lenders are encouraged to check whether the technology to be funded confers to the standards of the Bureau of Indian Standards (BIS). The ISI mark is a standards-compliance mark for industrial products in India since 1950. The mark certifies that a

<sup>47</sup> [Cover\\_Single\\_Pages.cdr \(sidbi.in\)](https://www.sidbi.in/Cover_Single_Pages.cdr)

<sup>48</sup> Govt. of India. 2010. Enabling Manufacturing Sector to be Competitive through Quality Management Standards and Quality Technology Tools (QMS/QTT). Available at: <https://msme.gov.in/sites/default/files/QMSQTT10.pdf>

product conforms to an Indian standard (IS) developed by the Bureau of Indian Standards (BIS), the national standards body of India. An equivalent international standard would be followed for the imported machines as well.

97. Activity 2.1.2 will address barriers related to: i) lack of awareness of climate technologies and potential sources of climate finance; ii) lack of a strong MSME climate technology OEM ecosystem; and iii) lack in understanding of on ground practical benefits of climate resilient technologies. As with other TA activities, Activity 2.1.2 will address the need for training bank officials, enhancing climate risk assessment processes, and integrating monitoring, ESG, and gender frameworks.

**Activity 2.1.3: Implement knowledge management and sharing on climate change adaptation and mitigation for MSMEs**

98. The learning of the program will be disseminated amongst stakeholders through knowledge products to enable large scale capacity building within MSME sector. SIDBI will document the success stories, best practices and the learning of adoption of climate change mitigation and adaptation technologies within the MSMEs through case studies, documentaries, assessment studies. These multiple knowledge products including case studies, AV documentaries, brochures, flyers, etc. will be published and disseminated for wider circulation among MSMEs, Financial institutions, policy makers, and other relevant stakeholders. The learning of the programme will also be disseminated through workshops, trainings, information campaigns that further contribute to the uptake of low carbon technologies supported by the programme. The knowledge products including the reports of baseline studies developed under the programme will be made available for MSME stakeholders. These Knowledge Products will also be made readily available online in public domain to ensure wider dissemination.

**Output 2.2: Enabling environment for a low-emission and climate-resilient MSME sector.**

99. The ecosystem for innovative and customized products and services for building adaptive resilience of communities and vulnerable sectors lacks adequate policy support. The programme aims to strengthen MSMEs understanding of the climate ecosystem and shall enable large scale adoption of relevant climate technologies through innovative business models, products and services.

One of the key enabling factors for increased adoption of climate technologies is the availability of a clear taxonomy that defines 'what is' versus 'what is not' a relevant climate mitigating/adapting technology.

In this regard SIDBI's experience in financing MSMEs specially for mitigation has resulted in development of robust list of 900+ technologies which is attached as annexure in the Operations Manual. Similarly, through this programme, one of the objectives is to develop a similar list of adaptation technologies and disseminate the same for wider adoption.

Having a robust database of mitigation and adaptation technologies based on the need assessment of specific MSMEs will also enable the PFIs to take an informed decision with regards to financing the MSMEs for adopting low carbon and climate resilient interventions

**Activity 2.2.1: Develop and implement MSME sectoral adaptation and mitigation benchmarking tools.**

100. The programme will contribute to the upgradation and further development of SIDBI's existing online MSME benchmarking tool (earlier developed by SIDBI under one of the GEF project <sup>49</sup> covering database of almost all sectors and 800+ project reports). The tool provides a comparison of existing climate mitigation practices and savings potential in a particular type of industry. In this regard, through this exercise the tool shall incorporate climate adaptation practices as well going forward. Ultimately, the tool will help the MSME industry make informed decisions on adoption of appropriate low carbon and climate-resilient climate change adaptation and mitigation technologies.
101. The Financial Institutions will also be sensitised to use this benchmarking tool to bridge the gap in their understanding of applicable climate technologies within the MSME sector. Currently, lack of understanding among the PFIs about the mitigation and adaptation technologies applicable for MSMEs is also a deterrent for the funding of MSMEs to uptake these technologies by the PFIs.

<sup>49</sup> <https://prsf.sidbi.in/user/pages/viewpage/unnatee>

**Activity 2.2.2: Digital Green Credit Rating Tool (DGCRT) includes integration of Green parameters (Energy and Environment) with Credit rating tool**

102. This activity will support the digitization of an integrated credit rating tool (to be used across national, sub-national and local levels) which includes integration of climate parameters (Energy and Environment) into credit rating tool. This tool will provide a fair assessment to a FI of a loan applicant's potential to invest in low carbon and climate-resilient technologies. In doing so, the tool will act as a risk mitigation measure for FIs — the higher the rating, the lower the risk. Basis our stakeholder conversations, it is clear that PFIs lack the ability to accurately determine the risk associated with climate initiatives within the MSME sector. Thus, such an integrated credit tool will reduce information asymmetry for PFIs allowing them to make more informed investment decisions.

**B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)**

*Provide a description of the project/programme implementation structure, outlining legal, contractual, institutional and financial arrangements from and between the GCF, the Accredited Entity (AE) and/or the Executing Entity(ies) (EE) or any third parties (if applicable) and beneficiaries.*

*Provide information on governance arrangements (supervisory boards, consultative groups among others) set to oversee and guide project implementation. Provide a composition of the decision-making body and oversight function, particularly for Enhanced Direct Access (EDA) proposals.*

*Provide information on the financial flows and implementation arrangements (legal and contractual) between the AE and the EE, between the EE or any third party and beneficiaries. For EEs that will administer GCF funds, indicate if a Capacity Assessment has been carried out. Where applicable, summarize the results of the assessment.*

*Describe the experience and track record of the AE and EEs with respect to the activities (sector and country/region) that they are expected to undertake in the proposed project/programme.*

*Provide a diagram(s) or organogram(s) that maps such arrangements including the governance structure, legal arrangements, and the flow and reflow of funds between entities.*

**Implementation Arrangements**

**(a) FAA**

The GCF and the Accredited Entity will enter into a single FAA (Loan and Grant), which will reflect the terms and conditions described in the Term Sheet and the AMA.

**(b) Executing Entity**

The Accredited Entity will act as an executing entity for all Activities of the Programme.

**(c) Flow of Funds**

Under the FAA, the GCF will make available to the Accredited Entity the GCF Proceeds, in the form of loan and grant, for the implementation of the Programme. The GCF will disburse the GCF Proceeds into the GCF Account pursuant to the Accredited Entity's Requests for Disbursement, subject to fulfillment of applicable conditions precedent to the Disbursements.

The GCF Reimbursable Funds shall be provided by the GCF to the Accredited Entity pursuant to the key terms and conditions set out in the Term Sheet.

The GCF Non-Reimbursable Funds shall be provided by the GCF to the Accredited Entity pursuant to the key terms and conditions set out in the Term Sheet.

**(d) GCF Reimbursable Funds (GCF Loan)**

The Accredited Entity, acting as Executing Entity, will lend the GCF Reimbursable Funds, together with the Co-financing, to the MSMEs via two modalities: (i) loans to MSMEs and (ii) refinance/on-lending to PFIs for subsequent on-lending to the MSMEs.

**(i) Direct Loans**

The Direct Loans (i.e. loans from SIDBI to MSMEs) shall be materially consistent with the key terms and conditions set out in the Term Sheet.

**(ii) Indirect Loans and Sub-Loans**

The Indirect Loans (i.e. loans from SIDBI to PFIs) shall be materially consistent with the key terms and conditions set out in the Term Sheet.

The Sub-Loans (i.e. loans from PFIs to MSMEs) shall be materially consistent with the key terms and conditions set out in the Term Sheet.

The Accredited Entity, acting as the Executing Entity, will refinance/on-lend the GCF Reimbursable Funds, together with the Co-financing, to the PFIs. The Programme will offer refinance against eligible assets created by the empaneled PFIs and originated no longer than six months prior to approval from the Accredited Entity for financing in accordance with FAA. In addition, on-lending to PFIs for creating eligible assets within stipulated time from the date of disbursement shall also be permitted on a case-by-case basis, given compliance with Eligibility Criteria and Investment Criteria as outlined in the Term Sheet.

It is proposed to follow a three-step approach. First step is to identify and empanel those eligible PFIs which have interest in climate financing and are willing to adopt new systems processes. One of the initial conditions for empanelment would be that the PFI should agree to leverage 3X of SIDBI's financing amount.

Second step is to provide sensitization and handholding support to these empanelled PFIs as also provide sanction of loan assistance at concessional terms along with the term sheet which will specify the terms as per the FMAP programme following the requirements set forth in the Operations Manual including but not limited to Eligibility Criteria and Investment Criteria as set out in the Operations Manual, Annexure 6 ESMS due-diligence and Annexure 22 climate impact. The term sheet of the Indirect Loan Agreements shall also specify that PFIs would need to create 3X (during the first two years of the programme leverage of 3X may not be insisted. However, leverage of 3X will be ensured on a programme basis) of the portfolio of new MSME loans meeting all the conditions of the term sheet under the programme. PFIs would be required to accept the sanction terms and enter into loan documentation with SIDBI.

The third stage would be disbursement wherein PFI would be required to submit (for Refinance) all the details, copy of documentation, etc. to satisfy SIDBI that the conditions stipulated in the term sheet at the time of sanction have been fully complied with. The disbursement under on-lending could be considered, upfront, based on a declaration/ commitment by the PFI borrower for generating eligible portfolio/ Qualifying Assets within an agreed timeline (maximum upto 120 days from the date of respective disbursement) and conforming to the end use as per SIDBI's stipulation.

For avoidance of doubt, the GCF is taking the credit risk for its portion of the loan, with full recourse to SIDBI. SIDBI will take on the FX risk arising out of GCF loan and the credit risk of sub-projects under the programme.

SIDBI will be managing the currency risks through appropriate hedging mechanisms and may charge risk premium to the end borrowers

#### ***(e) GCF Non-Reimbursable Funds (GCF Grant)***

The GCF Non-Reimbursable Funds (GCF Grant) will be used by the Accredited Entity, acting as the Executing Entity, in order to conduct baseline/feasibility studies, capacity building and awareness, knowledge management, contribute to policy initiatives, M&V, impact evaluation. SIDBI will engage services of established firms for respective activities via service contract.

The GCF Non-Reimbursable Funds shall also cover the expenses of the PMU, which scope includes capacity building and monitoring of PFIs, along with the Technical Assistance activities.

There will be no on-granting of the GCF Non-Reimbursable Funds by the Accredited Entity.

#### ***(f) Eligibility Criteria***

The eligibility criteria of all the technologies (mitigation and adaptation) financed under the programme is outlined in the Operations Manual.

#### ***(g) Contracting Structure***

For Direct Loans, the Accredited Entity, acting in its capacity as the Executing Entity, will enter into legally binding loan agreements with the MSMEs, which will pass down the relevant GCF requirements and obligations to the MSMEs, including policy requirements set out in the AMA and FAA, and which shall be based on the key terms and conditions set out in the Operations Manual.

For Indirect Loans, the Accredited Entity, acting in its capacity as the Executing Entity, will enter into legally binding loan agreements with the PFIs, which will pass down the relevant GCF requirements and obligations to the PFIs, including policy

requirements set out in the AMA and FAA, and which shall be based on the key terms and conditions set out in the Operations Manual

For Sub-Loans, the PFIs will enter into legally binding loan agreements with the MSMEs, which will pass down the relevant GCF requirements and obligations to MSMEs, including policy requirements set out in the AMA and FAA, and which shall be based on the key terms and conditions set out in the Operations Manual.

**(h) Host Country**

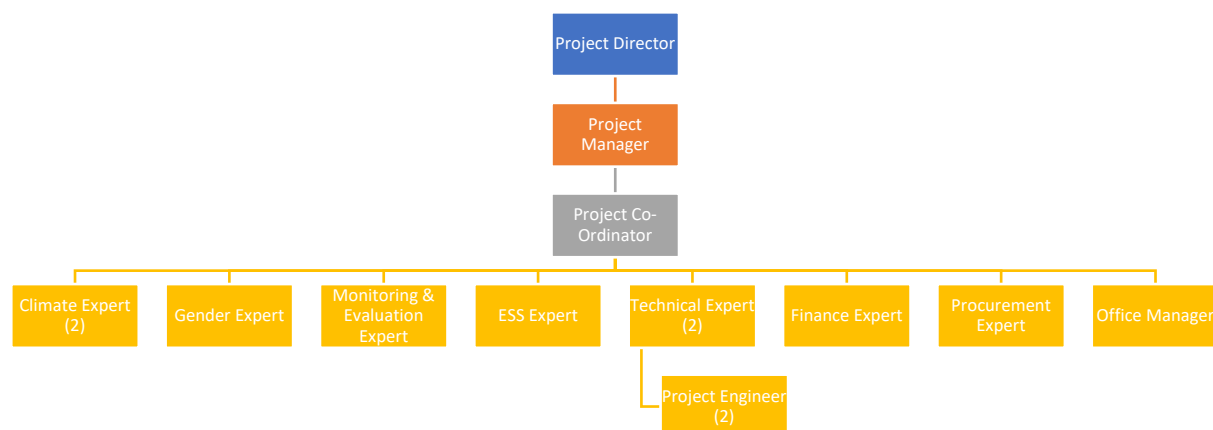
The host country for the Programme is the Republic of India.

**(i) Programme Management Unit**

Given the Pan- India expanse of the programme, it is necessary to have a centralised governance and monitoring for smooth implementation of the Programme. For the project's administration and implementation, SIDBI (Executing Entity) has identified a multi-vertical core team which will serve as the Programme Management Unit (PMU). The PMU, headed by the Project Director consists of total 15 experts and shall provide sensitisation / capacity building as well as technical hand holding support to EE and the PFIs regarding assessment / compliance of climate related aspects, technical eligibility norms, environment, social due diligence as per the management framework, monitoring, evaluation and reporting of the climate aspects and environment, social parameters of the sub loans. The PMU shall not be involved in origination, credit due diligence, credit appraisal, KYC, credit decision of sub- loans, disbursement of sub- loans, loan documentation, follow-up, monitoring, recovery, accounting, etc. The EE and PFIs shall utilise their own staff for all these activities. For better clarification, it may be mentioned that the PMU staff shall only be supporting/ providing technical hand holding for additional due diligence/compliances required for climate financing which are over and above the normal activities which are required to be undertaken for providing normal MSME loans. The PMU resources shall have adequate technical as well as fiduciary and safeguards experience. The E&S aspects will be coordinated and managed by an expert who shall be the part of the PMU. He/she shall be supported by an E&S specialist (ESS) dedicated to the safeguards function.

The PMU shall be responsible for the execution of day-to-day activities of the programme and report to the AE about the progress of the FMAP. The primary responsibility of the PMU includes providing assistance to the Direct Credit Vertical (for direct lending) and NBFC vertical (for indirect lending) for the execution of the component 1. The PMU shall be responsible for providing technical assistance to the PFIs in line with the GCF guidelines and the FAA. One of the core responsibilities of the PMU includes the capacity building of the PFIs & MSMEs and sensitising the sector about low-emission and climate-resilient technologies. The team shall engage in TA activities including capacity building and knowledge management etc as well as overall co-ordination for the component 1 (including due diligence aspects) in the initial preparatory and loan disbursement phase. In the Second phase i.e. the Loan repayment phase, the team shall be responsible for the monitoring and reporting of climate, environment and social aspects as per the M&E log frame, technical hand holding and other necessary engagements with the stakeholder. The PMU shall regularly report to the AE (as described in FAA) on the progress of the FMAP programme and also of the Grievances or issues related to ESS or gender (if any).

**Composition of PMU is as below:**

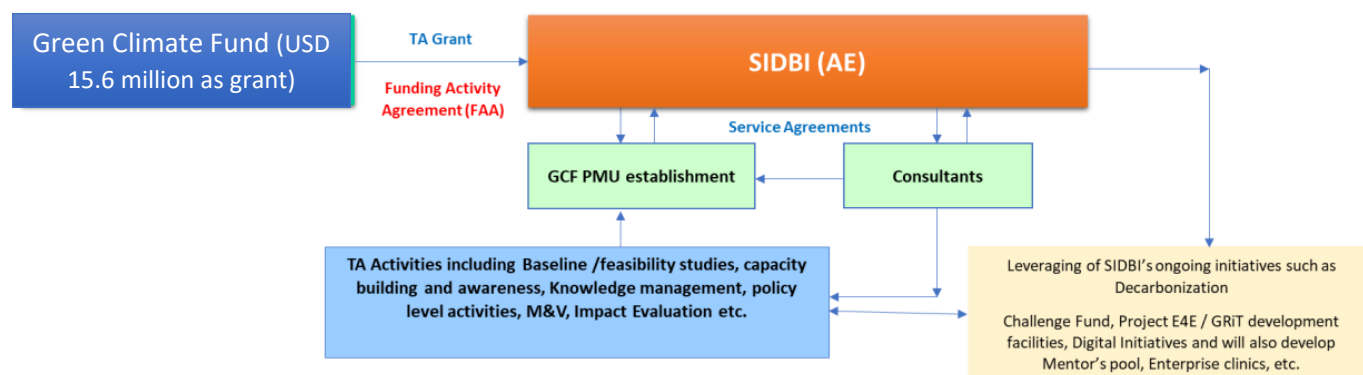




### Loan Approval Process-

103. The Loan Approval Process is a comprehensive three-stage procedure designed to rigorously evaluate loan applications and ensure adherence to the stringent criteria. The first stage, Application and Initial Screening, is conducted at the branch level by the Relationship Manager. During this phase, applications are meticulously screened for completeness, including the submission of mandatory documentation such as climate-related, environmental, and social safeguards documents. Incomplete applications are promptly rejected/ hand held to be completed, as the case may be, while only those deemed complete undergo initial screening against the exclusion list criteria. Applications that meet the criteria are forwarded to the second stage, Due Diligence, while those that fall under the exclusion list are rejected. The Due Diligence stage is executed in parallel at the branch level and involves several crucial assessments. The Processing Officer conducts evaluations related to Know Your Customer (KYC) requirements, financial viability, credit due diligence and checking of ESDD checklist. Concurrently, the Climate Expert, Technical Expert and ESS expert at the Project Management Unit (PMU) will perform climate-related, technical and ESS due diligence to ensure the soundness of the proposed project from a technical and climate perspective. In the third stage, technically sound and financially viable applications proceed to the Credit Appraisal Memorandum preparation phase. The Processing team comprising of Maker and Checker compiles a comprehensive Credit Appraisal Memorandum, which is then presented to the Loan Sanctioning Committee for final evaluation and decision-making. To maintain transparency and impartiality, no member involved in the Initial Application Screening and Due Diligence processes is permitted to be part of the Loan Sanctioning Committee.

### Technical Assistance Flow of Funds



104. GCF will enter into a single Funded Activity Agreement with SIDBI for implementation of grant worth USD 15.6 million for technical assistance and loan worth USD 200 million

105. The objective of the technical assistance would be to conduct baseline/feasibility studies, capacity building & awareness, knowledge management, policy level activities, M&V, impact evaluation and SIDBI will engage services of established firms for respective activities.

106. The grant shall cover the expenses of the experts/staff, Project Management Cost along with the Technical Assistance activities.

### Experience and track record of SIDBI in implementing climate schemes:

An assessment of SIDBI's previous experience in lending to MSMEs in climate financing is detailed out in section B2, further scheme specific details is given in this section. SIDBI acts as the Principal Financial Institution dedicated for the Micro, Small and Medium Enterprise (MSME) sector as well as for coordination of functions of institutions engaged in similar activities. Apart from the regular business of financing to MSME units, SIDBI has also developed a special division with a focus on sustainable financing for the MSME sector. SIDBI has been associated with different national and international development agencies for the last 15 years.

### B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

*Explain why the project/programme requires GCF funding to address mitigation or adaptation measures, i.e. Why is the project/programme not currently being financed by public and/or private sector? Which market failure is being addressed with GCF funding? Are there any other domestic or international sources of financing?*

*Explain why the proposed financial instruments were selected in light of the proposed activities and the overall financing package. i.e. What is the coherence between activities financed by grants and those financed by reimbursable funds? How were co-financing amounts and prices determined? How does the concessionality of the GCF financing compare to that of the co-financing? If applicable, provide a short market read on the prevailing of the pricing and/or financial markets for similar projects/programmes.*

*Justify why the level of concessionality of the GCF financial instrument(s) is the minimum required to make the investment viable. Additionally, how does the financial structure and the proposed pricing fit with the concept of minimum concessionality? Who benefits from concessionality?*

*In your answer, please consider the risk sharing structure between the public and private sectors, the barriers to investment and the indebtedness of the recipient. Please reference relevant annexes, such as the feasibility study, economic analysis or financial analysis when appropriate.*

Several reports have highlighted the gaps in climate finance as highlighted below:

- (a). India stares at a "funding gap" of \$10.1 trillion to achieve its net zero emissions commitment by 2070<sup>50</sup>
- (b). Uneven distribution of climate finance source and need to leverage private investments - Of the \$44 billion raised in climate financing in 2020, 40 per cent was funded directly or indirectly through public finance<sup>51</sup>
- (c). Over-reliance on domestic financing, with nearly 80 per cent of funds raised coming from domestic sources<sup>52</sup>

As per the findings of the survey by RBI vide its notification dated July 27, 2022 and also as indicated in RBI's Report on Currency and Finance 2022-23 issued on May 03, 2023, it is observed that the financing flows from the banking sector towards climate financing are presently quite low and there is a need for training, sensitisation and capacity building of the banking officials in this regard as also putting in place proper systems and processes towards climate risk assessments and integrating suitable framework for M&E, ESG & gender, etc. The report highlights the following:

- (a) Only 24% of respondent banks have set their timelines to achieve carbon neutrality
- (b) Few banks have taken initiative to expedite climate lending
- (c) Only 8/34 respondent banks have separate verticals for Sustainable Finance
- (d). 1 in 3 respondent banks- no internal accountability for managing climate change & sustainability
- (e). Only 53% of respondent banks have an ESG strategy in place

As highlighted by RBI in their survey, there is a need to bring in transformation in the Indian Banking System to boost climate financing by PFIs especially targeted at financing low emission, climate resilience interventions in MSMEs. This programme aims to bring in this transformation by building the capacities of PFIs to lend to MSMEs with increased confidence.

#### Catalyzing climate financing in the MSME space

107. Today, green lending towards MSMEs are restricted to a significantly small portion of the larger lending portfolios of financial institutions in India – limited to ~1% of the total book size

108. Indian FIs are reluctant to finance MSMEs in the climate space primarily due to their:

- a. Lack of understanding of tools/mechanism to underwrite the risks associated with deployment of climate technologies in the MSME space

<sup>50</sup> <https://economictimes.indiatimes.com/news/economy/finance/india-faces-10-trillion-funding-gap-in-bid-to-meet-net-zero-pledge-fm-nirmala-sitharaman/articleshow/106739207.cms?from=mdr>

<sup>51</sup> <https://www.downtoearth.org.in/blog/climate-change/scaling-up-private-finance-is-crucial-to-bridge-funding-gap-for-climate-change-projects-95042#:~:text=Apart%20from%20the%20staggering%20finance,or%20indirectly%20through%20public%20finance.>

<sup>52</sup> *ibid*

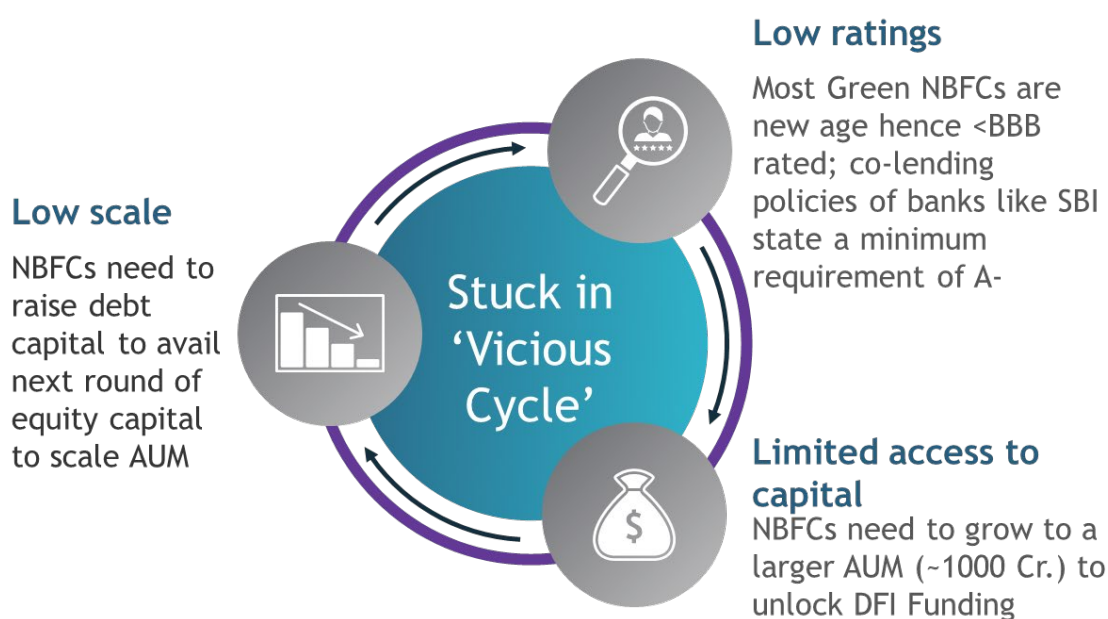
- b. Inability to create scale – individual MSMEs present very small/disaggregated opportunities – for effective adoption of various climate technologies (e.g. RE or EE)

Thus, highlighting a lack of alternative risk management/underwriting tools and innovative financing instruments that are more tailored towards the deployment of climate technologies within an MSME's operating environment

109. Under these circumstances, the landscape has witnessed the rise of medium-sized and smaller NBFCs (including focused Green NBFCs) that are dedicatedly financing MSMEs in the climate space. As can be observed from the figure above, these NBFCs are playing a critical market making role rather than larger commercial banks or general purpose NBFCs

- a. The MSME focused climate lending carried out by these new age BBB rated green NBFCs are primarily geared towards electric mobility but with increasing lending towards RE deployment particularly solar rooftop as well.

110. However, these NBFCs, particularly Green focused NBFCs are stuck in a vicious cycle, inhibiting their ability to scale, as showcased in the figure below:



111. Consultations with Green NBFCs has revealed the need for an ecosystem enabling environment that offers support across the following areas:

- Co-lending partnerships with medium/small NBFCs** to expand last mile reach
- On-lending partnerships with medium/small NBFCs against a defined use of proceeds** (e.g., *financing fleet aggregators*)
- Partnering with global DFIs to effect the development of a dedicated co-lending infrastructure** to aid discovery and execution of co-lending deals
- Devising a securitization policy** to play a market making role for securitization of climate assets all activities which unlock financing from larger banks and NBFCs who now start extending debt lines of capital to these medium/small NBFCs on the back of the market created by them

112. SIDBI has actively taken up the role of creating this ecosystem enabling environment to help scale medium-/small-sized NBFCs with dedicated climate lending towards MSMEs.

113. Through this programme, SIDBI intends to scale this catalytic effect for medium/small scaled NBFCs across the country, by providing

- a. requisite concessional lines of credit to finance climate interventions within MSMEs thus creating an MSME driven market of climate technologies
- b. further build capacity of these organizations to deploy climate technologies within MSMEs on the back of SIDBI's extensive knowledge, expertise and existing risk management/underwriting tools that can be used to effectively finance the Indian MSME sector

114. For MSMEs, deployment of climate technologies and interventions are primarily a question of their ability to drive long-term investments with large capital requirements – which most MSMEs in India, even with an understanding of the significance and necessity of these technologies, are unable to deploy primarily due to a lack of:

- a. Access to concessional capital at rates lower than traditional market rates (which is usually offered for general purpose projects in MSMEs) which makes climate investments unviable – for example, consultations with stakeholders across the MSME ecosystem has revealed that in absence of concessional finance, 60% of the customers are unlikely to purchase new EE machines and shall continue to resort to purchasing older second-hand machines which are less efficient but cheaper.
- b. Innovative financing models and business practices that circumvent the need for large CAPEX investments (as is the norm with infrastructure projects) or risk guarantees to FIs while providing more attractive norms for MSMEs in the form of longer tenors, lower TATs, easier processing cycles, etc.
- c. On a broader sectoral level, a clear understanding of the on-ground, real world benefits of any climate technology especially in terms of financial benefits (including a translation of cost savings into potential revenue/profits) which makes the use of the technology in question attractive thus activating the “Neighbourhood Effect” amongst the wider segments

115. SIDBI through this programme intends to tackle all of these challenges by creating a range of services through the concessional loans it offers, including:

- a. Direct lending to select MSMEs to create showcases of the real-world benefits of deploying climate interventions within the MSME landscape and disseminating success stories across the landscape to activate the “Neighbourhood Effect”
- b. Leverage wider FI network to create a much larger ecosystem of concessional lending towards MSMEs adopting climate technologies by leveraging the catalytic financing offered through the programme as well as creating new financial products that are designed specifically to meet the needs of the MSME sector
- c. Enable deployment of innovative facilities such as RESCOs, on-bill financing through DISCOMs, etc. designed to create large scale demand amongst MSMEs for climate technologies

#### **Concessionalities of the funding investments:**

- To encourage the industry and businesses to choose energy-efficient equipment, which in many cases, is more expensive compared to conventional technology. Hence, there is a need to reduce the cost differences by providing concessional lending for energy efficient equipment. Although energy-efficient equipment may become economical on lifecycle cost basis, equipment buyers are most likely to decide solely based on investment cost, resulting in choosing cheaper, conventional non- energy-efficient equipment. Against this background, the Project aims to provide a low interest loan to the buyers so that total payment amount for energy efficient equipment (equipment cost and financial cost) will break even with the cost of acquiring conventional equipment.
- It is felt that there is a need of 1% to 3% concessionalities in the interest rate in order to breakeven the additional cost of interest of EE and RE/PV equipment with that of cheaper traditional equipment as well as the additional compliance costs pertaining to environment and social management framework. The price difference between conventional and efficient machines ranges from 25%-100%.
- Similarly, for EV there is a need of 3% to 5% concessionalities in the interest rate in order to breakeven the additional cost of interest of EV with that of cheaper traditional vehicles as well as the additional compliance costs pertaining to environment and social management framework. The price difference between conventional and efficient vehicles ranges from 40 to 300% depending upon vehicle category<sup>53</sup>
- In case of adaptation, most Indian companies are yet to completely grasp the implications of climate change and how it will affect them. This is particularly true for the Indian MSME sector where managing the immediate challenges of

<sup>53</sup> [https://www.sidbi.in/head/uploads/greenpathwayseseries\\_documents/green-series-de-risking-lending.pdf](https://www.sidbi.in/head/uploads/greenpathwayseseries_documents/green-series-de-risking-lending.pdf)

everyday business remains a higher priority than addressing long term risks. At the same time, the MSME sector is also most vulnerable to climate risks since most within the sector lack the financial resources to endure even a single rupture in supply or demand, caused for instance by an extreme weather event or a new environmental or climate regulation. Concessional loans in this case would motivate the vulnerable MSME units to take up adaptation technologies to enhance climate resilience of their units.

- The financial model (Annexure 3 of the Funding Proposal) clearly demonstrates that to ensure the feasibility of the project, concessional support is required from GCF to SIDBI to enable SIDBI to extend loan at the low interest to MSMEs.
- Based on Capital Investment for each project ticket size – currently SIDBI is financing these loans from 75%-100% of total project cost. The loan amount typically ranges from INR 1 million to INR 500 million. The balance amount has to be borne by borrowers (say 0%-25% of the project cost).
- The concessional in terms of interest rate to MSMEs for their climate proposals with the help of blending other funds with GCF Funds is envisaged. The final prevailing concession in interest rate to each borrower would be decided based on credit rating of the MSME unit. This blending concept is a proven concept which has been earlier used by SIDBI.
- The Interest Rate shall be calculated as follows:
  - a. Determine the Accredited Entity's average cost of funding at prevailing market rates for similar loans.
  - b. Determine the blended funding cost for Direct and Indirect Loans provided under this Programme.
  - c. Subtract the blended funding cost from the average market rate cost to determine the Direct and Indirect Loan Concessionality Margin.
  - d. The concessional interest rate to be charged to the MSMEs shall be the blended funding cost plus any allowable administrative margin, ensuring the Concessionality Margin is passed on.

#### **B.6. Exit strategy (max. 500 words, approximately 1 page)**

*Explain how the project/programme will successfully exit once implementation is completed, including how results and benefits will continue beyond the project/programme period and how the contribution to paradigm shift will be maintained.*

*Include information pertaining to the longer-term ownership, project/programme exit strategy, operations and maintenance of investments (e.g. key infrastructure, assets, contractual arrangements). In case of private sector, please describe the GCF's financial exit strategy through Initial Public Offerings, trade sales, etc.*

*Provide information on additional actions to be undertaken by public and private sector or civil society as part of the project/programme to ensure sustainability of the results attained.*

#### **Creating a market for long-term climate lending to MSMEs**

The programme has been designed to encourage wider Indian FI landscape to increasingly finance climate technologies and initiatives within the MSME sector by creating a market for climate lending for MSMEs.

To do so, the programme shall create an enabling environment that helps FIs understand the Indian MSME sector, particularly its risks and the underwriting skills required to finance it effectively. A range of activities including capacity development and training, knowledge sharing and awareness building, provision of credit risk assessment tools, etc. will be deployed to help Indian FIs understand and manage the risks associated with the Indian MSME landscape. Furthermore, detailed databases containing lists of applicable climate-related technologies (mitigation and adaptation) will further ensure that FIs are completely aware of the risks and benefits of the climate technologies that they are financing.

In addition to this enabling environment, the concessional finance offered by the programme towards capitalizing small finance banks, NBFCs and MFIs, is expected to create a catalytic effect and potentially leverage upto an additional USD 2,400 million (to be ensured on a programme basis). Successful deployment of this large corpus towards climate technologies shall effectively showcase the bankability of the climate interventions within the MSME sector, thereby unlocking further climate financing from the wider Indian FIs (including large banks and NBFCs) far beyond the lifecycle of the current programme.

#### **Making climate technologies attractive for MSMEs**

The programme has been designed to tackle the challenges faced by MSMEs in adopting climate technologies, by creating a range of services covering:



- Concessional lending directly to select MSMEs to create showcases of the real-world benefits of deploying climate interventions within the MSME landscape and disseminating success stories across the landscape.
- Creating new financial products that are designed specifically to meet the needs of the MSME sector and leveraging wider FI network to create an ecosystem of concessional lending towards MSMEs adopting climate technologies.
- Enable deployment of innovative facilities such as RESCOs, on-bill financing through DISCOMs, etc. designed to create large scale demand amongst MSMEs for climate technologies.

All of which, when deployed together creates an environment which is designed to activate the neighbourhood effect to its fullest capacity within the MSME sector, making climate technologies a very attractive tool to benefit and drive revenue generation/profitability within the MSME. This ensures that going forward, beyond the life of the programme, the MSME sector remains dedicated towards the adoption of climate technologies in the long-term.

Finally, the activities proposed under this programme, for both financiers as well as MSMEs are designed to be sector and/or region agnostic which ensures that it is replicable, at scale within the country.

### **Sustainability and Replicability of the programme**

The activities suggested under the programme are designed considering their sustenance beyond programme period and is as described below:

a. The investment in around ~10000 climate loans under the programme would help in mainstreaming these amongst lenders for segments otherwise perceived as unviable or highly risky. Also, concessional loans is being used to trigger a dormant/slow moving market of climate technologies especially w.r.t MSMEs. According to several studies the following is required to stir the MSMEs to adopt climate technologies:

- Concessional funding which is possible through innovative financing model
- Capacity building/training/awareness generation for both the borrowers and lenders
- Economies of scale where implementation of a technology/ies at a large scale would lead to spill over effect. For example: large scale adoption of any technology would also ensure a market for its services
- Knowledge management and e-modules would address gaps in information asymmetry

Hence, sustainability and replicability will not just be attained by provisioning loans but all the other related activities of this programme is aimed at achieving the same.

b. Also, the proposed digital integrated climate credit rating tool as one of policy instruments will further strengthen the credibility of MSMEs proactive in their approach towards environment and social aspect thus bringing such activities as part of their business priorities. This will also help the PFIs to conduct loan appraisal/ financial appraisal of MSMEs by not just considering the credit worthiness aspects but also how the MSMEs are performing in various environmental and social aspects.

c. The knowledge repository developed during the project implementation will help other stakeholders including FIs and Banks, MSMEs etc. for taking decisions for the adoption of such clean and climate technologies and their financing. The knowledge management and capacity building activities under the programme will help overcome existing "Risk Averse" nature of MSMEs wherein they tend to "follow rather than taking a lead". The vendor development activity will help creating knowledge about the latest available technologies among MSMEs, at the same time help vendors in developing their local network which is quite essential for the penetration of technologies in MSMEs.

d. Training of Banker/FIs will help in appraisal of technical proposals for the proposed areas of interventions will support in overcoming of the perceived risk. E-modules and uploading the same will ensure that the modules are readily available for further trainings. Further, the banking association such as Indian Bank's Association (IBA) will also be encouraged to confer trainings for all the banks thus ensuring wide scale reach.

f. The programme will continue financing for climate change mitigation and adaptation for MSME sector beyond project duration to support other financial institutions in the country. This will help sustainability and replicability of the programme through entering into financing ecosystem of the country. This will also support other Indian financial institutions to develop standard loan products/instruments for MSME climate actions. SIDBI will seek additional resources from market to replenish the financing to fulfil the ultimate objective of mainstreaming the climate financing in MSME sector. The pilot projects developed as part of this programme will further be mainstreamed to enhance climate financing by banking sector for MSMEs.

## C. FINANCING INFORMATION

### C.1. Total financing

|                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                            |                     |                         |                                          |               |                   |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------|------------------------------------------|---------------|-------------------|
| (a) Requested GCF funding<br>(i + ii + iii + iv + v + vi + vii)                             | Total amount                                                                                                                                                                                                                                                                                                                                                                                                               |                     |                         | Currency                                 |               |                   |
|                                                                                             | 215.60                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |                         | million USD (\$)                         |               |                   |
| GCF financial instrument                                                                    | Amount                                                                                                                                                                                                                                                                                                                                                                                                                     | Tenor               | Grace period            | Pricing                                  |               |                   |
| (i) Loans                                                                                   | <u>200</u>                                                                                                                                                                                                                                                                                                                                                                                                                 | 13 years            | 6 years                 | <u>As per termsheet</u>                  |               |                   |
| (ii) Subordinated loans                                                                     | <u>Enter amount</u>                                                                                                                                                                                                                                                                                                                                                                                                        | <u>Enter years</u>  | <u>Enter years</u>      | <u>Enter %</u>                           |               |                   |
| (iii) Equity                                                                                | <u>Enter amount</u>                                                                                                                                                                                                                                                                                                                                                                                                        |                     |                         | <u>Enter % equity return</u>             |               |                   |
| (iv) Guarantees                                                                             | <u>Enter amount</u>                                                                                                                                                                                                                                                                                                                                                                                                        | <u>Enter years</u>  |                         |                                          |               |                   |
| (v) Reimbursable grants                                                                     | <u>Enter amount</u>                                                                                                                                                                                                                                                                                                                                                                                                        |                     |                         |                                          |               |                   |
| (vi) Grants                                                                                 | 15.6                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |                         |                                          |               |                   |
| (vii) Results-based payments                                                                | <u>Enter amount</u>                                                                                                                                                                                                                                                                                                                                                                                                        |                     |                         |                                          |               |                   |
| (b) Co-financing information                                                                | Total amount                                                                                                                                                                                                                                                                                                                                                                                                               |                     |                         | Currency                                 |               |                   |
|                                                                                             | 3577                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |                         | Options                                  |               |                   |
| Name of institution                                                                         | Financial instrument                                                                                                                                                                                                                                                                                                                                                                                                       | Amount              | Currency                | Tenor & grace                            | Pricing       | Seniority         |
| SIDBI                                                                                       | <u>Loans</u>                                                                                                                                                                                                                                                                                                                                                                                                               | <u>800</u>          | <u>million USD (\$)</u> | <u>TBD</u>                               | <u>TBD</u>    | <u>pari passu</u> |
| NBFCs/PFIs                                                                                  | <u>Loans</u>                                                                                                                                                                                                                                                                                                                                                                                                               | <u>2400</u>         | <u>million USD (\$)</u> | <u>TBD</u>                               | <u>TBD</u>    | <u>pari passu</u> |
| MSMEs                                                                                       | <u>Equity</u>                                                                                                                                                                                                                                                                                                                                                                                                              | <u>377</u>          | <u>million USD (\$)</u> | <u>TBD</u>                               | <u>TBD</u>    | <u>Options</u>    |
| Click here to enter text.                                                                   | <u>Options</u>                                                                                                                                                                                                                                                                                                                                                                                                             | <u>Enter amount</u> | <u>Options</u>          | <u>Enter years</u><br><u>Enter years</u> | <u>Enter%</u> | <u>Options</u>    |
| (c) Total financing<br>(c) = (a)+(b)                                                        | Amount                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |                         | Currency                                 |               |                   |
|                                                                                             | <u>3792.6</u>                                                                                                                                                                                                                                                                                                                                                                                                              |                     |                         | <u>million USD (\$)</u>                  |               |                   |
| (d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page) | <p>Please explain if any of the financing parties including the AE would benefit from any type of guarantee (e.g. sovereign guarantee, MIGA guarantee).</p> <p>Please also explain other contributions such as in-kind contributions including tax exemptions and contributions of assets.</p> <p>Please also include parallel financing associated with this project or programme (refer to the co-financing policy).</p> |                     |                         |                                          |               |                   |

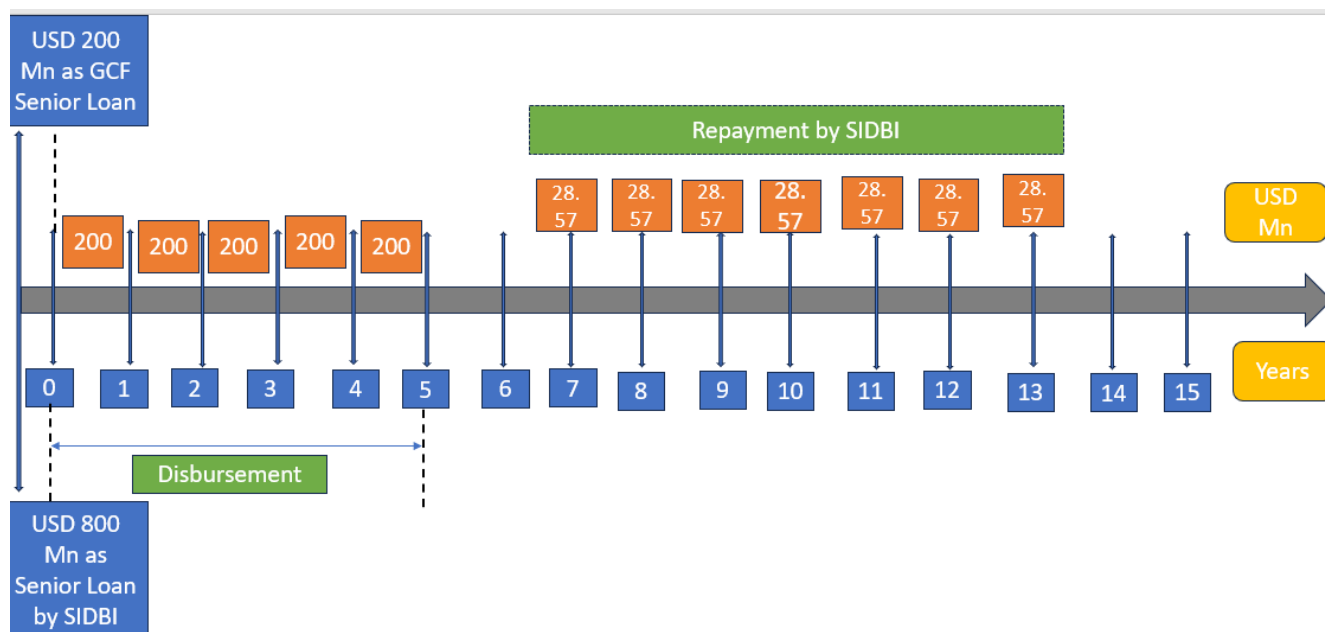
### C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below and the detailed budget plan as annex 4.

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

In case of a multi-country/region programme, specify indicative requested GCF funding amount for each country in annex 17, if available.

Proposed Financing Structure (tentative):



### C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

| Component                                 | Output                                                                                                        | Indicative cost Options | GCF financing  |                      | Co-financing      |                        |                              |  |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------|----------------|----------------------|-------------------|------------------------|------------------------------|--|
|                                           |                                                                                                               |                         | Amount Options | Financial Instrument | Amount Options    | Financial Instrument   | Name of Institutions         |  |
| Component-I<br>Concessional<br>Term Loans | Mitigation-Output 1.1: EE loan                                                                                | 944                     | 50             | Loan                 | 200<br>600<br>94  | Loan<br>Loan<br>Equity | SIDBI<br>NBFCs/PFIs<br>MSMEs |  |
|                                           | Mitigation-Output 1.2: RE Loan                                                                                | 944                     | 50             | Loan                 | 200<br>600<br>94  | Loan<br>Loan<br>Equity | SIDBI<br>NBFCs/PFIs<br>MSMEs |  |
|                                           | Mitigation-Output 1.3: EV loan                                                                                | 378                     | 20             | Loan                 | 80<br>240<br>38   | Loan<br>Loan<br>Equity | SIDBI<br>NBFCs/PFIs<br>MSMEs |  |
|                                           | Adaptation-Output 1.4: Adaptation Loan                                                                        | 1511                    | 80             | Loan                 | 320<br>960<br>151 | Loan<br>Equity         | SIDBI<br>NBFCs/PFIs<br>MSMEs |  |
| Component-II<br>Technical Assistance      | 2.1. Capacity built, and knowledge management improved to support climate change adaptation and mitigation in | 4.74                    | 4.74           |                      |                   |                        |                              |  |

|                                    |                                                                                                                       |               |       |       |      |  |  |  |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------|-------|-------|------|--|--|--|
|                                    | MSMEs and PFIs. Capacities of selected MSMEs and PFIs are enhanced to rapidly scale their business and climate impact |               |       |       |      |  |  |  |
|                                    | 2.2. Enabling environment for a low-emission and climate-resilient MSME sector                                        | 4.21          | 4.21  |       |      |  |  |  |
|                                    | 2.3. Monitoring and Reporting                                                                                         | 2.71          | 2.71  | Grant |      |  |  |  |
|                                    | 2.4. Operations Cost                                                                                                  | 0.48          | 0.48  | Grant |      |  |  |  |
| <b>Indicative total cost (USD)</b> |                                                                                                                       | <b>3792.6</b> | 215.6 |       | 3577 |  |  |  |

*If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and quantify the total requested GCF funding amount for these activities, to the extent possible*

**The programme will support capacity building activities through Component 2 (Output 2.1 and Output 2.2) as detailed below:**

***Output 2.1: Capacity built and knowledge management improved to support climate change mitigation and adaptation in MSMEs and PFIs. Capacities of selected MSMEs and PFIs are enhanced to rapidly scale their business and climate impact***

***Output 2.2: Enabling environment for a low-emission and climate-resilient MSME sector***

125. The program will extend technical support to MSMEs and PFIs to address knowledge gaps on low carbon and climate resilient technologies. MSMEs will also be facilitated to develop bankable business proposals and access finance through the Fund of the project.
126. Capacity building and strengthening of other financial institutions is a continuous process adopted by SIDBI under various programs such as PRSF wherein 15 banks/FIs/NBFCs are empanelled with SIDBI to support financing through performance contracting routes. While SIDBI conducts regular knowledge/capacity building programs for other banks/FIs to support them in enhancing their portfolios and to strengthen their capacities to appraise loans for MSMEs, focused climate specific capacity building for MSMEs is the need of the hour. Hence, large scale capacity building programs supporting the wider FI ecosystem in India to understand MSME climate financing landscape will be a critical part of the TA activities in this programme. Successful implementation of these activities is expected to play a key role towards the success of the larger programme.
127. The best practices and learning of the program will be shared with larger audiences including other MSMEs through e-modules on a digital platform. The TA component will support in development of a repository of knowledge and learning established under the proposed program in digital form which will be maintained by SIDBI beyond project period and will be accessible by all stakeholders.
128. The proposed program has been designed incorporating and building on learnings acquired by SIDBI over time (as showcased in preceding sections). Further, the program will build on the existing knowledge landscape developed by SIDBI under its ongoing schemes.
129. The necessity of these activities has been determined through multiple stakeholder conversations with players across the climate financing ecosystems including MSMEs, PFIs, local communities etc. Basis their inputs, we have considered numerous TA related activities that have been included in our proposal which are designed to address climate concerns in the long term.



## D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

*This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).*

### D.1. Impact potential (max. 500 words, approximately 1 page)

*Describe the potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas. As applicable, describe the envisaged project/programme benefits for mitigation and/or adaptation. Provide the intended outcomes for mitigation by elaborating on how the project/programme contributes to low-emission sustainable development pathways. Provide the intended outcomes for adaptation by elaborating on how the project/programme contributes to increased climate-resilient sustainable development. Calculations should be provided as an annex. This should be consistent with section E.3 reporting GCF's core indicators.*

130. The programme will potentially leverage USD 3577 million from public and private sector sources, having a mobilization ratio of 1:16.59 towards a dedicated investment in MSME sector for carbon emission reduction and promoting MSMEs for accelerating adaptation actions. Of this, 40% will address the adaptation actions and the remaining 60% will address mitigation efforts. The monitoring protocol shall describe the approach to obtain, record, compile, analyse and document monitoring data, including assumptions, references, activity data, calculation factors, quality management procedures and uncertainty, in a transparent manner that enables the reproduction of the determination of emissions by the Fund. The monitoring protocol shall also consist of a detailed, complete, and transparent documentation of the monitoring methodology for a specific technology.

#### Mitigation

131. Mitigation impact will be achieved through investments at scale in the RE solar, energy efficiency, and low-emission transport sectors. In particular, MSME's in these sectors suffer from several financial and non-financial barriers, such as limited access to finance due to high perceived risk, technology and capacity gaps, limited private sector involvement, and low awareness of impact potential.

132. The total mitigation impact of the project will be 35.34 million tCO<sub>2</sub>eq of GHG emissions over equipment lifecycle through the adoption of low carbon climate technologies in ~10000 climate loans in MSMEs. A summary of climate baseline and additionality assumptions and targeted impact is provided in the table below, with assumptions and calculations further detailed in Annexure 22.

| GCF Result Area                                          | Project activity                                                      | Methodologies                                                        | Analysis of baseline & additionality                                                                                                                                                                                                                                                                                                                                                          | Impact (tCO <sub>2</sub> Lifetime)                          |
|----------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>MRA 1</b><br>Energy generation and access             | Promoting renewable energy generation in 2352 climate loans in MSMEs. | CDM Methodology AMS-ID                                               | <p><b>Baseline:</b> CO<sub>2</sub> emissions from fossil fuel power energy generation in MSME sectors.</p> <p><b>Additionality:</b> The programme will target small to medium size solar PV projects (average annual installation planned is 16MW, including for use in agriculture and manufacturing).</p>                                                                                   | 26.99 million tCO <sub>2</sub> eq (Over equipment lifetime) |
| <b>MRA 2</b><br>Low-emission transport                   | Promoting EV in 941 climate loans in MSMEs.                           | CDM Methodology AMS-III C                                            | <p><b>Baseline:</b> Petrol-powered vehicles operating at current use and fuel efficiency levels.</p> <p><b>Additionality:</b> The project will replace petrol consumption of baseline vehicles, resulting in net emission reduction.</p>                                                                                                                                                      | 2.26 million tCO <sub>2</sub> eq. (Over equipment lifetime) |
| <b>MRA 3</b><br>Building, cities, industries, appliances | Promoting energy efficient appliances in 2352 climate loans in MSMEs. | CDM Methodology AMS-II.C<br>AMS- II.D<br>AMS- III.B<br>AMS- III A.N. | <p><b>Baseline:</b> Existing factories continue to operate at current levels of energy efficiency and consumption.</p> <p><b>Additionality:</b> Retrofitting various factories in manufacturing industry for increased energy efficiency. Increasing the efficiency of factory equipment cuts energy use and avoids emissions from the local grid supplying the factory with electricity.</p> | 6.08 million tCO <sub>2</sub> eq. (Over equipment lifetime) |

## Adaptation

133. Adaptation impact will be achieved through increased access to climate resilient energy supply, agriculture, livelihoods, water supply, and infrastructure. The programme will impact a total of **10.80 million beneficiaries**. Additionally, the project will create ~420000 new climate jobs directly related to climate change Mitigation. The detailed calculation sheet is provided in the Annexure 22.

| Sector                                                                                                                                                                                                                                                                                                                                                                                                         | Hazard                                                                                                                                                                                                                                                                                                                      | Climate risk/impact                                                                                                                                                                                                                                                                                                                                                                                                            | Adaptation interventions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Adaptation impact                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Water-intensive MSME sectors.</p> <p>Major Subsectors include:</p> <ul style="list-style-type: none"> <li>Textiles</li> <li>Food processing</li> <li>Paper production</li> <li>Tanneries</li> <li>Breweries and soft drinks production</li> <li>Dairy and dairy production</li> <li>Steel/ foundry mills</li> <li>Canning</li> <li>Roller and flour mills</li> <li>Rubber and plastic production</li> </ul> | <p>Reduced precipitation during the dry season and an increased trend toward short, intense rainfall events</p> <p>References:</p> <ul style="list-style-type: none"> <li>IPCC's AR6 Report<sup>54</sup></li> <li>Projected changes in precipitation based on the IPCC's CMIP6 near-term data sets<sup>55</sup>.</li> </ul> | <p>Leads to a decrease in ground water levels and an increase in the occurrence of drought events in vulnerable areas, including<sup>56</sup>.</p> <p>According to a recent report<sup>57</sup>, 39% of the portfolio of Indian banks is exposed to sectors that face high levels of operational water risk, including agriculture and allied activities (13.3%), power (6.8%), and basic metal and metal products (4.8%).</p> | <p>The project will increase water resilience for production (manufacturing sector) and service delivery (service sector) through the promotion of water recycling technologies, water resilient production technologies and increased rainwater harvesting, while simulates promoting better management practices to monitor and reduce freshwater usage. Water-efficient technologies and processes utilized under the project will include soft flow machines, PLC in jet dyeing machines, zero liquid discharge, effluent treatment plants, and reverse osmosis plants<sup>58</sup>. These water scarcity reduction processes will be further supported through the implementation of efficient wastewater treatment measures, assisting in the reduction of surface water contamination and the availability of clean water for MSMEs and households.</p> | <ul style="list-style-type: none"> <li>Water saved especially in water stress and drought prone regions</li> <li>More number of people in these regions will have access to safe drinking water</li> <li>Impact on Food security</li> <li>Improved cropping pattern by using weather information services</li> <li>Increased resilience of vulnerable population to natural hazards</li> </ul> |
| <p>MSMEs supporting agriculture and allied sectors.</p> <p>Subsectors include:</p> <ul style="list-style-type: none"> <li>Crop farming</li> <li>Agroforestry</li> <li>Livestock</li> <li>Agro- processing</li> <li>Storage operations</li> </ul>                                                                                                                                                               | <p>Increase in mean surface temperatures</p> <p>References:</p> <ul style="list-style-type: none"> <li>IPCC SREX Report<sup>59</sup></li> <li>Projected changes in temperature</li> </ul>                                                                                                                                   | <p>Leads to increased instances of heat waves in vulnerable areas, including<sup>61</sup>.</p> <p>Rising temperatures threatens crop yields through heat stress and exacerbates and on-farm food spoilage and waste, ultimately jeopardizing</p>                                                                                                                                                                               | <p>The project will invest in MSME enterprises that design, develop and propagate commercially available improved products, technologies and techniques related to climate resilient agriculture and food storage. Rollout of appropriate climate resilient technologies and products will be supported by</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                |

<sup>54</sup> [https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC\\_AR6\\_WGI\\_Full\\_Report.pdf](https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Full_Report.pdf)

<sup>55</sup> Source: <https://interactive-atlas.ipcc.ch/>

<sup>56</sup> Ground Water Yearbook, India 2017-18. <http://cgwb.gov.in/Ground-Water/Groundwater%20Year%20Book%202017-18.pdf>

<sup>57</sup> NITI Aayog, August 2019. <https://www.niti.gov.in/sites/default/files/2019-08/CWMI-2.0-latest.pdf>

<sup>58</sup> Pg. no. 76: Report on Adaptation and Indian MSMEs by GiZ and SIDBI: <https://www.adelphi.de/en/system/files/mediathek/bilder/adelphi-giz-2012-msme-climate-risk-study.pdf>

<sup>59</sup> IPCC, 2012: Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation. A Special Report of Working Groups I and II of the Intergovernmental Panel on Climate Change [Field, C.B., V. Barros, T.F. Stocker, D. Qin, D.J. Dokken, K.L. Ebi, M.D. Mastrandrea, K.J. Mach, G.-K. Plattner, S.K. Allen, M. Tignor, and P.M. Midgley (eds.)]. Cambridge University Press, Cambridge, UK, and New York, NY, USA, 582 pp. Available: [https://www.ipcc.ch/site/assets/uploads/2018/03/SREX\\_Full\\_Report-1.pdf](https://www.ipcc.ch/site/assets/uploads/2018/03/SREX_Full_Report-1.pdf)

<sup>61</sup> IMD, India [https://internal.imd.gov.in/section/nhac/dynamic/FAQ\\_heat\\_wave.pdf](https://internal.imd.gov.in/section/nhac/dynamic/FAQ_heat_wave.pdf)

|  |                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | based on the IPCC's CMIP6 near-term data sets <sup>60</sup> .                                                                                                                                                                                                                                                                   | food security in rural and connected urban areas <sup>62</sup> .                                                                                                                                                                                                                                                                                                                                                                                                                                                | technical capacity building, training and awareness raising through the project.<br><br>Appropriate intervention activities include: <ul style="list-style-type: none"> <li>• Drought resistant crop varieties</li> <li>• Improved soil, nutrient and pest management practices</li> <li>• Improved feed and shelter management for livestock</li> <li>• Precision irrigation systems and solutions</li> <li>• Agroforestry products and services</li> <li>• Weather forecasting and early warning systems</li> </ul>                                                                                                                                                                                                                                                                                                |  |
|  | <p>Reduced precipitation during the dry season and an increased trend toward short, intense rainfall events</p> <p>References:</p> <ul style="list-style-type: none"> <li>• IPCC's AR6 Report<sup>63</sup></li> <li>• Projected changes in precipitation based on the IPCC's CMIP6 near-term data sets<sup>64</sup>.</li> </ul> | <p>Decline in precipitation and rise in temperature both contribute to agricultural drought in the country. 17 states (out of the 27 states and UTs) reported having more than 50% area under rain-fed agriculture, highlighting inadequate coverage by irrigation systems in the country<sup>65</sup>.</p> <p>Two major crops, wheat (74% of the area under cultivation) and rice (65% of the area under cultivation) face significant levels of water scarcity and vulnerability to drought<sup>66</sup>.</p> | <p>The project will invest in enterprises that propagate climate resilient agriculture products/ services/ technologies to save water and reduce water stress and reduce fertilizer consumption (thereby reducing water contamination).</p> <p>Appropriate intervention activities include:</p> <ul style="list-style-type: none"> <li>• Drought resistant crop varieties</li> <li>• Improved soil, nutrient and pest management practices</li> <li>• Improved feed and shelter management for livestock</li> <li>• Precision irrigation systems and solutions</li> <li>• Agroforestry products and services</li> <li>• Weather forecasting and early warning systems</li> </ul> <p>Furthermore, afforestation and agroforestry can assist in reducing risk and damage from climate change induced flood events.</p> |  |

<sup>60</sup> Source: <https://interactive-atlas.ipcc.ch/>

<sup>62</sup> <https://www.pnas.org/doi/full/10.1073/pnas.1317275111>

<sup>63</sup> [https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC\\_AR6\\_WGI\\_Full\\_Report.pdf](https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Full_Report.pdf)

<sup>64</sup> Source: <https://interactive-atlas.ipcc.ch/>

<sup>65</sup> NITI Aayog, August 2019. <https://www.niti.gov.in/sites/default/files/2019-08/CWMI-2.0-latest.pdf>

<sup>66</sup> NITI Aayog, August 2019. <https://www.niti.gov.in/sites/default/files/2019-08/CWMI-2.0-latest.pdf>

|  |  |  |                                                                                                                                                                        |  |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  |  |  | Technical capacity building, training and awareness raising through the project will ensure the appropriate adoption and sustainable operation of these interventions. |  |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

## D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

*Paradigm shift potential is defined as 'degree to which the proposed activity can catalyse impact beyond a one-off project or programme investment'. In this section, elaborate on the contribution to paradigm shift and how the proposed project/programme aims to contribute towards it based on the theory of change described in section B2(a). Also describe how and to what extent the project/programme will be able to promote or contribute to paradigm shift through the below.*

*Potential for scaling up and replication*

*Potential for knowledge sharing and learning*

*Contribution to the creation of an enabling environment*

*Contribution to the regulatory framework and policies*

*Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans*

134. The programme will focus on MSME sector which is credit starved particularly in the climate space. The sub-projects typically will have low investment, and high emission reduction potential. The dedicated credit and technical support approach of the programme will result in crowding-in of private funds and unlock the potential of investment for climate change projects in MSMEs. The exclusive investment in climate technologies for adaptation and mitigation in MSMEs will accelerate industrial decarbonization trajectory of the country through (a) New innovations in climate change domain; and (b) Acceleration of the market for climate technologies. The capacity building and training of MSMEs / Industry Associations will generate awareness among the MSMEs on the need of climate technologies, as well as increase the focus of banks/ FIs on climate financing. The support from the programme will unleash climate innovations, and thereby facilitate climate low carbon industrialization. The support for business approach for adaptation actions by investing in MSMEs operating in the ecosystem of climate adaptation technologies will upscale adaptation actions and build ecosystem for adaptation private sector investment in the country. As detailed below, the project's paradigm shift potential is anchored in its comprehensive approach to integrating sustainable practices within the MSME sector through scalable, replicable, and knowledge-driven initiatives that are supported by robust regulatory frameworks and an enabling environment for private sector participation.

### Potential for Scaling Up and Replication

135. The project has potential to scale up its objectives within India's MSME sector by addressing prevalent challenges such as energy inefficiency, high GHG emissions, and excessive water use across sub-sectors such as manufacturing and services. It will introduce adaptable technologies and practices that are widely applicable, while leveraging SIDBI's financial structures (such as concessional loans and technical support funds) as scalable and replicable models for broader adoption in similar emerging subsectors and economies. The project's integration of advanced, resource-efficient technologies facilitated by SIDBI's extensive network will lay the basis for widespread technology transfer, ensuring dissemination and adaptation across India's industrial landscape. Furthermore, the project's alignment with national policies and institutional frameworks enhances its replicability, with potential for replication being bolstered by a strong emphasis on capacity building, data-driven approaches, and demonstration effects. Enhanced partnerships with technology providers and policymakers, alongside alignment with global sustainability goals and India's climate actions, position the project to catalyze a broad paradigm shift towards sustainability, attracting widespread support and fostering scalable and effective interventions across sectors. Moreover, mobilizing the Indian Banking system towards climate financing will trigger the market and more MSMEs will have access to climate finance for adopting climate and clean interventions related to mitigation and adaptation. The adoption of large scale climate technologies will also boost the market of service providers therefore ensuring that end of sales service for these technologies is not a constraint.

### Potential for Knowledge Sharing and Learning

136. The project will leverage SIDBI's network to enhance MSMEs' access to sustainable practices through knowledge hubs that adapt global innovations to local needs and collaborative platforms that engage multiple stakeholders. It will promote knowledge dissemination by documenting best practices and lessons learnt, which

will provide scalable templates for energy efficiency, environmental sustainability and climate adaptation in the MSME sector. Additionally, the project will link with national and international networks to leverage the latest available and innovative technologies. The documentation and dissemination of learnings through various platforms and collaborations with international bodies will further the project's reach and impact, creating a blueprint for knowledge sharing that can be emulated in other contexts. The integration of capacity-building programs and technical support, particularly in energy-efficient technologies and resource management practices, will further foster a community of practice that extends beyond the immediate project lifespan. Additionally, a robust Monitoring, Evaluation, and Learning framework to assess the effectiveness of interventions will facilitate real-time strategy adjustments. This comprehensive approach will aim to minimize the environmental impact of MSMEs while also bolstering their global competitiveness, aligning with the Green Climate Fund's goals for sustainable, low-emission development.

### **Contribution to the Creation of an Enabling Environment**

137. The project will contribute to creating an enabling environment for sustainable industrial practices, particularly in the MSME sector. By aligning with national priorities and frameworks such as India's NDC targets, the project will solidify the institutional and policy groundwork necessary for broad-based adoption of climate technologies. Through concessional loans and targeted financial products, the project will build institutional capacity within SIDBI and local financial entities to manage climate financing effectively. Market transformation will be fostered by demonstrating and promoting the adoption of low-emission technologies, thus catalyzing a broader shift toward sustainable practices. This comprehensive approach includes capacity building for MSMEs, fostering collaborations among stakeholders, and utilizing robust data for informed decision-making, which supports the integration of sustainable development goals. These efforts collectively facilitate scalable and replicable models of resource efficiency and emissions reduction, positioning the project as a catalyst for widespread industrial and environmental reforms aligned with both national and international climate action objectives.

### **Contribution to the Regulatory Framework and Policies**

#### **National Climate Change Strategies and Priorities targeting MSMEs**

138. The program is in line with the regulatory and policy frameworks governing the financial, environment, energy conservation and promotion of MSME sectors. Micro, Small & Medium Enterprises Development Act 2006, an exclusive legislation for micro, small and medium enterprises states under its Section.9 on Measures for promotion and development that the Central Government may, from time to time, for the purposes of facilitating the promotion and development and enhancing the competitiveness of micro, small and medium enterprises, particularly of the micro and small enterprises, by way of development of skill in the employees, management and entrepreneurs, provisioning for technological upgradation, marketing assistance or infrastructure facilities and cluster development of such enterprises with a view to strengthening backward and forward linkages, specify, by notification, such programmes, guidelines or instructions, as it may deem fit. The Act under Section.10 on Credit Facility<sup>67</sup> states that the policies and practices in respect of credit to the micro, small and medium enterprises shall be progressive, and such as may be specified in the guidelines or instructions issued by the Reserve Bank, from time to time, to ensure timely and smooth flow of credit to such enterprises, minimise the incidence of sickness among and enhance the competitiveness of such enterprises.

139. Government of India has been implementing overarching framework of National Action Plan on Climate Change (NAPCC) with its eight missions and State specific Action Plans on Climate Change targeting climate change mitigation and adaptation efforts. Under its National Mission on Energy Efficiency, Government of India is implementing Perform, Achieve and Track (PAT) scheme to achieve energy efficiency in selected Designated Consumers targeting only major high energy-consuming industries. The Performance Linked Incentive Scheme is promoting co-investment model for raising blended capital to finance start-ups for agriculture & rural enterprise relevant for farm produce value chain. MSME sector is also one of the priority sectors in national financial inclusion<sup>68</sup> Strategy of Government of India/ RBI which aims to ensure access to appropriate financial products and services needed by vulnerable groups such as weaker sections and low-income groups at an affordable cost by mainstream institutional players. The Zero Effect, Zero Defect, Make in India Campaign also focuses on to the Small and medium-sized enterprises. In line with ambitious climate targets for 2030 and net-

<sup>67</sup> [https://www.indiacode.nic.in/show-data?actid=AC\\_CEN\\_46\\_77\\_00002\\_200627\\_1517807324919&orderno=10](https://www.indiacode.nic.in/show-data?actid=AC_CEN_46_77_00002_200627_1517807324919&orderno=10)

<sup>68</sup> [https://www.ncfe.org.in/images/pdfs/reports/NSFE\\_20-25\\_ENG.pdf](https://www.ncfe.org.in/images/pdfs/reports/NSFE_20-25_ENG.pdf) 79 <https://www.industrytransition.org/>



zero pathways for 2070, India has also co-launched Leadership Group for Industry Transition (LeadIT)<sup>69</sup> and Industry Deep Decarbonization Initiative<sup>70</sup> in the last three years to facilitate climate finance and technology transfer for industry decarbonization and energy efficiency measures. However, these mainstream schemes and programs of the government in areas of energy efficiency and productivity of the industries with emission reduction as co-benefits often primarily target large industries. The proposed program will bridge the national priorities of enhancing the competitiveness of MSMEs and economy wide emission reduction strategies, by focusing on providing low-cost finance and support for adoption of low carbon technologies.

140. Govt. of India has carried out various pilot projects in association with Bilateral / Multilateral agencies including the World Bank, UNDP, UNIDO, etc. for enhancement of awareness and capacity building of MSMEs along with identification of energy savings potential and implementation challenges. SIDBI is also part of many of these projects as a development financial institution working with MSMEs. These projects have been successful in demonstrating the potential for GHG emission reductions and resource optimization including water, etc. on pilot scale and highlighted the need of customized solutions including technology support available at concessional rates for their uptake in MSMEs. The ongoing GCF investments in India are not targeted for the needs of MSMEs but may have spillover technology transfer impacts.

141. On adaptation side, the Government of India has adopted various sectoral approaches to address the climate vulnerabilities and enhance resilience of communities and the ecosystem. Five out of eight missions under NAPCC address sectoral adaptation actions in agriculture, water, forests, Himalayan region, and sustainable habitat. For example: The main objective of National Water Mission is the conservation of water, minimising wastage and ensuring equitable distribution both across and within states. The private sector including MSME is an important partner for achieving the Mission's objectives, both because it is a major consumer of water and because it is an important source of innovation for water efficient processes and technologies. While water scarcity and rising water prices already challenge Indian society and business, climate change is likely to put additional pressure on water supply. Enhancing water efficiency and sustainably managing water withdrawal therefore are key components of a business adaptation strategy. The FMAP programme will target financing the MSMEs in water stressed regions to install water efficient technologies in their units so as to reduce their dependence on groundwater. Similarly, The National Mission on Strategic Knowledge for Climate Change (NMSKCC) seeks to address a number of gaps in existing research in climate change impacts on India. The programme will contribute to a better availability of regionalized data on climate impacts on MSMEs, adaptation interventions specific to MSMEs, key preconditions for MSMEs to proactively and successfully implement the adaptation interventions etc. The National Mission on Sustainable Habitat seeks to strengthen the ability of habitats to adapt to climate change by improving the resilience of infrastructure, strengthening disaster management and improving warning systems for extreme weather events. The programme aims to finance the MSMEs who would either manufacture/supply/service climate resilient infrastructures like amphibious houses, climate proofing/insulation products which can enhance the resilience of habitats and building against climate hazards. However, there is huge gap in demand and supply of finance to be invested in adaptation actions. The gap in adaptation finance also needs to be addressed by engaging private sector investment with innovative business models, enterprises, products and services to accelerate adaptation actions in the country to reach its required goals.

142. The project will contribute to regulatory framework and policies by demonstrating sustainable practices within India's MSME sector. It will provide data to advocate for stringent environmental standards and enhance policy through insights gained from implementing advanced technologies focused on energy efficiency, climate house gas reduction, and water conservation. These efforts will support India's commitments under international accords like the Paris Agreement and align with national actions plans on climate change, reinforcing the regulatory basis for sustainability. Additionally, the project will introduce innovative financing models that incentivize adoption of cleaner technologies and serve as examples for institutional frameworks for climate finance, ultimately fostering comprehensive integration of environmental considerations into regulatory and financial governance. This integration will not only promote a shift towards sustainable development but will also ensure enhanced compliance mechanisms and stakeholder involvement in policy development, making environmental policies effective and broadly accepted across the industry. Through these multifaceted contributions, the project sets a foundational precedent for future sustainability initiatives within the regulatory landscape.

<sup>69</sup> <https://www.cleanenergyministerial.org/initiatives-campaigns/industrial-deep-decarbonisation-initiative/>

<sup>70</sup> <https://www.industrytransition.org/>

### Additional national legislation and priorities that the programme supplements

143. The Energy Conservation (EC) Act, 2001, accommodates productive utilization of energy and its preservation for matters associated therewith or incidental thereto. It was instituted by the Parliament in the Fifty-second Year of the Republic of India and was enacted on 29th September 2001 but came into practice from 1st March, 2002.

144. Environment Protection Act, 1986, and as amended thereof, relates to the protection and improvement of the human environment and the prevention of hazards to human beings, other living creatures, plants and property. The Act is an “umbrella” legislation designed to provide a framework for central government coordination of the activities of various central and state authorities established under previous laws.

145. In addition to the above, this programme is also aligned to the resolve of the Government of India to achieve the following:

- Reach 500GW non-fossil energy capacity by 2030.
- 50 per cent of its energy requirements from renewable energy by 2030.
- Reduction of total projected carbon emissions by one billion tonnes from now to 2030.
- Reduction of the carbon intensity of the economy by 45 per cent by 2030, over 2005 levels.
- Achieving the target of net zero emissions by 2070.

146. Further, in the very recent G20 Delhi declaration, the leaders unanimously accepted the recommendation of the Sustainable Finance Working Group (SFWG) to scale up climate financing through innovating finance models such as blended finance, to leverage private finance and multilateral climate funds.

### D.3. Sustainable development (max. 500 words, approximately 1 page)

*Describe the wider benefits and priorities of the project/programme in relation to the Sustainable Development Goals and provide the potential in terms of:*

- Environmental co-benefits
- Social co-benefits including health co-benefit
- Economic co-benefits
- Gender-sensitive development benefit

| Sustainable Development Goal                                                                                                          | Programme support and expected impact                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No Poverty (SDG-1)                                                                                                                    | There are studies to corroborate the fact that investments in climate energy (EE and RE) help in the growth of the economy as well as generate jobs <sup>71</sup> . The employment generation per Million\$ of investment in climate energy in India was evaluated to be at 116.1 direct jobs in Energy Efficiency and 215 direct jobs in Renewable Energy <sup>72</sup> .                                                                                       |
| Gender Equality (SDG-5)                                                                                                               | It is estimated that the male and female ratio in the above estimated jobs and lives will be equally distributed @ <b>80:20 ratio</b> <sup>73</sup>                                                                                                                                                                                                                                                                                                              |
| Clean water and sanitation (SDG-6)                                                                                                    | Water savings through adoption of water efficient technologies in the operations of MSME units and water recycling & treatment technologies for ZLD, ETP, RO plant etc.                                                                                                                                                                                                                                                                                          |
| Affordable & clean energy (SDG-7)                                                                                                     | The project will support financing for clean energy such as solar power initiative that could halve energy costs for MSMEs, thereby advancing India towards its net-zero target by 2070.                                                                                                                                                                                                                                                                         |
| Decent Work and economic Growth (SDG-8), Industry, innovation and infrastructure (SDG-9), Sustainable cities and communities (SDG-11) | The proposed project will invest in technology and infrastructure to boost economic growth and sustainability in cities by lowering GHG emissions and fossil fuel use. This will clean cities and support long-term community sustainability. Additionally, it will help MSMEs in the service sector to meet various climate standards like USGBC LEED, BEE Star Rating, TERI GRIHA, IFC Edge, Energy Conservation Building Code, Net Zero Energy Buildings etc. |
| Sustainable consumption & production (SDG 12), Climate action (SDG-13)                                                                | The programme follows the concept of “ <b>produce more with less consumption</b> ”. In addition to economic benefits in terms of increased productivity and enhanced product quality, the programme activities are also expected to bring other environmental and social co-benefits                                                                                                                                                                             |

<sup>71</sup> Max Wei, Shana Patadia, and Daniel M. Kammen. Putting renewables and energy efficiency to work: How many jobs can the clean energy industry generate in the US? Energy Policy, 38:919–931, February 2010. URL <https://doi.org/10.1016/j.enpol.2009.10.044>

<sup>72</sup> [https://cse.azimpremjuniiversity.edu.in/wp-content/uploads/2019/06/Annual\\_Report\\_PLFS\\_2018\\_19\\_HL.pdf](https://cse.azimpremjuniiversity.edu.in/wp-content/uploads/2019/06/Annual_Report_PLFS_2018_19_HL.pdf)

<sup>73</sup> Section 2.2.2.1: <https://msme.gov.in/sites/default/files/MSME-ANNUAL-REPORT-ENGLISH%202020-21.pdf>

such as improved working conditions, reduce negative health impacts due to reduction in fugitive emissions, reduced workplace temperature, etc. Further, the Programme is expected to create additional jobs either directly in the MSMEs or allied operations such as manufacturing, maintenance and services, etc. thus affecting the livelihood of associated work force and their dependents.

#### Co-benefits:

- **Job Creation / Employment opportunities through low-cost climate resilient financial assistance** - There are studies to corroborate the fact that investments in climate energy (EE and RE) help in the growth of the economy as well as generate jobs <sup>74</sup>. The employment generation per million\$ of investment in climate energy in India was evaluated to be at 116.1 direct jobs in Energy Efficiency and 215 direct jobs in Renewable Energy<sup>75</sup>.
  - (1) This Programme will cover ~10000 climate loans in MSMEs and will impact around 10.80 million lives indirectly.
  - (2) Partnerships with technical experts / ESCOs to bring in much needed expertise through joint ventures and will also create new job avenues in service segment.
  - (3) Further policy measures (a) Digital Integrated Climate Credit rating tool; (b) Assessment and recommendation for domestic carbon trading scheme for MSME sector, will create more jobs.
- **Environmental and climate benefits** - Energy efficiency in MSMEs can play a major role in reducing local air pollution, particularly outdoor concentrations of pollutants such as sulphur dioxide, particulate matter, unburned hydrocarbons and nitrogen oxides. In doing so, it can lead to a range of associated economic, environmental and health benefits. The Project will enable expected saving of more than 2.75 million MWh energy and ~ 1,458,641 million litres of water saving through the adoption of improved technologies in MSMEs.
- **Better productivity & Quality** - Advanced EE machines improves competitiveness, product quality, material efficiency and enhance demand in the market. Studies indicate that the value of the productivity and operational benefits derived can be up to 2.5 times the value of the energy savings (IEA, 2014b)<sup>76</sup>.
- **Improved bottom line 'profitability'** - Cost-effective investments in energy efficiency generate energy savings that have a direct impact on the profitability of MSMEs and provide net positive earnings in the long run. Energy cost reductions can free up resources that can be directed at more productive activities, such as expansion or developing new product lines or services.
- **Reducing Credit Risk** - Energy efficiency can also help mitigate risks for MSMEs. While risk is inherent in all business functions and activities, MSMEs are more vulnerable than large companies to changing conditions in credit markets, given their limited financial base. MSMEs often face more severe credit conditions than large enterprises. Banks may not be as motivated to engage with MSMEs, as the transaction costs of smaller loans are proportionately higher. Many banks are uneasy about the risks involved or fail to see quality or resilience in the small business segment. At the same time, most MSMEs do not have enough collateral to provide a strong alternative repayment source. MSMEs in many countries also pay more per unit of energy than larger companies and are hence more vulnerable to price increases. Despite a dip in energy prices across many countries, energy costs are increasingly restricting MSME profitability. MSMEs are also more sensitive to energy supply disruptions as they may not have back-up systems or on-site generation.
- **Future Entrepreneurs 'Innovation'** - MSMEs are important for climate growth as drivers of eco-innovation and key players in emerging climate industries. New and young companies are particularly important for radical climate innovations, as they often exploit technological or commercial opportunities that have been neglected by more established enterprises or even challenge the business models of existing companies.
- **Women Empowerment**- Various TA activities including capacity building and financial literacy of women workers/civil society representatives will be conducted so that they can become an independent entrepreneur. On the other hand, women entrepreneurs will be attracted for credit incentives in terms of Interest Subventions. These

<sup>74</sup> Max Wei, Shana Patadia, and Daniel M. Kammen. Putting renewables and energy efficiency to work: How many jobs can the clean energy industry generate in the US? Energy Policy, 38:919–931, February 2010. URL <https://doi.org/10.1016/j.enpol.2009.10.044>

<sup>75</sup> [https://cse.azimpremjiuniversity.edu.in/wp-content/uploads/2019/06/Annual\\_Report\\_PLFS\\_2018\\_19\\_HL.pdf](https://cse.azimpremjiuniversity.edu.in/wp-content/uploads/2019/06/Annual_Report_PLFS_2018_19_HL.pdf)

<sup>76</sup> <https://c2e2.unepdtu.org/wp-content/uploads/sites/3/2016/03/sme-2015.pdf>

initiatives will help in increasing the existing women employment ratio in MSMEs. As per the GCF gender policy, the Project will implement Gender Assessment and Action Plan as per Annexure 8 submitted with the Full Proposal.

#### D.4. Needs of recipient (max. 500 words, approximately 1 page)

*Describe the scale and intensity of vulnerability of the country and beneficiary groups and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc.).*

*Describe how the project/programme addresses the following needs:*

- Vulnerability of the country and/or specific vulnerable groups, including gender aspects (for adaptation only)
- Economic and social development level of the country and the affected population
- Absence of alternative sources of financing (e.g. fiscal or balance of payments gap that prevents government from addressing the needs of the country; and lack of depth and history in the local capital market)
- Need for strengthening institutions and implementation capacity

##### Country Needs

India is entering a dynamic phase in energy development post-Covid19, according to the India Energy Outlook 2021 report. It highlights the imminent surge in appliance, air conditioning, and vehicle purchases. Over the next two decades, to meet electricity demand, India must expand its power system to the size of the European Union's. Key to economic growth will be affordable, clean, and reliable energy. This requires strong policies, technological advancements, and increased clean energy investments. Plans include quadrupling renewable electricity by 2030, doubling natural gas use, enhancing energy efficiency and infrastructure, boosting coal production, and reducing oil import costs<sup>77</sup>. As the world's fourth-largest climate house gas emitter<sup>78</sup>, India's vulnerability to climate change was emphasized in the 2015 Paris pledge, given its large population, agricultural reliance, extensive coastal areas, and fragile Himalayan region<sup>79</sup>. A World Bank study<sup>80</sup> suggests that unchecked emission growth could impact 600 million people. Urban migration and population increases could significantly raise overall emissions, necessitating large investments to mitigate climate change effects like altered rainfall, frequent droughts, groundwater depletion, and rising sea levels, which threaten agriculture and food security. District-level vulnerability reports<sup>81</sup> by the Council on Energy, Environment, and Water (CEEW) show that over 80% of India's population resides in districts highly susceptible to extreme hydro-meteorological disasters. Southern India is notably vulnerable, followed by eastern, western, northern, north-eastern, and central zones. Several states face significant risks from floods, droughts, and cyclones. India has committed to reducing emissions intensity by 45% by 2030 from 2005 levels under the Paris Agreement. This could lead to 90% emissions increase from 2014 levels by 2030, yet India's efforts are aligned with the global 2°C warming limit, marking it as one of the few countries on track to meet its climate targets. Meeting these targets will require an estimated \$2.5 trillion<sup>82</sup>, spotlighting the investment gap with current climate finance flows significantly below the needed \$170 billion annually. This underscores the importance of innovative financing and technologies in advancing India's climate goals.

<sup>77</sup> <https://indiadialogue.net/2021/02/10/indias-energy-transition-to-play-key-role-in-global-climate-action/>

<sup>78</sup> <https://www.carbonbrief.org/the-carbon-brief-profile-india>

<sup>79</sup> <https://dst.gov.in/sites/default/files/Full%20Report%20%281%29.pdf>

<sup>80</sup> <https://www.worldbank.org/en/news/press-release/2018/06/28/climate-change-depress-living-standards-india-says-new-world-bank-report>

<sup>81</sup> <https://www.ceew.in/publications/mapping-climate-change-vulnerability-index-of-india-a-district-level-assessment>

<sup>82</sup> <https://www.climatepolicyinitiative.org/wp-content/uploads/2020/09/Landscape-of-Green-Finance-in-India.pdf>

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| <p><b>Needs of MSMEs</b></p> | <p>Meeting the India's aspirations related to climate change mitigation and adaptation through it's INDC and NAPCC requires prolonged approach and resource including easy and affordable finance to support its efforts. These efforts have been supported by various National and International agencies e.g., GEF, CTF, GiZ, ADB, The World Bank, JICA, KfW, AFD etc. by means of technical and financial assistance. The focus of these assistance has been diversified and not specific to MSMEs only. Further, many of these have focused on Climate Mitigation only and do not include Adaptation measures. Further, the cost of funds from these agencies is on the higher side presently, making such projects unviable.</p> <p><b>Specific needs of MSMEs in terms of Climate Change Mitigation:</b></p> <ul style="list-style-type: none"> <li>• <b>Reduction of Climate house Gas Emissions:</b> MSMEs are significant contributors to carbon emissions, particularly the energy-intensive subsectors. There is a critical need to adopt low-carbon technologies and improve energy efficiency.</li> <li>• <b>Technology Upgrade:</b> Many MSMEs use outdated technologies that are less efficient and more polluting. Upgrading these technologies is essential for reducing energy consumption and associated emissions.</li> <li>• <b>Access to Climate Financing:</b> MSMEs often face barriers in accessing finance for climate upgrades. There is a need for financial products tailored to the unique challenges and risks of the MSME sector.</li> <li>• <b>Capacity Building and Awareness:</b> MSMEs require training and awareness programs on the benefits of energy efficiency and information on available technologies and financing options.</li> <li>• <b>Policy Support:</b> Supportive policies and incentives are needed to drive the adoption of climate technologies and practices in the MSME sector.</li> </ul> <p><b>Climate Change Adaptation Needs:</b></p> <ul style="list-style-type: none"> <li>• <b>Risk Assessment and Management:</b> MSMEs need tools and support to assess, manage and forecast climate risks, particularly those related to natural hazards like water scarcity, floods, and extreme weather events.</li> <li>• <b>Infrastructure Resilience:</b> Investments are necessary to make infrastructure more resilient against climate impacts, which includes upgrading facilities and adopting resilient construction practices.</li> <li>• <b>Diversification of Water Sources:</b> Given the dependency on monsoon rains, there is a need for MSMEs to diversify their water sources and increase their storage capacities to buffer against droughts and irregular rainfall.</li> <li>• <b>Enhanced Information Access:</b> MSMEs need better access to climate forecasts and market information to make informed decisions regarding crop planting, water use, and other critical business operations.</li> <li>• <b>Community and Ecosystem Resilience:</b> Programs that enhance the overall resilience of the communities and ecosystems in which MSMEs operate are crucial. This includes protecting local water sources and ecosystems that support agriculture and other MSME activities.</li> </ul> <p>Further MSMEs can contribute towards the manufacturing/servicing of adaptation technologies, especially those which are appropriate and locally suited to the climatic vulnerable regions of Himalayas, coastal and eastern belt in the sectors of agriculture, water resources, disaster risks, agroforestry and services. The tentative list of technologies are outlined below:</p> <ul style="list-style-type: none"> <li>• Weather forecasting and early warning systems to identify the climate related natural hazards in advance and to prevent loss in lives and livelihood</li> <li>• Software application- Consumer software applications and enterprise software applications on mobile devices concerning climate related services.</li> </ul> |
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|                                                                                             | <ul style="list-style-type: none"> <li>Flood and other disaster response services,</li> <li>Climate-smart precision irrigation systems and solutions- MSME units to be supported for developing and commercial scale deployment of irrigation solutions that reduce water usage and enhance the resilience of crops to climate risks.</li> <li>Manufacturing of raw materials</li> <li>Manufacturing of Climate innovation related products</li> <li>Water Efficient technologies to reduce overuse of water in drought prone areas</li> <li>Supplying alternative raw materials for those that are at risk from climate change</li> <li>Glacier-covering tarpaulins to prevent melting of glaciers</li> </ul> <p>Technologies to cater to the impact of climate induced migration like establishing temporary camps, manufacturing of tents, solar boats etc</p> |
| <b>Project Developers, Equipment Manufacturer, Service Providers across the value chain</b> | <ul style="list-style-type: none"> <li>GCF fund of USD 215.6 million as loan and TA grant will potentially leverage USD 3577 million from SIDBI and private sector sources, hence making a total programme outlay of USD 3792.6 million which will be utilized for climate investments in Industry, technology, manpower, machines, innovation, infrastructure, services, etc. adding to economic growth and will lead to a development of the value chain in the MSME sector in the country.</li> <li>This programme will also lead to around 420000 employment generation.</li> </ul>                                                                                                                                                                                                                                                                           |

#### D.5. Country ownership (max. 500 words, approximately 1 page)

Please describe how the beneficiary country takes ownership of and implements the funded project/programme. Describe the following:

- Existing national climate strategy
- Existing GCF country programme
- Relevance to and alignment with existing policies such as Nationally Determined Contributions (NDCs), Nationally Appropriate Mitigation Actions (NAMAs), and National Adaptation Plans (NAPs)
- NDCs, NAMAs, and NAPs
- Capacity of Accredited Entities or Executing Entities to deliver
- Role of National Designated Authority

Engagement with civil society organizations and other relevant stakeholders, including indigenous peoples, women and other vulnerable groups

147. The Government of India has been implementing overarching framework of National Action Plan on Climate Change (NAPCC) with its eight missions and State specific Action Plans on Climate Change targeting climate change mitigation and adaptation efforts. Under its National Mission on Energy Efficiency, the Government of India is implementing Perform, Achieve and Track (PAT) scheme to achieve energy efficiency in selected Designated Consumers targeting only major high energy consuming industries. The Performance Linked Incentive Scheme is promoting co-investment model for raising blended capital to finance start-ups for agriculture & rural enterprise relevant for farm produce value chain. MSME sector is also one of the priority sectors in national financial inclusion Strategy of Government of India/ RBI which aims to ensure access to appropriate financial products and services needed by vulnerable groups such as weaker sections and low-income groups at an affordable cost by mainstream institutional players. The Zero Effect Zero Defect, Make-in-India Campaign also focuses on to the Small and medium-sized enterprises. India's climate actions, economic development and financial inclusion strategies are aligned with the proposed project's focus on MSME low carbon development pathways and accelerating climate adaptation actions.

148. In line with ambitious climate targets for 2030 and net-zero pathways for 2070, India has also co-launched Leadership Group for Industry Transition (LeadIT) and Industry Deep Decarbonization Initiative in the last three years to facilitate climate finance and technology transfer for industry decarbonization and energy efficiency measures. However, these mainstream schemes and programs of the government in areas of energy efficiency and productivity of the industries with emission reduction as co-benefits and often primarily targeting large industries. The proposed program will bridge the national priorities of enhancing the competitiveness of MSMEs and economy wide emission reduction strategies, by focusing on providing low-cost finance and support for adoption of low carbon technologies.

149. The proposed program is a great priority of the NDA and is aligned with several priorities included in India's NDC83, including: i) to put forward and further propagate a healthy and sustainable way of living based on traditions and values of conservation and moderation, including through a mass movement for 'LiFE' – 'Lifestyle for Environment' as a key to combating climate change; ii) to adopt a climate friendly and a cleaner path than the one followed hitherto by others at corresponding level of economic development; iii) to reduce Emissions Intensity of its GDP by 45 percent by 2030, from 2005 level; iv) to achieve about 50% cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030, with the help of transfer of technology and low-cost international finance including from Green Climate Fund (GCF); v) to create an additional carbon sink of 2.5 to 3 billion tonnes of CO2 equivalent through additional forest and tree cover by 2030; vi) to better adapt to climate change by enhancing investments in development programmes in sectors vulnerable to climate change, particularly agriculture, water resources, Himalayan region, coastal regions, health and disaster management; vii) to mobilize domestic and new & additional funds from developed countries to implement the above mitigation and adaptation actions in view of the resource required and the resource gap; and viii) to build capacities, create domestic framework and international architecture for quick diffusion of cutting-edge climate technology in India and for joint collaborative R&D for such future technologies.
150. India's NDC is ambitious, and it is a significant contribution towards achieving the goals of the Paris Agreement. Environmentally sustainable, low carbon initiatives are underpinning all key sectors of the Indian economy. This update to India's existing NDC is a step forward towards our long-term goal of reaching net-zero by 2070. To align with the NDC aims, the project is focusing on financing clean energy generation sources like solar power capacity of around 1.62 MWh/Yr which will not only provide clean energy to MSMEs but also reduce their power bills. Currently power generation cost on solar rooftop plants is around INR 3-4 per unit (kWh) which is almost 40-50% less than the average cost of conventional power.
151. SIDBI, being the government's DFI, is consulted on all matters related to the development and financing of MSMEs. Therefore, it is uniquely positioned to implement effective measures for the development of MSMEs. The organization is also consulted by MoEFCC which is the NDA for GCF on specific matters related to climate financing of MSMEs, hence it has a strong say in policy consultation of Government of India.
152. Several stakeholder consultations have taken place in the development of this program to ensure strong country ownership. These consultations have resulted in the following points to be considered:
- Accelerating the MSMEs to adopt climate technologies require incentivization and innovative financing models that would lead to provisioning loans to MSMEs at concessional rates.
  - A consultation conducted with NBFCs has shown an increasing interest among NBFCs in lending to MSMEs for adopting climate interventions.
  - Continued handholding in the form of technical assistance is required to address the gap of information symmetry among MSMEs with regards to climate technologies.

#### D.6. Efficiency and effectiveness (max` . 500 words, approximately 1 page)

*Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers, and providing the minimum concessionality to ensure the project is viable without crowding out private and other public investments. Refer to section B.5 on the justification of GCF funding requested as necessary.*

*Please describe the efficiency and effectiveness of the proposed project/programme, taking into account the total financing and mitigation/ adaptation outcome the project/programme aims to achieve, and explain how this compares to an appropriate benchmark.*

*Please specify the expected economic rate of return based on a comparison of the scenarios with and without the project/programme.*

*Please specify the expected financial rate of return with and without the Fund's support to illustrate the need for GCF funding to illustrate overall cost effectiveness.*

*Please explain how best available technologies and practices have been considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.*

<sup>83</sup> [Microsoft Word - V5 NDC submission to UNFCCC](#)

153. Funding climate technologies in MSMEs is currently perceived as a risky business. While the MSMEs can borrow from other private PFIs, the concessionality provided under this programme will drive the adoption of climate technologies in MSMEs. In addition, funding of climate technologies in remote/rural areas in the country is particularly challenging. To address this, the programme's key feature will be reaching MSMEs in rural areas through indirect lending through PFIs who have wider presence. Thus, the programme will not crowd out private sector investment as it is targeting MSMEs that do not have access to alternative funding.

154. The entire programme is structured by taking learnings from SIDBI's previous programs related to Renewable Energy and Cleaner Production.

1. The climate finance portfolio of SIDBI grew 16 times in FY 2023 which is expected to lead to energy savings of 330 GWh per annum, and GHG emission reduction of 0.23 million tCO<sub>2</sub> eq per annum in Indian MSMEs.
2. In FY 2023-2024, financing of MSMEs for RECP has led to Electricity Savings of > 1.3 billion units annually with equivalent GHG Emission Reduction of 1.07 million tCO<sub>2</sub> eq./Yr.
3. The list of 900+ mitigation technologies financed by SIDBI through its various programs is attached as annexure. Thus, the best available technologies and practices have been considered and applied by using SIDBI's past successes as models for this programme.

#### **Funding multiplier effect and Estimated cost per tCO<sub>2</sub>e**

155. GCF contribution by way of loan and TA Grant aggregating to USD 215.6 million (USD 200 million as loan, and USD 15.6 million as TA Grant) shall have a multiplier effect and leveraging a total investment of USD 3577 million in 5 years (SIDBI – USD 800 million; NBFC/PFIs – USD 2400 million; MSME – USD 377 million) for climate mitigation and adaptation measures, thereby improving the efficiency and effectiveness of each dollar invested by GCF.

#### **Co-financing Ratio of GCF loan to SIDBI's loan 1:4**

#### **Estimated cost per tCO<sub>2</sub>eq.**

USD 120 million (GCF contribution) / 35.34 million tCO<sub>2</sub> eq. = 3.39 USD / tCO<sub>2</sub> eq.

#### **Water Savings**

156. Through the programme, there will be substantial water savings through Adaptation measures by MSMEs, amounting to ~ 1,458,641 million litres of water (lifetime).

#### **GCF mobilization mechanism**

#### **Mobilization Ratio of 1:16.59**

157. Based on current estimations, the GCF fund of USD 215.6 million will help SIDBI to channelize a total investment of USD 3577 million for Mitigation and Adaptation measures resulting a mobilization rate of 1:16.59. Co-Financing of USD 800 million from SIDBI, USD 2400 million as loan from NBFCs/PFIs and another USD 377 million will be mobilized as investments by MSMEs (around 10% will be invested by MSMEs before disbursements of these loans).

158. SIDBI will lend around 20% of the loans directly and another 80% of the loans will be financed indirectly through PFIs from the USD 1000 million FMAP facility. SIDBI will provide technical handholding support to the selected Participating Financial Institutions (PFIs) to design and implement climate lending products within their asset portfolio, in line with the climate financial instruments of SIDBI. SIDBI will select the PFIs based on their asset portfolio and experience. A detailed selection criteria of PFIs is provided in the term sheet and operations manual. Further list of PFI, SFBs and NBFCs with whom SIDBI is working is attached as Additional supporting documents 2.

## E. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

### E.1. Project/Programme Focus

*Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.*

- ☒ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

### E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

*This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.*

*Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).*

The programme will exclusively focus on MSMEs which have poor credit access, low investment, and high emission reduction potential. The dedicated credit and technical support approach of the programme will result in crowding-in of private funds and unlock the potential of investment for climate change projects in MSMEs. The exclusive investment in climate friendly Clean Technologies in MSMEs will accelerate industrial decarbonization trajectory of the country through (a) New innovations in climate change domain; and (b) Acceleration of the market for climate technologies. The capacity building and training of MSMEs / Industry Associations will increase climate risk resilience, as well as increase the focus of banks/ FIs on green financing. The support from the programme to new and young companies and entrepreneurs who often exploit technological or commercial opportunities that have been neglected by more established enterprises or even challenge the business models of existing companies will unleash green innovations, and thereby facilitate green low carbon industrialization. The support for business approach for adaptation actions by investing in MSMEs operating in the ecosystem of climate adaptation technologies will upscale adaptation actions and build ecosystem for adaptation private sector investment in the country.

| Assessment Dimension | Current state (baseline)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            | Potential target scenario (Description)                                                                                                                                                                                                                                                                                                                                      | How the project/programme will contribute (Description)                                                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Rating     |                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                     |
| <b>Scale</b>         | According to one study, the MSME sector consumes about 25% of the total energy consumed by the industrial sector in India, and out of the total energy consumed in MSME, 15% is electricity consumption and 85% is thermal energy consumption (FMC and GIZ 2018). According to the report, the projected energy consumption shows a steep increase (by more than 100%) from CO2 equivalent of 30 million tonnes in 2016-17 to 72.17 million tonnes in 2029-30 (FMC and GIZ 2018). To overcome this MSMEs in India will require to use energy efficient technology and reduce their dependency on conventional energy and go for renewable energy generation. This will also lead to both mitigation and adaptation against climate effects. | <u>Low</u> | On demand side management, MSME's through energy conservation and energy efficiency measures can reduce energy intake and hence reduce the carbon footprint of their operations. On the energy supply side, one of the measures is that MSMEs will deploy solar rooftops and become a source of clean thus contributing to overall mitigation and net-zero efforts of India. | Through this project reduction of ~35 million tCO2e is envisaged.<br><br>Climate funds through PFIs/NBFCs will ensure adaptability and blending of private resources to the MSME sectors will scale up commercial efficiencies and transformational impact.                                                         |
| <b>Replicability</b> | As per study of International Energy Agency, it is indicated that technologies are readily available today to achieve most of the carbon emissions through 2030. It has been recorded that only 16 percent of MSMEs are being                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <u>Low</u> | Demonstration effects from the returns reaped by early-stage risk capital invested in energy efficient technologies would help attract more and varied capital towards green financing. With the launch of FMAP program private sector's appetite for investments in                                                                                                         | The investment by GCF into FMAP will unlock a healthy cycle of capital mobilisation and scaling-up of impact. The knowledge repository developed during the project implementation will help other stakeholders including FIs and Banks, MSMEs etc. for taking decisions for replicating the adoption of such clean |



|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | financed by the formal banking system in India.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            | renewable energy generation and use of energy efficient technology will increase. Paradigm shift can be achieved in terms of perception change of private investors. Eventually the financing structure can be replicated in other sectors – wind energy generation, energy efficiency, etc. - and/or in other countries facing similar climate change challenges.                                                                                                                                                  | and green technologies and their financing, Successful outcomes demonstrated by FMAP investments will draw in additional capital from private sector investors into MSMEs.<br><br>The project will have a high demonstration effect. Technical and financial success factors will be identified with the project team and stakeholders so that the project's financing structure can be scaled and/or replicated across other sectors and regions.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Sustainability</b> | Strong policy support for sustainability initiatives and transition to a low carbon economy, with NDC commitments and multiple national priority initiatives across EV, water, renewable energy, among others. MSME sector has emerged as a highly vibrant and dynamic sector of the Indian economy, with the share of 30% of GDP and 49.8% of total export in 2019-20. The sector accounts for 45% of total industrial production, 40% of total exports with over 6000 products ranging from traditional to high-tech items are being manufactured by the MSME Sector in addition to provide wide range of service | <u>Low</u> | It contributes significantly to the economic and social development of the country by fostering entrepreneurship and generating large employment opportunities at comparatively lower capital cost, next only to agriculture. Paradigm shift can be achieved by eliminating a range of barriers to low carbon and resilient development by ensuring that projects are able to progress smoothly and expeditiously from a concept stage to a sustainable investment beyond the end of the GCF implementation period. | The fund set up under the programme will continue as a dedicated fund for climate change mitigation and adaptation for MSME sector beyond project duration to support other financial institutions in the country. This will help sustainability and replicability of the programme through entering into financing ecosystem of the country. This will also support other Indian financial institutions to develop standard loan products/ instruments for MSME climate actions. SIDBI will seek additional resources from market to replenish the Fund to fulfil the ultimate objective of mainstreaming the climate financing in MSME sector. By demonstrating a replicable and adaptable approach to attract private sector financing to climate projects, local project developers and international equity funds may adopt a similar approach building on the FMAP experience. |

### E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

| GCF Result Area                                       | IRMF Indicator                                                       | Means of Verification (MoV)                                                                   | Baseline | Target                                                                                                                       |                                                                                                                                                                                                                                                                                                    | Assumptions / Note                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       |                                                                      |                                                                                               |          | Mid-term                                                                                                                     | Final <sup>84</sup>                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <u>All mitigation result areas (Core 1 indicator)</u> | <u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u> | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0        | GHG mitigation at mid-term<br>~7.49 million tCO <sub>2</sub> eq (Reduction from Solar PV, EV and Energy Efficiency measures) | GHG mitigation at the end of project implementation<br>~20.93 million tCO <sub>2</sub> eq (Reduction from Solar PV, EV and Energy Efficiency measures)<br><br>Lifetime target of GHG mitigation<br>~35.34 million tCO <sub>2</sub> eq (Reduction from Solar PV, EV and Energy Efficiency measures) | For Energy Efficient measures<br>Emission Factor: 0.608 tCO <sub>2</sub> /MWh and Lifespan: 15 years<br><br>For Solar PV<br>Emission Factor: 0.81 tCO <sub>2</sub> /MWh<br>Lifespan: 25 years<br><br>For EV-Emission factor for baseline vehicle category (g CO <sub>2</sub> / km)<br>2W- 17.37 g CO <sub>2</sub> / km<br>3W- 40.53 g CO <sub>2</sub> / km<br>4W- 46.31 g CO <sub>2</sub> / km<br>Bus Commercial- 548 g CO <sub>2</sub> / km<br>Lifespan for all type of vehicle is 15 years |

<sup>84</sup> The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

|                                          |                                                                      |                                                                                                         |   |                                                                                            |                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                         |
|------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>MRA1 Energy generation and access</u> | <u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u> | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term           | 0 | Mid-term GHG mitigation target ~4.94 million tCO <sub>2</sub> eq (Reduction from Solar PV) | End term GHG mitigation target ~13.83 million tCO <sub>2</sub> eq (Reduction from Solar PV)<br><br>Lifetime target for GHG mitigation ~ 26.99 million tCO <sub>2</sub> eq (Reduction from Solar PV, EV and Energy Efficiency measures) | For Solar PV<br>Emission Factor: 0.81 tCO <sub>2</sub> /MWh<br>Lifespan: 25 years                                                                                                                                                       |
| <u>MRA1 Energy generation and access</u> | <u>Supplementary 1.3: Installed renewable energy capacity</u>        | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term           | 0 | 1231 MWp                                                                                   | Final term target 1231 MW<br><br>Lifetime target 1231 MWp                                                                                                                                                                              | The project is envisaged to install create 1231 MW of solar energy capacity distributed among the 11 sectors by the end of the 6 <sup>th</sup> year of the program                                                                      |
| <u>MRA1 Energy generation and access</u> | <u>Supplementary 1.4: Renewable energy generated</u>                 | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term           | 0 | Mid-term target ~6.01 million MWh                                                          | End term target ~16.83 million MWh<br><br>Lifetime target ~32.83 million MWh                                                                                                                                                           | The project is envisaged to generate ~1.65 million MWh energy per year. Lifetime of solar plant has been taken as 25 years.                                                                                                             |
| <u>MRA2 Low-emission transport</u>       | <u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u> | Telematic data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0 | Mid-term target ~0.64 million tCO <sub>2</sub> eq (Reduction from EV)                      | End term target ~1.90 million tCO <sub>2</sub> eq (Reduction from EV)<br><br>Lifetime target ~2.26 million tCO <sub>2</sub> eq                                                                                                         | For EV-Emission factor for baseline vehicle category (g CO <sub>2</sub> / km)<br>2W- 17.37 g CO <sub>2</sub> / km<br>3W- 40.53 g CO <sub>2</sub> / km<br>4W- 46.31 g CO <sub>2</sub> / km<br>Bus Commercial- 548 g CO <sub>2</sub> / km |

|                                                          |                                                                                                     |                                                                                                         |   |                                                                                |                                                                                                                                                                      |                                                                                                                              |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                                                          |                                                                                                     |                                                                                                         |   |                                                                                | (Reduction from EV)                                                                                                                                                  | Lifespan for all type of vehicle is 15 years                                                                                 |
| <u>MRA2 Low-emission transport</u>                       | <u>Supplementary 1.5 Improved low-emission vehicle fuel economy</u>                                 | Telematic data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0 | Mid-term target 25,267 million megajoule energy saving                         | End term target 74,399 million megajoule energy saving<br><br>Lifetime target 88,436 million megajoule energy saving                                                 | The energy saving numbers depends on the actual number of vehicle being deployed.                                            |
| <u>MRA3 Buildings, cities, industries and appliances</u> | <u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>                                | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term           | 0 | Mid-term target ~1.90 million tCO <sub>2</sub> eq (Reduction from EE measures) | End term target ~5.18 million tCO <sub>2</sub> eq (Reduction from EE measures)<br><br>Lifetime target ~6.08 million tCO <sub>2</sub> eq (Reduction from EE measures) | For Energy Efficient measures<br>Emission Factor: 0.608 tCO <sub>2</sub> /MWh and Lifespan: 15 years                         |
| <u>MRA3 Buildings, cities, industries and appliances</u> | <u>Supplementary 1.1: Annual energy savings</u>                                                     | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term           | 0 | Mid-term target 521,989 MWh annual energy saving                               | Final term target 656,334 MWh annual energy saving<br><br>Lifetime target 667,120 MWh annual energy saving                                                           | Total 667,120 MWh annual energy saving from EE interventions is envisaged.<br><br>Lifetime of 15 years have been considered. |
| <u>MRA3 Buildings, cities, industries and appliances</u> | <u>Core 3: Value of physical assets made more resilient to the effects of climate change and/or</u> | Data gathered through MIS from MSMEs, Independent ex-post                                               | 0 | Mid-term target USD million 850 provided as loan to                            | Final target USD million 850 provided as loan to                                                                                                                     | The project envisaged to provide USD million 850 to as loan to MSMEs create climate resilient                                |

|                                                    |                                                          |                                                                                               |   |                                                                               |                                                                                                                                                                             |                                                                                                                                                                                                     |
|----------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <u>more able to reduce GHG emissions</u>                 | verification at mid and end of term                                                           |   | MSMEs for climate resilient machineries/appliances                            | MSMEs for climate resilient machineries/appliances<br><br>Life time target USD Million 850 provided as loan to MSMEs for climate resilient machineries/appliances           | assets/machineries/appliances to reduce GHG emissions                                                                                                                                               |
| <u>All adaptation result areas (Core 2)</u>        | <u>Core 2: Direct and indirect beneficiaries reached</u> | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0 | Mid term target (Indirect):<br>Total: 3,241,004<br>M: 2,592,804<br>F: 648,200 | Final term target (Indirect):<br>Total: 10,803,348<br>M: 8,642,679<br>F: 2,160,669<br><br>Life time target (Indirect):<br>Total: 10,803,348<br>M: 8,642,679<br>F: 2,160,669 | Indirect: 10,803,348<br>The assumption is different for different types of technology and is based on the impact that the technology creates to increase the resilience of the communities.         |
| <u>ARA1 Most vulnerable people and communities</u> | <u>Core 2: Direct and indirect beneficiaries reached</u> | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0 | Mid term target (Indirect):<br>Total: 3,241,004<br>M: 2,592,804<br>F: 648,200 | Final term target (Indirect):<br>Total: 10,803,348<br>M: 8,642,679<br>F: 2,160,669                                                                                          | The program is envisaged to provide support to indirect beneficiaries only. The assumption is different for different types of technology and is based on the impact that the technology creates to |



|                                                         |                                                                                                                      |                                                                                               |   |                                                                               |                                                                                                                                                                           |                                                                                                                                                                                                                                    |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         |                                                                                                                      |                                                                                               |   |                                                                               | Lifetime target (Indirect):<br>Total:<br>10,803,348<br>M: 8,642,679<br>F: 2,160,669                                                                                       | increase the resilience of the communities.                                                                                                                                                                                        |
| <u>ARA1 Most vulnerable people and communities</u>      | <u>Supplementary 2.5: Beneficiaries (female/male) adopting innovations that strengthen climate change resilience</u> | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0 | Mid term target (Indirect):<br>Total: 6,495<br>M: 5,196<br>F: 1,299           | Final term target (Indirect):<br>Total: 21,653<br>M: 17,323<br>F: 4,330<br><br>Lifetime (Indirect):<br>Total: 21,653<br>M: 17,323<br>F: 4,330                             | A total of 21,653 beneficiaries will be impacted through creation of 231 innovative technologies during its project period. These innovative products will be context specific and locally suited and would be used for adaptation |
| <u>ARA2 Health, well-being, food and water security</u> | <u>Core 2: Direct and indirect beneficiaries reached</u>                                                             | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0 | Mid term target (Indirect):<br>Total: 2,992,893<br>M: 2,394,315<br>F: 598,578 | Final term target (Indirect):<br>Total: 9,976,310<br>M: 7,981,048<br>F: 1,995,262<br><br>Life time target (Indirect):<br>Total: 9,976,310<br>M: 7,981,048<br>F: 1,995,262 | The assumption is different for different types of technology and is based on the impact that the technology creates to increase the resilience of the communities.                                                                |
| <u>ARA2 Health, well-being, food and water security</u> | <u>Supplementary 2.2: Beneficiaries (female/male) with improved food security</u>                                    | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0 | Mid term target (Indirect):<br>Total: 170,090<br>M: 136,072<br>F: 34,018      | Final term target (Indirect):<br>Total: 566,967<br>M: 453,574<br>F: 113,393                                                                                               | Introduction of pack houses will insure less post harvest food loss and will increase food security amongst the communities.                                                                                                       |

|                                                         |                                                                                                        |                                                                                               |   |                                                                               |                                                                                                                                                                          |                                                                                                                                                         |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         |                                                                                                        |                                                                                               |   |                                                                               | Life time target (Indirect):<br>Total: 566,967<br>M: 453,574<br>F: 113,393                                                                                               |                                                                                                                                                         |
| <u>ARA2 Health, well-being, food and water security</u> | <u>Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security</u>       | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0 | Mid term target (Indirect):<br>Total: 2,822,803<br>M: 2,258,242<br>F: 564,561 | Final term target (Indirect):<br>Total: 9,409,343<br>M: 7,527,475<br>F: 1,881,868<br><br>Lifetime target (Indirect):<br>Total: 9,409,343<br>M: 7,527,475<br>F: 1,881,868 | Introduction of water efficient technologies in water stress regions will insure less extraction of fresh water hence making the communities resilient. |
| <u>ARA2 Health, well-being, food and water security</u> | <u>Supplementary 2.4: Beneficiaries (female/male) covered by new or improved early warning systems</u> | Data gathered through MIS from MSMEs, Independent ex-post verification at mid and end of term | 0 | Mid term target (Indirect):<br>Total: 240,487<br>M: 192,390<br>F: 48,097      | Final term target (Indirect):<br>Total: 801,623<br>M: 641,299<br>F: 160,324<br><br>Life time target (Indirect):<br>Total: 801,623<br>M: 641,299<br>F: 160,324            | The assumption is that these technologies will increase the resilience of the MSMEs and related beneficiaries towards predicting climatic uncertainties |

#### E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

| Core Indicator                                                                                                                                                                                                  | Baseline context (description)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Rating for current state (baseline) | Target scenario (description)                                                                                                                                                                                         | How the project will contribute                                                                                                                                                                                                                                                                                                                          | Coverage                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u> | According to a report, total energy consumed in MSME, 15% is electricity consumption and 85% is thermal energy consumption. The projected energy consumption shows a steep increase (by more than 100%) from CO2 equivalent of 30 million tonnes in 2016-17 to 72.17 million tonnes in 2029-30 (FMC and GIZ 2018). To overcome this MSMEs in India will require to use energy efficient technology and reduce their dependency on conventional energy and go for renewable energy generation. This will also lead to both mitigation and adaptation against climate effects. | <u>low</u>                          | >500 MSMEs, specially manufacturing as well as solution / service provider addressing adoption of clean technologies towards emission reduction and providing solutions to enhance climate resilience of communities. | FMAP will set up a dedicated Mitigation and Adaptation Fund targeting the reduction of carbon emissions and enhancing the resilience to the impacts of climate change, in line with the requirements of MSMEs based on its past experience of working with MSMEs. A total reduction of ~34 million tco2 is envisaged to be attained through this project | <u>National level (one country)</u> |
| <u>Core Indicator 6: Degree to which GCF investments contribute to technology</u>                                                                                                                               | Energy efficient technologies exist for certain use cases in industry and                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <u>low</u>                          | Market for energy efficient technologies and sustainable transport through EV.                                                                                                                                        | SIDBI will identify the MSMEs which bring innovative technologies and business models to                                                                                                                                                                                                                                                                 | <u>National level (one country)</u> |

|                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                       |            |                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                        |                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <u>deployment, dissemination, development or transfer and innovation</u>                                                                                                             | transportation, novel technologies are needed for use cases such as replacement of energy efficient machinery process in line in industries.                                                                                                                                                                                          |            | This will involve not only increased adoption of existing technologies, but continuous innovation (and investments) in research and development across the sector.                                                                                                                                                                              | market, supporting them in business building and scale-up by leveraging its deep networks across innovation, industry and policy ecosystems.                                                                                                                                                                                           |                                     |
| <u>Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>                                   | India has immense potential to leapfrog its development towards energy efficient and low-carbon pathways.                                                                                                                                                                                                                             | <u>low</u> | Strong financial markets offering a range of products to support investment of capital across the sector, which increases the penetration of low-carbon and climate resilient technologies. This should in turn, put downward pressure on the cost curve and make these technologies even more attractive compared to traditional alternatives. | FMAP will provide financing to MSMEs adopting energy efficient solutions which will support scaling up of low-carbon and climate friendly technologies which have the potential to be replicated to further scale-up the market impact.                                                                                                | <u>National level (one country)</u> |
| <u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u> | SIDBI will perform the duty of Accredited Entity as well as the Executing Entity for the proposed project. SIDBI is the principal development financial institution set up by Government of India under an Act of Parliament in 1990 with the mandate of promotion, financing, and development, of MSMEs and co-ordination with other | <u>low</u> | Knowledge of financing products and best practices within each subsector are developed. Increased capacity among financial institutions to provide customized financing solutions to companies/customers in the space at attractive terms. Financiers provide knowledge and learnings to support measure outcomes and                           | FMAP will provide support for EE, EV and green technologies to the MSMEs that have the potential to be replicated. SIDBI will have monitoring and evaluation systems in place to measure the outcomes and impacts of these interventions and also identify lessons learned, it will also participate in forums of knowledge sharing to | <u>National level (one country)</u> |

|  |                                        |  |                                 |                                                                  |  |
|--|----------------------------------------|--|---------------------------------|------------------------------------------------------------------|--|
|  | entities engaged in similar activities |  | impacts of their interventions. | influence market behaviour, regulations and policy in the space. |  |
|--|----------------------------------------|--|---------------------------------|------------------------------------------------------------------|--|

### E.5. Project/programme specific indicators (project outcomes and outputs)

*This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the “Project/programme co-benefit indicators” section in table below. Add rows as needed.*

*Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).*

| Project/programme results (outcomes/ outputs)                                                             | Project/programme specific Indicator                                                       | Means of Verification (MoV)               | Baseline | Target     |            | Assumptions / Note                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------|----------|------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                           |                                                                                            |                                           |          | Mid-term   | Final      |                                                                                                                                                                                                                        |
| <b>Outcome 1:</b> Adoption of low-emission climate-resilient technologies across MSME sectors             | No. of loan disbursed/ sanctioned to MSMEs and they have adopted the proposed technologies | MIS/MPR and M&V through visits by RMs/M&E | 0        | 2963 MSMEs | 9876 MSMEs | Energy efficient technologies exist for certain use cases in industry and transportation, novel technologies are needed for use cases such as replacement of energy efficient machinery process in line in industries. |
| <b>Outcome 2:</b> Strengthened finance and market environment for upscaling commercialised adaptation and | No. of loan disbursed/ sanctioned to MSMEs and they have adopted the proposed technologies | MIS/MPR and M&V through visits by RMs/M&E | 0        | 2963 MSMEs | 9876 MSMEs | Energy efficient technologies exist for certain use cases in industry and transportation, novel technologies are needed for use cases such as replacement of energy                                                    |
|                                                                                                           | Number of electric vehicles financed                                                       | MIS/MPR and M&V through visits by RMs/M&E | 0        | 41974      | 139915     |                                                                                                                                                                                                                        |

|                                                                                                                     |                                                                                                       |                                           |   |                                                                   |                                                                   |                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------|---|-------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| mitigation technologies through MSMEs                                                                               | Number of solar projects financed                                                                     | MIS/MPR and M&V through visits by RMs/M&E | 0 | 370                                                               | 1231                                                              | efficient machinery process in line in industries.                                                                                                                                                           |
|                                                                                                                     | Volume of (USD millions) financing approved and disbursed to MSMEs, disaggregated by targeted sectors | MIS/MPR and M&V through visits by RMs/M&E | 0 | EE- USD 255 million<br>PV- USD 255 Million<br>EV- USD 102 million | EE- USD 850 million<br>PV- USD 850 million<br>EV- USD 340 million |                                                                                                                                                                                                              |
|                                                                                                                     | No. of FIs selected and participating in the program                                                  | MIS/MPR and M&V through visits by RMs/M&E | 0 | 8                                                                 | 25                                                                | The program will target empanelling 25 numbers of PFIs.                                                                                                                                                      |
|                                                                                                                     | No. of loans disbursed by FIs, disaggregated by sectors                                               | MIS/MPR and M&V through visits by RMs/M&E | 0 | EE- 706<br>PV- 706<br>EV- 282                                     | EE- 2352<br>PV- 2352<br>EV- 941                                   | 80% of the program financing will be through PFIs                                                                                                                                                            |
| Output 1.1: MSME mitigation technologies financed and implemented. Increased access to mitigation finance for MSMEs | No. of loan disbursed/ sanctioned to MSMEs                                                            | MIS/MPR                                   | 0 | 1693 MSMEs                                                        | 5645 MSMEs                                                        | Project envisaged to support a total of 5,645 mitigation projects in the following mitigation areas:<br>EE- 2352<br>EV- 941<br>Solar PV- 2352                                                                |
| Output 1.2: MSME adaptation technologies financed and implemented. Increased access to adaptation finance for MSMEs | No. of MSMEs adopted the proposed technologies                                                        | MIS/MPR and M&V through visits by RMs/M&E | 0 | 1269 MSMEs                                                        | 4232 MSMEs                                                        | The project is envisaged to support 4232 adaptation projects in the area such as Water, wastewater, cold supply chain, Food Processing including sea food, fisheries, beverages, juice, Solar Boats, Others) |
| <b>Outcome 3:</b> Improved resilience of the most vulnerable populations, reduced GHG                               | No. of MSMEs adopted the proposed technologies                                                        | MIS/MPR and M&V through visits by RMs/M&E | 0 | 2962 MSMEs                                                        | 9876 MSMEs                                                        | Energy efficient technologies exist for certain use cases in industry and                                                                                                                                    |



|                                                                                                                                                                                                                                             |                                                                 |                                                     |   |              |               |                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------|---|--------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| emissions in the MSME sector of India                                                                                                                                                                                                       |                                                                 |                                                     |   |              |               | transportation, novel technologies are needed for use cases such as replacement of energy efficient machinery process in line in industries.                                                                                                      |
| Output 2.1: Capacity built, and knowledge management improved to support climate change adaptation and mitigation in MSMEs and PFIs. Capacities of selected MSMEs and PFIs are enhanced to rapidly scale their business and climate impact. | No. of trainings organized                                      | MIS/MPR, training reports                           | 0 | 48 Trainings | 160 Trainings | SIDBI will build capacities of the MSMEs/PFIs. Further, various training on climate change mitigation and adaptation will be carried out for the prospective borrower from the PFI side.                                                          |
|                                                                                                                                                                                                                                             | No. of policy frameworks to support green financing developed   | Integrated Digital Green Credit Rating Tool (DGCRT) | 0 | 1            | 1             | An established firm will be hired to develop an Integrated Digital Green Credit Rating Tool (DGCRT) consisting of Green parameters (Energy and Environment). The tool will be implemented for five years with a cost estimate of USD 120000/year. |
| Output 2.2: Enabling environment for a low-emission and climate-resilient MSME sector                                                                                                                                                       | No. of units adopted the proposed energy efficient technologies | MIS/MPR                                             | 0 | 705 loans    | 1647 loans    | Strengthening environmental regulations and standards can drive demand for greener products and services from MSMEs while improving overall environmental performance. Gradually tightening emissions standards, mandating energy audits, and     |
|                                                                                                                                                                                                                                             | No. of units adopted Electric vehicle                           | MIS/MPR                                             | 0 | 282 loans    | 659 loans     |                                                                                                                                                                                                                                                   |
|                                                                                                                                                                                                                                             | No. of units adopted Solar power (PV) generation option         | MIS/MPR                                             | 0 | 705 loans    | 1647 loans    |                                                                                                                                                                                                                                                   |

|                                                       |                                                                 |                                           |   |                    |                    |                                                                                                                                                                                                                |
|-------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------|---|--------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       |                                                                 |                                           |   |                    |                    | introducing eco-labelling schemes are some ways to shift MSME behaviour                                                                                                                                        |
| <b>Project/programme co-benefit indicators</b>        |                                                                 |                                           |   |                    |                    |                                                                                                                                                                                                                |
| Co-benefit 1: Improved water supplies                 | No of beneficiary having access to water efficient technologies | MIS/MPR and M&V through visits by RMs/M&E | 0 | 2,822,802          | 9,409,343          | Deploying water efficient technologies in MSMEs located in water stress regions will save water and increase number of people residing in the periphery of the unit will have access to improved water supply. |
| Co-benefit 2: Enhanced job creation assistance        | No. of direct jobs created                                      | MIS/MPR and M&V through visits by RMs/M&E | 0 | 126,405 Jobs       | 421,350 Jobs       | The employment generation per million \$ of investment in green energy in India was evaluated to be at 116.1 direct jobs in Energy Efficiency and 215 direct jobs in Renewable Energy.                         |
| Co-benefit 3: Greater gender equality in MSMEs        | No. of women owned MSMEs                                        | MIS/MPR and M&V through visits by RMs/M&E | 0 | 150 units          | 500 units          | The program will benefit overall 5% of MSMEs to be financed under this program.                                                                                                                                |
| Co-benefit 4: Improved productivity and profitability | Assessment of productivity and operational benefits             | MIS/MPR and M&V through visits by RMs/M&E | 0 | USD 26.17 Million  | USD 87.24 Million  | Studies indicate that the value of the productivity and operational benefits derived can be up to 2.5 times the value of the energy savings (IEA, 2014b)                                                       |
| Co-benefit 5: Reduced resource use and pollution      | Water saving                                                    | MIS/MPR and M&V through visits by RMs/M&E | 0 | 0.43 million litre | 1.45 million litre | Deploying water efficient technologies in MSMEs located in water stress regions will save water and increase number of                                                                                         |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                      |                                                                                                                             |  |  |                                                                                                                                                | people residing in the periphery of the unit will have access to improved water supply. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>E.6. Project/programme activities and deliverables</b>                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                      |                                                                                                                             |  |  |                                                                                                                                                |                                                                                         |
| <p><i>All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.</i></p> <p><i>Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.</i></p> |                                                                                      |                                                                                                                             |  |  |                                                                                                                                                |                                                                                         |
| Activities                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Description                                                                          | Sub-activities                                                                                                              |  |  | Deliverables                                                                                                                                   |                                                                                         |
| <b>Activity 1.1.1:</b> Establish a loan instrument to provide concessional loans for MSME mitigation needs.                                                                                                                                                                                                                                                                                                                                                | FMAP aims to provide support to MSMEs through concessional loan for mitigation needs | Sub-activity 1.1.1.1: Carry out detailed technical assessment of potential MSME units                                       |  |  | Detailed technical assessment and credit risk rating evaluation of around 5957 potential MSME units                                            |                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                      | Sub-activity 1.1.1.2: Carry out credit risk rating evaluation of potential MSME units                                       |  |  | Detailed technical assessment and credit risk rating evaluation of around 5957 potential MSME units                                            |                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                      | Sub-activity 1.1.1.3: Selection of MSMEs according to Eligibility Criteria as well as technical and risk assessment results |  |  | Due diligence at GCFV using existing credit rating tool and creation of one due diligence checklist, which will be applicable in all the cases |                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                      | Sub-activity 1.1.1.4: Incentivize the units to create green MSME ambassadors and a group of champion units                  |  |  | 05 number of green MSME ambassadors and group of champion units created per year                                                               |                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                      | Sub-activity 1.1.1.5: Online loan application and speedy appraisal process setup                                            |  |  | Creation of one online loan application and speedy appraisal process                                                                           |                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                      | Sub-activity 1.1.1.6: End to end technical and handholding support to MSMEs, PFIs, NBFCs, and other stakeholders            |  |  | End to end technical and handholding support to 263 MSMEs, PFIs, NBFCs, and other stakeholders                                                 |                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                      | Sub-activity 1.1.1.7: Facilitate networking of technology providers                                                         |  |  | Facilitate around 20 networking of technology providers                                                                                        |                                                                                         |
| <b>Activity 1.2.1:</b> Establish a loan instrument to provide concessional loans for MSME adaptation needs.                                                                                                                                                                                                                                                                                                                                                | FMAP aims to provide support to MSMEs through concessional loan for adaptation needs | Sub-activity 1.2.1.1: Carry out detailed technical assessment of proposal of potential MSMEs                                |  |  | Detailed technical assessment and credit risk rating evaluation of around 3971 potential MSME units                                            |                                                                                         |

|                                                                                                                                                   |                                                                                                                                    |                                                                                                                                                                  |                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                   |                                                                                                                                    |                                                                                                                                                                  |                                                                                                                                                                                              |
|                                                                                                                                                   |                                                                                                                                    | Sub-activity 1.2.1.2: Carry out credit risk rating evaluation of potential MSMEs                                                                                 | Detailed technical assessment and credit risk rating evaluation of around 3971 potential MSME units                                                                                          |
|                                                                                                                                                   |                                                                                                                                    | Sub-activity 1.2.1.3: Selection of MSMEs according to Eligibility Criteria as well as technical and risk assessment results                                      | Due diligence at GCFV using existing credit rating tool and creation of one due diligence checklist, which will be applicable in all the cases                                               |
|                                                                                                                                                   |                                                                                                                                    | Sub-activity 1.2.1.4: Incentivize the units to create green MSME ambassadors and a group of champion units                                                       | 05 numbers green MSME ambassadors and group of champion units created per year                                                                                                               |
|                                                                                                                                                   |                                                                                                                                    | Sub-activity 1.2.1.5: Online loan application and speedy appraisal process setup                                                                                 | Creation of one online loan application and speedy appraisal process                                                                                                                         |
|                                                                                                                                                   |                                                                                                                                    | Sub-activity 1.2.1.6: End to end technical and handholding support to MSMEs, PFIs, NBFCs, and other stakeholders, facilitate networking of technology providers. | End to end technical and handholding support to 262 MSMEs, PFIs, NBFCs, and other stakeholders                                                                                               |
| <b>Activity 2.1.1:</b> Undertake baseline studies for 500 MSMEs                                                                                   | Conduct baseline and feasibility study                                                                                             | 2.1.1.1 Undertake Technology Need Assessments for 500 MSMEs                                                                                                      | Baseline and feasibility studies will be conducted in 500 number of MSMEs                                                                                                                    |
| <b>Activity 2.1.2:</b> Undertake awareness raising and trainings on climate change adaptation and mitigation for MSMEs and Financial Institutions | Conduct awareness raising trainings and trainings on climate change adaptation and mitigation for MSMEs and Financial Institutions | 2.1.2.1 Undertake awareness raising and trainings on climate change adaptation and mitigation for MSMEs and Financial Institutions                               | 32 No. of trainings will be conducted- (one training per quarter per cluster in initial five years of the project)                                                                           |
| <b>Activity 2.1.3:</b> Implement knowledge management and sharing on climate change adaptation and mitigation for MSMEs                           | Case studies and documentation of best practices                                                                                   | 2.1.1.3 Implement knowledge management and sharing on climate change adaptation and mitigation for MSMEs                                                         | Case study and research study reports- five case and research study per sectors i.e. EE, EV, RE generation and adaptation. A total of 20 no. of case research/case studies will be conducted |
| <b>Activity 2.2.1:</b> Develop and implement MSME sectoral adaptation and mitigation benchmarking tools                                           | The program will contribute to the upgrading and further development of the online assessment / benchmarking tools for MSMEs.      | 2.2.1.1 Develop and implement MSME sectoral adaptation and mitigation benchmarking tools                                                                         | Tool- - One tool prepared at the start of the program                                                                                                                                        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                  |                                                                              |                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------|
| Activity 2.2.2: Digital Green Credit Rating Tool (DGCRT) includes integration of Green parameters (Energy and Environment) with Credit rating tool                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | This activity will support the digitization of an integrated credit rating tool which includes integration of green parameters (Energy and Environment) into credit rating tool. | 2.2.2.1 Contribute to Policy Initiatives to support Green Financing of MSMEs | 2.2.2.1 One comprehensive credit rating tool will be developed |
| <b>E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                  |                                                                              |                                                                |
| <p>Monitoring and verification (M&amp;V) for verification of project indicators will be undertaken once in two years, starting from the 3rd year onwards, on a sample basis. Indicators tracked under M&amp;V activities will include implementations of sub-projects, energy savings, environmental benefits e. g. CO2 emission reductions, and other co-benefits at programme level.</p> <p>As an Accredited Entity, SIDBI will be responsible to Oversee, Supervise, Manage and Monitor the overall implementation of the programme and periodically submit progress reports to the GCF as agreed in the Funded Activity Agreement (FAA) and the Accreditation Management Agreement (AMA). Executing Entity headed by the Project Director consists of experts shall be responsible for the execution of day-to-day activities of the programme for direct loans as well as for the loan sectioned through PFIs and report to the AE about the progress of the FMAP. Experts placed at Executing Entity (EE) will be responsible for M&amp;V of sub projects sanctioned directly and through PFIs. EE will ensure in house M&amp;V as well as M&amp;E through independent agencies.</p> <p>Additionally, a Management Information System (MIS) will be developed to assist in project tracking, including dashboards for loan cases and online regular monitoring of proposals and related outcomes indicators. AE will collate the data and will further report to NDA and GCF.</p> <p>Moreover, an Impact Evaluation at the end of the project will be Undertaken. Lastly, as part of the project monitoring process, internal financial audits will be conducted annually. Measurement, Reporting and Verification (MRV) of Climate Impacts- Mitigation and Adaptation will be achieved through the use of energy efficient technologies by MSMEs.</p> |                                                                                                                                                                                  |                                                                              |                                                                |

## F. RISK ASSESSMENT AND MANAGEMENT

### F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

#### Selected Risk Factor 1

| Category                         | Probability   | Impact        |
|----------------------------------|---------------|---------------|
| <u>Technical and operational</u> | <u>Medium</u> | <u>Medium</u> |

#### Description

Please describe the risk to the best of your knowledge at this point in time.

The technical risk may be owing to lack of understanding among the end beneficiaries regarding a particular Resource Efficient and Cleaner Production (RECP) technology which may hinder its adoption and may affect market demand for these technologies. The operational risk may be related to after end of sale services wherein servicing of these equipment/technologies may be hindered by lack of service providers in the market. In addition, MSMEs may be unable to invest 10%, resulting in weak commercial appetite from MSMEs in the FMAP facility.

#### Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

- Technical Assistance component includes various activities such as awareness creation, capacity building of various stakeholders to motivate industries and enabling them to make an informed decision.
  - Further, the programme envisages providing end to end handholding support to participating units. The approach has been very fruitful during the implementation of similar programmes in the past by SIDBI.
  - The proposed activities will also support “development of mentors” (Cluster champions or stalwarts) and will also perform various cluster development activities under “Project GRiT (Climate inclusiviTy)” managed by SIDBI.
- The operational risks related to end of sale services will be mitigated by the market itself. It is understood that end of sale services is difficult to achieve for limited number of technologies, however, due to large scale technologies funded under this programme, there is bound to be spillover effect in the market which would incentivise service providers to scale up their operations in the market.

#### Selected Risk Factor 2

| Category      | Probability | Impact        |
|---------------|-------------|---------------|
| <u>Credit</u> | <u>High</u> | <u>Medium</u> |

#### Description

Please describe the risk to the best of your knowledge at this point in time.

The risk perceived under this category relates to the MSMEs those may become non performing over the period of time due to various factors including market crunch, loss of revenue, or any other situation beyond their control. Credit risk is the possibility of a loss happening due to a borrower's failure to repay a loan or to satisfy contractual obligations which may result in the MSMEs becoming non performing over the period of time due to various factors including market crunch, loss of revenue, or any other situation beyond their control.

#### Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?



In line with the RBI guidelines, SIDBI has developed a detailed assessment process for risk measurement following which investment decisions are taken. Some of the Risk Mitigation Measures are –

- First charge over all assets created under the programme.
- Personal guarantee of promoter(s)
- Corporate Guarantee of group companies, if available and feasible
- Any other security as offered by the borrower / other unencumbered assets of the Borrower, to meet asset coverage norms.
- PRSF / CGTMSE / NCGTC cover may be obtained, wherever felt necessary.
- Further, as per SIDBI's annual report for FY22-23, the Gross NPA and Net NPA of the Bank stood at nominal 0.01% and 0.00% during the year. There was no NPA reported in the climate lending portfolio of SIDBI. The low rate of NPA is owing to various risk management policies of the organization, rating tools, usage of block chain for security information services, focused monitoring of potential units that might defer etc.
- SIDBI commits USD 800 million, through co-financing commitment letter and documented in the term sheet.

### Selected Risk Factor 3

| Category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Probability   | Impact        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|
| <u>Forex</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <u>Medium</u> | <u>Medium</u> |
| <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |               |
| <p><i>Please describe the risk to the best of your knowledge at this point in time.</i></p> <p>Dollar value may change with respect to time. Foreign exchange risk is the possibility for an entity to be affected by a variation in the exchange rate between its local currency and the currency used in a transaction with a foreign country.</p>                                                                                                                                                                                          |               |               |
| <b>Mitigation Measure(s)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |               |               |
| <p><i>Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?</i></p> <p>The funds requested from GCF will be in USD while SIDBI will lend in INR. SIDBI will be managing the currency risks through appropriate hedging mechanisms and may charge risk premium to the end borrowers. The indicative annual hedging cost is taken as 2.85 percent per annum and is indicated in assumptions given in Annexure 3- Financial model.</p> |               |               |

### Selected Risk Factor 4

| Category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Probability   | Impact        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|
| <u>ML/FT</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <u>Medium</u> | <u>Medium</u> |
| <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |               |               |
| <p><i>Please describe the risk to the best of your knowledge at this point in time.</i></p> <p>Banks and other financial institutions may be used as a channel for Money Laundering (ML)/ Terrorist Financing (TF) hence there is an associated risk related to ML/FT</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |               |
| <b>Mitigation Measure(s)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |               |               |
| <p><i>Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?</i></p> <p>SIDBI has a well formulated policy to prevent ML/FT with well laid out procedure to conduct risk assessment which is attached as Annexure. Yes, the mitigation measure can well reduce the possibility of a risk originating from ML/FT. a) SIDBI shall carry out 'Money Laundering (ML) and Terrorist Financing (TF) Risk Assessment' exercise periodically to identify, assess and take effective measures to mitigate its money laundering and terrorist financing risk for clients, countries or geographic areas, products, services, transactions or delivery channels, etc. The assessment process should consider all the relevant risk factors before determining the level of overall risk and the appropriate level and type of mitigation to be applied. While preparing the internal risk assessment, SIDBI shall take cognizance of the overall sector-specific vulnerabilities, if any, that the regulator/supervisor may share with SIDBI from time to time. (b) The risk assessment by SIDBI shall be properly documented and be proportionate to the nature, size, geographical presence, complexity of activities/structure, etc. of SIDBI. Further, the periodicity of risk assessment exercise shall be</p> |               |               |

determined by the Board or any other Committee of the Boards of SIDBI to which power in this regard has been delegated, in alignment with the outcome of the risk assessment exercise. However, it should be reviewed at least annually. (c) The outcome of the exercise shall be put up to the Board or Audit Committee of the Board and should be available to competent authorities and self-regulating bodies. (d) SIDBI shall apply a Risk Based Approach (RBA) for mitigation and management of the identified risk and would have Board approved policies, controls and procedures in this regard. Bank shall implement a Customer Due Diligence exercise, having regard to the ML/FT risks identified and the size of business. Further, SIDBI shall monitor the implementation of the controls and enhance them if necessary.

| Selected Risk Factor 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------|
| Category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Probability | Impact     |
| <u>Other</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <u>Low</u>  | <u>Low</u> |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |            |
| <i>Please describe the risk to the best of your knowledge at this point in time.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |            |
| Security risk may be related to law and order in any jurisdiction, it may also be due to international territorial disputes.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |            |
| Mitigation Measure(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |            |
| <i>Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?</i>                                                                                                                                                                                                                                                                                                                                                                                                       |             |            |
| In this programme, though we will be focusing on Himalayan region for our adaptation interventions, not all areas within Himalayan region have security risk. Also, the industrial units to be financed are located in regions with low or minimal security risks with the law and order enforcement being managed by the state and central government. So, mitigations related to security risks (if any) are managed by the respective states and central government. The programme will not target areas which are affected by international territorial disputes/issues. |             |            |

## G. POLICIES AND STANDARDS

*Provide the environmental and social risk category assigned to the proposal as a result of screening and the rationale for assigning such category. Present also the environmental and social assessment and management instruments developed for the proposal (for example, Environment and Social Impact Assessment (ESIA), Environment and Social Management Framework (ESMF), Environment and Social Management Plan (ESMP), Environmental and Social Management System (ESMS), environmental and social audits, etc.). Provide a summary of the main outcomes of these instruments. Present the key environmental and social risks and impacts and the measures on how the project/programme will avoid, minimize and mitigate negative impacts at each stage (e.g. preparation, implementation and operation), in accordance with GCF's Environmental and Social Safeguard (ESS) standards. If the proposed project or programme involves investments through financial intermediations, describe the due diligence and management plans by the Executing Entities (EEs) and the oversight and supervision arrangements. Describe the capacity of the EEs to implement the ESMP and ESMF and arrangements for compliance monitoring, supervision and reporting. Include a description of the project/programme-level grievance redress mechanism, a summary of the extent of multi-stakeholder consultations undertaken for the project/programme, the plan of the Accredited Entity (AE) and EEs to continue to engage the stakeholders throughout project implementation, and the manner and timing of disclosure of the applicable safeguards reports following the requirements of the GCF [Information Disclosure Policy](#) and [Environmental and Social Policy](#).*

*Describe any potential impacts on indigenous peoples and the measures to address these impacts including the development of an Indigenous Peoples Plan and the process for meaningful consultation leading to free, prior and informed consent, pursuant to the GCF [Indigenous Peoples Policy](#).*

*Attach the appropriate assessment and management instruments or other applicable studies, depending on the environmental and social risk category as annex 6.*

### **Environmental and social risk categorization:**

The Environmental and Social Management System (ESMS) is prepared referring to policies, act, rules of the government of India and Green Climate Fund (GCF) of Project/Program Financing.

The existing national policies and legislations, international frameworks on environment, social and energy efficiency issues along with the ESMP of Accredited Entity (SIDBI) and GCF's environmental and Social Safeguard Policies were reviewed for the ESS categorization of the proposal. Further, the ESMS has been aligned with IFC's Performance Standards. The potential adverse environmental and/or social risks and impacts related to the Programme activities are individually or cumulatively generally site-specific, largely reversible, and readily addressed through adequate policy implementation and workplace standards. Further, the initial screening based on Exclusion List will ensure that Category A/I1 projects are screened out and will not be financed under this programme. The activities and sub-activities under this programme therefore would be classified as a Category 'B' under the GCF Risk Categories. However, the program will be categorized as Medium level of intermediation, or I2 according to GCF guidelines as financial intermediation will be involved in the investment process

### **Compliance with National Legislation:**

As a responsible financier, SIDBI has adopted various institutional mechanisms to ensure its compliance with the relevant environmental and social laws/norms. Under this, SIDBI has:

- Adopted an Environmental and Social Management Policy (ESMP) which ensures that it's initiatives are aimed at reducing environmental and social risks. The ESMP outlines SIDBI's engagement with PFIs in the case of indirect financing.
- Adopted the Sexual Harassment of Women (Prevention, Prohibition & Redressal) at Workplace as per the Sexual Harassment of Women (Prevention, Prohibition & Redressal) at Workplace Act, 2013
- Adopted a Customer Grievance Redressal Policy to ensure prompt and efficient service to its customers.

These detailed policies are placed in Annexure 1, Attachment 2 & 3 respectively at the end of Annexure 6.

In compliance with GCF's ESMS framework, the ESMS framework for this programme aims to achieve the following outcomes/meet performance standards:

1. **List of projects supported/Screening:** Based on the risk categorization, SIDBI's ESMP identifies the type/nature of projects to be supported.

2. **Identify and assess environmental & social risks:** Projects are assessed for their potential environmental and social impacts. This includes both negative impacts that need to be mitigated and potential benefits that can be enhanced.
3. **Manage & Monitor Risk, Reporting & Compliance:** Based on the assessments, management plans are developed to address the risks through various measures including mitigation strategies, monitoring plans, and stakeholder engagement activities. Further, it ensures compliance with national regulations and international standards, and reporting on environmental and social performance.
4. **Grievance Redressal:** This also includes a Grievance Redressal Mechanism to ensure all grievances are addressed in line with applicable policies and standards.
5. **Public Disclosure and Consultation:** SIDBI's through its website, will publicly disclose relevant reports in compliance with its Information Disclosure policy. The results of the preliminary environmental and social screening, and draft environmental and social assessment, shall be made available for public consultations.
6. **Zero Tolerance of SEAH:** SIDBI has ZERO TOLERANCE towards any sexually harassment in any form. SIDBI has adopted the Sexual Harassment of Women (Prevention, Prohibition & Redressal) at Workplace as per the Sexual Harassment of Women (Prevention, Prohibition & Redressal) at Workplace Act, 2013.
7. **Indigenous People:** The COP decision adopting the Paris Agreement recognized the need to strengthen practices and efforts of local communities and Indigenous Peoples related to addressing and responding to climate change and operationalized the local communities and Indigenous Peoples platform to help do this. However, given that MSMEs are restricted to areas of industrial clusters/special economic zones, SIDBI's project is not anticipated to have impacts on indigenous people, their lands, culture, livelihood or way of life.

Details on each of the above elements of the ESMS framework are shared below:

### **Project Screening**

SIDBI itself, or the PFIs that are engaged, will screen the projects to ensure that the proposed projects are not in the Negative list/ Exclusion list prepared by SIDBI as per their ESMP (**Appendix I of Annexure 1 of the ESMS (annexure 6)**). The negative list includes projects/ programmes involving harmful or exploitative form of child labour, production or trade in weapons or munitions, production or trade in alcoholic beverages and tobacco, etc. The screening based on Exclusion List will ensure that Category A/I1 projects are screened out and will not be financed under this programme.

### **Identification & assessment of E&S risks**

Further to the initial screening of the projects, SIDBI or the PFIs are required to screen the potential risks and environmental and social compliance of the project through document verification, field visits, and environmental and social checklist (**Refer Appendix II of Annexure 1 of the ESMS (Annexure 6)**). Further, each project is evaluated for its E&S risk using a proprietary ESG rating tool. This allows SIDBI to characterize the risk being considered while financing and accordingly plan for the mitigation.

### **Manage & Monitor Risk, Reporting & Compliance**

SIDBI's ESMP (**Refer Section 9.2, 9.3, 9.4 of Annexure 1 of the ESMS (Annexure 6)**) states that it is the responsibility of the PFIs/Host entity to address all environmental and social risks identified during the screening of environmental and social risks. The implementing entities' programme reports shall include a section on status of implementation of any environmental and social management plan, including those measures required to avoid, minimize, or mitigate the risks.

The reports shall also include, if necessary, a description of any corrective actions that are deemed necessary. Accordingly, the mid- term and terminal evaluation reports shall also include an evaluation of Programme performance with respect to environmental and social risks.

The formats and periodicity of submitting such reports for effective monitoring is decided during the commencement of the Programme.

### **Grievance Redressal Mechanism**

SIDBI has a comprehensive 3-tier Grievance Redressal Mechanism (GRM) (**Refer Attachment 3 of ESMS Annexure 6**). It has put in place a web- enabled registration form for complaints/ grievances, whereby the project affected stakeholders can log their grievances.

SIDBI further mandates that the implementing agencies/ PFIs shall also have in place an effective GRM that provides for stakeholders affected by the projects supported by SIDBI. Such a mechanism shall be accessible, transparent, fair, and effective for receiving and addressing complaints about potential or actual environmental or social harm caused by any such project.

### **Public Disclosure & Consultation**

SIDBI mandates that the PFIs identify stakeholders as early as possible in planning any programme. The results of the preliminary environmental and social screening, and draft environmental and social assessment, shall be made available for public consultations.

Further, the scope and frequency of stakeholder engagement depends on the nature and scale of the programme and its potential environmental and social risks.

Additionally, SIDBI's existing ESMP states that SIDBI, through its website, will publicly disclose relevant reports in compliance with its Information Disclosure policy.

Further, the Annexure on Stakeholder Engagement (**Annexure 7 of Funding Proposal**) has details on the plan to consult stakeholders at a regular frequency. It also has key findings from extensive stakeholder consultations conducted by SIDBI.

### **Zero tolerance of SEAH**

SIDBI has adopted the Sexual Harassment of Women (Prevention, Prohibition & Redressal) at Workplace as per the Sexual Harassment of Women (Prevention, Prohibition & Redressal) at Workplace Act, 2013. The Act aspires to ensure women's right to workplace equality, free from sexual harassment through compliance with the mentioned three elements viz. "Prevention, Prohibition and Redressal."

SIDBI has ZERO TOLERANCE towards any sexually harassment in any form.

To give effect to the principle of zero-tolerance, FMAP recognizes the importance of its financed activities to prevent and respond effectively to SEAH in a survivor-centered and gender-responsive way.

SIDBI will further ensure that the implementing agencies/ financial institutions shall also have in place an effective SEAH guidelines, norms acts in place as per requirement.

### **Indigenous People**

Given that MSMEs are restricted to areas of industrial clusters/special economic zones, SIDBI's project is not anticipated to have adverse impacts on indigenous people, their lands, culture, livelihood or way of life.

However, SIDBI through its PFIs will ensure sub-projects through FMAP, include measures to (a) avoid potentially adverse effects on the Indigenous Peoples' communities, or (b) when avoidance is not feasible, minimize, mitigate, or compensate for such effects. The sub-projects are also designed to ensure that the Indigenous Peoples receive social and economic benefits that are culturally appropriate and gender and inter-generationally inclusive.

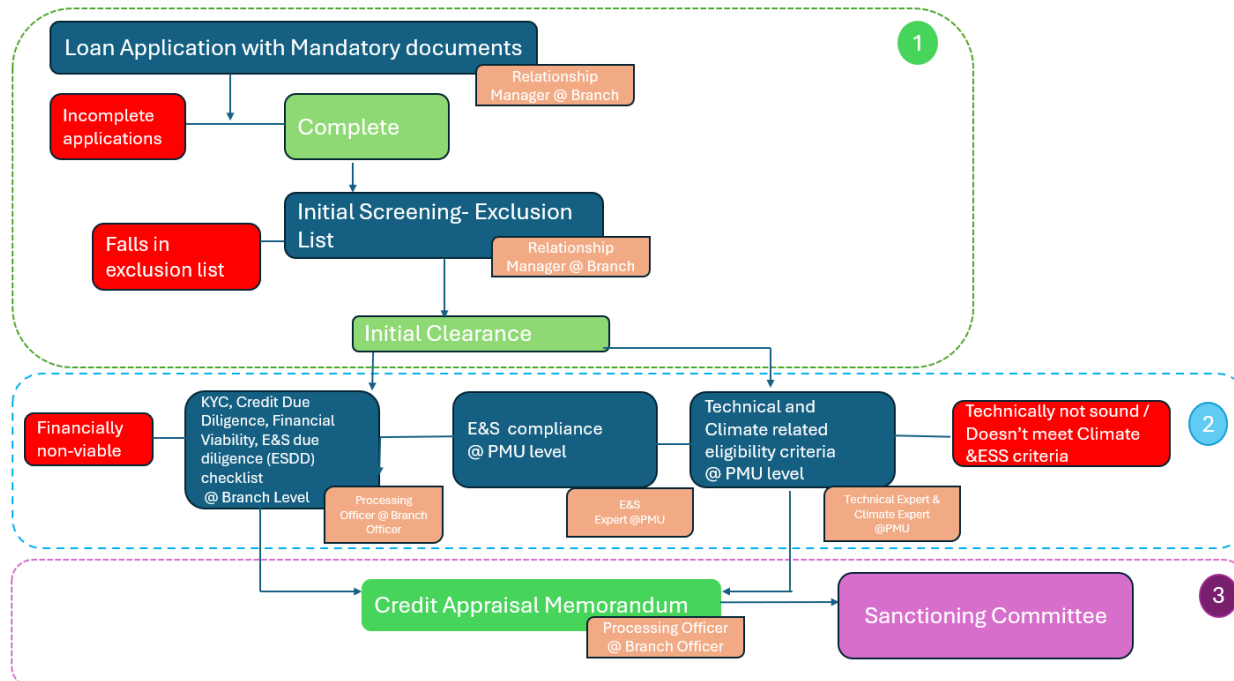
An Indigenous People Framework (IPPF) has been developed (**Refer Attachment 6 of the Annexure on ESMS**) to mitigate any potential risk arising out of the activities of the programme that may have adverse impacts on indigenous people. Further, an Indigenous People Plan (IPP) will be developed for all projects affecting indigenous communities and Indigenous Peoples where impacts are identified, whether they are direct or indirect, negative or positive, if applicable with necessary disclosure for IPPs.

### **Resettlement Action Plan:**

In case of greenfield projects where there is any likelihood of compulsory land acquisition and involuntary resettlement, a resettlement action plan has been developed which is aimed at the following:

- (a). To identify the risks at the sub-project level which may lead to compulsory land acquisition, involuntary resettlement affecting physical and/or economic displacement of people (loss of structures, livelihoods and access to common resources
- (b). To mitigate the identified risks associated with compulsory land acquisition and involuntary resettlement and to check the adequacy and implementation of the mitigation measures
- (c). To develop a monitoring framework to check the adequacy and implementation of mitigate measures.

**Flow Chart depicting E&S Risk Management Process:**



The ESMS also ensures compliance with India's legislative & regulatory landscape across relevant themes such as:

- Environment Protection
- Land Acquisition & Resettlement
- Labour & Working Conditions
- Social Safeguards on engagement & inclusion, cultural heritage and safety

**A comprehensive ESMS is attached in Annexure 6.**

**G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)**

*Provide a summary of the gender assessment and project/programme-level gender action plan that is aligned with the objectives of GCF's [Gender Policy](#). Confirm a gender assessment and action plan exists describing the process used to develop both documents. Provide information on the key findings (who is vulnerable and why) and key recommendations (how to address the vulnerability identified) of the gender assessment. Indicate if stakeholder consultations have taken place and describe the key inputs integrated into the action plan, including: how addressing the vulnerability will ensure equal participation and benefits from funds investment; key gender-related results to be expected from the project/programme with targets; implementation arrangements that the AE has put in place to ensure activities are implemented and expected outcomes will be achieved, monitored and evaluated.*

*Provide the full gender assessment and project-level gender action plan as annex 8.*

**Approach**

In preparing the Gender Assessment & Action Plan for the Programme, following approaches have been leveraged:

**1. Baseline assessment of participation of women in the MSME sector in India as entrepreneurs**

- The literature reviewed from secondary sources to understand the baseline situation and the broad scope for EE applications in the sector considering gender dimensions. In this regard, relevant reports, about national policies and guidelines (environmental, social, gender equality), historical studies and research on Energy Efficiency (EE), international best practices, potential EE interventions etc. considering gender equity were analysed.



- b. This assessment has been done based on SIDBI's existing initiatives to promote women entrepreneurship and based on research studies supported by SIDBI.

## 2. Assessment of challenges faced by women entrepreneurs in the MSME sector

- a. Stakeholder consultation i.e. "Strengthening Green Financing and Mainstreaming Gender" organised under the banner 'Green Indian Financial System (GIFS)<sup>85</sup> – Greening of Finance by Women (Grow)<sup>86</sup> Initiatives, founded by SIDBI, Shakti Sustainable Energy Foundation (SSEF) and French Development Agency (AFD) has identified the key challenges faced by women entrepreneurs in the MSME sector

## 3. Review of SIDBI's existing capabilities to promote women entrepreneurship/gender equality

- a. This assessment has been done based on a review of SIDBI's existing initiatives to promote women entrepreneurship and participation of women in the MSME sector.
- b. Further, SIDBI's gender policies have been reviewed in line with GCF's gender policies

### Key Findings & Assessment

- **Gender Parity in Workforce:** Only ~24% of the workforce in MSMEs is women. This is lower than the national labour force participation rates for females (~28.7% in 2023). This suggests that MSMEs have a scope to improve their gender parity in the workforce.
- **Gender Parity in Ownership:** As per the statistical data provided on the Udyam Registration Portal ("URP") under the Ministry of MSME<sup>87</sup>, women owned MSMEs constitute 20.37 per cent of the total number of registered MSMEs
- Following are some of the key challenges faced by women entrepreneurs in India.
  - **Not Sustainable:** ~73% women-led enterprises have faced a reduction in revenues post COVID-19. Most female entrepreneurs are concentrated in sectors and industries highly affected by the pandemic and digitization like micro and small businesses in non-professional services, such as tailor shop owners, dressmakers, shopkeepers and street vendors, arts and handicrafts vendors, and beauty-parlor owners, among others. **This points to the fact that women led businesses are not financially sustainable/resilient enough to recover from external shocks.**
  - **Not Autonomous:** In ~20% cases, the business-related decisions are taken by the respective spouses and not the women entrepreneur running the business. Often, women do not truly own or control many women-led enterprises. **They depend considerably on men for key business decisions, procurement of raw materials, pricing information, and market access.**
  - **Not Formal:** ~95.6% of the women entrepreneurs in India are informal and not registered with relevant national/state institutions. A high level of informality harms the growth and resilience of their enterprises and their entrepreneurial rights. The level of effort involved in running such enterprises go unaccounted for and expose women to occupational health hazards. **Further, informality leads to challenges in accessing finance, market, etc.**
  - **Not sufficient employment generating:** ~95% of women proprietors run with less than six workers, most being own-account or single person enterprises. Only ~1.9% proprietors run with six to 10 employees. **This points to the fact that women entrepreneurs often lack skills in scaling businesses i.e. in areas such as business management, market access, access to finance, etc.**
  - **Lack of awareness and skills on green jobs and technologies:** Women in Indian MSMEs have been confined to low-paying, low-skilled jobs. The ongoing transitions present them with an opportunity to realign their skill sets and adjust to the evolving nature of work. Digital & Green Technologies proficiency is one of the most important skills that women need to acquire to keep pace with the ongoing transitions. This will enable them to enter traditionally male-dominated sectors that rely heavily on digital technologies, such as healthcare services, automobiles, and more advanced roles within the textiles sector with green technologies.

<sup>85</sup> <https://www.gifsinitiative.in/>

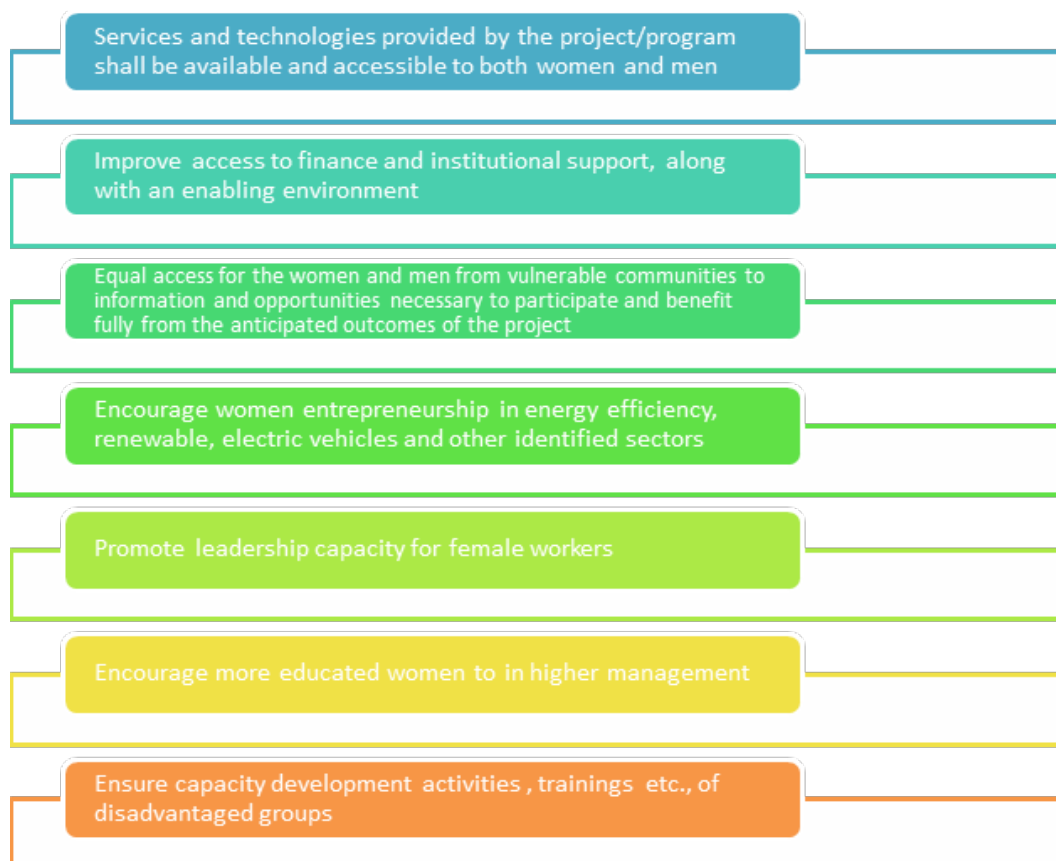
<sup>86</sup> <https://grownetwork.in/>

<sup>87</sup> <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2002574>

**SIDBI has undertaken several structural reforms/adopted policies to promote women's welfare and entrepreneurship as detailed in Annexure 8.**

#### Alignment of the Project with GCF Gender Assessment Guideline

- SIDBI as an AE shall take the necessary measures to ensure that proposed projects, as well as proposed activities under the FMAP, submitted to GCF for financing are aligned with national policies and priorities on gender.
- The project activities shall be framed around principals of GCF and will ensure the following issues in the gender action plan:



#### **Gender Action Plan**

Based on the programme design, the broad aim is to ensure adequate representation for women as stakeholder and beneficiary under component (1) supporting affordable financing of mitigation and adaptation activities; targeted technical support under component (2), for capacity development, awareness generation, etc., as well as in M&E frameworks; and leveraging financing mechanisms for greater and more equitable outreach and coverage in the MSME sector going ahead.

A stated endeavor of the programme is to secure 80:20 coverage (i.e. 20% financing towards women), broadly in line with the selected baselines for women's ownership and labour force participation. In the longer term, the aim is to contribute to a paradigm shift in terms of women's capacity and participation in ownership/ leadership/ management and technical roles/workforce participation in, as well as other benefits from climate mitigation and adaption activities taken up in the MSME sector in India. This will ensure that the current challenges on women in MSMEs, especially in the area of climate, will be addressed and thus the landscape of women-led MSMEs in India will be significantly transformed.

Following are the outcomes that the program aims to achieve:

- **Number of women led MSMEs:** Increase the number of women owned MSMEs through necessary interventions and ensure enhanced participation of women workforce in MSMEs.
- **Business Sustainability through improve access of finance:** Make women led enterprises/MSMEs more sustainable & resilient to external shocks, especially climate shocks by providing access to affordable finance for mitigation and adaption technologies. Services and technologies provided by the project/program shall be available and accessible to both women and men.

- **Autonomy:** Increased skilling for women entrepreneurs to ensure their businesses are more climate resilient by offering handholding support on the adoption of better technology. Encourage women entrepreneurship in energy efficiency, renewable, electric vehicles and other identified sectors
- **Formalization:** Increased formalization of women-led MSMEs, leading to enhanced access to credit.
- **Employability & Employment Generation:** Ensuring the opportunities of creating more employment from Women own MSMEs by capacity building on climate finance and awareness on energy efficient technology. The measures will also be taken to equip the women workforce for green jobs helping in just transition.
- **Encouraging gender-focused practices among MSMEs:** The project would encourage those MSMEs with women-centric workplace or are willing to create the one. These will include having a gender action plan, separate rest spaces, better lighting etc.

A detailed gender action plan outlining the baseline assessment, investment amounts, M&E processes, etc. has been outlined in **Annexure 8**.

### G.3. Financial management and procurement (max. 500 words, approximately 1 page)

*Describe the project/programme's financial management including the financial monitoring systems, financial accounting, auditing, and disbursement structure and methods. Refer to section B.4 on implementation arrangements as necessary.*

*Articulate any procurement issues that may require attention, e.g. procurement implementation arrangements and the role of the AE under the respective proposal, articulation of procurement risk assessment undertaken and how that will be managed by the AE or the implementing agency. Provide a detailed procurement plan as annexure 10.*

159. In line with the Govt. of India Procurement guidelines, SIDBI has an established Procurement Manual which is approved by the Board (enclosed). The manual provides guidance on the following (including but not limited to):
- categorization of procurement as Goods, Works and Services.
  - procurement methods/approach for Goods, Works and Services
  - thresholds/ limits under various methods of procurement
  - Mandatory procurement through Govt e-marketplace (GeM) which is an electronic platform by GoI for transparency and to reduce manual intervention.
  - Purchase preference policies of Govt of India like MSE, Make in India (MII), Khadi and Village Industries Commission (KVIC) etc.

Procurement activities under the Project would be undertaken in line with the Bank's procurement manual. Detailed draft of procurement plan is enclosed at **Annexure 10**.

The procurement plan majorly consists of procurements of services and goods with small values and thus of lower risk. The same would be carried out in line with the thresholds defined in the procurement manual which have been categorized duly addressing the potential risk and to maintain transparency in the process. Further SIDBI has an established mechanism for financial and administrative review and approval powers delegated to the Competent Authorities. The same would be followed for seeking approvals prior to initiating the procurements under the project. Since the procurements envisaged are of smaller values and SIDBI's established review mechanism which gives due attention to economy, efficiency and transparency, no major procurement challenge/risk is envisaged. Further, assessed Procurement Risks and corresponding Mitigation Measures are tabulated below:

| Risk Factor                                                                                                                        | Risk Category | Mitigation Measure                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Limited capacity and inefficiencies resulting in delays in procurement processes</li> </ul> | Lower         | <ul style="list-style-type: none"> <li>Use of skilled procurement staff for handling procurement</li> <li>Monitoring through Regular Review meetings</li> <li>Adequate delegation for procurement related reviews, decision making/approvals.</li> <li>Use of e-platforms (viz; GeM, CPPP, etc.)</li> <li>Regular trainings, workshops</li> <li>Disclosure of procurement-related information</li> <li>Appropriate handling and addressal of complaints</li> </ul> |

Although the procurement plan enlists the procurements envisaged to be undertaken during the initial 60 months for Consultancy and Non-Consulting Services and whole duration of the of project/program implementation in case of Goods, the same would be an ever-evolving. Changes/modifications, if any, to the approved procurement plan (if needed) during the currency of the of project/program implementation would be conveyed to GCF.

#### G.4. Disclosure of funding proposal

*Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.*

*Indicate below whether or not the funding proposal includes confidential information.*

☐ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

## ANNEXES

### H.1. Mandatory annexes

- ☒ **ANNEX 1** NDA no-objection letter(s) [\(template provided\)](#)
- ☒ **ANNEX 2** Feasibility study - and a market study, if applicable
- ☒ **ANNEX 3** Economic and/or financial analyses in spreadsheet format
- ☒ **ANNEX 4** Detailed budget plan [\(template provided\)](#)
- ☒ **ANNEX 5** Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ **ANNEX 6** E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):  
[\(ESS disclosure form provided\)](#)
  - ☐ Environmental and Social Impact Assessment (ESIA) or
  - ☐ Environmental and Social Management Plan (ESMP) or
  - ☒ Environmental and Social Management System (ESMS)
  - ☒ Others (Please Specify – E.G. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, Etc.)
- ☒ **ANNEX 7** Summary of consultations and stakeholder engagement plan
- ☒ **ANNEX 8** Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ **ANNEX 9** Legal Due Diligence (Regulation, Taxation and Insurance)
- ☒ **ANNEX 10** Procurement plan [\(template provided\)](#)
- ☒ **ANNEX 11** Monitoring and evaluation plan [\(template provided\)](#)
- ☒ **ANNEX 12** AE fee request [\(template provided\)](#)
- ☒ **ANNEX 13** Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ **ANNEX 14** Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

### H.2. Other annexes as applicable

- ☐ **ANNEX 15** Evidence of internal approval (template provided)
- ☐ **Annex 16** Map(s) indicating the location of proposed interventions
- ☐ **Annex 17** Multi-country project/programme information [\(template provided\)](#)
- ☐ **Annex 18** Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ **Annex 19** Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ **Annex 20** First level AML/CFT (KYC) assessment

- |                                     |                 |                                                                                                                                    |
|-------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | <b>Annex 21</b> | Operations manual (Operations and maintenance)                                                                                     |
| <input checked="" type="checkbox"/> | <b>Annex 22</b> | Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects) <sup>88</sup> |

*\* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

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<sup>88</sup> Annex 22 is mandatory for mitigation and cross-cutting projects.



**No-objection letter(s) issued by the national designated authority(ies) or focal point(s)**



**राजश्री रे**  
**RAJASREE RAY**

**आर्थिक सलाहकार**  
**पर्यावरण, वन एवं जलवायु परिवर्तन मंत्रालय**  
**भारत सरकार**  
**ECONOMIC ADVISER**  
**MINISTRY OF ENVIRONMENT, FOREST**  
**AND CLIMATE CHANGE**  
**GOVERNMENT OF INDIA**

No. CC- 13008/240/2021-CC

New Delhi, 25<sup>th</sup> April 2024

To,

**The Green Climate Fund ("GCF")**

**Subject: Funding Proposal for the GCF by Small Industries Development Bank of India (SIDBI) regarding "Financing Mitigation and Adaptation in Indian MSMEs".**

Dear Madam/Sir,

We refer to the programme titled **"Financing Mitigation and Adaptation in Indian MSMEs"** in India as included in the funding proposal submitted by Small Industries Development Bank of India (SIDBI) to us on 5<sup>th</sup> April 2024.

2. The undersigned is the duly authorized representative of Ministry of Environment, Forest and Climate Change, the National Designated Authority of India.
3. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.
4. By communicating our no-objection, it is implied that:
  - (a) The Government of India has no-objection to the programme as included in the funding proposal;
  - (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of India;
  - (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.
5. We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.
6. We also confirm that our no-objection applies to all activities to be implemented within the scope of the programme.
7. We acknowledge that this letter will be made publicly available on the GCF website.

Yours Sincerely,

*Rajasree*

(Rajasree Ray)

Economic Adviser to the Government of India,  
GCF Focal Point for India

## Environmental and social safeguards report form pursuant to para. 17 of the IDP

| Basic project or programme information                                                          |                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project or programme title</b>                                                               | Financing Mitigation and Adaptation Projects (FMAP) in Indian MSMEs                                                                                                                                                                                                              |
| <b>Existence of subproject(s) to be identified after GCF Board approval</b>                     | Yes                                                                                                                                                                                                                                                                              |
| <b>Sector (public or private)</b>                                                               | Private                                                                                                                                                                                                                                                                          |
| <b>Accredited entity</b>                                                                        | Small Industries Development Bank of India (SIDBI)                                                                                                                                                                                                                               |
| <b>Environmental and social safeguards (ESS) category</b>                                       | Category I-2                                                                                                                                                                                                                                                                     |
| <b>Location – specific location(s) of project or target country or location(s) of programme</b> | Republic of India                                                                                                                                                                                                                                                                |
| Environmental and Social Impact Assessment (ESIA) (if applicable)                               |                                                                                                                                                                                                                                                                                  |
| Date of disclosure on accredited entity's website                                               | N/A                                                                                                                                                                                                                                                                              |
| Language(s) of disclosure                                                                       | N/A                                                                                                                                                                                                                                                                              |
| Explanation on language                                                                         | N/A                                                                                                                                                                                                                                                                              |
| Link to disclosure                                                                              | N/A                                                                                                                                                                                                                                                                              |
| Other link(s)                                                                                   | N/A                                                                                                                                                                                                                                                                              |
| Remarks                                                                                         | N/A                                                                                                                                                                                                                                                                              |
| Environmental and Social Management Plan (ESMP) (if applicable)                                 |                                                                                                                                                                                                                                                                                  |
| Date of disclosure on accredited entity's website                                               | N/A                                                                                                                                                                                                                                                                              |
| Language(s) of disclosure                                                                       | N/A                                                                                                                                                                                                                                                                              |
| Explanation on language                                                                         | N/A                                                                                                                                                                                                                                                                              |
| Link to disclosure                                                                              | N/A                                                                                                                                                                                                                                                                              |
| Other link(s)                                                                                   | N/A                                                                                                                                                                                                                                                                              |
| Remarks                                                                                         | N/A                                                                                                                                                                                                                                                                              |
| Environmental and Social Management (ESMS) (if applicable)                                      |                                                                                                                                                                                                                                                                                  |
| Date of disclosure on accredited entity's website                                               | Friday, June 14, 2024                                                                                                                                                                                                                                                            |
| Language(s) of disclosure                                                                       | English and Hindi                                                                                                                                                                                                                                                                |
| Explanation on language                                                                         | Hindi is the official language of India.                                                                                                                                                                                                                                         |
| Link to disclosure                                                                              | English:<br><a href="https://www.sidbi.in/assets/front/pdf/ESMS-FMAP-E.pdf">https://www.sidbi.in/assets/front/pdf/ESMS-FMAP-E.pdf</a><br><br>Hindi:<br><a href="https://www.sidbi.in/assets/front/pdf/ESMS-FMAP-H.pdf">https://www.sidbi.in/assets/front/pdf/ESMS-FMAP-H.pdf</a> |
| Other link(s)                                                                                   | N/A                                                                                                                                                                                                                                                                              |
| Remarks                                                                                         | An ESMS consistent with the requirements for a Category I-2 project is contained in the 'Environment and Social Management System (ESMS)'.                                                                                                                                       |

| Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable) |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description of report/disclosure on accredited entity's website                                                                                                                                       | Indigenous Peoples Planning Framework (IPPF) and Resettlement Action Plan (RAP)/ Friday, June 14, 2024                                                                                                                                                                                                                         |
| Language(s) of disclosure                                                                                                                                                                             | English and Hindi                                                                                                                                                                                                                                                                                                              |
| Explanation on language                                                                                                                                                                               | Hindi is the official language of India.                                                                                                                                                                                                                                                                                       |
| Link to disclosure                                                                                                                                                                                    | <p>English:<br/> <a href="https://www.sidbi.in/assets/front/pdf/ESMS-FMAP-E.pdf">https://www.sidbi.in/assets/front/pdf/ESMS-FMAP-E.pdf</a></p> <p>Hindi:<br/> <a href="https://www.sidbi.in/assets/front/pdf/ESMS-FMAP-H.pdf">https://www.sidbi.in/assets/front/pdf/ESMS-FMAP-H.pdf</a></p>                                    |
| Other link(s)                                                                                                                                                                                         | N/A                                                                                                                                                                                                                                                                                                                            |
| Remarks                                                                                                                                                                                               | <p>An IPPF consistent with the requirements for a Category I-2 project is contained in the Attachment 6 of Annexure 6 ESMS.</p> <p>In addition, RAP consistent with the requirements for a Category I-2 project is contained in the Attachment 8 of Annexure 6 ESMS</p>                                                        |
| Disclosure in locations convenient to affected peoples (stakeholders)                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                |
| Date                                                                                                                                                                                                  | Friday, June 14, 2024                                                                                                                                                                                                                                                                                                          |
| Place                                                                                                                                                                                                 | <p>Hard copies will be made available in SIDBI Delhi Office<br/> Address: ATMA RAM HOUSE, 1 Tolstoy Marg, New Delhi-110001</p> <p>Hard copies will also be made available in SIDBI Branch Offices as per the list indicated here:<br/> <a href="https://www.sidbi.in/branch-office">https://www.sidbi.in/branch-office</a></p> |
| Date of Board meeting in which the FP is intended to be considered                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                |
| Date of accredited entity's Board meeting                                                                                                                                                             | TBD                                                                                                                                                                                                                                                                                                                            |
| Date of GCF's Board meeting                                                                                                                                                                           | Monday, July 15, 2024                                                                                                                                                                                                                                                                                                          |

**Note: This form was prepared by the accredited entity stated above.**

## Independent Technical Advisory Panel's assessment of FP241

|                         |                                                                     |
|-------------------------|---------------------------------------------------------------------|
| Proposal name:          | Financing Mitigation and Adaptation Projects (FMAP) in Indian MSMEs |
| Accredited entity:      | Small Industries Development Bank of India (SIDBI)                  |
| Country/(ies):          | India                                                               |
| Project/programme size: | Large                                                               |

### I. Assessment of the independent Technical Advisory Panel

#### 1.1 Overview

1. The Financing Mitigation and Adaptation Project (FMAP) aims to accelerate the adoption of low-emission and climate-resilient practices among micro, small and medium-sized enterprises (MSMEs) in India while promoting the greening of the country's financial sector. Led by Small Industries Development Bank of India (SIDBI) as the accredited entity (AE) and executing entity (EE), the programme focuses on financing resource-constrained sectors. A key benefit of FMAP is its emphasis on financing adaptation projects, which have so far seen limited uptake by both public and private financial institutions. In addition, the programme seeks to boost the effectiveness of primary financial institutions (PFIs), including small finance banks (SFBs), microfinance institutions (MFIs) and non-banking financial companies (NBFCs), by enabling them to build a track record that supports scaling up climate finance in India.

2. FMAP consists of two components. The first is a USD 1 billion lending facility managed by SIDBI, capitalized by an USD 800 million contribution from SIDBI and an additional USD 200 million from the GCF. These funds will support direct lending (20 per cent) to MSMEs for climate-aligned initiatives and indirect financing (80 per cent) of MSMEs through PFI partners. The indirect financing will be channelled to MSMEs via PFIs, which are required to co-finance three times the amount received, mobilizing an additional USD 2.4 billion. This three times commitment is measured on a portfolio basis rather than at the level of the individual loans. Furthermore, participating MSMEs are expected to contribute equity ranging from 10 to 25 per cent of the total loan for eligible projects. The concessionality provided by the GCF funds enables FMAP to adopt a flexible approach (e.g. in terms of grace periods and tenors) in structuring sub-loans and to offer concessional capital costs to incentivize uptake among MSMEs.

3. The second component is a USD 16 million grant facility to be implemented by SIDBI. These grants will support MSMEs and PFIs in greening their operations by identifying the technical needs of MSMEs, promoting knowledge-sharing, building capacity among stakeholders and contributing to policy initiatives. SIDBI will prioritize conducting technology needs assessments to identify key solutions for MSMEs and capacity-building for PFIs, and will implement due diligence on their loan books through a dedicated project management unit. Leveraging its experience in knowledge-sharing and capacity-building, SIDBI will lead training sessions and workshops to raise awareness and address barriers. The grant component is sized at approximately 1.6 per cent of the loan facility component.

4. The objective of FMAP is to address market gaps in access to finance for resource-constrained sectors, particularly in adaptation, by providing concessional lending and technical assistance. By working through PFIs in addition to direct lending, the programme seeks to support

the greening of the Indian financial sector. In addition, FMAP aims to induce positive market signals on the commercial viability of climate projects and attract local financial institutions and private sector finance.

## 1.2 Impact potential

*Scale: Medium to high*

5. India is a major emitter of greenhouse gases (GHGs). Given its rapid economic development and substantial population growth, it is imperative for India to pursue a development path that leads to a low-carbon, more resilient future. According to the third national communication and initial adaptation communication submitted by India under the Convention in December 2023, India's net national GHG emissions in 2019 amounted to 2.6 billion tonnes of carbon dioxide equivalent (t CO<sub>2</sub> eq), reflecting a 4.56 per cent increase from the 2016 level and a 115 per cent increase since 1994. CO<sub>2</sub> remains the main GHG, accounting for 79 per cent of total emissions in 2019, primarily due to increased fossil fuel combustion, and rising aluminium and cement production.

6. The FMAP funding proposal explains that India's energy-intensive MSME subsectors are substantial contributors of GHG emissions. However, MSMEs face several barriers to becoming more climate-aligned, include their use of conventional processes and technologies, coupled with unskilled or semi-skilled workers, leading to higher resource consumption and associated waste generation and emissions. There are over 63 million MSMEs in India, employing about 110 million people and accounting for about 45 per cent to the country's manufacturing output. MSMEs consume about 25 per cent of the total energy used by the industrial sector in India, representing a significant opportunity to reduce resource consumption, including electricity, fossil fuels and water, in a cost-effective manner. FMAP aims to support this transition by improving access to finance and facilitating MSME access to greener technologies.

7. Considering the above, the independent Technical Advisory Panel (iTAP) takes a positive view of FMAP's main goal of sensitizing MSMEs towards adopting green and cleaner production technologies, and developing and deploying adaptation technologies. The iTAP also concurs with the programme's aim to encourage banks and financial institutions to develop dedicated green instruments and lend with increased confidence to meet MSMEs' demand for adopting and developing the necessary green technologies. Transitioning MSMEs in India to lower-carbon, more resilient pathways could be a crucial contributor to helping India meet its climate goals.

8. FMAP will provide loans to MSMEs targeting the renewable energy, energy efficiency and low-emission transport sectors. In addition, it will support adaptation by helping MSMEs in the water-intensive and agriculture sectors to implement climate solutions in their businesses, thereby increasing resilience and adaptive capacity against climate change. Adaptation investments will be directed to highly vulnerable regions such as coastal areas, the Himalayas and the eastern belt of India. According to the funding proposal, FMAP is expected to avoid 43.86 million t CO<sub>2</sub> eq and benefit 10.3 million people. The iTAP notes that estimating GHG reductions over such a large programme and long tenor (13 years) is challenging due to uncertainties in the composition of underlying loans. It is therefore difficult to put very much stock on specific projections in the context of FMAP; however, the iTAP acknowledges that the targeted loan areas (e.g. energy efficiency, low-emission transport, renewables) coupled with climate-aligned screening mechanisms mean that there is the potential for significant impact. Importantly, the sheer scale of India and the role of MSMEs in the economy support an argument that the overall impact will be positive from a climate standpoint.

9. Moreover, FMAP is expected to mobilize USD 3.5 billion from public and private sector sources, with co-financing at a ratio of 1:18 towards dedicated investments in MSMEs for GHG emission reductions and accelerating MSME adaptation actions. Of this, the funding proposal estimates that 40 per cent of the funding will address adaptation actions, and the remaining 60



per cent will focus on mitigation efforts. The iTAP agrees that the potential mobilization impact could be significant, and the potential to redirect the lending of PFIs in India towards climate-aligned interventions is a valuable aspect of this funding proposal.

10. The iTAP also concurs that the well-sized and tailored use of grants alongside commercial lending can help to develop a pipeline for the lending facility and, in turn, increase confidence that MSMEs will take up loans and effect operational changes that have beneficial climate outcomes. Moreover, the grant component has the potential to accelerate the greening of the local financial sector as envisaged in the funding proposal, which is critical to the programme's success.

11. The iTAP considers that the programme is well-positioned to mitigate barriers to accessing finance, by directly supporting MSMEs through green lending at concessional rates while incentivizing local financial institutions through wholesale lending. Direct lending will target medium-sized enterprises, while indirect lending will support micro and small enterprises, ensuring complementarity between both channels. For indirect lending, SIDBI will work with eligible local financial institutions, MFIs, NBFCs and SFBs. This approach could allow SIDBI to reach a broader range of MSMEs through these institutions' networks and help them increase their capacity to extend green loans for implementing or developing climate solutions. This inclusive structure addresses the immediate need for finance availability.

12. FMAP intends not only to increase the availability of finance for climate-related loans to MSME but also to use concessionalities to accelerate the transition to climate-friendly technologies. The AE has made a credible case for the levels of concessionalities needed to shift PFIs towards climate-aligned sectors and, more importantly, to encourage MSMEs to adopt technology changes that enhance their resilience to climate impacts and reduce GHG emissions. However, the iTAP notes a concern over how levels of concessionalities will be determined over time. The iTAP recommends that SIDBI regularly evaluate and adjust concessionalities on a sector-by-sector basis to avoid over-subsidization, which could generate windfall gains for some PFIs while crowding out other players.

13. In addition, the iTAP has some qualms concerning the risk that a substantial portion of FMAP funding could end up refinancing existing PFI portfolio loans instead of financing new and additional MSME efforts to adopt climate-friendly technology. There is very little climate-directed utility in using GCF concessionalities to refinance mitigation efforts that have already been funded commercially. SIDBI has indicated that it may wish to fund some portfolio refinance in the early stages of the programme as a way to attract PFI partners and facilitate their co-financing capacity. The iTAP is willing to accept this argument but strongly recommends that the Secretariat and SIDBI develop safeguards to ensure that the bulk of concessional funding is truly for additional projects. The iTAP encourages SIDBI to develop a plan for FMAP that minimizes portfolio refinancing over time on a trajectory to zero over the programme's tenure.

14. Notwithstanding the above concerns, the iTAP regards FMAP's impact potential as **medium-high**. The significant potential for mobilization of substantial local finance into a high climate impact sector has strongly influenced this assessment.

### 1.3 Paradigm shift potential

*Scale: High*

15. The funding proposal convincingly demonstrates that MSMEs in India are crucial to the local economy and that any significant shift in India's climate trajectory must address their GHG emissions and vulnerability to climate impacts. The proposal also effectively argues that a key barrier to climate action is the lack of access to affordable finance. FMAP will provide concessional loans to MSMEs with limited credit access, enabling them to adopt and develop climate solutions. By enhancing the capacity of MSMEs and financial institutions for green financing



practices, FMAP seeks to build institutional capacity within SIDBI and local financial entities to manage climate financing effectively. Market transformation will be driven by demonstrating and promoting low-emission technologies, thereby catalysing a broader shift towards sustainable practices.

16. The iTAP views favourably the effort to pull the existing financial system into climate lending, noting that the most effective approach is to adapt current systems to align better with climate goals. FMAP recognizes that MSMEs in India already engage in financial practices, albeit inadequately scaled, which can be steered towards a greener future through a mix of co-finance, concessionality and technical support. This is a pragmatic approach and avoids the pitfall of one-off innovations that often plague climate finance thinking. The iTAP also notes that the GCF paradigm shift potential is defined as the “degree to which the proposed activity can catalyse impact beyond a one-off project or programme investment”. By creating conditions that incentivize Indian PFIs to finance climate technologies for MSMEs, it is expected that PFIs will develop a comfort level in these sectors. This should spur additional private capital flows from the Indian financial system into the sector, scaling up climate finance and accelerating the country’s decarbonization efforts. Potential for replication and overall market uptake is a key factor in achieving a paradigm shift.

17. SIDBI has demonstrated its past success in providing local financial institutions with technical support and concessional loans. This track record gives the iTAP some confidence in the AE’s ability to scale and replicate models for addressing energy inefficiency and high GHG emissions across various MSME sectors. FMAP’s capacity-building initiatives are expected to increase awareness of available climate solutions among MSMEs and help participating PFIs to enhance their ability to extend climate financing. The programme will promote knowledge dissemination by documenting best practices and lessons learned, providing scalable templates for energy efficiency, environmental sustainability and climate adaptation in the MSME sector.

18. The iTAP also notes favourably that FMAP will focus on MSMEs with poor credit access, low investment potential and high emission reduction potential. Investing exclusively in climate-friendly technologies for MSMEs will have the added benefit of accelerating the growth of the product and service providers essential to India’s decarbonization trajectory. The paradigm shift benefits extend beyond direct interventions at the MSME level or getting PFIs comfortable with climate lending; they also support the expansion of the ecosystem of climate-friendly solutions in India. The scale of lending and concessional support has the potential to drive demand for low-carbon technologies and services (and climate adaptive interventions), which in turn could spur greater supply and investment into providers of these products and services.

19. Overall, the iTAP rates the potential for FMAP to effect paradigm shift as **high**.

## 1.4 Sustainable development potential

*Scale: Medium to high*

20. The FMAP funding proposal estimates that the programme will directly reduce GHG emissions and benefit 10.3 million people through approximately 10,000 green loans to MSMEs over its 13-year lifespan. These loans will facilitate the adoption of climate technologies, which are expected to help MSMEs save over 100,000 GWh of energy and 1,241 million cubic metres of water over the life time of the technologies and equipment. In addition to the anticipated climate benefits, the proposal projects ancillary environmental benefits, such as cleaner air through reduced air pollution. For instance, energy efficiency improvements implemented by MSMEs can significantly reduce local air pollution, particularly outdoor concentrations of pollutants such as sulphur dioxide, particulate matter, unburned hydrocarbons and nitrogen oxides.

21. The funding proposal credibly demonstrates FMAP’s potential to contribute to India’s sustainable development beyond mitigation and adaptation impacts, as follows:

22. **Economic co-benefits:**

- (a) The proposed climate financing is expected to stimulate economic activity and create over 400,000 jobs. Low investment costs for implementing green technologies are anticipated to directly enhance the profitability of MSMEs through increased productivity and product quality;
  - (b) The programme will also target innovative entrepreneurs to drive MSME growth, positively impacting India's gross domestic product, given the sector's economic significance; and
  - (c) Cost-effective investments in energy efficiency will generate energy savings that directly improve the profitability of MSMEs and provide net positive earnings in the long run. Reduced energy costs can free up resources for more productive activities, such as expansion or developing new product lines or services; and
23. **Social and gender co-benefits:**
- (a) The adoption of climate technologies and best practices will support increased access to affordable clean energy and water-efficient technologies, improving the health and well-being of the Indian population; and
  - (b) The programme will empower women and ensure their adequate representation as stakeholders and beneficiaries. The funding proposal specifies that 20 per cent of the total annual FMAP lending will be allocated to women.
24. Overall, the iTAP rates the sustainable development potential of FMAP as **medium-high**.

## 1.5 Needs of the recipient

*Scale: Medium to high*

25. India is experiencing rapid economic growth and entering a crucial phase of energy development, which will shape its emissions profile for many years to come. This pivotal moment affects both the supply and demand sides of energy. Economic growth is driving increased demand for air conditioning, vehicle purchases and other symbols of rising affluence. Over the next two decades, to meet electricity demand, India must expand its power system to a scale comparable with that of the European Union.

26. It is essential that this economic growth is fuelled by clean, reliable energy. To achieve this, climate-aligned technologies need to be adopted across the economy, particularly by MSMEs. MSMEs are significant contributors to GHG emissions (over 110 million t CO<sub>2</sub> eq in 2018), especially in energy-intensive subsectors. Achieving a low-carbon future for India will necessitate rapid adoption of low-carbon technologies and improved energy efficiency by MSMEs. However, MSMEs often face barriers in accessing finance for green upgrades. There is a pressing need for financial products tailored to the unique challenges and risks of the MSME sector.

27. Under the Paris Agreement India has committed to reducing emissions intensity by 45 per cent by 2030 compared with the 2005 level. Meeting these targets will require an estimated USD 2.5 trillion, highlighting the significant investment gap, with current green finance flows well below the USD 170 billion needed annually. This underscores the critical importance of innovative financing and technologies in advancing India's climate goals.

28. India is also highly vulnerable to climate change, as emphasized in its nationally determined contribution (NDC), due to its large population, agricultural dependence, extensive coastal areas and fragile Himalayan region. India ranks seventh on the Climate Risk Index and thirty-first on the INFORM Risk Index, both of which measure climate change vulnerability. The funding proposal demonstrates that the MSME sector is particularly vulnerable to extreme weather events such as floods, droughts and cyclones, especially MSMEs dependent on agriculture and located in vulnerable regions. This sector faces a number of barriers, such as limited

access to technology, limited capacity to adopt new technologies and lack of affordable financing options.

29. In India, MSMEs need tools and support to assess, manage and forecast climate risks, particularly those related to natural hazards (e.g. water scarcity, floods and extreme weather events). Investments are needed to make infrastructure more resilient to climate impacts, including upgrading facilities and adopting resilient construction practices.

30. The funding proposal describes how FMAP aims to alleviate financing barriers for MSMEs' mitigation and adaptation interventions by providing long-term concessional lending. In addition, technical assistance will be provided to MSMEs to increase awareness of available climate solutions, addressing market gaps and delivering climate impact projects that are unlikely to be implemented without FMAP.

31. The iTAP rates the needs of the recipient as **medium-high**.

## 1.6 Country ownership

*Scale: Medium to high*

32. The FMAP aligns with the implementation of India's NDC and is consistent with national adaptation plans and the national climate strategy. The focus on sectors such as renewable energy, energy efficiency, transportation, water and agriculture align with high-priority areas identified in India's NDC. The iTAP considers the proposed programme's emphasis on MSME low-carbon development pathways and accelerating climate adaptation efforts to be in harmony with India's climate actions, economic development and financial inclusion strategies.

33. India has issued a non-objection letter for FMAP. In addition, the AE (SIDBI) is an agency of the Government of India, with over 30 years of experience working with Indian MSMEs and local financial institutions. These factors collectively indicate strong government engagement and national ownership.

34. The iTAP finds it commendable that SIDBI will contribute a substantial portion (80 per cent) of the finance in the lending facility from its own balance sheet, demonstrating confidence in the programme's potential to achieve its desired outcomes. However, it is disappointing that SIDBI's financial commitment is limited to the loan facility and does not extend to the grant component for capacity-building. While SIDBI intends to contribute in-kind assistance (personnel) to some capacity-building efforts, the lack of any SIDBI budget at stake in the grant component increases the risk of quality control issues over time. When grant facilities rely solely on external funding there is a higher risk that managers prioritize fund deployment over ensuring the funds achieve the best impact outcomes.

35. The iTAP also notes positively that FMAP builds on SIDBI's past experiences with direct lending to MSMEs and collaboration with PFIs to promote MSMEs' adoption of green technologies. This includes providing concessional loans, partnering with NBFCs and offering technical assistance to bridge the information gap. We observe approvingly that SIDBI has incorporated several lessons learned from these past efforts into FMAP, particularly the requirement for PFIs to co-lend alongside FMAP. The iTAP also appreciates that the FMAP development process included stakeholder consultations with both potential borrowers and the banking community in India.

36. The iTAP considers FMAP to be aligned with host country priorities and assesses involvement on the part of India to be significant. Overall, country ownership of the funding proposal is rated as **medium-high**.

## 1.7 Efficiency and effectiveness

*Scale: Medium to high*

37. iTAP concurs that focusing on transitioning MSMEs to a low-carbon and more resilient pathway is an effective strategy, given the overall size of the Indian economy, the central role of MSMEs and the rapid changes occurring in India. The proposal's aim to deliver high-impact mitigation and adaptation projects by enhancing the capacity of local financial institutions to provide climate loans to MSMEs is viewed by iTAP as positive, and greening the existing financial systems is an efficient and effective approach. Moreover, FMAP plans to fund green technologies in remote/rural areas of India, which can be particularly challenging. To address this, the programme's key feature will be reaching MSMEs in rural areas through indirect lending via local financial institutions with a wider presence. This inclusive structure is crucial for developing the capacity and sustainability of climate financing in India.

38. Challenges may arise in ensuring that underlying loans deliver the expected climate impacts. This will require the AE to maintain strict screening criteria for its direct loans and effective oversight of its PFI partners. The iTAP assessed the criteria and processes described in the funding proposal and notes that they meet a minimal standard of acceptability but could be improved. In particular, standards are only as effective as their implementation. In this case, SIDBI serving as both AE and EE dilutes one of the usual checks on misapplication that the GCF relies on, namely the AE acting as the GCF's fiduciary eyes and ears. The iTAP encourages the Secretariat to consider requesting additional safeguards to ensure that GCF funds and concessionality are used effectively, minimizing the risk of merely supporting PFI balance sheets or financing interventions without a strong climate rationale.

39. The USD 200 million senior loan provided by GCF will be blended with SIDBI's USD 800 million and further leveraged by 16 times through participating local financial institutions, supplemented by USD 377 million in equity at the MSME level, and this is viewed as positive by the iTAP. Specifically, the funding proposal states that GCF funding will help SIDBI channel a total investment of approximately USD 3.5 billion for mitigation and adaptation measures, resulting in a mobilization rate of 1:20.71. This represents significant co-finance by multiple partners, including the MSMEs themselves. This structure aligns the interests of all parties with those of the GCF. By requiring financial inputs from all relevant parties, FMAP has the potential to scale up green financing for MSMEs, delivering climate impact projects. In addition, partnering with local financial institutions will leverage existing banking relationships to improve scaling and speed of deployment.

40. Funding green technologies in MSMEs is currently perceived as risky. While MSMEs can borrow from other private financial institutions, the concessionality provided under this programme will drive the adoption of green technologies in MSMEs. The GCF funds will enable FMAP to offer concessionality as needed, depending on the relevant MSME sector and technology. This flexible approach allows lenders to provide financing that is fit for purpose. However, there is a risk that over the long tenor of FMAP, the provided concessionality may not keep up with market needs. The iTAP encourages the Secretariat and the AE to explore ways to ensure that GCF concessionality does not end up over-subsidizing sectors as technologies become more accepted and additional financing entrants join the market. As the AE and EE are the same entity, the iTAP suggests having an external process to periodically evaluate the levels of concessionality and the additionality of FMAP funding. This would help mitigate the risk of crowding out other funders, or of PFIs capturing windfall profits without corresponding climate benefits.

41. The iTAP agrees on the effectiveness of combining a grant component with the lending programme to provide technical assistance and capacity-building for both MSMEs and participating financial institutions, and the grant programme is well-sized and focused on providing technology assessments for MSMEs. These assessments can increase awareness of climate technologies and dispel outdated perceptions about the costs and effectiveness of many clean technologies. Capacity support for participating financial institutions, particularly underserved NBFCs, MFIs and SFBs, is also crucial in increasing their ability to extend climate finance.

42. Overall, the iTAP rates the efficiency and effectiveness of this programme as **medium-high**.

## II. Overall remarks from the independent Technical Advisory Panel

43. The iTAP considers that the FMAP programme aligns closely with the GCF private sector strategy and overall mission of supporting developing countries to pursue growth pathways that are aligned with a low-carbon and resilient future for the planet.

44. If executed well, FMAP will facilitate access to capital for vulnerable MSMEs and create incentives for these businesses to adopt climate-aligned technologies and processes. Importantly, FMAP will also aim to green the existing financial system by building capacity and establishing a track record for PFIs in financing mitigation and adaptation investments among the most vulnerable economic segments. In doing so, FMAP should mobilize large-scale funds from these local financial institutions by de-risking private sector investments and supporting the greening of the financial system.

45. The iTAP considers that the concessional loans and technical assistance provided through FMAP can play a pivotal role in reducing the costs and risks – both real and perceived – associated with investments in climate change mitigation and adaptation. SIDBI has long been a market maker for MSMEs in India and has comprehensive access to the MSME landscape, including those not significantly serviced by traditional lenders. The iTAP notes that SIDBI is well-positioned to be catalytic across MSME segments for both mitigation and adaptation activities.

46. Significantly, the commitment of GCF funding will be complemented by substantial financial support from SIDBI, signalling that this programme is well-suited for the Indian context. The programme also incorporates co-finance obligations from PFIs and the MSMEs themselves to ensure proper buy-in from all key actors. At the heart of FMAP is the recognition that addressing the climate finance challenge in larger and middle-income countries is best achieved by catalysing a green shift in local financial markets, rather than relying on cross-border funding. Local finance can operate more efficiently due to a better understanding of localized implementation risks, legal frameworks and market challenges, and doing all this in the right currency. In any investment analysis, the journey from perceived risk to real risk understanding is much shorter for local financial actors.

47. However, the iTAP has noted some areas of concern, primarily related to the operationalization of FMAP over time. The FMAP concept is strong, but it relies heavily on SIDBI's capacity to underwrite loans and oversee participating PFIs' underlying loans to achieve the target mitigation and adaptation impacts. Areas of FMAP that can be strengthened are noted throughout this assessment. FMAP may face challenges associated with pipeline risks and MSMEs' willingness to adopt climate-resilient technologies and solutions. In such circumstances, there could be a temptation to loosen climate screening standards or oversight of PFI portfolios. It is important to consider additional safeguards in the operational plan to mitigate this risk and uphold the climate imperative of FMAP.

48. Considering the foregoing considerations, the iTAP recommends that the Board approve this funding proposal.



## Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP241)

|                         |                                                                     |
|-------------------------|---------------------------------------------------------------------|
| Proposal name:          | Financing Mitigation and Adaptation Projects (FMAP) in Indian MSMEs |
| Accredited entity:      | Small Industries Development Bank of India (SIDBI)                  |
| Country/(ies):          | India                                                               |
| Project/Programme size: | Large                                                               |

AE duly notes the comments of iTAP and appreciates the efforts of the team in reviewing the Funding Proposal package in detail. The comprehensive feedback would help the AE in further strengthening the proposal and programme. It may be noted by the iTAP that the Funding Proposal has undergone significant revisions as suggested by the secretariat and most of the concerns/feedback/recommendation given by the iTAP members has already been factored in the FP package. However, AE acknowledges the valuable feedback and would like to give below clause wise explanation on the comments by iTAP.

### Impact potential

**iTAPs feedback (Point 8):-** FMAP will provide loans to MSMEs targeting the renewable energy, energy efficiency and low-emission transport sectors. In addition, it will support adaptation by helping MSMEs in the water-intensive and agriculture sectors to implement climate solutions in their businesses, thereby increasing resilience and adaptive capacity against climate change. Adaptation investments will be directed to highly vulnerable regions such as coastal areas, the Himalayas and the eastern belt of India. According to the funding proposal, FMAP is expected to avoid 43.86 million t CO<sub>2</sub> eq and benefit 10.3 million people. The iTAP notes that estimating GHG reductions over such a large programme and long tenor (13 years) is challenging due to uncertainties in the composition of underlying loans. It is therefore difficult to put very much stock on specific projections in the context of FMAP; however, the iTAP acknowledges that the targeted loan areas (e.g. energy efficiency, low-emission transport, renewables) coupled with climate-aligned screening mechanisms mean that there is the potential for significant impact. Importantly, the sheer scale of India and the role of MSMEs in the economy support an argument that the overall impact will be positive from a climate standpoint.

### SIDBI Response:

While estimating the GHG emissions, AE has considered the following:

- (i) AE has used the approved methodologies (CDM and SEC) for calculation for energy savings and GHG emissions of different sub-sectors.
- (ii) AE has experience of working in the area of Energy Efficiency in MSME since 2005-06. A technology list comprises of 891 clean, green, energy efficient technologies have been developed with the support of technical expert agencies, Bureau of Energy Efficiency etc., AE has used the data from the projects assisted in the past, by AE for the baseline calculations.



However, considering the various limitations as well as vast heterogeneity of the MSME sector, AE agrees with the observations of the iTAP that although realistic estimations of GHG emissions is difficult for such a long tenor but the programme has a huge potential for significant impact on the most vulnerable segment i.e MSMEs.

**iTAPs feedback (Point 12)**- FMAP intends not only to increase the availability of finance for climate-related loans to MSME but also to use concessionality to accelerate the transition to climate-friendly technologies. The AE has made a credible case for the levels of concessionality needed to shift PFIs towards climate-aligned sectors and, more importantly, to encourage MSMEs to adopt technology changes that enhance their resilience to climate impacts and reduce GHG emissions. However, the iTAP notes a concern over how levels of concessionality will be determined over time. The iTAP recommends that SIDBI regularly evaluate and adjust concessionality on a sector-by-sector basis to avoid over-subsidization, which could generate windfall gains for some PFIs while crowding out other players.

#### **SIDBI Response-**

The concerns regarding determination of levels of concessionality and passing of the same over a period is duly noted and the explanation is as follows:

(a). The concessionality provided by GCF on blended basis shall be passed on to the borrowers (MSMEs as well as PFIs) by AE by way of reducing the spread by atleast an amount equal to the concessionality worked out on blended basis towards each tranche of drawal from GCF. Further, AE shall suitably bind the PFIs to pass on the concessionality on a blended basis to their end borrowers.

(b). AE duly notes the suggestions of iTAP regarding the necessity to evaluate and adjust concessionality to avoid over-subsidization. In this regard, AE submits that all the sub-loans will undergo credit rating and the effective interest rate for the end borrower would be the interest rate based on the credit rating of the borrower. The credit rating parameters include various factors on obligor and facility, as well as sectoral outlook.

(c). The programme has also inbuilt suitable checks and balances which includes M&E on sample basis of sub-loans, periodical reporting of the climate benefits, pricing and passing on the concessionality to the ultimate borrowers etc., to avoid over-subsidisation or PFIs getting windfall profits without any climate benefits. The programme proposes to work not with one two PFIs but with around 20 to 25 PFIs and it is expected that over the period gradually there will be sufficient competition among these PFIs for climate financing. This would obviate the risk of windfall profits to the PFIs and shall ensure further benefits to the ultimate borrowers i.e. MSMEs.

**iTAPs feedback (Point 13)**- In addition, the iTAP has some qualms concerning the risk that a substantial portion of FMAP funding could end up refinancing existing PFI portfolio loans instead of financing new and additional MSME efforts to adopt climate-friendly technology. There is very little climate-directed utility in using GCF concessionality to refinance mitigation efforts that have already been funded commercially. SIDBI has indicated that it may wish to fund some portfolio refinance in the early stages of the programme as a way to attract PFI partners and facilitate their co-financing capacity. The iTAP is willing to accept this argument but strongly recommends that the Secretariat and SIDBI develop safeguards to ensure that the bulk of concessional funding is truly for additional projects. The iTAP

encourages SIDBI to develop a plan for FMAP that minimizes portfolio refinancing over time on a trajectory to zero over the programme's tenure.

**SIDBI Response-** AE wish to state that iTAP concerns are addressed in the revised Funding Proposal and the Term Sheet. Further, AE would like to supplement with the following w.r.t refinancing of existing portfolio:

- (A) The programme will not cover any sub-loans prior to the date of effectiveness of FAA.
- (B) The Programme will offer onlending/ refinance support against eligible assets created by the empanelled PFIs and originated no longer than six months prior to approval from the Executing Entity for financing in accordance with FAA. The concerned PFI would be required to submit (for Refinance) all the details, copy of documentation, etc. to satisfy AE that the conditions stipulated in the term sheet at the time of sanction have been fully complied with.

The above will ensure that only new/additional projects (which were not already support by PFIs before launch of the programme) are covered under the programme.

#### Paradigm shift potential

AE acknowledges the positive feedback from iTAP on the paradigm shift potential of the FMAP proposal.

#### Sustainable development potential

AE acknowledges the positive feedback from iTAP on the sustainable development potential of the FMAP proposal.

#### Needs of the recipient

AE duly notes and appreciates that iTAP recognises the objectives of the FMAP programme in alleviating the financing barriers of MSMEs in adopting mitigation and adaptation interventions.

#### Country ownership

**iTAP feedback (Point 34)-** The iTAP finds it commendable that SIDBI will contribute a substantial portion (80 per cent) of the finance in the lending facility from its own balance sheet, demonstrating confidence in the programme's potential to achieve its desired outcomes. **However, it is disappointing that SIDBI's financial commitment is limited to the loan facility and does not extend to the grant component for capacity-building. While SIDBI intends to contribute in-kind assistance (personnel) to some capacity-building efforts, the lack of any SIDBI budget at stake in the grant component increases the risk of quality control issues over time. When grant facilities rely solely on external**

**funding there is a higher risk that managers prioritize fund deployment over ensuring the funds achieve the best impact outcomes.**

**SIDBI Response-** It is worth noting that iTAP acknowledges that the FMAP aligns with the implementation of India's NDC and is consistent with national adaptation plans and the national climate strategy. AE appreciates the positive feedback received in this section.

In this regard, AE wish to submit the following:

- (i) AE is ensuring 16X leverage of the funds contributed by the GCF.
- (ii) AE is taking the credit risk on NBFCs, SFBs, MFIs as well as the MSMEs. GCF is taking credit risk on SIDBI which is a AAA rated quasi sovereign entity.
- (iii) It may be noted that the TA supported by GCF has been estimated on a highly conservative basis and it is not even 2% of the total cost i.e. USD 400 Mn (around 10% of total climate financing expected, USD 4 Bn) which will be incurred by the EE (SIDBI) as well as the PFIs towards manpower and administrative costs towards deployment of the climate loans.

As regards comment on prioritising fund deployment of the donor, AE may like to mention again that AE has managed several technical assistance(s) which were part of a project involving LOC for being channelised and TA for capacity building. There has been only counterpart contribution in-kind (from SIDBI) and still AE has achieved the intended outcome set by the donor(s).

### **Efficiency and effectiveness**

**iTAP feedback (Point 38)-** Challenges may arise in ensuring that underlying loans deliver the expected climate impacts. This will require the AE to maintain strict screening criteria for its direct loans and effective oversight of its PFI partners. The iTAP assessed the criteria and processes described in the funding proposal and notes that they meet a minimal standard of acceptability but could be improved. In particular, standards are only as effective as their implementation. In this case, SIDBI serving as both AE and EE dilutes one of the usual checks on misapplication that the GCF relies on, namely the AE acting as the GCF's fiduciary eyes and ears. The iTAP encourages the Secretariat to consider requesting additional safeguards to ensure that GCF funds and concessionality are used effectively, minimizing the risk of merely supporting PFI balance sheets or financing interventions without a strong climate rationale.

**SIDBI Response-** SIDBI duly acknowledges iTAP's comment in maintaining strict screening criteria. In this regard, we would like to highlight the Funding Proposal has been revised to accommodate the following:

1. A comprehensive eligibility criteria and investment process for the mitigation and adaptation technologies that would be financed has been laid down and the compliance of the same shall be ensured through a robust MRV mechanism for both direct as well as indirect lending.
2. The revised Funding Proposal also has details of the loan approval process in which at the very initial stage, due diligence related to technical viability, climate impact, ESS due diligence would be checked by dedicated professional

at the branch level and at the PMU level. For the PFIs, the required check to factor in climate criteria is placed at the very first stage of the loan approval which details out the condition for empanelment which necessitates that the PFIs which have interest in climate financing and are willing to adopt new systems processes would be empanelled.

SIDBI's total balance sheet size is around USD 57 billion (FY 2024). Out of the above around 90% approximately USD 53.4 billion is towards providing on-lending/ Refinance support to 50 Banks, 78 NBFCs, 41 SFB/MFIs. This depicts SIDBI's strong relationship/ operating capabilities with Financial Institutions. Further, with regard to climate financing, SIDBI has been actively pursuing energy efficiency financing in MSMEs since 2005 -06 and have channelised around USD 2 billion to over 16,000+ MSMEs. The present outstanding of direct green loans is USD 850 million as on March 31, 2024. Hence, AE has strong capacity to underwrite loans as well as to encourage MSMEs and the PFIs to participate in the programme.

AE again reassure the achievements of envisaged impact of the programme through our development and credit plus engagement. This is validated by AE's continued top ranking on multilateral supported programmes. It may be mentioned that in many such multilaterals programmes, SIDBI has been playing both the roles i.e AE as well as EE.

**iTAP feedback (Point 40)-** Funding green technologies in MSMEs is currently perceived as risky. While MSMEs can borrow from other private financial institutions, the concessionality provided under this programme will drive the adoption of green technologies in MSMEs. The GCF funds will enable FMAP to offer concessionality as needed, depending on the relevant MSME sector and technology. This flexible approach allows lenders to provide financing that is fit for purpose. However, there is a risk that over the long tenor of FMAP, the provided concessionality may not keep up with market needs. The iTAP encourages the Secretariat and the AE to explore ways to ensure that GCF concessionality does not end up over-subsidizing sectors as technologies become more accepted and additional financing entrants join the market. As the AE and EE are the same entity, the iTAP suggests having an external process to periodically evaluate the levels of concessionality and the additionality of FMAP funding. This would help mitigate the risk of crowding out other funders, or of PFIs capturing windfall profits without corresponding climate benefits.

**SIDBI Response-** The programme has been designed on the blended finance concept to offer concessional loans to MSMEs either directly or indirectly through PFIs (NBFCs/SFBs/MFIs). The programme envisages a leveraging of more than 16x of the GCF support by way of concessional loan. The programme shall create an ecosystem for financing climate technologies in order to ensure that results are achieved at a scale. Through this, the programme intends to encourage the PFIs to create an ecosystem of climate financing which would allow them to lend to the MSMEs with an increased understanding of the associated risks and underwriting capabilities required. SIDBI's direct lending to the MSMEs sets an example for the PFIs to follow to successfully lend to the MSMEs for adopting low emission climate resilient technologies.

This will be achieved by :

- (i). Offering concessional finance to MSMEs directly encouraging them to adopt relevant climate resilient technologies and interventions. However, at the same time minimum energy savings of 10% will be ensured on a sub-loans basis and 20% at the programme level.
- (ii). Offering concessional lines of capital and blended financial products to FIs which will help in reducing their risk of lending towards MSMEs thus encouraging them to offer additional financing.

The programme has also inbuilt suitable checks and balances which includes M&E on sample basis of sub-loans, periodical reporting of the climate benefits and passing on the concessionality to the ultimate borrowers etc., to avoid over-subsidisation or PFIs getting windfall profits without any climate benefits. The programme proposes to work not with one two PFIs but with around 20 to 25 PFIs and it is expected that over the period gradually there will be sufficient competition among these PFIs for climate financing. This would obviate the risk of windfall profits to the PFIs and shall ensure further benefits to the ultimate borrowers i.e. MSMEs.

As regard independent evaluation, it may be mentioned that the programme has suitable provisions for interim and final evaluations through independent third-party agencies.

### **Overall remarks from the independent Technical Advisory Panel:**

**iTAPs feedback (Point 47)-** However, the iTAP has noted some areas of concern, primarily related to the operationalization of FMAP over time. The FMAP concept is strong, but it relies heavily on SIDBI's capacity to underwrite loans and oversee participating PFIs' underlying loans to achieve the target mitigation and adaptation impacts. Areas of FMAP that can be strengthened are noted throughout this assessment. FMAP may face challenges associated with pipeline risks and MSMEs' willingness to adopt climate-resilient technologies and solutions. In such circumstances, there could be a temptation to loosen climate screening standards or oversight of PFI portfolios. It is important to consider additional safeguards in the operational plan to mitigate this risk and uphold the climate imperative of FMAP.

**SIDBI response-** SIDBI submits the following:

1. SIDBI's total balance sheet size is around USD 57 billion (FY 2024). Out of the above around 90% approximately USD 53.4 billion is towards providing on-lending/ Refinance support to 50 Banks, 78 NBFCs, 41 SFB/MFIs. This depicts SIDBI's strong relationship/ operating capabilities with Financial Institutions. Further, with regard to climate financing, SIDBI has been actively pursuing energy efficiency financing in MSMEs since 2005 -06 and have channelised around USD 2 billion to over 16,000+ MSMEs. The present outstanding of direct green loans is USD 850 million as on March 31, 2024. Hence, AE has strong capacity to underwrite loans as well as to encourage MSMEs and the PFIs to participate in the programme.
2. It may be mentioned that there are more than 63 million MSMEs in India and the programme is targeting only around 10,000 MSMEs. Notwithstanding the above, we

fully concur with the views of the iTAP regarding willingness of MSMEs to adopt climate -resilient technologies and solutions. Accordingly, the programme envisages to provide suitable technical handholding, capacity building, awareness as well as other TA activities to encourage MSMEs to participate in the programme. The concessionality will also play a major role in encouraging MSMEs to adopt climate-resilient technologies.





## **GENDER ASSESSMENT AND ACTION PLAN**

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### **‘Financing Mitigation & Adaptation Projects (FMAP) in Indian MSMEs’**

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## 1. GENDER ANALYSIS & ASSESSMENT STUDY

### 1.1 INTRODUCTION

In the past few years, India has been increasingly faced with the challenge of mitigating the detriments of and adapting to the adverse impact of climate change and balancing the same with the need of economic growth. While the ever-increasing demand of energy resources, energy systems and infrastructure, concomitant with the needs of a vast and rapidly developing country, climate related vulnerabilities are now in desperate need of mitigation and adaptation actions.

Micro, Small and Medium Enterprises (“**MSMEs**”), forming a vital part of the Indian economy and playing a critical role in the sector of employment generation in the country, today face considerable operational challenges, energy costs being one of the key causes. Challenges such as insufficient infrastructure, limited access to credit and investments, as well as limited financial, technical knowledge and capacities further render MSMEs vulnerable to adverse impacts of climate change.

It is in this context that SIDBI, India proposes to address such deficiencies, through a programme named ‘**Financing Mitigation & Adaptation Projects (FMAP) in MSMEs**’, that is aimed at facilitating MSMEs to become more climate resilient, sustainable and competitive in the long run.” (“**Programme**”) This programme will be upscale ongoing projects, through **affordable and accessible finance and technical support for projects pertaining to climate change mitigation and adaptation projects**.

Among other aspects, the design of the programme takes into consideration, the existing gender issues, and concerns, in conformity with international norms and national priorities.

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#### 1.1.1 RATIONALE OF THE ASSESSMENT

The Gender Assessment will help to understand relationship between men and women, their access to resources, their activities, and the constraints they face relative to each other.

**The objective of the present document is to showcase findings basis a Gender Assessment Study and formulate a Gender Action Plan for the Programme, with respect to its’ targeted sectors.** The projects in MSMEs under this Programme would be related to mitigation specific interventions such as Energy Efficiency/Technology upgradation, Electric Mobility, Renewable Energy (including Solar). Further, the interventions will also include adaptation specific interventions

across a variety of themes including Climate Smart technologies, Water, Wastewater, Innovative Technologies etc.

The scope of work under the Programme entails, conducting a gender assessment study, preparing a gender action plan, as well identifying the gender impact and benefits accruing from the said programme.

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### 1.1.2 METHODOLOGY NOTES

The assessment and the action plan under the Programme is prepared in alignment with the Green Climate Fund (“GCF”) Gender Policy, in consonance with the instructions provided by the United Nations Framework Convention on Climate Change (“UNFCCC”), the Sustainable Development Goals (“SDGs”) and other related international agreements and normative frameworks. The exercise of acknowledging disproportionate gender impact, aids in examining the inter-sectional constraints and issues prevalent in the national and sectoral context; and thereby aims to incorporate a course to enhance the ‘agency’ of women in the change process (in contributing to mitigation and adaptation activities, enhancing resilience and well-being, supporting livelihoods in volume and diversity, etc).<sup>1</sup>

It is to be noted that GCF has placed gender perspectives as a “key element of its’ programming architecture” and deems it to Be an “essential decision-making element for the deployment of its resources” towards a thoroughly “gender-responsive” approach for its projects/ programmes.

In preparing the Gender Action Plan for the Programme, following approaches have been leveraged:

#### 1. Baseline assessment of participation of women in the MSME sector in India as entrepreneurs

- a. The literature reviewed from secondary sources to understand the baseline situation and the broad scope for EE applications in the sector considering gender dimensions. In this regard, relevant reports, about national policies and guidelines (environmental, social, gender equality), historical studies and research on Energy Efficiency (EE), international best practices, potential EE interventions etc. considering gender equity were analysed. SIDBI’s Gender Policy attached at **Appendix-II**.
- b. This assessment has been done based on SIDBI’s existing initiatives to promote women entrepreneurship and based on research studies supported by SIDBI.
- c. SIDBI in collaboration with Dun and Bradstreet introduced a quarterly Sustainability Perception Index (SIDBI - D&B SPeX)<sup>2</sup> - the green pulse

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<sup>1</sup> UNWomen and Green Climate Fund, (2017), Mainstreaming Gender in Green Climate Fund Projects, available at [https://www.greenclimate.fund/sites/default/files/document/guidelines-gcf-toolkit-mainstreaming-gender\\_0.pdf](https://www.greenclimate.fund/sites/default/files/document/guidelines-gcf-toolkit-mainstreaming-gender_0.pdf)

<sup>2</sup> <https://www.sidbi.in/spex>

indicator. SPeX is a pioneering effort to assess the sustainability perceptions of MSMEs across various industries and locations. The index is designed to provide a comprehensive overview of the sustainability landscape in the Indian MSME sector.

## 2. **Assessment of challenges faced by women entrepreneurs in the MSME sector**

- a. Stakeholder consultation i.e. “Strengthening Green Financing and Mainstreaming Gender” organised under the banner ‘Green Indian Financial System (GIFS)<sup>3</sup> – Greening of Finance by Women (Grow)<sup>4</sup> Initiatives, founded by SIDBI, Shakti Sustainable Energy Foundation (SSEF) and French Development Agency (AFD). The proceedings of stakeholder consultation attached at **Annexure-III**.

## 3. **Review of SIDBI’s existing capabilities to promote women entrepreneurship/gender equality**

- a. This assessment has been done based on a review of SIDBI’s existing initiatives to promote women entrepreneurship and participation of women in the MSME sector.
- b. Further, SIDBI’s gender policies have been reviewed in line with GCF’s gender policies

The above assessments feed into the Gender Action Plan proposed for the Programme and the specific outcomes from the Gender Action Plan for the Programme.

## 1.2 BACKGROUND AND CONTEXT: GENDER IN INDIA

### **Gender and Employment in India – Data Trends**

There remains no doubt in the growing potential of women in terms of their economic participation. It is to be noted that the difference between the overall literacy rate for adult males and females aged 15 and above have significantly reduced, with the current participation of women estimated at 65.79 per cent of the total female population, against the male population comprising of 82.37 per cent.

However, it is pertinent to note that there was a significant **disproportionate gender impact because of the COVID - 19 pandemic**, in sectors of labour, employment, financial inclusion and with reference to other socio-economic considerations. In the

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<sup>3</sup> <https://www.gifsinitiative.in/>

<sup>4</sup> <https://grownetwork.in/>



global context, the WEF Gender Gap Report 2021<sup>5</sup>(“**WEF Report 2021**”) highlighted the following pointers:

1. 5 per cent of all employed women lost their jobs, compared with 3.9 per cent of employed men (ILO projection)
2. marked decline of women’s hiring into leadership roles across multiple industries<sup>6</sup>
3. increase in care work for women.
4. automation, digitalization as compounding factors in ‘labour market disruption’ – with implications for ‘gender parity in further of jobs’ and ‘occupational gender segregation

The WEF Report 2021 further identifies the increased likeliness of gender gaps in sectors that require disruptive technical skills. Further, in fields where women are under-represented, they are likely to face discrimination and other institutional or social barriers in job transition, particularly in emerging areas of growth potential, with widening disparities in access to resources and capacities.

In case of India, reports furnished by UN Women (2021)<sup>7</sup> and Centre for Sustainable Employment, Azim Premji University (2021)<sup>8</sup> point to certain alarming trends, that threaten the national and international investments in gender empowerment and long-term gains in this regard. The latter recognizes the impact of the pandemic, rendering a large section of the workforce without access to employment or social protection. These reports also identify the increased informality and decline in earnings for majority of workers, with women and younger workers disproportionately affected, because of the pandemic.

For women, it is observed that wage inequality and the burden of unpaid care has pushed more women out of employment falling into depths of poverty, with women losing jobs during COVID-19, and therein engaging more in unpaid care work, both in India and globally.

This section attempts to provide an overview of gender related workforce allocation vis a vis the MSME sector in India. Given the scope of the present study, the focal point of this sections is to furnish national level data, in respect to the participation of women as workforce in the economy and the current scenario of women workforce and ownership vis a vis the MSME sector.

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<sup>5</sup> Global Gender Gap Report, WEF 2021, [https://www3.weforum.org/docs/WEF\\_GGGR\\_2021.pdf](https://www3.weforum.org/docs/WEF_GGGR_2021.pdf)

<sup>6</sup> Ibid. It is noted that while this trend is countered in industries as Software and IT Services, Financial Services, Health and Healthcare, and Manufacturing; the gains are offset by high dislocation in Consumer sector, Non-profits, and Media and Communication – industries with higher participation of women – based on Linked in data.

<sup>7</sup> <https://www.unwomen.org/en/news/stories/2021/7/faq-women-and-covid-19-in-india>

<sup>8</sup> Centre for Sustainable Employment (2021), State of Working India 2021, Azim Premji University, [https://cse.azimpremjiversity.edu.in/wp-content/uploads/2021/08/SWI2021\\_August\\_WEB.pdf](https://cse.azimpremjiversity.edu.in/wp-content/uploads/2021/08/SWI2021_August_WEB.pdf)

### Female population (per cent of total)<sup>9</sup>

| 2018   | 2019   | 2020   | 2021   | 2022   |
|--------|--------|--------|--------|--------|
| 48.34% | 48.35% | 48.36% | 48.38% | 48.40% |

### Labor force, female (per cent of total labor force)<sup>10</sup>

| 2018    | 2019   | 2020   | 2021   | 2022   | 2023   |
|---------|--------|--------|--------|--------|--------|
| 25.04 % | 25.09% | 24.98% | 25.39% | 25.98% | 28.70% |

### Gender-Wise Distribution of Workers in Rural and Urban areas (in lakhs)<sup>11</sup>

| Sector         | Female | Male   | Total   | Share (per cent) |
|----------------|--------|--------|---------|------------------|
| Rural          | 137.50 | 360.15 | 497.78  | 45               |
| Urban          | 127.42 | 484.54 | 612.10  | 55               |
| Total          | 264.92 | 844.68 | 1109.89 | 100              |
| Share per cent | 24     | 76     | 100     |                  |

More recently, a gender assessment of enterprise ownership has also been included in the Annual Report of 2022-23 by the Ministry of MSMEs, as has been furnished hereinbelow:

Out of 1109.89 lakh employees in the MSME sector, males comprise of 844.68 lakh employees totalling to approximately 76 per cent of the total workforce in India, whereas the remaining 264.92 lakh workers i.e. 24 per cent of the total workforce in India comprises of females. The sectoral distribution of workers in male and female categories have been provided as follows:

### Participation of Women in MSME sector as entrepreneurs

As per the statistical data provided on the Udyam Registration Portal (“**URP**”) under the Ministry of MSME<sup>12</sup>, women owned MSMEs constitute 20.37 per cent of the total number of MSMEs registered on the said portal since its inception in 1<sup>st</sup> July 2020.

<sup>9</sup> World Bank Data/ WDI, <https://data.worldbank.org/indicator/SP.POP.TOTL.FE.ZS?locations=IN>

<sup>10</sup> Derived using data from International Labour Organization, ILOSTAT database. The data retrieved on June 15, 2021, <https://data.worldbank.org/indicator/SL.TLF.TOTL.FE.ZS?locations=IN&view=chart>

<sup>11</sup> Ministry of Micro, Small and Medium Enterprises, GoI, Annual Report 2022-23. [Untitled-1 \(msme.gov.in\)](#)

<sup>12</sup> <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2002574>

These women owned enterprises are spread across several states, with Eastern region of India having the highest share of women-owned MSMEs.

### Nature of women owned MSMEs in India

- **Based on type & scale of organization<sup>13</sup>:** Most women-owned enterprises in India are a single-person micro enterprise. Only ~17% of women-owned enterprises have employed workers, of which ~95% operate with less than 6 workers. As the size of the organization increases, the proportion of women-led enterprises decreases. There is a need to increase the share of women-owned MSME in the small and medium enterprise.

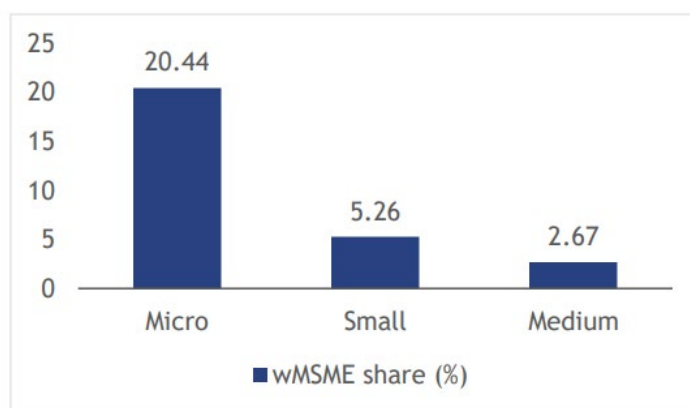


Figure 1: Women owned MSME across different categories of MSME

### Social & Financial Impact of the women owned MSMEs:

- These women-owned MSMEs' encompasses 18.73 per cent of the total employment generated by the total Udyam registered MSMEs, further, generating 11.15 per cent of the total investment therein<sup>14</sup>.
- The contribution of women owned MSMEs vis a vis the total turnover of URP registered MSMEs totals to 10.22 per cent.
- As per the data provided on the Udyam Assist Platform ("UAP")<sup>15</sup>, the contribution of women-owned Informal Micro enterprises (IMEs) to the total number IMEs since the inception of Udyam Assist portal on 11.01.2023 is 70.49 per cent, and their contribution to employment stands at 70.84 per cent.

### Women participation in total workforce:

<sup>13</sup> MoMSME annual report 2021-22

<sup>14</sup> <https://pib.gov.in/PressReleaseIframePage.aspx?PRID=2002574>

<sup>15</sup> The Udyam Assist Platform of Ministry of MSME. GoI facilitates online registration of Informal Micro Enterprises ("IME") through Assist Process for generation of Udyam Registration Number and Udyam Assist Certificate. (<https://udyamassist.gov.in/>)

- As per SIDBI's SPeX conducted survey of 250+ MSMEs across India during October-December 2023<sup>16</sup> through a digital questionnaire containing questions aimed to elicit their perception, understanding and implementation of sustainability practices/measures. Out of total MSMEs surveyed only 38% of MSMEs cited the participation of higher women in total workforce.

### 1.3 DATA REFERENCES & TRENDS

The details of the total number of women owned MSMEs and their contribution to employment, investment and turnover as available on the URP and UAP is provided hereinbelow<sup>17</sup>:

| Category                               | Total         | Employment     | Investment (INR In Crore) | Turnover (INR In Crore) |
|----------------------------------------|---------------|----------------|---------------------------|-------------------------|
| <b>Women MSME</b>                      | 4,667,278     | 28,407,069     | 126845.12                 | 1714992.98              |
| <b>MSMEs</b>                           | 22,819,417    | 151,668,034    | 1,137,237                 | 16,784,358              |
| <b>Percentage of Women owned MSMEs</b> | 20.5 per cent | 18.73 per cent | 11.15 per cent            | 10.22 per cent          |

Contribution of women-owned Informal Micro Enterprises (IMEs) to total IMEs registered on the UAP since Inception of the portal (11.01.2023 to 31.01.2024) is furnished hereinbelow:

| Category           | Total (nos.)   | Employment (no of persons) |
|--------------------|----------------|----------------------------|
| <b>Women IME</b>   | 9,108,058      | 11,023,945                 |
| <b>IMEs</b>        | 12,920,177     | 15,561,967                 |
| <b>per centage</b> | 70.49 per cent | 70.84 per cent             |

<sup>16</sup> [https://www.sidbi.in/head/uploads/spexthegreenpulse\\_documents/D&B%20SIDBI%20SPeX%20Oct-Dec%202023.pdf](https://www.sidbi.in/head/uploads/spexthegreenpulse_documents/D&B%20SIDBI%20SPeX%20Oct-Dec%202023.pdf)

<sup>17</sup> <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2002574>

## 1.4 CHALLENGES RELATED TO WOMEN ENTREPRENEURSHIP IN MSME SECTOR IN INDIA

The MSME sector in India contributes to ~30% of India's GDP and employs around 110 million people. In India, about 89% of the women-owned enterprises operate all year round, 9% are seasonal, and the remaining 2% operate casually, which are neither perennial nor seasonal<sup>18</sup>.

The stakeholder consultation highlights the efforts needed to support the greening of Indian financial systems and the role of MSMEs in initiatives scaling up of renewables, energy efficiency, electric vehicles and sustainable business. The event brings together stakeholders from different sectors to share learnings and drive sustainable practices. The consultation also focuses on promoting women's participation in the green and climate finance sector. Women entrepreneurs share their experiences and insights in sustainable business, highlighting the challenges they face and the potential solutions. The experts also deliberated on challenges faced by sustainable businesses in scaling their products, the need for government support, and the importance of user-friendly technology and clear communication for making sustainable technologies commercially viable.

Following are some of the key challenges faced by women entrepreneurs in India.

- **Not Sustainable:** ~73% women-led enterprises have faced a reduction in revenues post COVID-19. Most female entrepreneurs are concentrated in sectors and industries highly affected by the pandemic and digitization like micro and small businesses in non-professional services, such as tailor shop owners, dressmakers, shopkeepers and street vendors, arts and handicrafts vendors, and beauty-parlor owners, among others. **This points to the fact that women led businesses are not financially sustainable/resilient enough to recover from external shocks.**
- **Not Autonomous:** In ~20% cases, the business-related decisions are taken by the respective spouses and not the women entrepreneur running the business. Often, women do not truly own or control many women-led enterprises. **They depend considerably on men for key business decisions, procurement of raw materials, pricing information, and market access.**
- **Not Formal:** ~95.6% of the women entrepreneurs in India are informal and not registered with relevant national/state institutions. A high level of informality harms the growth and resilience of their enterprises and their entrepreneurial rights. The level of effort involved in running such enterprises go unaccounted

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<sup>18</sup> Niti Aayog – “Decoding support to women entrepreneurs in india” - 2023

for and expose women to occupational health hazards. **Further, informality leads to challenges in accessing finance, market, etc.**

- **Not sufficient employment generating:** ~95% of women proprietors run with less than six workers, most being own-account or single person enterprises. Only ~1.9% proprietors run with six to 10 employees. **This points to the fact that women entrepreneurs often lack skills in scaling businesses i.e. in areas such as business management, market access, access to finance, etc.**
- **Lack of awareness and skills on green jobs and technologies:** Women in Indian MSMEs have been confined to low-paying, low-skilled jobs. The ongoing transitions present them with an opportunity to realign their skill sets and adjust to the evolving nature of work. Digital & Green Technologies proficiency is one of the most important skills that women need to acquire to keep pace with the ongoing transitions. This will enable them to enter traditionally male-dominated sectors that rely heavily on digital technologies, such as healthcare services, automobiles, and more advanced roles within the textiles sector with green technologies.

Currently, there is limited evidence on how climate change affects women in the MSME sector. The impact of these disruptions presents both a challenge and an opportunity for mainstreaming gender inclusion and empowerment through the proposed project.

## 1.5 GENDER LENS ON MSMES AND CLIMATE (MITIGATION AND ADAPTATION)

The role of women entrepreneurs in driving inclusive economic growth cannot be ruled out. Women entrepreneurs not only generate income for themselves and their families but also reinvest in their communities, thereby creating a ripple effect of positive socioeconomic impact. By fostering entrepreneurship opportunities for women, we can reduce gender disparities in employment and help alleviate poverty. Targeted interventions to address the specific needs of women in different regions and sectors, especially in clean energy and climate-resilient agriculture, would be the best next step.

Women entrepreneurs are not just agents of change; they are the architects of a sustainable future. To create a more inclusive, equitable, and sustainable future for generations to come, women-led endeavours must be supported. Further, an enabling environment should be created for women to flourish and share their learning's and insights while tackling climate change.

A recent report for “**improving women’s productivity and incomes**” through **energy/ climate initiatives** has also been conducted in 2021 by the Council on



Energy, Environment and Water.<sup>19</sup> Citing a direct link or contribution to SDGs 5, 7 and 8, women's engagement as producers, customers, and suppliers in this area is seen to have potential for expanding markets and achieving scale.

## Opportunities:

In order to address women's specific and systemic vulnerabilities, there is a need for collective efforts that support women in the sector.

- There is a need for national and state level governments to ensure effective skill training in addition to training provided by the employer.
- **Drawing lessons from past experiences:** Women are also more susceptible to economic fluctuations and other shocks that affect the MSME sector. For Instance, during the first lockdown in 2020, only 7 per cent of men lost their jobs, compared to 47 per cent of women who lost their jobs and did not return to work by the end of the year. In the informal sector, women fared even worse. Between March and April 2021, rural Indian women in informal jobs accounted for 80 per cent of job losses.”
  - ➔ “Indian women also spend more time doing unpaid care work at home than men. On an average, they spend 9.8 times more time than men on unpaid domestic chores and 4.5 hours a day caring for children, elders and the sick. During the pandemic, their share of unpaid care work grew by nearly 30 per cent.” Centre for Sustainable Employment, State of Working India (2021)
  - ➔ “More girls than boys were left out of school during the pandemic and 65 per cent of parents surveyed were reluctant to continue the education of girls and resorting to child marriages to save costs. This can create an entire generation of young women without education and employment opportunities.” UN Women (2021)

*As women-led businesses continue to recover from the pandemic-induced shocks, there is an emerging need to adopt cleaner energy technologies, digitalization, and other transitions in the value chain. As these transitions necessitate newer and more specialized skills, they further compound the existing challenges that women in MSMEs currently face.*

- **New skills and training for women in green jobs:** Women are more inclined to embrace technologies and mechanisms that offer cost and energy savings, leading to increased cash flows. It is imperative to develop suitable business cases and models to promote the adoption of renewable technologies, such as solar energy and biogas. There is, therefore, a need to make such innovations

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<sup>19</sup> Improving Women's Productivity and Incomes Through Clean Energy in India Sasmita Patnaik, Shaily Jha, and Tanvi Jain, CEEW Report July 2021, <https://www.ceew.in/sites/default/files/Improving-Womens-Incomes-and-Productivity-through-Clean-Energy-in-India.pdf>

accessible to female entrepreneurs through training and research provisions. It is crucial to educate women about the long-term implications of climate change for their businesses and highlight the potential opportunities arising from the adoption of clean energy and climate-resilient practices. Additionally, on-the-job training programs should be designed to develop relevant skill sets among female workers. These programs should include modules on financial management, digital and information technology, leadership, and management. This approach will not only support female workers in making informed choices when selecting from various career paths but also contribute to their professional development and decision-making capabilities.

- **Establishing Institutional support:** Women workers and entrepreneurs may find themselves in the minority as they strive to enter roles and sectors traditionally dominated by men. Ensuring that women are not left behind during the low-carbon transition and placing them at the forefront of climate adaptation and resilience require comprehensive support at all levels. Creating opportunities for women requires innovative strategies and policies that effectively address social, cultural, and economic biases. The formation of networks of female workers that provide mutual support and training for new and multiple roles is vital. Even as women enter sectors such as aviation, engineering, and construction, they require targeted support and skills for technical roles where they are currently underrepresented.
- **Gender-responsive financing for MSMEs:** Access to credit is a primary barrier for women in MSMEs. Improving access to formal and affordable financial services can serve as a catalyst in supporting women during the transition to clean energy and climate change. The financial inclusion of women can lead to social and economic empowerment by improving access to a range of financial products, including savings, credit, insurance, and payment services, and by building financial awareness. Most women-owned MSMEs operate as unregistered enterprises and continue to face issues in accessing formal financing to meet their business requirements. The credit needs of women entrepreneurs remain largely unmet owing to social, economic, cultural, and legal factors.
- **Promote decent and dignified workplaces for women:** Business sustainability involves striking a balance between social, economic, and environmental considerations. From a gender perspective, this implies providing conducive and safe working conditions, access to resources, training, and capacity building for female workers. To ensure comprehensive support for women across different life stages, including marriage and motherhood, there is a need for innovative and flexible mechanisms that facilitate and encourage an effective return to work.

The current mechanisms for accessing climate funds offer ad hoc solutions. There is a need to move towards systems approaches that provide financial assistance and

offer support to entrepreneurs in sectors undergoing multiple transitions, such as low-carbon and technological transitions. It is crucial to consider systemic benefits and institutional mechanisms for a just transition.

In this regard, encouraging women entrepreneurs in the “green” economy represents an opportunity to make women-led enterprises more resilient and sustainable, thus enabling a “just” transition through FMAP.

## 1.6 INITIATIVES TAKEN BY GOVERNMENT TO PROMOTE & SUPPORT WOMEN ENTREPRENEURSHIP IN MSME SECTOR IN INDIA

There are several Gender focused legal regulations in India, that aim to address the vulnerabilities and cultural biases associated with gender, through the provision of certain measures that support a safe and equal professional platform for women. These regulations are presented below.

**Equal Remuneration Act, 1973** provides for payment of equal remuneration to men and women workers for the same work of similar nature without any discrimination. In order to ensure social security to the workers including women in the unorganised sector, the Government has enacted the Unorganised Workers' Social Security Act 2008.

**The Maternity Benefit Act, 1961 and The Maternity Benefit (Amendment) Act, 2017** regulate employment of women in certain establishments for a certain period (26 weeks) before and after childbirth and provides for maternity and other benefits.

**The Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act (POSH), 2013** has been enacted, which covers all women, irrespective of their age or employment status and protect them against sexual harassment at all workplaces both in public and private sector, whether organised or unorganised.

The G20 New Delhi Leaders' Declaration 2023 focuses on “Enhancing Economic and Social Empowerment,” “Bridging the Gender Digital Divide,” “Driving Gender Inclusive Climate Action,” “Securing Women’s Food Security, Nutrition, and Well-Being, Creation of a Working Group on the Empowerment of Women.

The Government of India had launched programmes like Beti Padhao, Beti Bachao campaign (launched in 2015) for improving the Gender Parity Index (GPI)<sup>20</sup> by promoting survival, protection and education of girl child. The schemes aim to address the issue of declining Child Sex Ratio through a mass campaign targeted at changing social mind set and creating awareness about the criticality of the issue. Furthermore, Mahila E-Haat has been launched by the Government in 2016 for promoting women entrepreneurs, with the objective to provide an e-marketing platform by leveraging technology for showcasing product made/manufactured/sold by women entrepreneurs/SHGs/NGOs.

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<sup>20</sup> Gender Parity Index (GPI) in education is a valuable indicator which reflects the discrimination against girls in access to educational opportunities

**Further, The Ministry of MSME, Government of India has taken a number of steps to promote and support women-owned MSMEs<sup>21</sup>**

1. Special drives have been undertaken for registration of women owned MSMEs under the URP.
2. In order to endorse women entrepreneurs, the Public Procurement Policy was amended in 2018 mandating Central Ministries/Departments/ Undertakings to procure **at least 3 per cent of the annual procurement from women-owned MSMEs.**
3. To further support women entrepreneurs under the Credit Guarantee Scheme for MSMEs, the following two provisions have been introduced for w.e.f. 01.12.2022.
  - a. **10 per cent concession in annual Guarantee fees; and**
  - b. **10 per cent additional Guarantee coverage of up to 85 per cent**, as against the 75per cent for other entrepreneurs.
4. In order to encourage entrepreneurship among women, the Ministry of MSME has implemented the 'Skill Upgradation & Mahila Coir Yojana' under 'Coir Vikas Yojana', which is an exclusive training program aimed at the skill development of women artisans engaged in the Coir sector.
5. The Ministry of MSME has also implemented Prime Minister's Employment Generation Programme ("PMEGP"), which is a major credit-linked subsidy programme aimed at generating self-employment opportunities through establishment of micro enterprises in the non-farm sector.
6. A higher subsidy is being provided to beneficiaries belonging to special categories such as Scheduled Caste/Scheduled Tribe/OBC /minorities/women/ ex-serviceman/ physically handicapped/ NER/ Hill and Border areas, etc.
7. For purposes of encouraging participation of women entrepreneurs in trade fairs under the Procurement & Marketing Support Scheme, a higher subsidy is provided for women owned MSMEs.
8. "SAMARTH" initiative was launched to support women owned MSMEs with the objective of providing Skill Development and Market Development Assistance to women and to train more than 7,500 women candidates from rural and sub-urban areas. Aspiring and existing women entrepreneurs have been provisioned with 20 per cent seats in free Skill Development Programs organized under Skill Development Schemes of the Ministry of MSME; 20 per cent of MSME Business Delegations for domestic & international exhibitions under the schemes for Marketing Assistance implemented by the Ministry of MSME; and 20 per cent discount on annual processing fee on National Small Industries Corporation's Scheme.

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<sup>21</sup><https://pib.gov.in/PressReleaselframePage.aspx?PRID=2002574>

9. MSME Sustainable Zero Defect Zero Effect (“ZED”) Certification Scheme is an initiative to provide a roadmap to global competitiveness for Indian MSMEs having economic and social impacts in terms of capacity building of MSMEs. The aim of the scheme is that the goods being manufactured by the MSME sector have zero recall from market at the same time has zero effect on environment. Under this scheme to support and empower women entrepreneurs, there is a provision of 100 per cent subsidy on cost of ZED Certification for Women owned MSMEs.
10. Under the incubation component of MSME Innovative Scheme, a special MSME Idea Hackathon 3.0 has been organized for women entrepreneurs wherein 18,888 ideas have been received yet.

The Ministry of MSME implements the following capacity building programmes for capacity building and skill enhancement for MSMEs, including women entrepreneurs in MSMEs:<sup>22</sup>

1. Entrepreneurship Skill Development Programme (“ESDP”): The objective of ESDP is to promote new enterprises, to build capacity of existing MSMEs and to inculcate entrepreneurial culture in the country. About 40 per cent of the targeted beneficiaries of ESDPs are stated to belong from weaker sections of the society (SC/ST/Women/Physical Handicapped). As per the said programme, no participation fee will be charged from SC, ST, Physical Handicapped, Below Poverty Line (BPL) participants and women.
2. Mahila Coir Yojana (“MCY”) aims at empowering women artisans in the coir sector, by providing these women with a stipendiary two-month training programme. These women artisans are encouraged to avail assistance under Prime Minister’s Employment Generation Programme (“PMEGP”).

These initiatives are expected to spur the growth of women entrepreneurs in the MSME sector in India.

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<sup>22</sup><https://pib.gov.in/PressReleaseframePage.aspx?PRID=2002574>

## 2. SIDBI'S EXISTING INITIATIVES/MECHANISMS TO SUPPORT WOMEN ENTREPRENEURS IN MSME SECTOR

Small Industries Development Bank of India (SIDBI) plays a proactive role in the promotion and development of Micro, Small and Medium Enterprises (MSMEs) to address various financial and non-financial challenges faced by them. It aims at innovating for wider impact and to benefit MSMEs in both “setting up” and “stepping up”, thereby leading to economic growth of the country. SIDBI under its Mission Swavalamban aims at promoting the aspiring, livelihood, micro and small entrepreneurs. SIDBI promotes entrepreneurship culture and encourages women, youth and other aspirants to become “Job creators” from “Job seekers”, which restricts rural migration to urban areas and promotes sustainable livelihood opportunities with thrust on the bottom of the pyramid, unserved/underserved pockets/ segments of Micro and Small Enterprises (MSEs) through various novel initiatives.

SIDBI has been undertaking several impactful interventions for promotion of entrepreneurship and development of enterprises at the grass root level. The Programmes for Development & Impact (PDI) Vertical of the Bank through its Swavalamban Resource Facility (SRF) with support from FCDO, UK, aims to promote inclusive and innovative entrepreneurship in low-income pockets of India, with a special focus on women entrepreneurs.

### **Institutional Capacity of SIDBI**

SIDBI has been a pioneer in supporting women-led MSMEs in India. SIDBI has created and nurtured (for 25+ years) the micro-finance ecosystem in India. For example, SIDBI's program to support microfinance institutions has created 150 micro-finance institutions to support women's self-help group (SHGs). This has reached 30-40 lakh women – this segment has grown such that 9 institutions have evolved to small finance banks (e.g. Bandhan, Utkarsh).

SIDBI continues to actively promote financial inclusion and entrepreneurship for women. SIDBI aims to support 20% of its financing towards women led MSMEs. **Following are some of the key programmatic initiatives undertaken by SIDBI to further this cause<sup>23</sup>.**

- Jal Sparsh-Waste to Wealth: The uncontrolled growth of water hyacinths is a problem for a lot of water bodies in the country. It blocks waterways and limits boat traffic, recreation, flood control and wildlife use. It produces a dense canopy at the water surface, and the pest plant shades out native submerged plant species and can uproot native emergent species that are important to wildlife. Eco-roots Foundation floated a solution to make productive use of the uncontrolled growth of water hyacinth, by its extraction from the water bodies. The solution has also

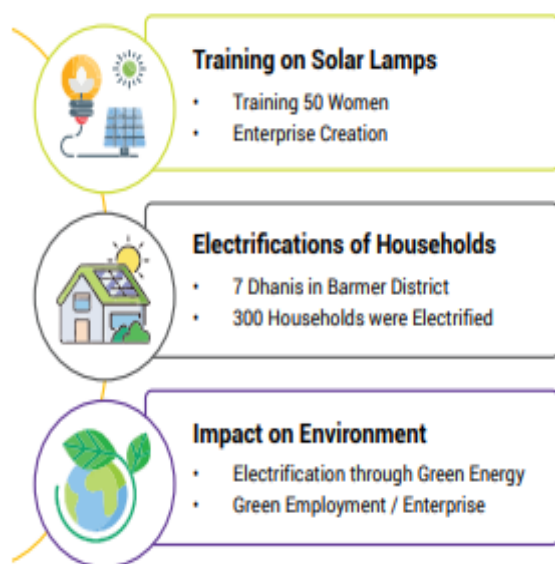
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<sup>23</sup> SIDBI Annual Report, FY 23



helped the local women earn sustainable and meaningful livelihoods. Hence, the intervention has a positive effect on the environment as well as the livelihood. The rural women were provided with skills to sustainably extract the water hyacinth vines and the methods to process it to form useful products such as bags, mats, stands, and other decorative. The activity mainly engages poor women for the entire process. Eco-roots have also marketed the finished products through various online and offline avenues such as e-listing on commerce platforms, exhibitions, and fashion shows in school. The project is present in 5 states in India.

- Women Entrepreneurship -Livelihood Initiatives on Green Energy and Rural Household Electrification (WE-LIGHT): Climate change is a massive issue to address, and women are most affected by climate change problems. Non-electrification and limited access to power supply further adds to the problems. It is an integrated approach to address climate change and poverty reduction through a solar-focused energy access program by enabling rural communities to become self-sufficient and resilient. The knowledge and utility of solar is thus transferred to the community for sustainable access to electricity at an affordable cost for the rural population. This program aims to combine technology, market strengthening and skills to effectively deploy capital that delivers impacts for the environment, women, communities, and the rural poor. It brings together the most holistic program of training on the design and fabrication of solar lanterns, marketing, entrepreneurship, and leadership which allows rural women to have access to capital to scale the distribution of renewable energy products in the rural market in India.



- Prayaas Initiative: IN FY 23, loans upto Rs. 5 Lakhs for livelihood activities were extended to ~80,000 micro enterprises, most of which are women led.
- ARJANA: This plan includes various financial benefits such as reduced credit guarantee fees, no prepayment charges, additional advantages for ZED

certification, discounts for timely repayment, lower processing fees, and lower promoter margins (only for women entrepreneurs).

- **Swavalamban Initiative**: SIDBI's Mission Swavalamban has collaborated with Usha International Limited to promote women entrepreneurship in underprivileged areas through the Swavalamban Silai (i.e. Tailoring Homepreneurs Programme. This has helped over 3000 marginalized women become Silai Homepreneurs in 23 states over 5 phases.
- **SWAS**: SIDBI and 'Mukti' collaborated to launch the 'Swavalamban Accelerator in Sunderbans' (SWAS) program to train SHG women in establishing alternative livelihoods and rebuild the Sundarbans, which was affected by Super Cyclone Amphan. 1000 women were trained in food processing and handicrafts (500 each) and received start-up kits to begin their microenterprises after the training. 417 beneficiaries established new microenterprises and 771 beneficiaries were provided with market linkage.
- **SAATHI**: The Aga Khan Rural Support Programme India is working to create 200 model villages in Western and Central India, with a focus on training tribal youth, especially women, in micro-enterprise development. Under this program, the goal is to establish 650 enterprises in these villages within three years. As of FY 2023, 275 people have been trained and 200 have started their own enterprises.
- **WE-LEAD**: The project being carried out by OSAFII in Eastern India, aims to support 12,000 women entrepreneurs, amongst SHGs/ JLGs members, by providing them with capacity building, skill training, and business development services. In FY 2023, 7,970 women were trained, 2,859 women were given credit, and 2,832 enterprises were established. The project is replicated in partnership with the Association of Microfinance Institutions, with the aim of promoting 10,000 women entrepreneurs.
- **TOWER**: Mahabhinishkraman Unnayan Sansthan is implementing the TOWER project to help 100 tribal women entrepreneurs in a district in Eastern India develop their businesses using non-timber forest products. A modern machinery powered by solar energy has been established to produce high-quality Saal leaf plates and bowls which is managed and owned by the community.
- **Empowering Women and Enhancing their Business through E-Mobility (EWEE)**: The rural India is critical for the nation's development and the women are the backbone of it. The empowerment of women at the grassroots level is important to build the nation. One of the many ways to empower women is to improve the mobility options in rural parts to encourage them to start their businesses, job opportunities, etc. SIDBI launched E-WEE Programme to ease the mobility of women entrepreneurs by providing access to affordable finance through blended finance facility to purchase electric two-wheelers with the support of Non-banking financial companies.
- SIDBI is also one of the signatory of Finance in Common (FiCS) gender coalition to the **Paris Development Banks Statement on Gender Equality and Women's Empowerment**.

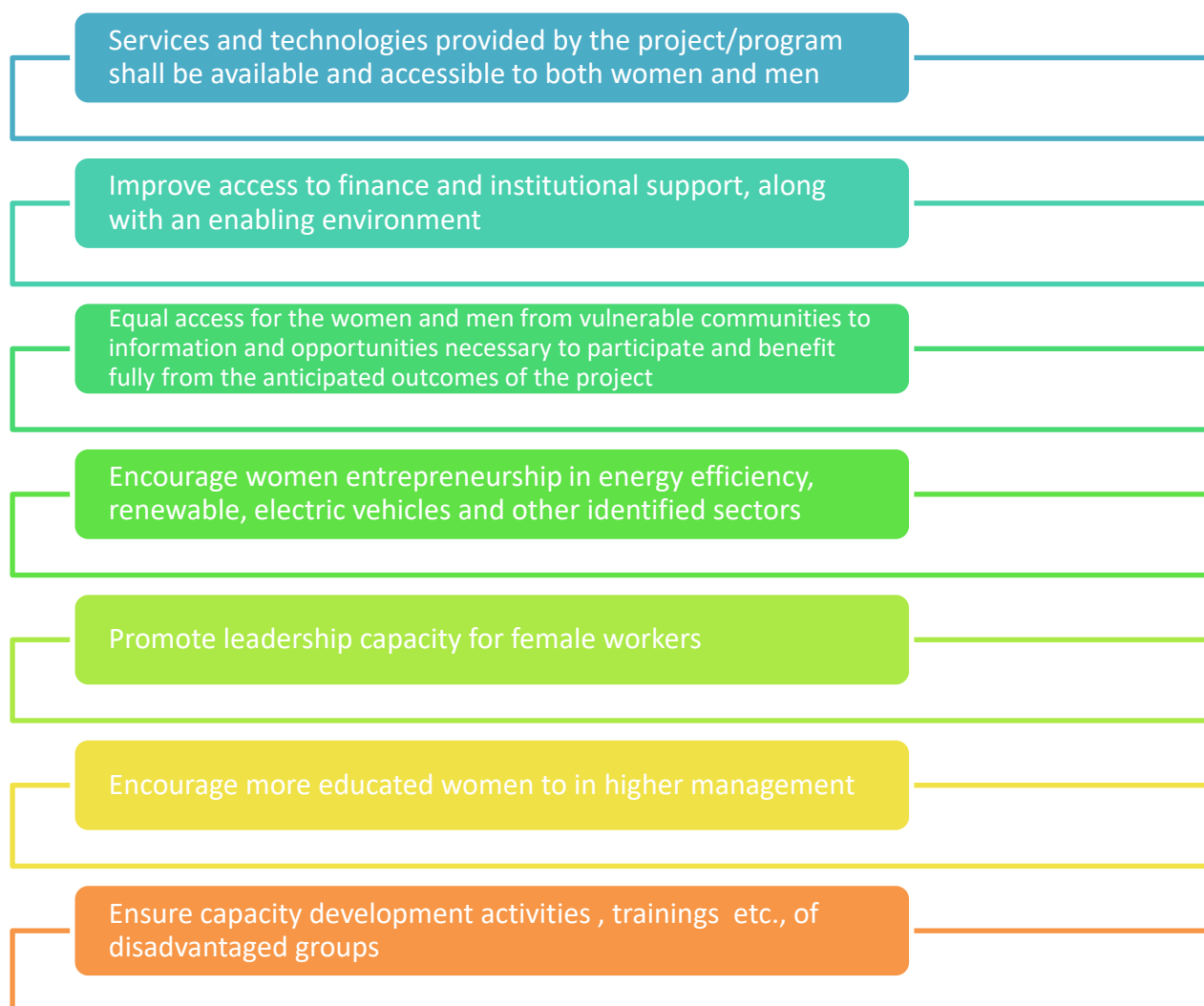
**Beyond these programmatic initiatives, SIDBI has undertaken several structural reforms/adopted policies to promote women's welfare and entrepreneurship.**

- Strength of women employee base & equal opportunity employer: As of FY23, SIDBI had 227 women employees (21.89% of the total strength). SIDBI is an equal opportunity provider and has implemented fair practices for enabling career progression. For example, in FY 23, SIDBI made 1607 nominations for various employee training programs. Of this, 322 nominations were women employees (~20% of overall nominations).
- Gender Policy: As a GCF accredited National Implementation Entity (NIE), SIDBI has adopted a Gender Policy at an organizational level (attached below). This policy governs SIDBI's organizational policies with respect to gender equality and inclusion. Further, the policy also governs SIDBI's engagement with their PFIs with respect to gender equality and inclusion.
- Further, SIDBI is mandated to comply with Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013
- Further, gender-related considerations have been factored in the E&S checklist that every financing proposal has to adhere to.

Based on the above information, SIDBI is adequately placed to implement gender awareness activities. SIDBI has already implemented multiple schemes encouraging women entrepreneurship across different geographies in India. The previous experience of these schemes has enriched SIDBI staff to undertake these activities. At a program level, SIDBI has employed Gender experts to drive the implementation of the Gender Policy/Action plan and ensure effective monitoring & reporting. For FAMP, SIDBI will employ suitably qualified Gender experts.

## Alignment of the Project with GCF Gender Assessment Guideline

- SIDBI as an AE shall take the necessary measures to ensure that proposed projects, as well as proposed activities under the FMAP, submitted to GCF for financing are aligned with national policies and priorities on gender.
- The project activities shall be framed around principals of GCF and will ensure the following issues in the gender action plan:



### 3. GENDER ACTION PLAN

#### 3.1 ESTABLISHING THE CONTEXT/ BASELINE TO THE ACTION PLAN

The present proposal prioritizes the mainstreaming of gender concerns, with focus on leveraging the funding to improve gendered coverage of programme financing and benefits. A stated endeavour of the programme is to secure 80:20 coverage (i.e. 20% financing towards women), broadly in line with the selected baselines for women's ownership and labour force participation. In the longer term, the aim is to contribute to a paradigm shift in terms of women's capacity and participation in ownership/ leadership/ management and technical roles/workforce participation in, as well as other benefits from climate mitigation and adaption activities taken up in the MSME sector in India.

As a critical support to MSME financing and development in the country, SIDBI is also called upon to provide momentum for creating an enabling environment, contributing to public/ stakeholder awareness, as well as identifying and setting operational and regulatory standards to this end.

The overall direction of recommended actions to this end would be in terms of:

- a. Building on existing mechanisms and provisions supporting financial inclusion and incentives/ concessionality for women entrepreneurs, with a view to enhancing their agency in climate mitigation and adaptation initiatives through this programme.
- b. Collaborating with existing industry associations and women entrepreneur/ other networks on targeted capacity-building programmes on women and energy efficiency, resource management.

In addition, and particularly relevant for an institution such as SIDBI, given its niche status in the MSMEs sector are aspects of strengthening the enabling environment for gender mainstreaming. These pertain to issues such as:

- operational learning to sustain gender mainstreaming efforts
- technology for outreach
- sector-wide awareness campaigns, capacity building and entrepreneurship development programmes
- innovations as gender sensitive credit ratings, support with registration and paperwork, etc

#### 3.2 MAPPING OF PROPOSED GENDER-SENSITIVE PROGRAMME MEASURES AGAINST PROGRAMME CYCLE AND PROJECT ACTIVITIES

Based on the programme design, the broad aim is to ensure adequate representation for women as stakeholder and beneficiary under component (1) supporting affordable financing of mitigation and adaptation activities; targeted technical support under component (2), for capacity development, awareness generation, etc., as well as in M&E frameworks; and leveraging financing mechanisms for greater and more equitable outreach and coverage in the MSME sector going ahead.

Specific, gender sensitive programme measures can then be given by the following categories:

- Screening Parameters for Beneficiary M/SMEs
- Actions by/ Performance Indicators for SIDBI
- Incentives
- SIDBI Niche/ Enablers

In the case of **screening parameters**, the suggestion is to give weightage and priorities to

- (i) sectors/ geographies with higher representation of women (top 10 states with women entrepreneurs/women led MSMEs is given in Section 1.2 of this document), as well as to
- (ii) firms owned by women, and those with significant representation in the management and work force. These weightages would ideally translate into/ align with a points system on eligibility criteria, and link with a higher degree of concessionality in financing terms as well as other support services. A suggestion, going beyond the individual firm focus in the programme, is to include MSME clusters comprising women-led enterprises and/or with significant representation, as eligible beneficiaries under this programme. It is believed that this would be especially relevant to extend programme benefits and impacts given the current distributions in this regard, and to secure/ advance beyond given targets.

While appraising the cases for MSMEs, certain commitments/undertakings would be required as a pre-requisite for providing financial assistance as under:-

- 1) Availability of a gender policy or adherence to the gender policy of SIDBI.
- 2) Existence of committee to perform Grievance redressal for the cases pertinent to gender based violence (GBV), Sexual Exploitation Abuse and Harassment (SEAH) as per the (POSH), Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013.

### **Actions by SIDBI**

In the implementation of the programme in line with its core functions and operating context would essentially include bringing together its gender inclusive provisions,



services and targeted outreach mechanisms in support of this programme. It is also recommended that a certain proportion of the entire planned corpus be earmarked for firms/ clusters with representation of women as owners/ management/ leaders accompanied by higher concessions (e.g. in eligibility criteria, application paperwork, better rates, etc.) and facilitative support. There is also a provision to (iii) improve gender sensitivity and responsiveness of SIDBI management and frontline staff, as a basis for continual learning and improving outreach in this regard. Actions under this category also translate directly into performance criteria and monitoring indicators for SIDBI.

The nature of **incentives**, as already highlighted, include (i) reductions and concessionality in terms of fees and rates for women entrepreneurs, which may be extended to firms/ clusters/ sectors and geographies with higher female participation. Another important area of incentivisation will be through the (ii) targeting of capacity building programmes, which are a critical pathway to increase women's ownership and participation not only in the MSME sector but also in climate initiatives and actions. In this case, it is suggested that the overall participation of female entrepreneurs/ technical and other workers be ensured at a minimum of 30 per cent of overall annual participation in such programmes. There may also be more intensive programmes aimed at young women/ youth in this regard – a segment of high potential given the baseline and general trends, as well as from a forward-looking perspective.

The SIDBI niche as an MSME enabler and as an institution that will set standards for the country in this area, is reiterated here. Monitoring and data analysis to inform operational processes as well as industry standards, research and knowledge capacities and outreach, as well as convening and consultative functions of the financial institutions are critical to taking this programme from business-as-usual to one with the potential to bring about a paradigm shift in women's financial inclusion and employment, also nurturing potential for leadership in the area of climate response. This nature of shift may be seen as the aspirational co-benefit of the programme. It is noted here that with this niche, SIDBI is also well placed to build exchanges and consultations with other institutions and frameworks, there is convergence of work on some or all aspects of women's entrepreneurship/ employment, MSMEs and development, and climate solutions/ services.

### 3.3 M&E, KNOWLEDGE, EVIDENCE AND LEARNING

With focus on M&E, knowledge, evidence and learning, the programme proposes to build on SIDBI's unique operational niche to enhance existing knowledge and capacities in at the intersection of gender, MSMEs and climate initiatives in India, building on data/ evidence from the implementation of this programme – through.

- Mandating monitoring and reporting standards for collection of gender disaggregated data – employment, entrepreneurship, capacities, compensation etc. in the selected sectors and geographies.
- Undertake and provide trend analysis, maintain a database to support ongoing collaborative learning with partner organisations, industry organisations and networks on priorities for women entrepreneurs and professionals in climate mitigation and adaptation initiatives in India.

### 3.4 OUTCOMES FROM PROGRAM

Reskilling women workers who are already employed in the MSME to adopt, adapt, and apply best practices and providing relevant skills to women entering the workforce is essential to ensure sustainability. Increasing women's share in MSME sector is essential to harness equitable economic gains and facilitate social transformation at scale<sup>24</sup>.

It is important to ensure that women in the MSME sector are supported in the context of climate change and low-carbon transition. The overarching goal is to equip them with the necessary capacities—the key skills required for green jobs, climate finance, and essential technologies—thereby disrupting the traditional occupational segregation of women in the sector and transitioning to a sustainable and just economy.

Ensuring decent and dignified workplaces for women business sustainability involves striking a balance between social, economic, and environmental considerations. From a gender perspective, this implies providing conducive and safe working conditions, access to resources, training, and capacity building for female workers.

The programme aims to significantly transform the landscape of women led MSMEs in India. The following are the outcomes that the program aims to achieve:

- **Number of women led MSMEs:** Increase the number of women owned MSMEs through necessary interventions and ensure enhanced participation of women workforce in MSMEs.
- **Business Sustainability through improve access of finance:** Make women led enterprises/MSMEs more sustainable & resilient to external shocks, especially climate shocks by providing access to affordable finance for mitigation and adaption technologies. Services and technologies provided by the project/program shall be available and accessible to both women and men.
- **Autonomy:** Increased skilling for women entrepreneurs to ensure their businesses are more climate resilient by offering handholding support on the adoption of better technology. Encourage women entrepreneurship in energy efficiency, renewable, electric vehicles and other identified sectors.

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<sup>24</sup> <https://wri-india.org/publication/building-ecosystem-gender-responsive-and-climate-resilient-msmes->

- **Formalization:** Increased formalization of women led MSMEs, leading to enhanced access to credit.
- **Employability & Employment Generation:** Ensuring the opportunities of creating more employment from Women own MSMEs by capacity building on climate finance and awareness on energy efficient technology. The measures will also be taken to equip the women workforce for green jobs helping in just transition.
- **Encouraging gender-focused practices among MSMEs:** The project would encourage those MSMEs with women-centric workplace or are willing to create the one. These will include having a gender action plan, separate rest spaces, better lighting etc.

### 3.6 GENDER ACTION PLAN: TEMPLATE (PROJECT/PROGRAM LEVEL)- INDICATIVE FIGURES

The resulting framework is created to achieve the intended objectives of promoting gender inclusion in MSMEs across different levels i.e., leadership, workforce, etc., and at the Participation Financial Institution level. The details are below in tabular form:

1. Number of women led MSMEs
2. Business sustainability through improved access
3. Autonomy
4. Formalization
5. Employability and employment generation
6. Encouraged gender-focused practices among MSMEs

The programme gender activities (as mentioned in gender action plan) will support to bridge the gender gap by increasing the percentage of women MSMEs in the country from currently at 20%.

The gender action plan is given below:

| Indicator*                                                                      | Proposed activities                                                                                                         | Project Baseline i.e., the number of units financed under the project | Targeted value                             | Responsibility | Frequency of reporting | Budget estimates     | Outcome mapping                                 |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------|----------------|------------------------|----------------------|-------------------------------------------------|
| <b>Women-owned MSMEs sensitized on mitigation and adaption technologies</b>     | Capacity building program on different technologies applicable in various sectors                                           | Nil                                                                   | 5% of the MSMEs supports under the program | PFI/ SIDBI     | Annual basis           | USD 50,000           | Business sustainability through improved access |
| <b>Number of borrowers sensitized to improve facilities for women workforce</b> | Capacity building on the facilities required for women's workforce through blended learning methodologies (Online/ Offline) | Nil                                                                   | 2000                                       | PFI/ SIDBI     | Annual basis           | USD 50,000           | Encouraged gender-focused practices among MSMEs |
| <b>Number of women beneficiaries of the programme</b>                           | Awareness on product features for women led MSMEs and concessionality on loans                                              | Nil                                                                   | 15% of total beneficiaries                 | PFI/ SIDBI     | Annual basis           | USD 30,000           | Increase in number of women-led MSMEs           |
| <b>Number of young women to</b>                                                 | Training Program on climate finance including                                                                               | Nil                                                                   | 5000                                       | PFI/ SIDBI     | Annual basis           | USD 10,000 & In-kind | Employability and employment generation         |

|                                                                      |                                                                                                         |     |       |            |              |                      |                                                                            |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----|-------|------------|--------------|----------------------|----------------------------------------------------------------------------|
| <b>be trained in entrepreneurship</b>                                | technology for college, Industry Training Institute, etc. through digital and other activities of SIDBI |     |       |            |              |                      | Autonomy                                                                   |
| <b>Number of women led MSMEs brought on digital lending platform</b> | Through digital lending activities                                                                      | Nil | 10000 | PFI/S IDBI | Annual Basis | USD 10,000 & In-kind | Formalization                                                              |
| <b>Number of PFI staff trained on gender awareness</b>               | Awareness program for PFIs on SEAH and POSH                                                             | Nil | 300   | PFI/ SIDBI | Annual basis | USD 40,000           | Encouraged gender-focused practices among MSMEs                            |
| <b>Number of Media Outreach products on Gender</b>                   | Publications and events on project progress                                                             | Nil | 10    | SIDBI      | Annual basis | USD 30,000           | Increased participation of women in MSME, job creation, formalization etc. |

\*The Programme endeavor to support more women beneficiaries, the indicators in the action plan to be revisited/reviewed every two years (during the implementation of the Programme) from the effectiveness of the Programme.



**CHECKLIST**

- UNWomen and Green Climate Fund, (2017), Mainstreaming Gender in Green Climate Fund Projects, available at [https://www.greenclimate.fund/sites/default/files/document/guidelines-gcf-toolkit-mainstreaming-gender\\_0.pdf](https://www.greenclimate.fund/sites/default/files/document/guidelines-gcf-toolkit-mainstreaming-gender_0.pdf)
- GCF (2019), Gender Analysis/Assessment and Gender and Social Inclusion Action Plan Templates available at [https://www.greenclimate.fund/sites/default/files/document/form-09-gender-assessment-and-action-plan-template\\_0.pdf](https://www.greenclimate.fund/sites/default/files/document/form-09-gender-assessment-and-action-plan-template_0.pdf)
- Examples of Gender Action Plans - <https://www.greenclimate.fund/projects/gender>
- UNFCCC Gender and Climate Change Guidelines & Tools (annotated bibliography) available at <https://unfccc.int/topics/gender/resources/guidelines-or-other-tools-for-integrating-gender-considerations-into-climate-change-related-activities-under-the-convention>
- UNIDO GENDER MAINSTREAMING CHECKLIST FOR PROJECTS available at [https://www.unido.org/sites/default/files/2015-09/GM\\_Checklist\\_for\\_Projectspdf\\_0.pdf](https://www.unido.org/sites/default/files/2015-09/GM_Checklist_for_Projectspdf_0.pdf)

## Gender Policy

### 1. Introduction

Gender equality and female empowerment are now universally recognized as core development objectives, fundamental for the realization of human rights, and key to effective and sustainable development outcomes. No sector including the micro, small and medium enterprises (MSMEs) can develop sustainably without transforming and increasing the distribution of opportunities, resources, and choices for females and males so that they have equal responsibilities and ability to shape their own lives, make choices and contribute to. Aligning with Government of India's efforts for women empowerment and gender equality, India's apex financial institution, SIDBI has been taking pioneering initiatives for women's financial inclusion and economic empowerment through some of its initiatives.

### 2. Need for Gender Policy

- As we know, MSME sector is the backbone of the Indian economy with a strong impact on creating jobs, stable income and reducing poverty. Further, it has largely been observed that one of the underlying causes of poverty, *inter alia*, can be attributed to the very unjust distribution of power and resources. Gender inequity is a key part of this.
- Gender discrimination cannot be simply eliminated by recruiting more women; it requires targeted efforts across all levels of the organization and developing a critical mass of support for furthering inclusion through organizational culture and its work.
- SIDBI's pioneering role in nurturing the microfinance sector in the country has been an important pillar to fight poverty, women empowerment and working towards achieving inclusive growth. SIDBI aligns with the constitutional mandate of Government of India for promoting equity and equality, as testified in SIDBI's Code of Commitment<sup>1</sup> to MSMEs, 2008<sup>2</sup>, which declares that banks "will not discriminate on the basis of age, race, gender, marital status, place, religion or disability"
- Promotion of gender equality and environment protection are the two major pillars of SIDBI's contribution to an equitable and sustainable society. The Article provisions under Constitution of India are provided in Box 1 below.



### Box 1: Constitution of India provisions

- ❖ Article 14 - Men and women to have equal rights and opportunities in the political, economic and social spheres.
- ❖ Article 15(1) - Prohibits discrimination against any citizen on the grounds of religion, race, caste, sex etc.
- ❖ Article 15(3) - Special provision enabling the State to make affirmative discriminations in favour of women.
- ❖ Article 16 - Equality of opportunities in matter of public appointments for all citizens.
- ❖ Article 39(a) - The State shall direct its policy towards securing all citizens men and women, equally, the right to means of livelihood.
- ❖ Article 39(d) – Equal pay for equal work for both men and women.
- ❖ Article 42 - The State to make provision for ensuring just and humane conditions of work and maternity relief.
- ❖ Article 51 (A)(e) –To renounce the practices derogatory to the dignity of women.

- A policy is needed to (i) reflect the changing environment where it is imperative to take special measures for equitable growth (ii) codify SIDBI's commitment to gender equality and sustainable development goals (iii) increase women entrepreneurship in the MSME sector (iv) introduce the institutional mechanisms to fully operationalize gender mainstreaming in banking culture and operations with the objective of improving the status of the society.
- Currently, SIDBI is accredited as National Implementing Entity (NIE) of the Green Climate Fund (GCF) of the UNFCCC. Development of a Gender Policy was one of the conditions for accreditation of SIDBI as NIE to GCF.

### 3. Objective of the Policy

The objective of the policy is to promote gender equality, women empowerment, capacity building of men & women on gender equity, gender responsive financial services and to be equally sensitive to the challenges and concerns of women entrepreneurs. The policy will also help SIDBI to implement strategies to mainstream gender equity in SIDBI's policies, products, processes and services.

### 4. Broad Framework of the Policy

Gender Policy of SIDBI is based on five principles:

#### 4.1 Equity and Equality:

- a. Promote equal rights, opportunities, power and responsibilities and opportunities within the operations and functions of SIDBI (gender equality);
- b. Promote equal and fair access to credit (gender equity) for income generating activities;



**4.2 Inclusiveness:** Work towards reducing social and economic barriers, addressing issue of wage discrimination against women, productive use of time, equal access to and control over productive assets and opportunities including institutional credit, right of decision making in economic, commercial etc. and promoting gender-responsive budgeting.

**4.3 Promoting women's economic empowerment through financial inclusion and entrepreneurship**

Actively contribute to furthering women's financial inclusion and entrepreneurship through the key pillars of its work viz, (a) Attending to the base of the pyramid with a focus on the most marginalized women (b) Funding innovative start up's by women entrepreneurs and fostering innovation directly or through its subsidiaries, (c) Using technology and innovation with credit plus approach to reduce gender gaps and empower next generation women entrepreneurs (d) Promoting sustainable finance which lays thrust on gender responsive, green, clean and energy efficient enterprises (d) Mentoring and networking support to women entrepreneurs to start and scale up their enterprises.

**4.4 Accountability and Impact**

Making institutional more sensitive for delivery of non-discriminative projects and programs to achieve progress without any gender discrimination.

**4.5 Alignment with national policies and priorities:**

SIDBI's policy is in consonance with the national policies and priorities of the Government of India for all the activities, projects / programmes related to gender equity and equality.

**5. Policy Governance & Organisation role**

**5.1 SIDBI's mandate**

In terms of Government of India's policy as applicable to public financial institutions, SIDBI provides equal opportunities for all women staff members, good working conditions for personal growth, equal opportunity for promotion, capacity building and trainings to enhance equitable representation of women in decision making at various levels, and affirmative decision in favour of women officers in transfers. Accordingly, SIDBI's human resource systems and policies are made gender-sensitive and responsive.

**5.2 Financial Intermediaries / Executing Entities**

- (i) SIDBI enthuses its financial intermediaries / executing entities to promote gender equality, women empowerment, capacity building of men & women on gender equity, gender responsive financial services and to create a sustainable society. Suitable conditions would be stipulated in the Loan Agreement / MoU of the financial intermediaries / executing entities for smooth implementation of the policy.
- (ii) Financial intermediaries / implementing entities will be responsible for overall implementation of various covenants of Gender Policy under the projects assisted with the funds from GCF.



- (iii) SIDBI as a national implementing entity of the GCF, will oversee the implementation of the gender specific action plan, through periodic monitoring & review.

### 5.3 Resource allocation

SIDBI through its various promotion & development activities allocates financial assistance to various implementing entities or partners to promote gender sensitive projects / programmes. SIDBI may also handhold the financial intermediaries / implementing entities for implementing gender sensitive projects either through its own staff and/or through consultants and the cost for the same will be a part of the project / P & D activities -

### 6. Operational Guidelines

SIDBI may modify/amend its operational guidelines under various developmental and business activities, wherever required, to include:

- a) The projects and activities that shall endeavour to have specific gender components, both in physical and financial terms; provide for project templates and review criteria, wherever required, to include specific section to capture information on equitable benefits and approaches for men and women.
- b) Gender analysis, wherever, may be incorporated into SIDBI's outreach activities and a suitable communication strategy may be developed to spread knowledge on gender equality and sensitization of SIDBI employees, the partners and public.
- c) Gender assessment & gender action plan, wherever required (at entity level, fund level and programme/ project level, in respect of climate change adaptation/mitigation projects for creating base-line data and promoting adequate participation of women in such projects.
- d) Gender-responsive procedures and requirements, gender assessment and action plan, gender responsive stakeholders consultation, gender budget allocation, gender-sensitive-and-gender-responsive indicators etc., at the entity level, fund level and at the project/programme levels shall be ensured appropriately for GCF supported projects/programmes referring to GCF updated gender assessment and action plan and gender policy of GCF.

### 7. Review Mechanism

Based on gaining experience in the implementation of the gender policy activities and operational modalities, SIDBI will be able to align its policies, processes, procedures, and project and programme design. In the light of this, the SIDBI shall review its gender policy once in a two (2) years.



# STRENGTHENING GREEN FINANCE AND MAINSTREAMING GENDER GIFS-GroW Annual Event



Green Indian  
Financial System



## CONFERENCE PROCEEDINGS

23 January 2024 | Mumbai

An Initiative by





## GIFS-GROW ANNUAL EVENT

The GIFS Initiative is the first Indo-European platform which supports the greening of the Indian financial system, through knowledge exchanges, trainings, and networking, targeting like-minded finance practitioners from banks, investment funds and financial institutions. The GIFS Initiative expects to widen and shape the discourse around greening the financial ecosystem and support its operationalization through the Indian institutions. It has been founded by Shakti Sustainable Energy Foundation (SSEF), Agence Française de Développement (AFD), and Small Industries Development Bank of India (SIDBI) in December 2021. In 2022 and 2023, more than 500 participants participated in the different events organized by the GIFS Initiative. GIFS gathered again the financial community to reflect on the recent developments on green finance from the financial market side and invited more stakeholders to discuss how the financial system and the Medium, Small and Micro Enterprises (MSMEs) can find a common ground to ensure a resilient and low-carbon transition in India. The conference also included the GroW summit. GroW is the professional network for women in green and climate finance, which was created as part of the GIFS Initiative.

## OBJECTIVES OF THE ANNUAL EVENT

- Platform to share updates on the 2022 and 2023 activities (newly created regulations, schemes, tools, and working groups) which have contributed to the greening of the Indian financial system
- GIFS visibility, capitalization and presentation of its next developments, which include (i) setting up the “Greening the Finance-to-Finance Green” (G2FG) Technical Assistance platform for MSMEs and financial institutions, and (ii) network for Greening of Finance by Women (GroW)
- Networking of the financial and MSME communities
- Present key updates, enable collaboration and advancing engagement of women in green and climate finance through GroW

## SESSION 1: VOICES ON GROW BY WOMEN ENTREPRENEURS

## OPENING REMARKS

- **Ms. Kruthika Jerome**

Programme Manager, Climate Finance, Shakti Sustainable Energy Foundation

## PANELISTS

- **Ms. Chaitsi Ahuja**

Founder, Brown Living

- **Ms. Kalyani Shinde**

Founder, Godaam Innovations

- **Ms. Sahar Mansoor**

Co-Founder, Bare Necessities

- **Dr. Vanita Prasad**

Co-Founder and CTO, REVY Environmental Solutions

## INTRODUCTION

The GIFS-GroW Annual Summit's opening session, themed "Voices on GroW by Women Entrepreneurs" featured visionary female climate entrepreneurs discussing sustainability. The session had Ms. Chaitsi Ahuja of Brown Living, who is operating an innovative plastic- and chemical-free online marketplace; Dr. Vanita Prasad of REVY Environmental Solutions that focuses on waste reduction and resource reuse; Ms. Kalyani Shinde, from Godaam Innovations, aiming to transform the sector of agricultural technologies (agri-tech) and add value to each step of the crop cycle; and Ms. Sahar Mansoor, of Bare Necessities, who is championing a people- and earth-centered approach to create zero-packaging footprint products. In the session, the attendees shared their respective journeys and obstacles faced while running environmentally friendly businesses. They discussed the achievements and challenges that

define the landscape of eco-conscious companies in the present world. The session highlighted the unique difficulties faced by these pioneers in obtaining funding, creating brand recognition, and balancing ecological impact with financial sustainability. All of this occurred in the context of a worldwide trend towards sustainable practices.

As the conversation unfolded, a common thread emerged on the pervasive funding gap for women-led green and climate enterprises. This disparity underscores the urgency of reshaping investment paradigms to nurture and propel women entrepreneurs striving to make meaningful contributions to environmental solutions. The session prompted a reassessment of investment strategies and a shared commitment to backing enterprises genuinely devoted to advancing global sustainability.

## Key Insights

### Funding Disparity in Sustainability

Studies show that female founders receive a disproportionately low amount of total start-up funding, with only a slight increase observed over the years. However, if start-ups with both male and female co-founders are considered, there is a small uptick in the share of funding received, although it remains below 20%. These findings indicate the need to address the gender gap in funding and provide more support and resources to female entrepreneurs. Overcoming this challenge requires a fundamental shift in investment practices, recognizing the indispensable role women entrepreneurs play in driving innovative and sustainable solutions.

### Language Barrier between Innovators and Investors

The need for a shared language between innovators and investors in the environmental sector was emphasized. Bridging this communication gap is vital for translating groundbreaking innovations into viable business models. Enhancing mutual understanding fosters an environment where environmentally impactful solutions align seamlessly with financial sustainability.

### Technological Solutions for Agri-Tech

Potential of the agri-tech was showcased with innovative IoT-based solutions addressing significant post-harvest wastage. Traditional storage practices often lead to significant losses

ranging from 40-50% in the agricultural sector. One transformative example is gas monitoring technology. Unlike the conventional method, where wastage is identified by smell, this IoT-based electronic device detects it at a much earlier stage, improving early identification from 25% to an impressive 1-2%. This demonstrates the revolutionary power of technology in reshaping the agri-tech storage ecosystem.

### Educating Investors on Sustainability

Women entrepreneurs face challenges in educating investors on the unique aspects of sustainability. Overcoming this hurdle is crucial for dispelling misconceptions and garnering support. To enhance this educational process, there's a growing need for targeted initiatives, workshops, and platforms that facilitate direct interactions between women entrepreneurs and investors. Establishing mentorship

programs and industry-specific training sessions can bridge the knowledge gap and foster a better understanding of the nuances of sustainable ventures. By actively promoting such educational avenues, the ecosystem can create a more informed and supportive environment for women entrepreneurs pursuing sustainability-driven initiatives.

### Role of Platforms in Increasing Visibility

Entrepreneurs expressed gratitude for platforms like GroW, recognizing them as essential for visibility, networking, and increased awareness. Such platforms play a pivotal role in advocating for increased funding for women-led initiatives, fostering an enabling environment, and collectively raising awareness about the impactful work of women entrepreneurs in a predominantly male-dominated sector.

## SESSION 2: POLICY PERSPECTIVE ON CLIMATE FINANCE LANDSCAPE INDIA: DEVELOPMENTS IN THE BANKING SECTOR, AND REGULATION FOR THE SAME

### PANELISTS

- **Mr. Sivasubramanian Ramann**

Chief Managing Director (CMD), SIDBI

- **Mr. Sunil Nair**

Chief General Manager, Reserve Bank of India (RBI)

- **Mr. V. Chandrasekar**

Senior Advisor, Indian Banks' Association (IBA)

## INTRODUCTION

The second session at the GIFS-GroW Annual Summit brought together esteemed speakers to address the pivotal role of policy perspectives in navigating India's evolving climate financing landscape. The overarching theme centered on the banking sector's initiatives on integrating climate and channeling funds for climate investments effectively. The session witnessed stellar speakers in Mr. Sivasubramanian Ramann, the Chief Managing Director (CMD) of SIDBI; Mr. Sunil Nair, the Chief General Manager of the Reserve Bank of India (RBI); and Mr. V.

Chandrasekar, the Senior Advisor to the Indian Banks' Association (IBA). They set the tone by acknowledging the commendable initiatives undertaken by women entrepreneurs in the sustainability space. They underscored the prevailing challenge that it is not the lack of funds but the misdirection of available finances. Despite India's substantial financial resources, the funds often fail to reach the areas demanding immediate attention for sustainable development.

## Key Insights

### Financial Challenges in Electric Vehicle (EV) Adoption

The challenges hindering the economic viability of EVs in India was stressed on. Despite the monumental aspiration of achieving 30% of new sales of electric vehicles by 2030, only 5 million out of the 90 million targeted electric vehicles have been realized. The financial obstacle lies in the fact that EVs cost 1.5 times that of Internal Combustion Engine (ICE) vehicles, coupled with high-interest rates reaching 18-20%. This financial landscape poses hindrances to consumers and necessitates a reevaluation of financing models to make the adoption of EVs more economically viable.

emphasizing the need for a financing framework and third-party verification. It showcased how a clear regulatory framework encourages responsible lending practices, mitigating the risk of greenwashing.

### Incentivizing Green Deposits and Capital Incentives

The debate on incentivizing green deposits and exploring differential interest rates for such deposits was a focal point. Complexities emerged concerning capital maintenance requirements for banks and the potential penalization or incentivization based on lending to green or non-green sectors. The ongoing discourse revolves around determining the ideal capital maintenance for climate-related activities. The Basel Committee is actively exploring possibilities to strike a balance between risk and incentivization in the realm of green financing.

The central bank has not mandated or endorsed any differential interest rates to be implemented at this time. The Basel Committee is reportedly in the process of discussing the

### RBI's Green Deposits Framework

The Reserve Bank of India's (RBI) response to the evolving landscape with the introduction of a green deposits framework in April 2023 was highlighted. The absence of a taxonomy in India prompted the need for a framework to ensure funds raised under green deposits are genuinely directed to green initiatives. The framework defined criteria for raising and deploying funds,



integration of climate factors into Pillar 1, 2 and 3, but it may be too early to make any changes to the capital requirements or risk weights for green and climate projects. As more knowledge is gained about the risk levels associated with different green and climate projects, supervisory guidelines will continue to evolve. However, it was emphasized that banks should not avoid financing transition and climate projects, and they must build significant capacity to support these activities.

### IBA's Advocacy Role in Green Finance

The Indian Banks' Association (IBA) was recognized for its pivotal role in advocating green finance. As an unregistered association, IBA champions issues on behalf of banks, collaborating with international agencies and conducting seminars to enhance awareness. The focus is on integrating climate considerations into the banking framework, emphasizing the importance of taxonomy, disclosure, and scenario analysis for attracting green investments. IBA's involvement in task forces related to the greening of steel and the National Green Hydrogen Mission showcased its commitment to steering the banking sector towards sustainable practices.

### Challenges in Greening MSMEs

The challenges in introducing green technologies to the Micro, Small, and Medium Enterprises (MSMEs) sector were underscored. The informal and production-oriented nature of MSMEs requires continuous handholding, awareness-building, and simplifying government benefit processes. Efforts to streamline processes under the Ease of Doing Business initiative and introducing clusters to comprehensive schemes from various ministries were discussed. The session emphasized the importance of providing MSMEs with a holistic package of schemes to effectively transition towards sustainable practices.

### Facilitating Energy Efficiency in Clusters

Discussions focused on facilitating a comprehensive understanding of the support available for MSMEs to adopt energy-efficient measures seamlessly. This aligns with the broader agenda of improving the ease of doing business and promoting sustainable practices in the MSME sector. An example of the establishment of the Facilitation Centre for Energy Efficiency was highlighted as a crucial initiative. This center aims to provide a consolidated platform for MSMEs, offering information on technologies, financing options, and government schemes.

## SESSION 3: SHARING OF GIFS ACHIEVEMENTS

## PANELISTS

- **Mr. Nitin Desai**

Board Member, Shakti Sustainable Energy Foundation

- **Mr. Sivasubramanian Ramann**

Chief Managing Director, SIDBI

- **Ms. Audrey Brulé**

Deputy Head, Financial Systems Division, AFD

## INTRODUCTION

The third session of the GIFS-GroW Annual Summit provided a nuanced exploration of the evolving landscape where climate action, Micro, Small, and Medium Enterprises (MSMEs), and gender inclusivity converge. The distinguished speakers, including Mr. Nitin Desai the Board Member of the Shakti Sustainable Energy Foundation; Mr. Sivasubramanian Ramann, the CMD of SIDBI; and Ms. Audrey Brulé, the Deputy Head of the Financial Systems Division from the AFD, collectively highlighted the pathways for sustainable finance, the pivotal role of MSMEs in climate resilience, and strategies to bridge gender gaps in the climate and financial sectors. This session encapsulated a robust dialogue on aligning financial mechanisms with environmental goals, propelling MSMEs beyond networking into tangible action, and the imperative of integrating gender-focused initiatives into the broader climate finance narrative.

The session, over and above all tried to underscore the initiatives undertaken by the Green Indian Financial System. The proposition of the G2FG (Greening the Finance-to-Finance Green Technical Assistance) platform surfaced as a catalyzing force to foster collaboration

between MSMEs and financial institutions, aiming to provide technical assistance and support. This initiative seeks to leverage collective efforts in greening financial practices and expanding avenues for sustainable finance, thereby creating an ecosystem conducive to innovation and resilience. To further its goal, GIFS is currently undertaking a feasibility study for setting up a technical assistance center that would help with carrying out systematic interventions to bridge green financing needs of MSMEs.

Secondly, the Greening of Finance by Women (GroW) Network, a part of the GIFS initiative, aims to promote the participation of women in the green and climate finance. The network plays an important role in recognizing the expertise of women in the field and helps strengthen and take advantage of the gender aspect of the sector. The main objectives of the network include increasing the visibility of women professionals in green and climate finance, fostering collaboration among different stakeholders in the field, and providing targeted knowledge enhancement opportunities for women professionals in green and climate finance.

## Key Insights

### MSMEs: Beyond Networking to Action

There was a demand for MSMEs to move away from networking and instead focus on taking action. This involved combining technology, entrepreneurship, and finance. Drawing parallels with the success of the Silicon Valley, this insight emphasized the need for a holistic ecosystem supporting MSMEs in climate change mitigation, innovation, and sustainability. The

narrative brought forth the urgency for actionable steps and collaboration to empower MSMEs to play a central role in sustainable finance.

### Role of Finance in Climate Resilience

A compelling case study, spotlighting the impact of finance in transforming traditional coal-based furnaces to gas-based ones in Moradabad, a city in the state of Uttar Pradesh, was underscored.

This not only underlined the transformative power of financial intervention in MSMEs but also shed light on the intricate challenges, including regulatory hurdles and the need for tailored solutions such as energy audits. These audits, happening in various clusters, emerged as strategic tools guiding MSMEs towards environmentally acceptable practices and aligning them with global environmental standards.

### Gender-Centric Climate Finance Strategies

A comprehensive approach to integrate gender-centric strategies into climate finance was articulated. This involved not only internal practices but also mainstreaming gender considerations across operations and

developing dedicated financial tools. The emphasis on partnerships, peer-to-peer exchanges, and global networks underscored the interconnectedness of gender diversity and financial success in green landscapes.

### Challenges and Opportunities in Climate Financing

Speakers acknowledged the intricate challenges faced by MSMEs, especially at the micro-level, and navigated through the ongoing debate on incentivizing green investments. The emphasis on addressing legal issues, building resilience, and adapting to environmental standards highlighted the multi-faceted nature of challenges and opportunities in the climate financing landscape.

## SESSION 4: INDUSTRY ASSOCIATION PERSPECTIVES: WHAT ARE THEIR GREEN ASPIRATIONS?

## PANELISTS

- **Mr. Rajiv Chawla**

Founder and Chairman, Integrated Association of SMEs

- **Anil Bhardwaj**

Secretary General, Federation of Indian Micro, Small, and Medium Enterprises (FISME)

- **Ms. Madhu Tyagi**

Director and National President, Confederation of Women Entrepreneurs (COWE)

## MODERATOR

- **Mr. R. K. Singh**

Chief General Manager (CGM), Green Climate Finance Vertical, SIDBI

## INTRODUCTION

The Industry Association Perspective session at the GIFS-GroW Annual Summit convened a distinguished panel to delve into the green aspirations shaping the landscape of MSMEs. The panel featured Rajiv Chawla, Founder and Chairman of the Integrated Association of SMEs; Anil Bhardwaj, Secretary General of the Federation of Indian Micro, Small, and Medium Enterprises (FISME); and Ms. Madhu Tyagi,

Director and National President of the Confederation of Women Entrepreneurs (COWE). It was moderated by Mr. R. K. Singh, the Chief General Manager (CGM) of the Green Climate Finance Vertical of SIDBI. The intrinsic role played by industrial associations in steering the MSME ecosystem towards sustainable practices was highlighted with key perspectives shared on fostering green initiatives.

## Key Insights

### On Collaborative Green Revolution

The session emphasized the need for collaborative efforts and cited example of SIDBI, World Bank, Bureau of Energy Efficiency, and GIZ in initiating a green revolution in India. The concerted endeavors sought to incentivize financial institutions and NBFCs, integrating them into regulatory frameworks to ensure the widespread embrace of environmentally conscious practices among entrepreneurs.

### On Navigating Regulatory Measures

The discussion shed light on the array of regulatory measures imposed by the Government of India, including bans on single-use plastic, firecrackers, and specific fuel sources. Acknowledging the compliance burden on MSMEs, the need for supportive measures during this transition was underscored. These measures include institutional backing,

specialized credit lines, and concessional policies.

### On Appeal for Transition Support

With India's commitment to a net-zero future and an anticipated surge in compliance and regulatory measures, the session advocated for institutional backing. The necessity of a framework that supports entrepreneurs during the intricate phase of transitioning towards greener business practices was stressed.

### Symbolic Tree Plantation

The session's unique approach of presenting certificates instead of physical plants held a symbolic gesture. Participants learned that SIDBI committed to planting 10 trees on behalf of each speaker and attendee. This initiative underscored the summit's overarching theme of environmental consciousness and sustainability.



## SESSION 5: LESSONS LEARNED FROM G20 PRESIDENCY FOR THE INDIAN FINANCIAL SECTOR & COP28 MESSAGES & TAKEAWAYS IN INDIA CONTEXT

### PANELISTS

- **Mr. Dhruba Purkayastha**

India Director, Climate Policy Initiative (CPI)

- **Mr. Vivek Adhia**

Head of Climate & Sustainability, Boston Consulting Group (BCG)

- **Ms. Neha Kumar**

Head, South Asia Programme, Climate Bonds Initiative

- **Mr. Shubhashis Dey**

Director, Climate Policy Program, Shakti Sustainable Energy Foundation

- **Mr. Mukund Govindarajan**

ECube Investment

- **Anil Bhardwaj**

Secretary General, Federation of Indian Micro, Small, and Medium Enterprises (FISME)

- **Ms. Madhu Tyagi**

Director and National President, Confederation of Women Entrepreneurs (COWE)

### MODERATOR

- **Mr. R. K. Singh**

Chief General Manager (CGM), Green Climate Finance Vertical, SIDBI

## INTRODUCTION

The session on Lessons learned from G20 Presidency for the Indian Financial sector & COP28 messages at the GIFS-GroW Annual Summit brought together seasoned experts to dissect global trends in green finance and chart out strategies for an effective transition. Moderated by Mr. R. K. Singh, the CGM of the Green Climate Finance Vertical at SIDBI, the panel included Mr. Dhruva Purkayastha, the India Director of the Climate Policy Initiative (CPI); Mr. Vivek Adhia, the Head of Climate & Sustainability at the Boston Consulting Group (BCG); Ms. Neha Kumar from the Head of South

Asia Programme at the Climate Bonds Initiative; Mr. Shubhashis Dey, the Director of Climate Policy Program at the Shakti Sustainable Energy Foundation; and Mr. Mukund Govindarajan from ECube Investment. This dynamic discussion aimed at evaluating COP28 outcomes, examined the complexities of global green finance and additionally, outlined critical challenges and opportunities, emphasizing the need for precision, collaboration, and innovative financial mechanisms in the pursuit of sustainable transitions.

## Key Insights

### Global Climate Finance Dynamics

The global climate finance landscape, particularly focusing on COP28 outcomes was discussed. Despite effective communication strategies, the lack of quantitative targets in phasing out fossil fuels was criticized. The commitment to triple renewable energy was deemed insufficient, especially considering the negligible pipelines in several countries, emphasizing the need for strategic allocation of international resources for a global public good.

### Mathematics of Transition Finance

The session delved into the fundamental mathematics of transition finance, where light was shed on the cost disparity of mitigating one unit of CO<sub>2</sub> in OECD countries versus developing nations. The failure of COP28 to recognize this distinction raises questions about the equitable utilization of international resources. The discussion emphasized the significance of accurate mathematical models and international collaboration in achieving sustainable transitions.

### Challenges in Climate Communication

Drawing parallels between COP28 and educational testing processes, the communication challenges in conveying transitional fuel nuances were discussed. The panel recognized the need for precise communication, avoiding the oversimplification of complex concepts. This insight resonates beyond climate conferences, accentuating the importance of effective communication in climate action initiatives and garnering global support.

### ESG Governance in MSMEs

The importance of incorporating Environmental, Social, and Governance (ESG) principles in MSMEs was also addressed. Best practices observed in larger companies, such as governance-focused committees, circularity, internal carbon pricing, and research & development (R&D), were highlighted. The challenge of MSMEs lies in accessing technology and financing, where institutions such as SIDBI could play a pivotal role by showcasing

exemplary MSMEs and providing additional support.

### Facilitating a Just Transition

The necessity of a rapid, orderly, and just transition, acknowledging that decarbonization is critical was underlined. While large corporations have taken the lead, it was noted that the financial sector's delayed entry needs acceleration. The session underscored the importance of a synergistic approach at global summits, advocating for increased accessibility and affordability of capital between developing countries.

### Role of Private Capital and Blended Finance

The evolving role of private capital in addressing climate aspects was acknowledged, indicating the need for innovative instruments. Blended finance emerged as a potential solution, but the emphasis shifted to converting projects into bankable propositions. Technical assistance was identified as crucial to enhancing bankability, ensuring the successful conversion of projects and attracting philanthropic funds for impactful market-making initiatives.

## SESSION 6: THE NEXUS OF WOMEN AND GREEN FINANCE ACROSS STAKEHOLDERS

## PANELISTS

- **Ms. Ekta Mehra**

Senior Sector Specialist, KfW

- **Ms. Smitha Hari**

President – India, auctusESG

- **Ms. Shruti Srivastava**

Investment Director at Avaana Capital

## MODERATOR

- **Ms. Neetu Chitkara**

Managing Director and Partner, BCG

## INTRODUCTION

The sixth session of the GIFS-GroW Annual Summit brought together a distinguished all-women panel to explore the interconnection of women and green finance across diverse stakeholders. The session heard insights from Ms. Ekta Mehra, Senior Sector Specialist at KfW; Ms. Smitha Hari, President at auctusESG; and Ms. Shruti Srivastava, Investment Director at Avaana Capital. It was moderated by Ms. Neetu Chitkara, Managing Director and Partner at

BCG. The session engaged in a comprehensive discussion, aiming to uncover challenges and opportunities in empowering women entrepreneurs within the climate and sustainability sphere. There was a particular focus on the role of women in green finance, examining how financial support could be channeled towards initiatives contributing to climate goals.

### Key Insights

#### Blending Climate and Gender in Financing

The effective integration of climate and gender perspectives in financing initiatives can yield positive outcomes, especially in sectors like urban mobility and energy. Urban mobility projects, for instance, can be designed with a gendered lens, enhancing workforce participation rates for women. However, challenges persist in sectors like energy, necessitating tailored approaches for blending climate and gender considerations.

#### Data Challenges in Gender-Related Metrics

Obtaining gender-related data for reporting and analysis poses a significant challenge despite reporting standards emphasizing gender metrics. The need for sensitizing organizations on reporting gender-related information was highlighted. Building internal capacities and creating structures supporting gender-friendly policies are essential to navigate the complexities of data collection.

#### Investment Philosophy and Gender Mainstreaming

Mainstreaming gender in investment processes involves considerations like sensitization, flexible post-maternity policies, and intentional efforts in organization design. Gender considerations can eliminate biases in investment processes, it has been increasingly observed. This includes creating roles that are women-friendly and supporting women founders through mentorship and networking.

During a panel discussion, a proposal was made for SIDBI to build a gender and climate-focused portfolio at the grassroots level. The panel discussed the financial parameters necessary for such initiatives with a specific focus on gender and climate objectives. Perspectives were shared on regulatory costs associated with supporting women in the workforce could be offset through philanthropic blended capital. They emphasized the potential benefits of initiatives like setting up daycare facilities and proposed framing these costs as R&D investments to showcase long-term advantages. While on the other hand, concerns were raised about global competition for funds and questions why companies would invest in such

initiatives abroad when they could use the funds domestically. The debate reflects the intricacies of balancing financial considerations, regulatory support, and global competition in advancing gender and climate goals.

### Role of Blended Finance in Gender and Climate Initiatives

Blended finance, involving concessional finance, plays a crucial role in supporting gender and climate initiatives, but aligning financial parameters poses challenges. The panel highlighted that while blended finance offers support, practical challenges need addressing to align expectations of financial and gender experts. Sustainability-linked products and strong dialogues between borrowers and lenders were deemed effective.

### Performance Impact of Women Borrowers

Studies suggest that financial products perform better when women are borrowers, emphasizing the importance of reporting mandates and investor focus on gender metrics. The potential advantages of reporting mandates, where investors prioritize gender metrics, acting as a driving force for institutions to utilize gender-related information effectively are significant.

### Future Expectations and Aspirations

The session concluded with aspirations for a future where women and men share equal vulnerability to climate change, leading to an overall reduction in vulnerability, particularly for economically vulnerable sections. The panel envisioned a future with increased integration

of women into diverse fields, particularly in Science, Technology, Engineering, and Mathematics (STEM), and improved working conditions to sustain their participation. Building capacities across both rural and urban women and fostering platforms like GroW for collaborative efforts were highlighted as essential steps in this vision for the future.



## ANNEXURE – AGENDA

| Timing            | Topic                                                                                                                                                                                                                    | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10am - 10:30am    | Opening                                                                                                                                                                                                                  | Registration, coffee, networking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 10:30am - 11:00am | Session I: Voices from GRoW Entrepreneurs                                                                                                                                                                                | <p>Introduction by Kruthika Jerome, Shakti Sustainable Energy Foundation (5 mins)</p> <p><i>What access for women entrepreneurs in financial institutions? (5 mins each)</i></p> <ul style="list-style-type: none"> <li>Ms. Kalyani Shinde, Godaam Innovations (climate smart post-harvest solutions)</li> <li>Ms. Chaitsi Ahuja, Founder of Brown Living</li> <li>Ms. Sahar Mansoor, Bare Necessities (sustainable home and personal care products)</li> <li>Ms. Vanita Prasad, REVY Environmental Solutions (waste management services)</li> </ul> |
| 11:00am - 11:45am | Session II: Policy Perspective on Climate Financing Landscape India: developments in the banking sector and regulation                                                                                                   | <p>Proposed Panelists:</p> <ul style="list-style-type: none"> <li>Mr. V Chandrasekar, Senior Advisor, IBA</li> <li>Mr. Sunil Nair, CGM, RBI</li> </ul> <p><i>Moderated by - Mr. Sivasubramanian Ramann, CMD, SIDBI</i></p>                                                                                                                                                                                                                                                                                                                           |
| 12:00pm - 12:45pm | <p>Session IV: Sharing of GIFS achievements by Founding Partners?</p> <ul style="list-style-type: none"> <li>GIFS Report Card</li> <li>Introduction to the G2FG Platform (Shakti)</li> <li>GRoW Network (AFD)</li> </ul> | <p>Speakers: (15 mins each)</p> <ul style="list-style-type: none"> <li>Mr. Nitin Desai, Member of Board, SSEF</li> <li>Mr. Sivasubramanian Ramann, IA&amp;AS, Chairman &amp; Managing Director, SIDBI</li> <li>Ms. Audrey Brulé - Françoise, Deputy Head Financial Systems Division, AFD</li> </ul>                                                                                                                                                                                                                                                  |
| 12:45pm - 2.00 pm | Networking Lunch                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Theme Sessions    |                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2:00pm - 2:45pm   | Session V: Industrial association session: what is their green aspiration?                                                                                                                                               | <p>Feedback from Companies' representatives regarding access to climate finance: aspirations, successes and difficulties</p> <p><i>Potential questions to be answered:</i></p> <p>(i) <i>What are they looking for (in terms of products, scale, level of involvement from the banker)?</i></p> <p>(ii) <i>What are the difficulties they encounter? can they share a success story?</i></p> <p>Speakers: (15 Min each)</p> <ol style="list-style-type: none"> <li>Mr. Rajiv Chawla, Chairman, IamSMEofIndia</li> </ol>                              |

|                    |                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |                                                                                                                                                       | <p>2. Mr. Anil Bhardwaj, Secretary General, FISME</p> <p>3. Ms. Madhu Tyagi, President, COWE India</p> <p><i>Moderated by - Mr. R. K. Singh, CGM, SIDBI</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2:45pm - 3:30pm    | <p>Session VI - lessons learnt of G20 Presidency for the Indian financial sector and perspective + COP28 messages and takeaways in Indian Context</p> | <p>Proposed Panelists:</p> <ul style="list-style-type: none"> <li>- Mr. Dhruva Purkayastha, Director, CPI</li> <li>- Mr. Vivek Adhia, Associate Director, BCG</li> <li>- Ms. Neha Kumar, Climate Bond Initiative</li> <li>- Mr. Shubhashis Dey, Director of Climate Policy Program at the Shakti Sustainable Energy Foundation</li> <li>- Mr. Mukund Govindarajan, ECube Investment</li> </ul> <p><i>Moderated by - Mr. R. K. Singh, CGM, SIDBI</i></p>                                                                                                                                                               |
| Break - 15 Minutes |                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3:45pm - 4:30pm    | <p>Session VII: GROW Session – Panel Discussion: The Nexus of Women and Green Finance across Stakeholders</p>                                         | <p><i>Discuss how climate outcomes can be improved from two lenses: (i) increased access to finance for women climate entrepreneurs; (ii) increased participation from women in green finance. Focus on key challenges to be addressed and what GROW is doing to bridge the gaps.</i></p> <p>Proposed Panelists:</p> <ul style="list-style-type: none"> <li>- Ms. Ekta Mehra (Senior Sector Specialist), KFW</li> <li>- Ms. Smitha Hari, President, AuctusESG</li> <li>- Ms. Shruti Srivastava, Investment Director, Avaana Capital</li> </ul> <p><i>Moderated by – Ms. Neetu Chitkara, MD &amp; Partner, BCG</i></p> |
| 4:30pm - 4.45pm    | Summing Up                                                                                                                                            | By GIFS Partner(s)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 4.45pm - 6.00pm    | Networking Tea                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |